PSU SportsTrack: A Web-Based Sports Tournament Management System for Palawan State University

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CHAPTER 1

INTRODUCTION

Sports play an important role in student growth and school identity at Palawan State University (PSU). Aside from physical fitness, sports teach discipline, teamwork, and school pride. However, even as PSU's sports events had grown bigger over the years, the way they were managed had remained manual, paper-based, and disorganized.

PSU is made up of eighteen campuses across Palawan. The university regularly holds major sports events, including the **Binturungan** (the official PSU Intramurals, where student-athletes from all colleges and campuses compete) and the **Faculty and Staff Friendly Games** (a yearly sports event for PSU employees). Each of these events required a lot of work — registering players, forming teams, setting up brackets, scheduling games, recording scores, and making reports — all of which were previously done by hand.

Without a proper digital system, this process caused many problems. Coaches and coordinators spent too much time collecting and checking player information. Tournament brackets and game results were tracked manually, which led to errors. After each event, reports were put together inconsistently, making it hard to keep accurate records over time.

This study proposes **PSU SportsTrack**, a web-based sports management system made specifically for Palawan State University. The system allows users to register as athletes, form teams, manage tournament brackets, score matches court-side, follow live scores, and generate reports — all in one place. By replacing

the manual process with a digital system, PSU SportsTrack aims to make PSU sports management faster, more accurate, and more organized across all eighteen campuses, while giving the wider PSU community a transparent, real-time view of ongoing competitions.

Project Context

Managing sports events at Palawan State University has long relied on manual processes, physical forms, paper brackets, and informal group chats. While this approach has been used for years, it has become increasingly inadequate as PSU's sports programs continue to grow across multiple campuses, involving students, coaches, faculty, and administrative staff who all require consistent coordination and documentation.

To understand these challenges firsthand, the researchers conducted a formal interview with a representative from the PSU Sports Department, who provided a detailed account of how sports events were organized. According to the representative, the management of events like the Binturungan Intramurals followed a layered chain of coordination from the sports director down to cluster sports coordinators, team coaches, and tournament managers (TMs), each assigned to handle a specific sport. Player eligibility was verified through physical submission of requirements before athletes were allowed to compete. Scores were recorded manually by a designated table management team and submitted to the tournament manager after each game. Post-event, all results were compiled on paper and passed up the chain, with no permanent digital record kept across seasons.

The representative also confirmed that while a paid third-party website was occasionally used for basketball statistics, no centralized digital system existed for

managing PSU sports events as a whole. The interview further revealed that the biggest pain point for the Sports Department was the volume of paperwork involved — collecting, tracking, and filing physical requirements and results for every event. When asked about transitioning to a digital platform, the representative expressed clear support, noting that younger coaches would adapt easily, and that even older staff could manage with minimal assistance.

Based on this consultation, the researchers identified the core needs that PSU SportsTrack had to address: digital player registration and eligibility verification, coach-led team formation, tournament bracket generation and management, live score tracking, and structured post-event reporting. The system was designed directly around these validated needs, with the Sports Department serving as its primary client, covering the Binturungan Intramurals and the Faculty and Staff Friendly Games across all eighteen PSU campuses.

Purpose and Description

PSU SportsTrack is a web-based sports management system made specifically for Palawan State University. It serves as the main centralized platform for organizing, managing, and recording PSU's official sports events — particularly the Binturungan (PSU Intramurals) and the Faculty and Staff Friendly Games — and for allowing the entire PSU community to follow these events as they happen across all eighteen campuses.

The system is built around four main functions:

First, **Athlete Registration and Verification**. Students and university personnel register their profiles using a secure dual-path email verification (6-digit

OTP or direct magic link). Students upload their Certificate of Registration (COR), while faculty and non-teaching staff upload their respective employee identification. For COR submissions, the system performs an automated text-extraction check on the uploaded document and displays a confidence score, whereas faculty and staff image uploads bypass this automated extraction for direct manual review. To ensure data privacy and comply with institutional security standards, all uploaded requirements are strictly safeguarded using 60-second expiring signed URLs. The administrator utilizes the confidence score and document previews to make the final decision to approve or reject each submission. This digital workflow replaces the traditional paper-based process and ensures that only legitimate, verified PSU members can participate in athletic events.

Second, **Coach-Led Team Formation and Lineup Management**. Coaches create teams under a college department or a Binturungan tribe or cluster, search the pool of verified athletes, and send recruitment invitations. Athletes may accept or decline these invitations. Coaches then manage the roster by assigning jersey numbers, positions, team captains, and starting lineups, alongside tracking athlete availability. This removes the need for informal recruiting and makes the entire roster management process highly organized and transparent.

Third, **Tournament Management and Live Match Operations**. Administrators engineer tournaments by setting the sport, event type, team affiliation, and format. The platform features an automated bracket generator supporting Single Elimination, Double Elimination, Round Robin, and two-stage hybrid (pool stage to playoffs with an automated bronze match) formats. During active games, appointed court-side facilitators operate a dedicated scoring console guarded by an automated

Anti-Conflict of Interest (Anti-COI) mechanism, which strictly prevents them from officiating matches involving their own affiliated academic college. Scores and match events are broadcast in real time to spectators through a live match viewer, a full-screen stadium display for gym projectors, and floating scoreboard widgets, completely eliminating the need for paper score sheets.

Fourth, **Rankings, Awards, and Reporting**. The system recognizes outstanding athletes through the Gold Star (Tournament MVP) and Red Star (Game MVP) awards, and ranks colleges and Binturungan tribes or clusters through leaderboards and Gold, Silver, and Bronze medal standings. Post-event, a dedicated institutional reporting engine provides multi-discipline interactive auditing of squad standings, top individual scoring statistics, and collegiate participation footprints. Administrators can toggle report sections and instantly export publication-ready records as structured CSV datasets or vector-rendered A4 landscape PDF documents for university accreditation, sports council presentations, and historical archiving.

The system enforces strict role-based access control across **seven** distinct user classifications: the Public Spectator, who can view published schedules and scores without an account; the Student Athlete, Faculty Athlete, and Staff Athlete, who verify their eligibility and join teams; the Coach, who forms teams and manages rosters; the Facilitator, who officiates live matches; and the Administrator, who holds executive oversight. Crucially, the platform implements a **dual-role architecture** that allows verified athletes and coaches to simultaneously hold court-side officiating privileges without losing access to their primary player profiles or team rosters. Spectator access is read-only, while active participation is restricted exclusively to verified PSU community members across all eighteen campuses. PSU SportsTrack is

universally accessible through any modern web browser on desktop, laptop, tablet, and smartphone devices.

Objectives

General Objectives

The general objective of this study aims to design, develop, and deploy PSU SportsTrack — a web-based sports management system for Palawan State University — that brings together athlete registration, team formation, tournament operations, live court-side scoring, and institutional sports reporting in one platform.

Specific Objectives

Specifically, this study aimed to:

1. Develop an athlete registration and verification module that allows PSU students, faculty, and non-teaching staff to register using a secure dual-path email verification (6-digit OTP or direct magic link), and submit their Certificate of Registration (COR) or employee identification for administrator verification, supported by an automated document check that produces a confidence score exclusively for student CORs (with faculty and staff image uploads undergoing direct manual review), while utilizing 60-second expiring signed URLs to protect user data privacy, ensuring only eligible PSU members can join sports events;
2. Build a coach-led team formation and lineup module that lets coaches create teams under colleges or Binturungan tribes and clusters, recruit verified athletes through invitations, and manage jersey numbers, positions, team

captains, starters, and athlete availability, replacing the current informal recruitment process;

3. Create a tournament management module that allows administrators to set up tournaments for the Binturungan and the Faculty and Staff Friendly Games, featuring an automated bracket generator that supports Single Elimination, Double Elimination, Round Robin, and two-stage hybrid (pool stage to single-elimination playoff with an automated third-place match) formats, alongside game scheduling, venue assignment, and facilitator assignment;
4. Develop a live match and court-side scoring suite with sport-specific scoring rules, an automated Anti-Conflict of Interest (Anti-COI) match reservation mechanism, score correction with rollback, game clock, foul, timeout, and substitution controls, and offline event buffering, whose results are delivered in real time to spectators through a live match viewer, a stadium display, and floating scoreboard widgets;
5. Design an institutional rankings and reporting module that generates leaderboards, Gold, Silver, and Bronze medal standings per college and per tribe or cluster, MVP award records (Gold Star and Red Star), and features multi-discipline interactive auditing with one-click export to structured CSV datasets and vector-rendered A4 landscape PDF formats for university accreditation and historical archiving;
6. Implement secure role-based access control for **seven** distinct user classifications (including separated Faculty and Staff roles, alongside a dual-role architecture for officiating privileges) using database-level Row Level Security, brute-force login protection, an immutable audit trail, and a recovery mechanism for soft-deleted records;

7. Evaluate the system through User Acceptance Testing (UAT) with PSU sports administrators, coaches, facilitators, and players, targeting a mean satisfaction rating of at least 4 out of 5 based on ISO 25010 quality standards.

Significance of the Study

PSU SportsTrack benefited several groups within the Palawan State University community:

PSU Physical Education Department and Sports Administrators The system directly solved the recurring problems caused by manual sports management. By centralizing player registration, team formation, and tournament operations, it significantly reduced the workload of the Sports Department. Furthermore, the integration of the new institutional reporting engine allows administrators to instantly generate and audit multi-discipline outcomes, exporting publication-ready CSV datasets and vector-rendered A4 landscape PDFs. This ensures sports records remain consistent, accurate, and readily available for university accreditation every year.

Coaches PSU SportsTrack gave coaches a clear and organized way to find and recruit players. Instead of relying on personal connections, coaches could securely browse the pool of verified players, view their profiles, and seamlessly build their rosters and game-day lineups through the system. This was especially helpful for new coaches or those managing sports where they do not yet know many players, making the entire roster management process highly transparent.

Student-Athletes, Faculty, and Non-Teaching Staff Participants get a secure and straightforward way to register for sports events using a dual-path email

verification (6-digit OTP or magic link). They can confidently submit their credentials knowing their data is protected by 60-second expiring signed URLs. Beyond registration, participants can monitor their verification status, respond to team invitations, and view interactive brackets, standings, personal statistics, and unlocked MVP achievements (Gold Star and Red Star) — making their experience as competitors more informed and engaging.

PSU as an Institution Across all eighteen campuses, PSU SportsTrack acts as a digital foundation that supports the long-term growth of university sports. The system's reporting capabilities — detailing winning colleges, top individual scoring statistics, squad win-loss standings, and collegiate participation footprints — give university administrators and the sports council structured, verifiable data for institutional planning, resource allocation, and the recognition of outstanding athletic programs.

Academic Contribution This project added to the growing field of applied technology research in Philippine state universities. It demonstrated that digital systems built specifically for a university's unique administrative and operational needs — and developed with close, iterative input from stakeholders — could greatly improve how non-academic and extracurricular operations are managed.

Court-Side Facilitators Game officials replace paper score sheets with a responsive digital scoring console designed for tablets and smartphones. Their workflow is protected by an automated Anti-Conflict of Interest (Anti-COI) mechanism that prevents them from officiating matches involving their own college. Crucially, the system features offline event buffering, meaning facilitators can

continue recording points, clock controls, fouls, and substitutions seamlessly even during courtside network dropouts, completely eliminating transcription errors.

Spectators and the PSU Community Fans, students, and families can follow live scores, dynamic brackets, schedules, and medal standings as Unauthenticated Public Spectators. Through a real-time live match viewer, full-screen stadium displays for gym projectors, and floating scoreboard widgets, the PSU community can stay updated on any device. This unprecedented transparency strengthens school pride and brings the university closer to its sports programs.

Scope and Limitations

Scope

PSU SportsTrack covers the design, development, and testing of a web-based sports management system that includes the following:

- Athlete registration for students, faculty, and non-teaching staff using a secure dual-path email verification (6-digit OTP or direct magic link), with Certificate of Registration (COR) upload for students and employee identification upload for faculty and staff. This includes an automated advisory confidence score exclusively for student COR submissions (with faculty and staff image uploads undergoing direct manual review), all safeguarded by 60-second expiring signed URLs to protect user privacy, followed by administrator verification (approval or rejection with feedback notes);
- Seven user classifications — Public Spectator, Student Athlete, Faculty Athlete, Staff Athlete, Coach, Facilitator, and Administrator — each with defined route access and database-level permissions enforced through Row Level Security,

including a dual-role architecture that allows verified users to hold court-side officiating privileges without losing primary profile access;

- Coach-led team formation under Collegiate Squads (college departments) and Binturungan Tribes or Clusters, including athlete recruitment through invitations, jersey number and position assignment, captain and starter designation, and athlete availability tracking (Active, Injured, Excused);
- Tournament management for the Binturungan (PSU Intramurals) and the Faculty and Staff Friendly Games across multiple sports, with sport-specific scoring rules for nine ball sports;
- A three-phase tournament lifecycle (Draft, Slotting, and Launched), followed by automated bracket generation supporting Single Elimination, Double Elimination, Round Robin, and two-stage hybrid (pool stage to single-elimination playoff with an automated third-place match) formats, displayed through an interactive bracket;
- Game scheduling with venue, date, and facilitator assignment;
- A court-side scoring suite for facilitators, an automated Anti-Conflict of Interest (Anti-COI) match reservation mechanism, score attribution by jersey number, score correction with rollback, game clock and period control, fouls and timeouts, substitutions, court swap, and a mobile-optimized kiosk mode;
- Court-side offline resilience through local buffering of scoring events and automatic synchronization once the connection is restored;
- Real-time spectator coverage, including a live match viewer with play-by-play commentary, a full-screen stadium display with broadcast

overlays, a Picture-in-Picture scoreboard, a floating score widget, and procedurally generated audio cues;

- A hardened MVP award governance system — featuring the Gold Star (overall tournament MVP) assigned exclusively by the Administrator upon tournament closure, and the Red Star (Match MVP) assigned by the Facilitator upon match conclusion. To mitigate officiating bias, Red Star assignments are secured by an automated Conflict of Interest (COI) alert mechanism, mandatory 10-character written justifications for anomalous nominations (e.g., losing team or zero points), and an unskippable Fair Play & Neutrality Attestation;
- Leaderboards and medal standings (Gold, Silver, and Bronze) per college and per tribe or cluster, with filters by sport, tournament season, academic year, win rate, and MVP star count;
- An institutional reporting engine (FR-41) providing multi-discipline interactive auditing of squad standings, individual scoring statistics, and collegiate participation footprints, with export capabilities to structured CSV datasets and vector-rendered A4 landscape PDF documents for institutional accreditation;
- Administrator console for executive metrics, athlete verification, tournament engineering, facilitator appointment, audit trail review, recovery of soft-deleted records, master override of live matches, and an in-place MVP Administrative Override Console allowing administrators to review, revoke, or reassign disputed awards with all actions permanently logged;

- Account security features, including strict domain restrictions requiring an *@psu.palawan.edu.ph* email, a 5-attempt/5-minute brute-force login lockout, and secure password recovery;
- Read-only public access to published schedules, scores, and standings, while athlete, coach, facilitator, and administrator functions are limited to registered PSU community members across all eighteen campuses; and
- Responsive web access through any modern browser on desktop, laptop, tablet, and smartphone, with light and dark display modes.

Limitations

The following are not included in this study:

- Native mobile application development — the system is a responsive web application only;
- Subscription, ticketing, or payment features
- AI-based or biometric identity verification — the automated confidence score is an advisory aid exclusively for uploaded student Certificates of Registration (with faculty and staff employee identifications undergoing direct manual review), and eligibility is always decided manually by the administrator;
- Direct integration with the PSU AIMS Student Portal — the system provides only a link for manual cross-referencing;
- In-app messaging or chat — team invitations and notifications are system-generated only;
- Real-time GPS tracking or live video or audio streaming of matches — live coverage is limited to scores, clocks, play-by-play text, and graphics;

- Offline operation beyond court-side scoring — only scoring events are buffered locally during network dropouts, and all other functions require an internet connection;
- Sports facility accreditation or external verification;
- Management of professional or commercial sports leagues outside PSU;
- Automated content moderation beyond what the administrator handles; and
- STRASUC inter-university competition management. The system covers only PSU internal sports events, specifically the Binturungan and the Faculty and Staff Friendly Games.

Operational Definition of Terms

For the purpose of this study, the following terms are defined operationally:

PSU SportsTrack – The web-based sports management system developed in this study, designed to centralize player registration, team formation, tournament operations, and reporting for Palawan State University across all eighteen campuses.

Binturungan – The official PSU Intramurals, a yearly multi-sport competition among student-athletes from the different colleges and campuses of PSU.

Faculty and Staff Friendly Games – A yearly sports event for PSU personnel, structurally separated into Faculty Athlete and Staff Athlete divisions and managed within the system.

Tournament – An organized sports competition created by the administrator in PSU SportsTrack, featuring a set event type, sport, teams, schedule, and an automated bracket format (Single Elimination, Double Elimination, Round Robin, or a two-stage hybrid pool-to-playoff).

Event Lifecycle Phase – The current stage of a tournament within PSU SportsTrack, consisting of three sequential phases — Draft, Slotting, and Launched — that determine what actions administrators and coaches are permitted to perform.

Draft (Phase 1) – The initial status of a tournament immediately after creation by the administrator. A tournament in Draft status is not yet visible to coaches and cannot accept team registrations.

Slotting (Phase 2) – The status of a tournament that has been opened by the administrator for team registration. During Slotting, coaches may claim available team slots and submit or edit their team rosters until the registration deadline passes or all slots are filled.

Launched (Phase 3) – The status of a tournament once the administrator has closed slot drafting and triggered bracket generation. A Launched tournament has its team registrations locked, its bracket generated, and its schedule and live scoring features made active.

Slot – An individual team position within a tournament, made available by the administrator during the Slotting phase and claimed by a coach through team roster submission.

Bracket – The tournament matchup chart automatically generated by the system after all teams have registered. It determines the sequence of games and routes winners and losers automatically based on match results.

Player – A registered PSU student (Student Athlete), teaching personnel (Faculty Athlete), or non-teaching employee (Staff Athlete) who has registered via a 6-digit OTP or magic link, submitted the required verification document, and been approved by the administrator as eligible to compete.

Coach – A registered user who creates a team, recruits verified athletes, and manages the roster and game-day lineup of that team in a sports event. A coach represents either a college department (Collegiate Squad) or a Binturungan tribe or cluster.

Facilitator – The appointed court-side match official who operates the live scoring console. Due to the system's dual-role architecture, verified players or coaches can simultaneously hold this privilege without losing their primary profile access.

Administrator – The executive system manager who verifies athlete documents, engineers tournaments and brackets, appoints facilitators, finalizes standings and medals, assigns Gold Star awards, reviews audit logs, and generates institutional reports. The Administrator holds unrestricted access to all system routes, including a master override of live matches.

Certificate of Registration (COR) – An official document from the PSU Registrar confirming a student's enrollment. It is evaluated by an automated advisory confidence score to assist the administrator during verification.

Employee Identification – The official institutional ID uploaded by Faculty and Staff Athletes during registration, which bypasses automated extraction and undergoes direct manual review by the administrator.

Expiring Signed URL – A 60-second secured link generated on-demand to preview uploaded student CORs and employee IDs, protecting users' personally identifiable information (PII) from unauthorized access.

MVP Award – Recognition given to outstanding players on their profiles. The **Gold Star** is the overall tournament MVP assigned exclusively by the Administrator upon tournament closure. The **Red Star** is the game MVP assigned exclusively by the Facilitator upon match conclusion.

Live Scoring Feed – A real-time, text-based score update broadcast during active matches. Scores are inputted court-side by a designated Facilitator, completely replacing paper score sheets.

Conflict of Interest (Anti-COI) – An automated governance check preventing Facilitators from officiating or operating the scoring console for matches involving their own affiliated academic college or cluster.

Offline Scoring Buffer – An internal queue that saves points, clock controls, fouls, and substitutions locally during venue network dropouts, synchronizing automatically once the Facilitator's internet connection is restored.

Institutional Report – A structured data compilation generated by the system's reporting engine (FR-41) after a season, covering multi-discipline squad standings, individual scoring statistics, and collegiate participation footprints. It is exportable as

a structured CSV dataset or a vector-rendered A4 landscape PDF for university accreditation.

Ranking – The system-generated ordering of teams or players based on match results, points, or standings, used to determine seedings and final placements within a tournament.

Season History – A yearly, per-category archive of past tournament records, including final standings, results, and player statistics, allowing users to view performance across previous PSU sports seasons.

Player Statistics – A record of a player's individual performance data (e.g., points, fouls, games played) tracked and updated per game, viewable on the player's profile.

Match Schedule – A system-generated timetable showing the date, time, venue, assigned Facilitator, participating teams, and round/stage for each game within a tournament.

Magic Link Authentication – A secure, time-limited token sent directly to a user's *@psu.palawan.edu.ph* email, enabling instantaneous login and password recovery without transmitting or storing passwords in clear text.

Soft-Delete Retention State – A secure database holding period where decommissioned, banned, or deactivated user accounts and records are retained for 365 days before permanent purging, allowing the Administrator to recover data if necessary.

Revalidation Workflow – A recurring administrative process requiring registered Student, Faculty, and Staff Athletes to submit updated enrollment or employment

credentials at the start of every academic year to maintain their eligibility on the platform.

Row-Level Security (RLS) – A strict database-level security protocol that isolates data access based on a user's assigned classification, ensuring, for example, that Coaches cannot manipulate brackets and Players cannot edit match scores.

Scorekeeper Lease – A 90-second temporary, exclusive lock granted to an appointed Facilitator operating the court-side console to prevent concurrent conflicting score inputs, renewed automatically via a 15-second heartbeat.

Dual-Path Email Verification – A secure authentication mechanism utilized during user registration and login. It allows users to verify their identity via either a system-generated 6-digit One-Time Password (OTP) or a direct cryptographic Magic Link, sent exclusively to their official *@psu.palawan.edu.ph* institutional email address.

Dual-Role Architecture (Dual-Role Facilitator) – A specialized role-based access control (RBAC) framework within the system that permits verified users (such as Athletes and Coaches) to concurrently hold court-side Facilitator privileges. It allows them to officiate matches not involving their affiliated college without losing access to their primary player or coach profiles.

Red Star (Game MVP) – An official system accolade awarded exclusively by the Facilitator at the conclusion of a match to recognize the most valuable player of that specific game. Its assignment is strictly governed by the Hardened Match MVP Governance Engine to mitigate officiating bias.

Yellow Star (Top Performer) – A secondary system accolade designated by the Facilitator upon match conclusion to recognize athletes who delivered outstanding or highly notable court performances, serving as a supplementary honor alongside the primary Red Star award.

Scorekeeper Desk (Kiosk) – A touch-first, zero-scroll, mobile-optimized digital scoring console operated by the assigned Facilitator. It completely replaces traditional paper score sheets, featuring physical-style buttons for point attribution, game clock controls, foul tracking, and two-tap substitutions.

Play-by-Play Commentary – A real-time, timestamped text feed generated autonomously by the system based on the Facilitator's kiosk inputs. It translates raw scoring events, fouls, and substitutions into a readable broadcast pushed instantly to the public spectator dashboard via WebSockets.

Leaderboards – The dynamic, system-generated rankings displayed on the public platform. They showcase the overarching performance of colleges and Binturungan clusters by automatically aggregating Gold, Silver, and Bronze medal tallies, overall point differentials, and win rates from finalized tournaments.

PKCE (Proof Key for Code Exchange) – A cryptographic security protocol implemented within the system's Magic Link and password recovery workflows. It mitigates token interception attacks by mathematically verifying that the specific browser or device requesting the authentication link is the exact same device that exchanges the token for access, ensuring secure login without transmitting passwords in clear text.

CHAPTER II

REVIEW OF RELATED LITERATURE OR SYSTEMS

Local

Sports Management System Development (Tarlac State University): A capstone project by students at Tarlac State University focused on developing a computerized system to conveniently manage sports events and eliminate manual paperwork. Much like PSU SportsTrack, this study emphasized evaluating the system's performance and efficiency through user surveys based on standardized software quality metrics. This aligns directly with the researchers' goal to evaluate usability and system reliability via ISO 25010 quality characteristics, specifically targeting the seven distinct user classifications in PSU SportsTrack, including the unique workflow requirements of Coaches and Court-Side Facilitators.

DigiTally Digital Score and Stats System for Local Leagues (2025): This recent Philippine study addresses the operational challenges of local sports leagues that rely on manual methods for recording scores and player statistics. The researchers developed a web-based application to streamline scorekeeping and statistical tracking. While DigiTally mirrors the intent of PSU SportsTrack's live match operations, PSU SportsTrack heavily innovates upon this foundation by introducing enterprise-grade courtside resilience. Unlike standard web scorekeepers, an automated Anti-Conflict of Interest (Anti-COI) reservation mechanism, and local IndexedDB offline buffering that preserves scoring events during venue network dropouts.

Sports Equipment Inventory and Profiling Management System (Ramon Magsaysay Technological University): This study tackled the inefficiencies of physical filing cabinets and manual tracking for athlete profiles and sports activities.

By implementing a computerized system, the university was able to efficiently manage athletic records and monitor events. PSU SportsTrack significantly advances this concept within its registration module. Rather than just basic digital filing, PSU SportsTrack implements a secure dual-path authentication (6-digit OTP or Magic Link) and introduces an automated advisory confidence score for uploaded student Certificates of Registration (COR). Furthermore, it addresses critical data privacy standards absent in older systems by protecting all uploaded PII (Personally Identifiable Information) behind 60-second expiring signed URLs.

Foreign

Football League Management System (Malaysia): This study focused on developing a web-based system to replace the traditional filing methods used during Malaysian football tournaments. The system successfully managed the registration process, fixtures, matches, and automated the compilation of league-standing tables. This strongly supports the necessity of PSU SportsTrack's Institutional Report Generation engine (FR-41). However, PSU SportsTrack scales this capability across nine different ballgame disciplines, allowing administrators to interactively audit multi-sport participation footprints, MVP registries, and squad standings, with one-click export capabilities to structured CSV datasets and vector-rendered A4 landscape PDFs designed specifically for university accreditation.

Online Tournament Management Application (React & PHP): This foreign literature outlines the development of an application designed to overcome the problems faced by organizers in creating and managing demographic-based tournaments. It features a robust database structure for handling teams, brackets, and real-time score tracking. While sharing a similar frontend philosophy, PSU

SportsTrack replaces the legacy PHP backend with a modern, decoupled PostgreSQL/Supabase architecture utilizing WebSocket real-time replication. Furthermore, PSU SportsTrack expands bracket generation beyond basic tracking to automatically engineer four distinct topologies: Single Elimination, Double Elimination, Round Robin, and two-stage hybrid formats, supplemented by procedurally generated Web Audio cues for live spectators.

Tournament Management System Database Architecture: A foreign research report detailing the relational database design necessary for a centralized tournament platform. The study highlights how digitizing entities like participants, teams, and matches streamlines communication and team formation for organizers. This serves as an excellent foundational study for PSU SportsTrack's coach-led team formation and overall backend schema. However, PSU SportsTrack extends this architectural theory by implementing strict Row-Level Security (RLS), a 365-day soft-delete retention state for data recovery, and a highly specialized "dual-role architecture" that permits verified athletes and coaches to simultaneously hold officiating privileges without data collision.

Matrix of Related Literature and Systems

The following matrix summarizes the reviewed local and foreign literature, highlighting their key features, identified gaps, and how PSU SportsTrack directly addresses these limitations.

Title / System	Institution / Source / Year	Key Features & Contributions	Identified Gaps & Limitations	Relevance & Application to PSU SportsTrack
Sports Management System Development	Tarlac State University (Local, 2024)	Computerized sports management designed to eliminate manual paperwork; evaluated using ISO 25010 software quality metrics.	Focuses on basic operational features; lacks specialized governance workflows for court-side officials.	Validates the use of the ISO 25010 framework to evaluate system efficiency and usability, specifically targeting PSU SportsTrack's 7 distinct user classifications.
DigiTally: Digital Score and Stats System	Ladera et al. (Local, 2025)	Web-based application built to streamline live scorekeeping and statistical tracking for local leagues.	Standard web scorekeeper lacking enterprise-grad e offline resilience and officiating bias prevention.	Serves as a baseline for live operations. PSU SportsTrack innovates by adding an automated Anti-COI reservation mechanism and IndexedDB offline buffering.
Sports Equipment Inventory and Profiling Management	Ramon Magsaysay Technological University (Local)	Digitized athletic records, profiles, and sports activities to replace physical filing cabinets.	Basic digital filing lacking strict modern data privacy controls and automated document verification.	Highlights the necessity of digital profiling. PSU SportsTrack advances this via secure dual-path authentication, automated OCR confidence scores, and 60-second expiring URLs.
Football League Management System	Fingoweb / Malaysia (Foreign, 2026)	Automates player registration, match fixtures, and the compilation of league-standin g tables.	Focused strictly on commercial football; lacks multi-disciplin e executive auditing and accreditation-r	Supports the necessity of automated standings. PSU SportsTrack scales this capability across 9 sports via the FR-41

			eady exports.	institutional reporting engine (CSV/PDF exports).
Online Tournament Management Application	Foreign Study	Features a robust relational database for demographic-based tournaments, brackets, and real-time tracking (React & PHP).	Utilizes legacy PHP architecture; features limited bracket topologies and basic spectator engagement.	Informs frontend philosophy. PSU SportsTrack modernizes the backend using PostgreSQL/Supabase WebSockets, expands to 4 automated bracket topologies, and adds procedural Web Audio cues.
Tournament Management System Database Architecture	Foreign Research	Detailed relational database design streamlining entities like participants, teams, and matches.	Lacks institutional security protocols, database-level data isolation, and complex officiating frameworks.	Forms the baseline architectural theory. PSU SportsTrack heavily extends this by implementing strict Row-Level Security (RLS) and a specialized dual-role officiating architecture.

Synthesis

The reviewed local and foreign studies demonstrate proven implementations of individual sports management features, such as bracket generation, automated league standing compilation, basic player profiling, and real-time score tracking. However, these existing systems remain focused on specific, isolated functionalities or commercial sports contexts; they are not architected for the strict governance and institutional workflows of a state university sports program.

While earlier systems provide foundational database architectures, they critically lack the institutional compliance mechanisms that a university like PSU

requires—specifically, automated Conflict of Interest (Anti-COI) governance for match officials, data privacy enforcement via expiring signed URLs, offline venue resilience for uninterrupted courtside scoring, and an executive auditing engine (FR-41) capable of generating accreditation-ready PDF reports. This highlights a significant gap in the literature: the absence of a fully integrated, multi-role sports management platform that marries athletic tournament mechanics with stringent academic administrative protocols. PSU SportsTrack directly bridges this gap by delivering a secure, seven-role platform tailored explicitly to the scale and security demands of Palawan State University.

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CHAPTER III

TECHNICAL BACKGROUND

This chapter presents the technical background of PSU SportsTrack, a web-based sports tournament management system developed for Palawan State University. It covers the previous state of sports event management at PSU, the operational bottlenecks that necessitated this study, and the technical approach used to address them. It also details the system's physical architecture, organizational hierarchy, system flowcharts for both the previous manual and proposed digital workflows, and the hardware and software specifications required for development and implementation.

Present Operations

Prior to the implementation of PSU SportsTrack, Palawan State University managed its sports events through a highly layered, manual process. Coordination cascaded from the sports director down to cluster sports coordinators, who then relayed instructions to their respective coaches and tournament managers (TMs). Each sport was assigned its own tournament manager—typically a Sports Department employee or a qualified staff member from another cluster—who was responsible for manually facilitating and overseeing the entire event for that specific discipline.

Before any competition began, players were required to submit physical requirements, including Certificates of Registration (COR) and school IDs to confirm their eligibility. These sensitive documents were collected, stored in physical folders, and reviewed manually by the tournament manager, which posed significant data privacy risks and logistical delays. Furthermore, coaches identified and recruited players strictly through personal networks and physical tryouts, as there was no

centralized digital platform to securely browse or verify the available pool of student-athletes.

During the events, designated table management teams handled score recording using traditional paper score sheets, submitting the final written results to the tournament manager only after each game concluded. This manual handover frequently resulted in transcription errors, conflicting score tallies, and significant delays in updating the public. Tournament brackets were constructed by hand, and schedules were determined manually based on the number of teams and available playing days. For the Faculty and Staff Friendly Games, the sports director organized the event entirely independently, grouping participants by department or preference based on manual headcounts.

After each event, results were compiled manually and passed up the chain of command from the table management team to the sports director. There was no permanent digital record of multi-discipline outcomes, squad standings, or MVP award distributions across seasons. While the Sports Department occasionally relied on a paid third-party website solely for basketball statistics, no unified, institutional system existed for managing PSU sports events as a whole. This left the Sports Department burdened with significant paperwork, vulnerable to data loss, and entirely lacking the structured, historical sports data required for university accreditation and resource planning.

Proposed Solution

PSU SportsTrack was developed in direct response to the operational challenges confirmed during the consultation with the PSU Sports Department. The system provided a centralized, secure web-based platform that digitized the full

lifecycle of PSU sports event management, replacing the vulnerable paper-based workflows that had burdened the Sports Department, tournament managers, coaches, and players.

Instead of collecting physical requirements before every event, player registration and eligibility verification were handled digitally using a secure dual-path email authentication (6-digit OTP or direct magic link). Student-athletes registered their profiles and uploaded their Certificate of Registration (COR), while faculty and non-teaching staff uploaded their employee identification. To ensure data privacy and institutional compliance, all uploaded requirements were strictly safeguarded behind 60-second expiring signed URLs. The system evaluated each uploaded COR and produced an advisory confidence score exclusively for student submissions—whereas faculty and staff IDs bypassed this for direct manual review. The administrator utilized these document previews and scores to approve or reject submissions. Coaches, rather than relying solely on personal networks and physical tryouts, could securely browse the pool of verified athletes directly in the system, send recruitment invitations, and build their team rosters and game-day lineups digitally.

Tournament managers, operating under the Administrator role, engineered tournaments, scheduled games, assigned court-side facilitators, and generated automated brackets supporting Single Elimination, Double Elimination, Round Robin, and two-stage hybrid formats. During games, the appointed facilitators replaced paper score sheets by operating a dedicated digital scoring console on a tablet or smartphone. Their access was strictly governed by an automated Anti-Conflict of Interest (Anti-COI) mechanism that prevented them from officiating matches involving their own affiliated academic college. Scores were broadcast in real time to

a live match viewer, a stadium display for gym projectors, and floating scoreboard widgets, allowing the community to follow the game on any device. Results were recorded automatically and reflected instantly in the bracket, eliminating the need for manual compilation.

At the conclusion of each event, PSU SportsTrack updated the leaderboards and Gold, Silver, and Bronze medal standings. Outstanding players were recognized through an MVP award system, featuring the Gold Star (overall tournament MVP) assigned exclusively by the Administrator upon tournament closure, and the Red Star (game MVP) assigned exclusively by the Facilitator upon match conclusion. Finally, the system's institutional reporting engine (FR-41) generated structured reports covering multi-discipline tournament outcomes, standings, and collegiate participation footprints. These reports could be exported instantly as structured CSV datasets and vector-rendered A4 landscape PDF documents—providing the Sports Department with a reliable, accreditation-ready digital archive spanning the Binturungan Intramurals and the Faculty and Staff Friendly Games across all eighteen campuses of Palawan State University.

Physical Architecture

PSU SportsTrack used a decoupled architecture of three layers: a presentation layer, a Backend-as-a-Service layer, and a data persistence layer. The presentation layer was a React 19 and TypeScript single-page application (SPA) styled with Tailwind CSS and shadcn/ui, with Next-Themes handling the dark and light modes. The SPA was hosted on the Vercel edge network, which acted as a global CDN and SPA host. Browser capabilities ran inside this layer: a Web Audio FX engine, an

HTML5 Canvas with Picture-in-Picture for the floating scoreboard, a PDF.js canvas viewer for verification documents, and an IndexedDB queue that buffered scoring events during network dropouts.

The presentation layer reached the backend through two paths. Queries, authentication and file operations went over HTTPS using the PostgREST client, and live updates arrived over WebSockets from the Realtime publish/subscribe engine. The backend was provided by Supabase, which supplied Auth with JWT sessions, Storage buckets for COR documents, and Realtime through logical replication (change data capture).

Supabase persisted all data in a PostgreSQL 15 relational database. Row-Level Security policies isolated each user's data, PL/pgSQL triggers such as `trg_sync_user_email` kept auth and profile emails in sync, and strict check constraints and foreign keys preserved data integrity.

Furthermore, to preserve institutional integrity against unauthorized post-match tampering, PostgreSQL migration 20261031_harden_mvp_star_governance.sql enforces strict Row-Level Security (RLS) invariants on award assignments. The database restricts court-side Facilitators exclusively to the initial insertion of Red Star (Game MVP) awards. Any update, modification, or deletion authorities for these records are strictly revoked from Facilitators and reserved solely for Accredited Administrators. Database-level triggers ensure that any administrative execution of OVERRIDE_STAR or REVOKE_STAR actions automatically commits a permanent, tamper-proof record into the public.audit_logs table.

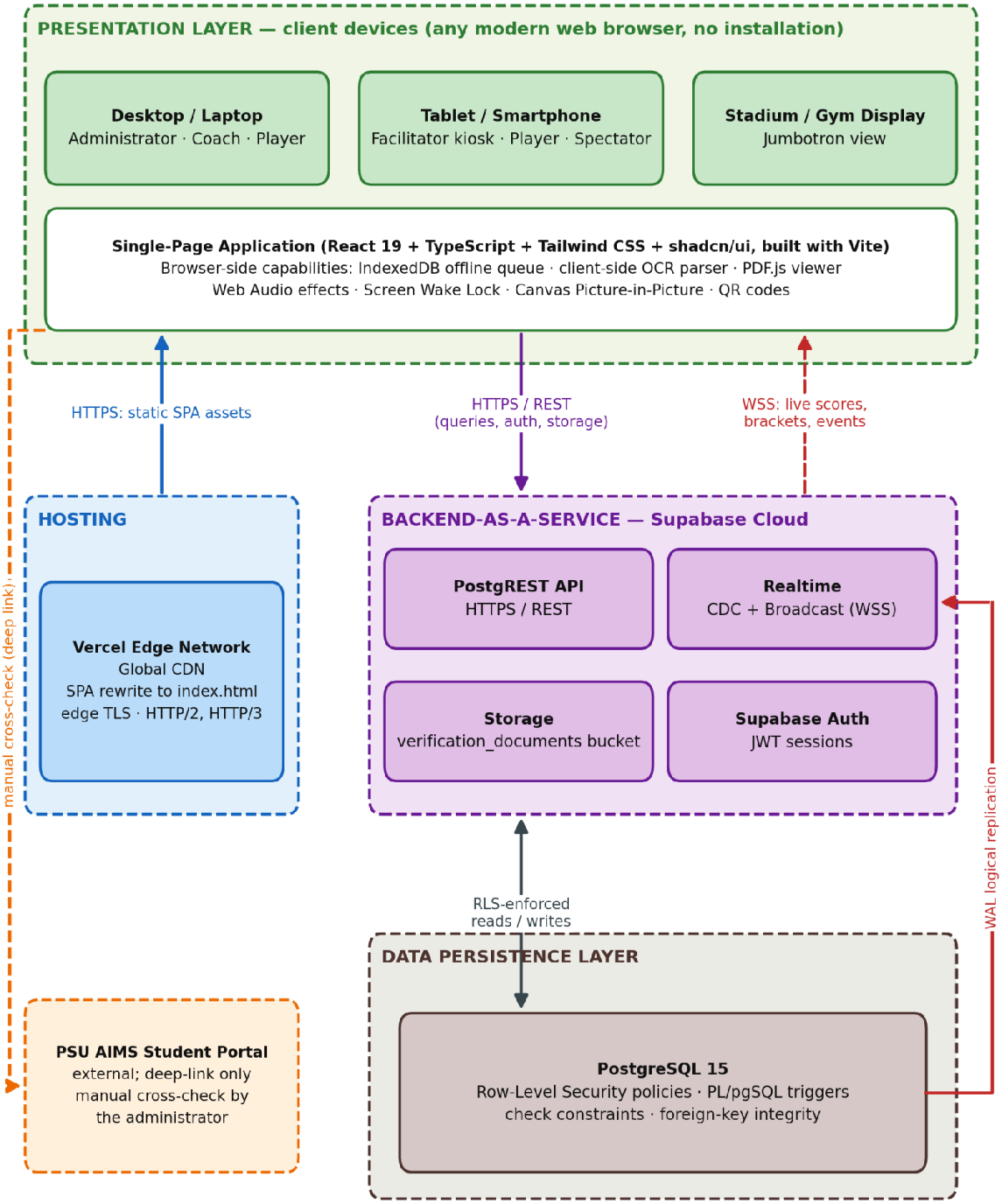


Figure 1. Physical Architecture Diagram of PSU SportsTrack

Organizational Chart

Figure 2 presents the governance structure of PSU SportsTrack, which defined the roles, responsibilities, and levels of authority within the system. At the apex of the structure was the System Administrator, who served as the executive and tournament head and held ultimate administrative and system control. The Administrator managed the complete tournament lifecycle, including tournament creation, bracket configuration, termination, and cascade deletion. The Administrator also conferred the Gold Star (Tournament MVP) upon the outstanding athlete of each tournament and oversaw the system audit logs, user compliance, and demonstration staging and resets. In addition, the Administrator held fallback authority over live scoring and could override or forcibly claim the 90-second scorekeeper lease whenever a match became locked.

Two operational roles stood directly below the Administrator. The Court Facilitators, who served as referees and official scorers, carried out court-side operations and held exclusive authority over live match scoring. Facilitators were also responsible for offline buffering and batch synchronization of scoring events through IndexedDB, and conferred the Red Star (Match MVP) at the conclusion of each fixture.

The Coaches served as delegated team managers. They managed the rosters of their assigned teams, including jersey numbers and sports positions, submitted pre-match starting lineups, and updated the game-day availability of players (Active, Injured, or Excused). Coaches were bound by the recruitment rules of the system, which prohibited the recruitment of an athlete who already belonged to another team in the same sport.

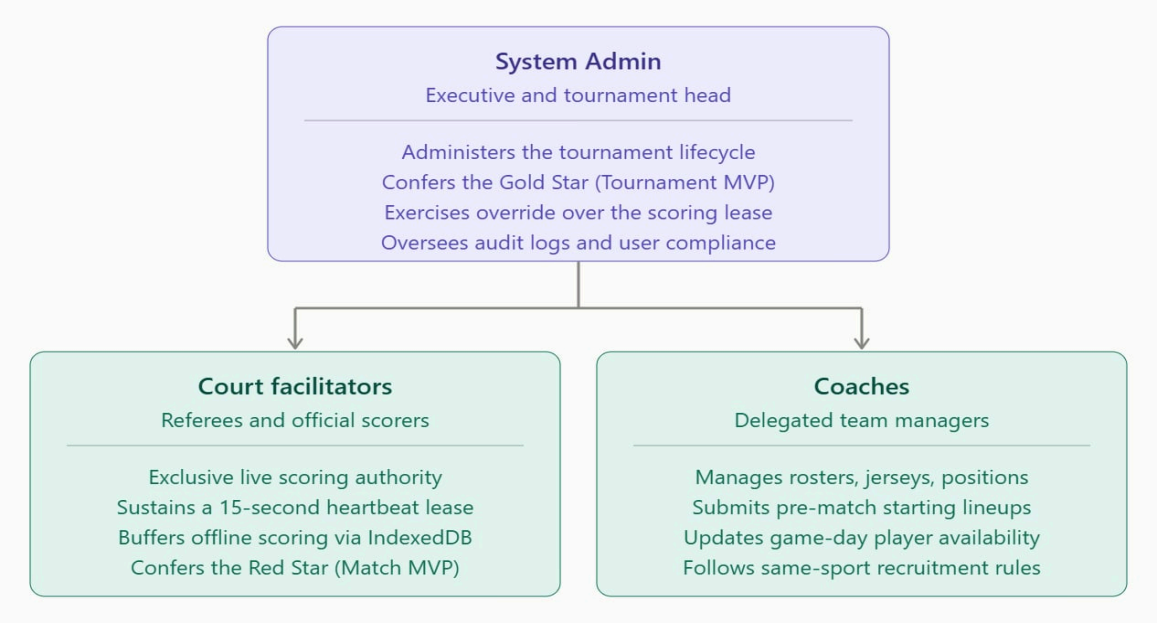


Figure 2. Organizational Chart of PSU Sports Management

System Flowchart

This section presents the system flowcharts of PSU SportsTrack. Figure 3 shows the manual process that PSU used before the system. Figures 4A to 4F show the proposed process in six stages: sign-up, sign-in, and role routing; athlete document verification; tournament setup and bracket generation; team recruitment and lineups; live match scoring; and standings and spectator views. Together they follow seven groups of user classifications: athletes (students, faculty, and non-teaching staff), coaches, facilitators, the Administrator, and unauthenticated spectators.

Existing System Flowchart

In the manual system PSU had previously used, a student-athlete would submit a physical registration form with a COR and valid ID to the Sports Department or college coordinator, who would manually check eligibility and record approved players on a paper roster. Coaches would recruit players through personal contacts, the Sports Department would build the bracket by hand, and schedules were

distributed through printed copies or group messages. Scores were written on paper sheets, and post-event reports were compiled manually — often inconsistently — after each event.

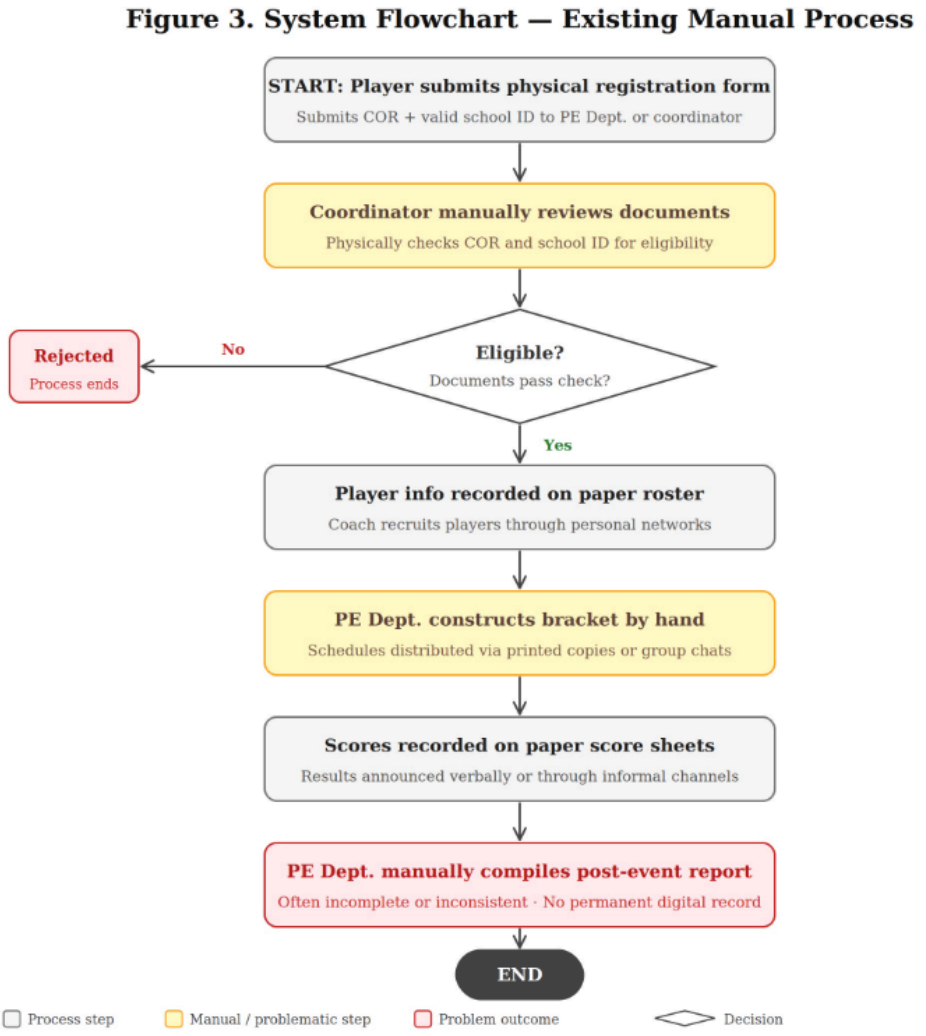


Figure 3. System Flowchart — Existing Manual Process

Proposed System Flowchart

The proposed PSU SportsTrack process is presented in six stages, each shown as its own flowchart with a narrative that describes it step by step.

Stage 1: Sign-Up, Sign-In, and Role Routing

Every session began when a user opened PSU SportsTrack. The application

first checked whether the user was already signed in, and a signed-in user went straight to the role lookup. Anyone else was shown the sign-in modal or the standalone */register* page and chose between signing up and signing in. When the user came from a match page or the facilitator kiosk, the application saved the current page path so the user could be returned to the game after signing in.

A new user signed up by entering their name, college or department, student number (if applicable), institutional email, password, and role. To enforce institutional security, the email was strictly validated against the *@psu.palawan.edu.ph* domain. A Zod schema validated the form before anything was sent, and an invalid form returned the user to correct the input. Supabase Auth then created the account, triggering a dual-path verification: a **6-digit OTP** or a direct PKCE magic link sent to their email. A database trigger, *trg_sync_user_email*, synced the email from *auth.users* to the *public.users* profile. The new account was initially registered with an Unverified status. Once the OTP or link was confirmed, the user was authenticated and the profile creation was finalized via *trg_handle_auth_user_confirmed*

A returning user signed in by entering an email and password, which the same Zod schema validated. The application tracked failed sign-in attempts in a dedicated database table (auth_rate_limits), not just browser storage. Once the credentials were valid, it checked the lockout status via an RPC (check_login_rate_limit). The third consecutive failure triggered a five-minute cooldown with a live countdown and disabled inputs. If the sign-in was not locked, the credentials went to Supabase Auth. A failed attempt added to the failure count, and a successful attempt reset the lockout and issue. Both flows then reached the same role lookup. The application read the user's role from the *public.users* table through token claims and enforced route access

via Row-Level Security (RLS) and ProtectedRoutes. Users were sent to their matching landing pages: /admin for the admin, /coach and /coach/teams for the coach, /player for player_student, player_faculty, and player_staff, and the scoring desk/kiosk for the facilitator.

A user who forgot their password could request a reset from the login modal, prompting Supabase Auth to send a magic link to /reset-password, where the token exchange occurred and the new password was securely saved.

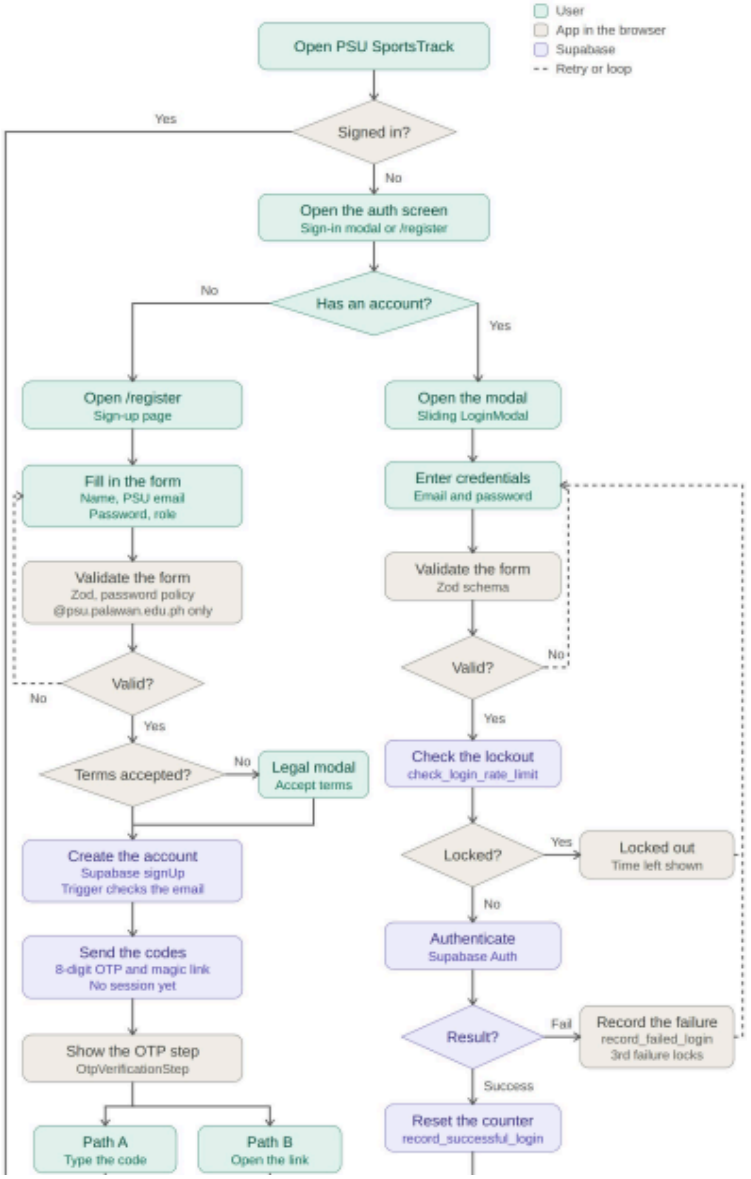


Figure 4.1. System Flowchart — Sign-Up, Sign-In, and Role Routing

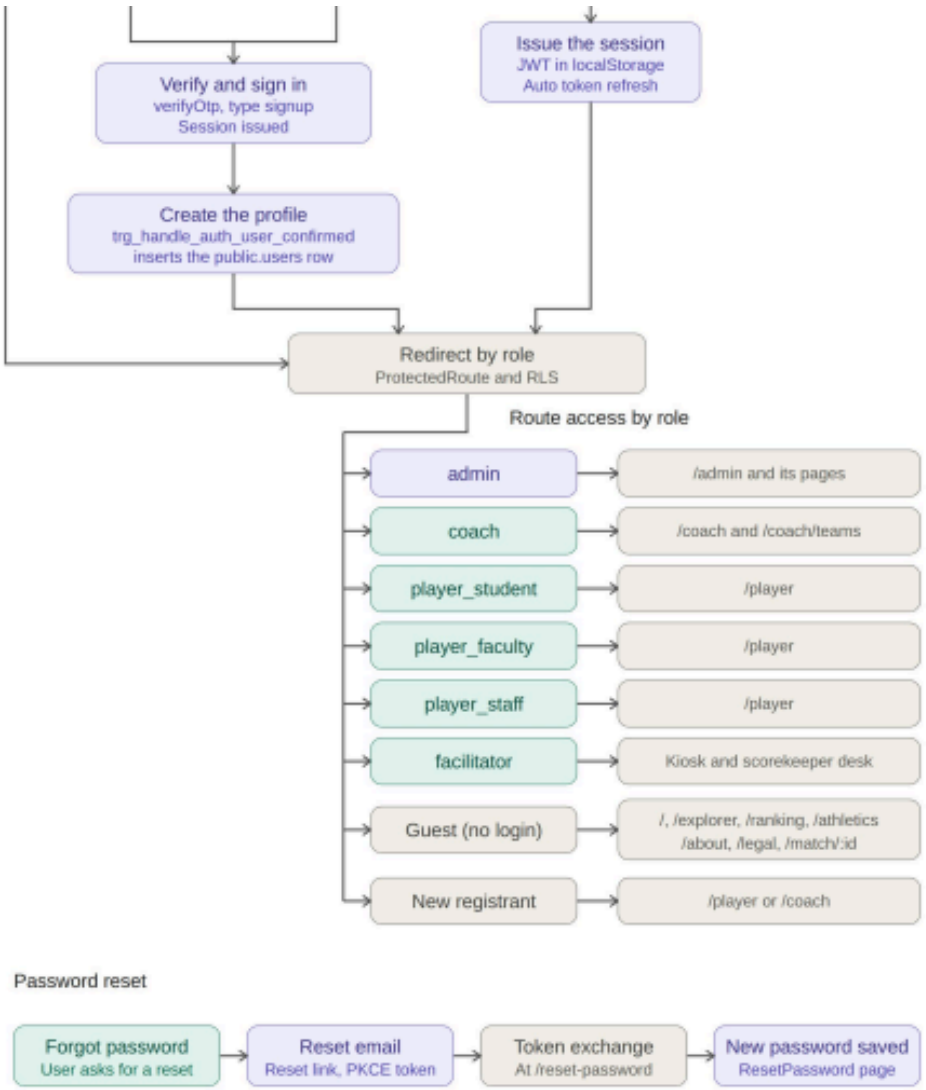


Figure 4.1 (continued). System Flowchart — Sign-Up, Sign-In, and Role Routing

Stage 2: Athlete Document Verification

A student athlete, faculty athlete, staff athlete, or coach uploaded a verification document. Students uploaded a Certificate of Registration (COR), faculty and staff uploaded their respective employee identification, and unverified coaches uploaded a valid institutional ID.

For a student COR, the browser first extracted the text of the PDF using

PDF.js and parsed it with *corParser.ts*. The parser evaluated the student number, name similarity, email format, college affiliation, and the official seal, returning an advisory confidence score from 0 to 100 alongside the extracted student number and college. A scanned, image-only COR lacking a text layer automatically scored 0 and required manual review. Faculty, staff, and coach IDs bypassed this automated text extraction and defaulted to direct manual review by the administrator.

The file was then uploaded to the private verification_documents bucket in Supabase Storage. For a COR, the storage path was structured as *user_id/cor_timestamp.pdf*. A corresponding row was inserted into the verification_documents table with the status marked as Pending, including the confidence score for student CORs.

The Administrator opened the */admin/verifications* dashboard, which listed all pending documents. To ensure strict data privacy and protect Personally Identifiable Information (PII) in compliance with institutional standards, the application generated a signed URL for each document that automatically expired after 60 seconds. The file was rendered on a secure canvas alongside the OCR score breakdown, ensuring sensitive information was never exposed through a permanent, public link. The Administrator then clicked "Verify on PSU AIMS Portal" to manually cross-check the details externally before making a final decision.

Upon approval, the document status updated to Approved, the *users.is_verified* flag was set to true, and an *APPROVE_DOCUMENT* entry was written to the immutable audit log. Upon rejection, the status updated to Rejected, the specific rejection reason was saved, and a REJECT_DOCUMENT entry was logged. Database Row-Level Security (RLS) policies strictly enforced this verification flag across the system, actively blocking unverified coaches from creating teams or

sending recruitment invitations, and preventing unverified athletes from participating in active rosters.

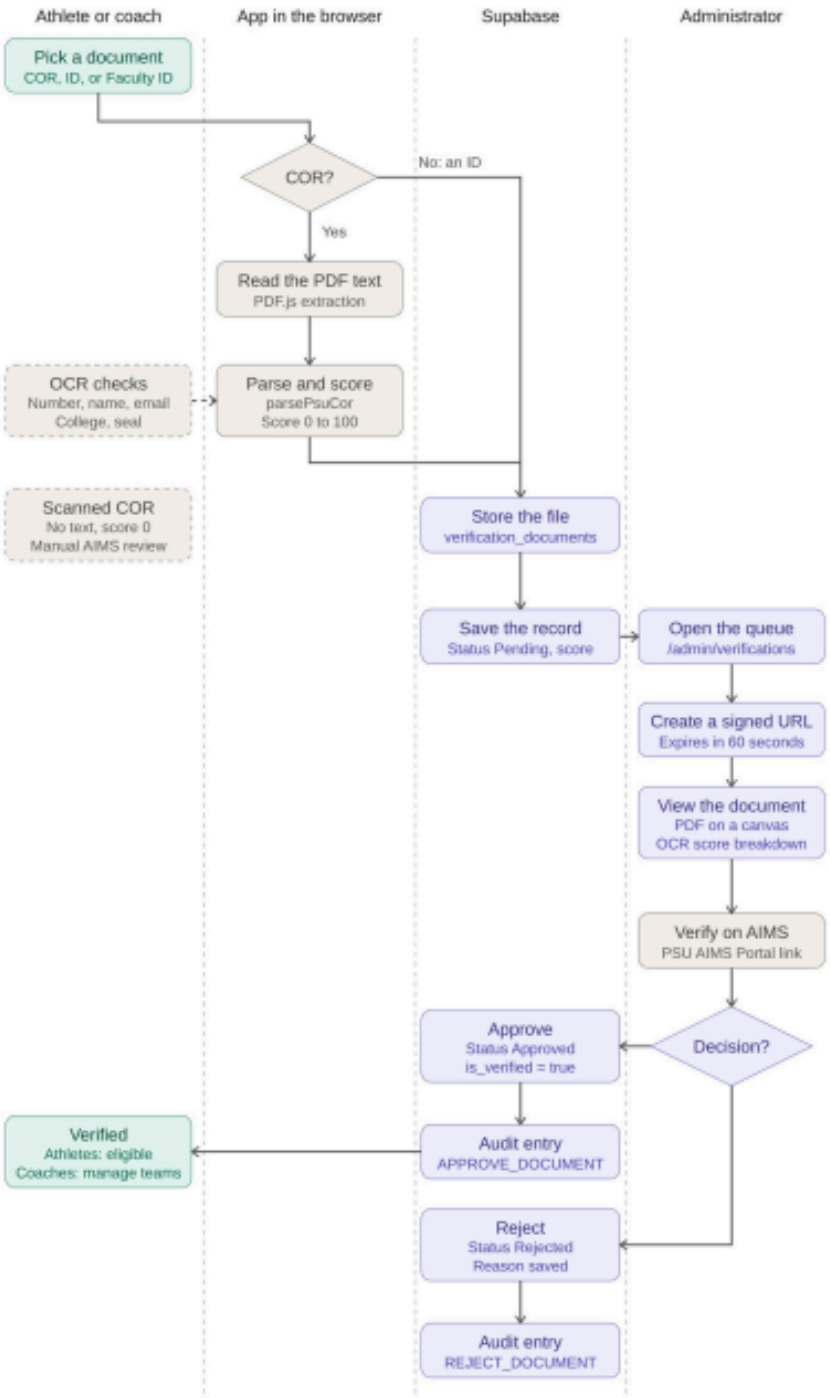


Figure 4.2. System Flowchart — Athlete Document Verification

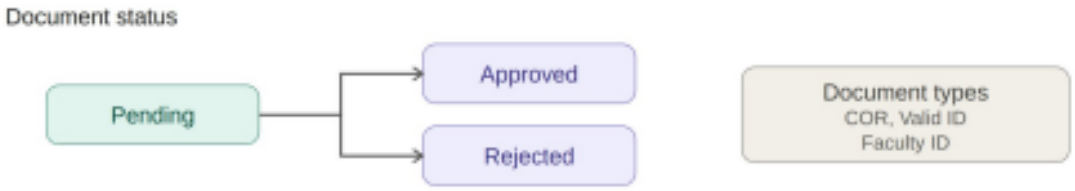


Figure 4.2 (continued). System Flowchart — Athlete Document Verification

Stage 3: Tournament Setup and Bracket Generation

The Administrator started tournament setup at */admin/tournaments* by creating a tournament with a name, a sport, a type (Binturungan or Faculty and Staff Friendly Games), and start and end dates. The tournament was saved with the status Draft. The Administrator then selected it and clicked Generate Bracket, which opened the bracket generation wizard. The wizard offered four automated formats: single elimination, double elimination with upper and lower brackets, round robin only, and a two-stage hybrid of round robin with playoffs.

The wizard calculated the number of rounds as the ceiling of the base-2 logarithm of the number of teams and seeded byes where needed, and the Administrator could swap or shuffle the seeds. Bracket creation ran in two passes: the first batch-inserted the match fixtures, and the second batch-updated each fixture with its *next_match_id* and *loser_next_match_id.* The Administrator approved the teams and set the tournament status to Ongoing.

While the tournament ran, every time a match was updated to Completed, the database trigger *advance_bracket_on_match_completed* automatically moved the winner into the *next_match_id* slot and the loser into the *loser_next_match_id* slot, such as routing to an automated third-place (bronze) match. A knockout match could not be finalized while its scores were tied. When the Administrator concluded the

tournament, its status became Completed.

Deleting a tournament, a team, or a user did not remove the record immediately. The deletion modal soft-deleted it, marking it as deleted with a timestamp, and the Administrator could restore it, or delete it permanently, from the recovery center at */admin/recovery*.

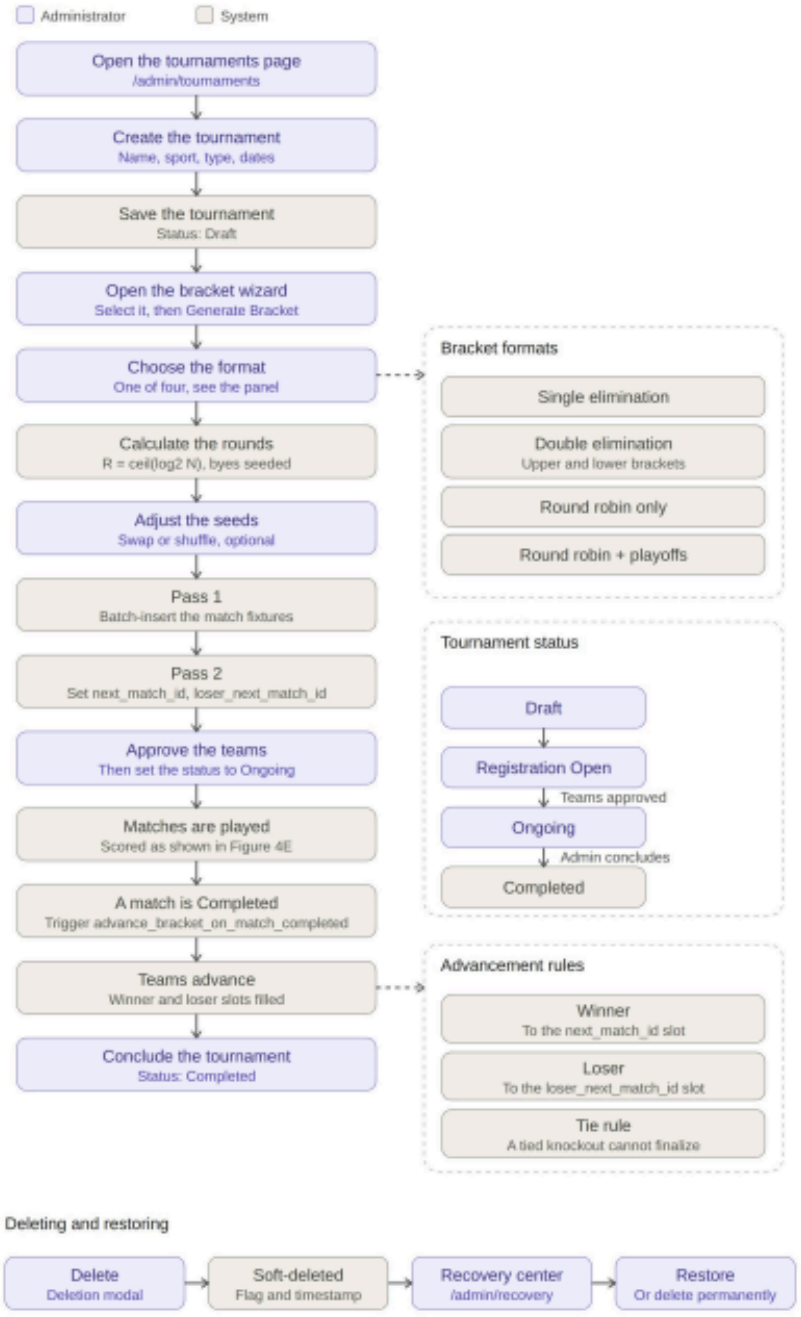


Figure 4.3. System Flowchart — Tournament Setup and Bracket Generation

Stage 4: Team Recruitment and Lineups

Coaches operated from the */coach* and */coach/teams* dashboards. To enforce institutional security, database Row-Level Security (RLS) actively blocked unverified coaches from creating teams or sending recruitment invitations. A verified coach created a team under a specific Collegiate Squad or Binturungan Tribe/Cluster for an active tournament, and the team was saved with a Pending status until the Administrator approved it.

Utilizing the recruitment dashboard, the coach securely browsed the pool of verified athletes (students, faculty, or staff) and sent squad invitations, with each invite generating a roster entry marked as Pending. The invited athlete received this notification on their */player* console and could choose to accept or decline it, instantly updating the roster entry's status to Approved or Rejected.

Once the roster was assembled, the coach configured the game-day lineup: designating a team captain (is_captain), assigning jersey numbers and specific positions, marking athletes as starters or bench players (is_starter), and setting real-time availability states (active, benched, injured, or excused). While managing their team during active matches, coaches were strictly blocked by RLS from writing to match scores or events. However, under the system's **dual-role architecture**, a verified coach could simultaneously hold Facilitator privileges and utilize the scoring kiosk for *other* matches, provided they successfully cleared the automated Anti-Conflict of Interest (Anti-COI) governance check ensuring their college was not competing.

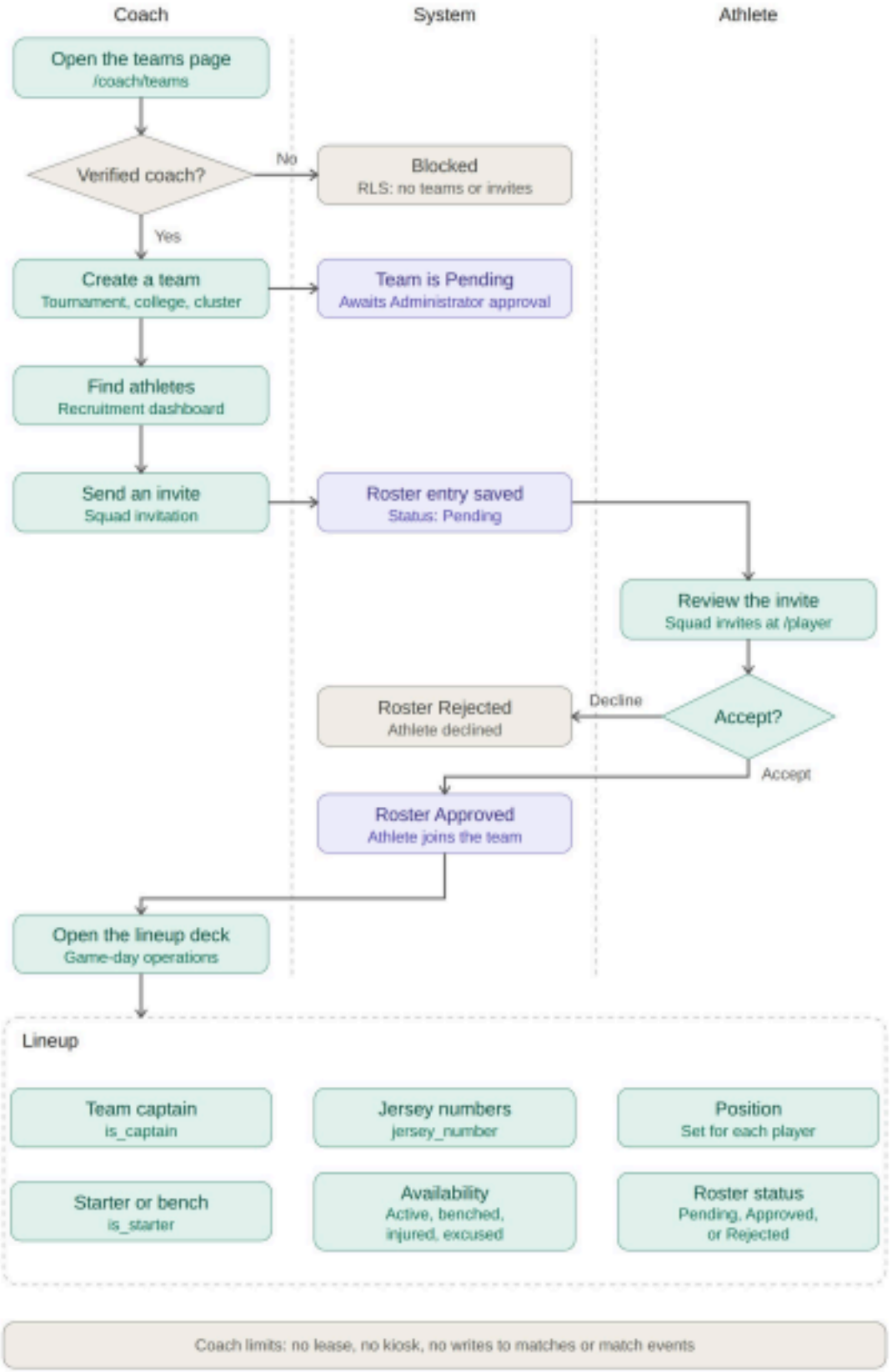


Figure 4.4. System Flowchart — Team Recruitment and Lineups

Stage 5: Live Match and Courtside Scoring

On match day, an appointed facilitator opened the kiosk at */match/:matchId/kiosk* and clicked Claim Scorekeeper Desk. Before granting access, the system ran an automated Anti-Conflict of Interest (Anti-COI) governance check, blocking the facilitator if their affiliated college matched either competing team. Once cleared, the match was updated with the facilitator's ID as *active_scorekeeper_id* and a lease that expired 90 seconds later (lease_expires_at).

The kiosk was a touch-first, zero-scroll scoring deck with physical-style buttons, and its scoring increments, periods, and timeouts followed the rules of the sport. For each scoring tap, such as +2 Field Goal for jersey #7, the kiosk inserted a row in *match_score_events* and increased the team's score on the match. The kiosk also kept team foul tallies and let the facilitator make a substitution in two taps. Undo Last Action marked the score event as undone, lowered the match score, and broadcast a negative delta.

After each change, the kiosk broadcast a score_update on the Supabase Realtime channel, and the live matchroom, the live widgets, and the stadium display updated in under 300 milliseconds. The live matchroom showed the scoreboard, period clocks, team fouls, timeouts, the commentary feed, and picture-in-picture, and the stadium display was an edge-to-edge arena view with keyframe animations and audio horns. If the Wi-Fi dropped, the kiosk buffered the actions in IndexedDB and, when the connection returned, called the *sync_offline_match_events* function to save them as one atomic batch.

At the end of the match, the facilitator or an Administrator concluded the match and awarded the Red Star to the Game MVP. A knockout match could not be

finalized while its scores were tied. When the match was finalized as Completed, the trigger *advance_bracket_on_match_completed* moved the winner to the *next_match_id* slot and the loser to the *loser_next_match_id* slot, for example the bronze match.

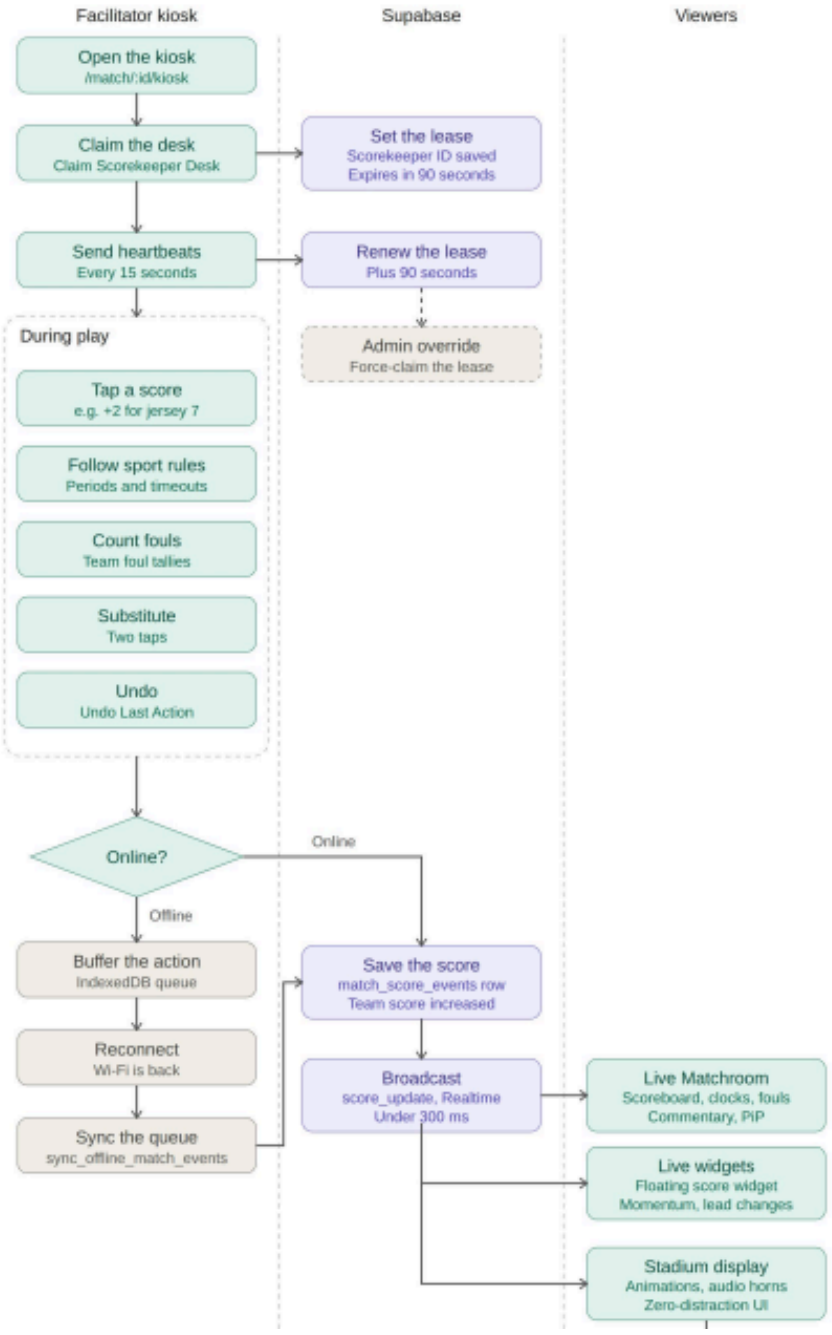


Figure 4.5. System Flowchart — Live Match and Courtside Scoring

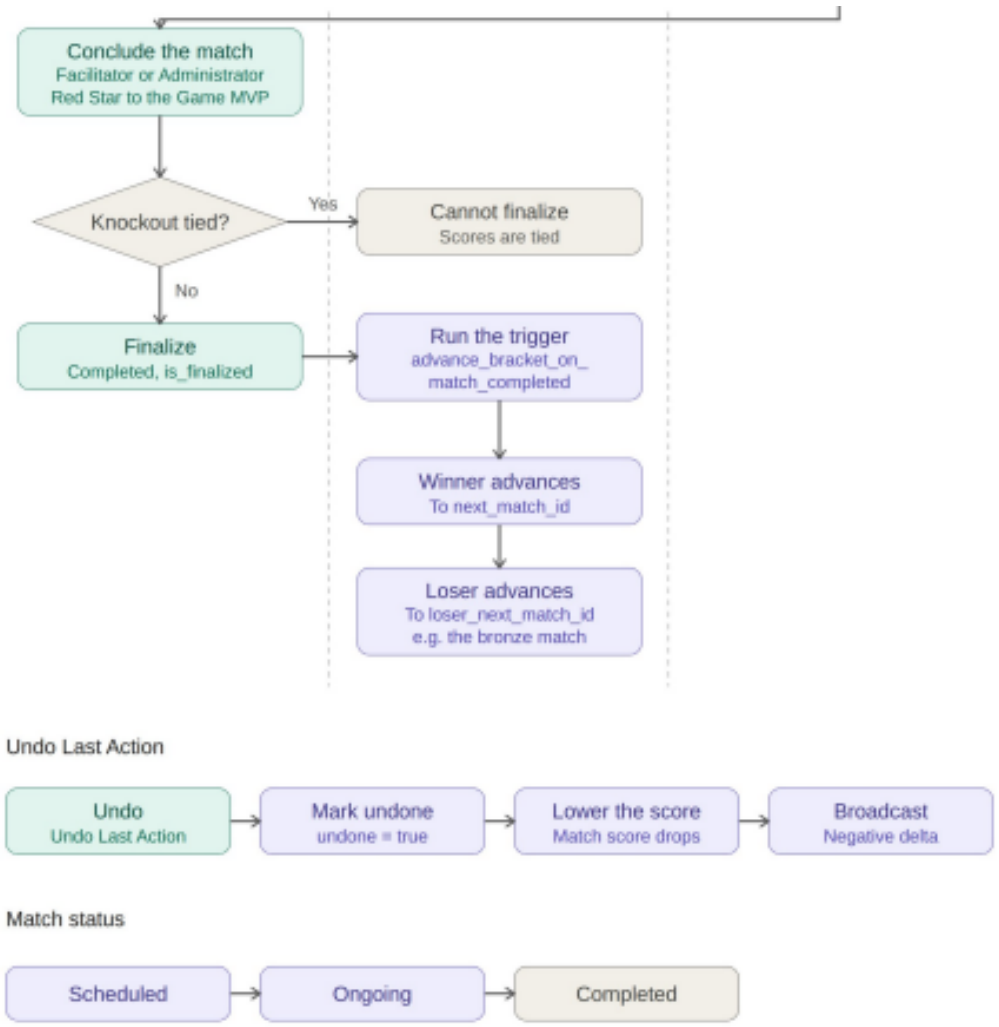


Figure 4.5. (continued). System Flowchart — Live Match and Courtside Scoring

Stage 6: Standings, Medals, and Spectator Views

Spectators did not need an account. The public portal at / showed a hero carousel, a ticker of live games, upcoming games, and the university standings. The Tournament Explorer at */explorer* showed the brackets in single, double, or round-robin form, with schedule filters and a match modal. The Athletics Hub at */athletics* offered a searchable athlete directory with player dossiers, career stats, and team rosters. The live matchroom at */match/:matchId* and the stadium display at */match/:matchId/display* showed matches in real time, featuring picture-in-picture,

floating widgets, momentum runs, lead-change banners, and procedurally generated Web Audio cues (such as horns and buzzers) synthesized directly in the browser.

When a match was completed, its result automatically fed the standings at */ranking*, which displayed college and cluster medal tallies (Gold, Silver, and Bronze) together with point differentials and win rates. The Red Star (Game MVP, awarded exclusively by a facilitator) and the Gold Star (Tournament MVP, awarded exclusively by an Administrator) appeared as permanent accolades on player profiles.

For executive oversight, the Administrator could review the immutable audit trail at */admin/audit-logs*, which securely logged the timestamp, action, actor, and JSON payload differences of each administrative event. From the recovery center at */admin/recovery*, they could restore or permanently delete soft-deleted records. Finally, utilizing the institutional reporting engine (FR-41), the Administrator could generate multi-discipline summaries of tournament outcomes, MVP records, and collegiate participation footprints, instantly exporting them as structured CSV datasets or vector-rendered A4 landscape PDF documents for university accreditation and historical archiving.

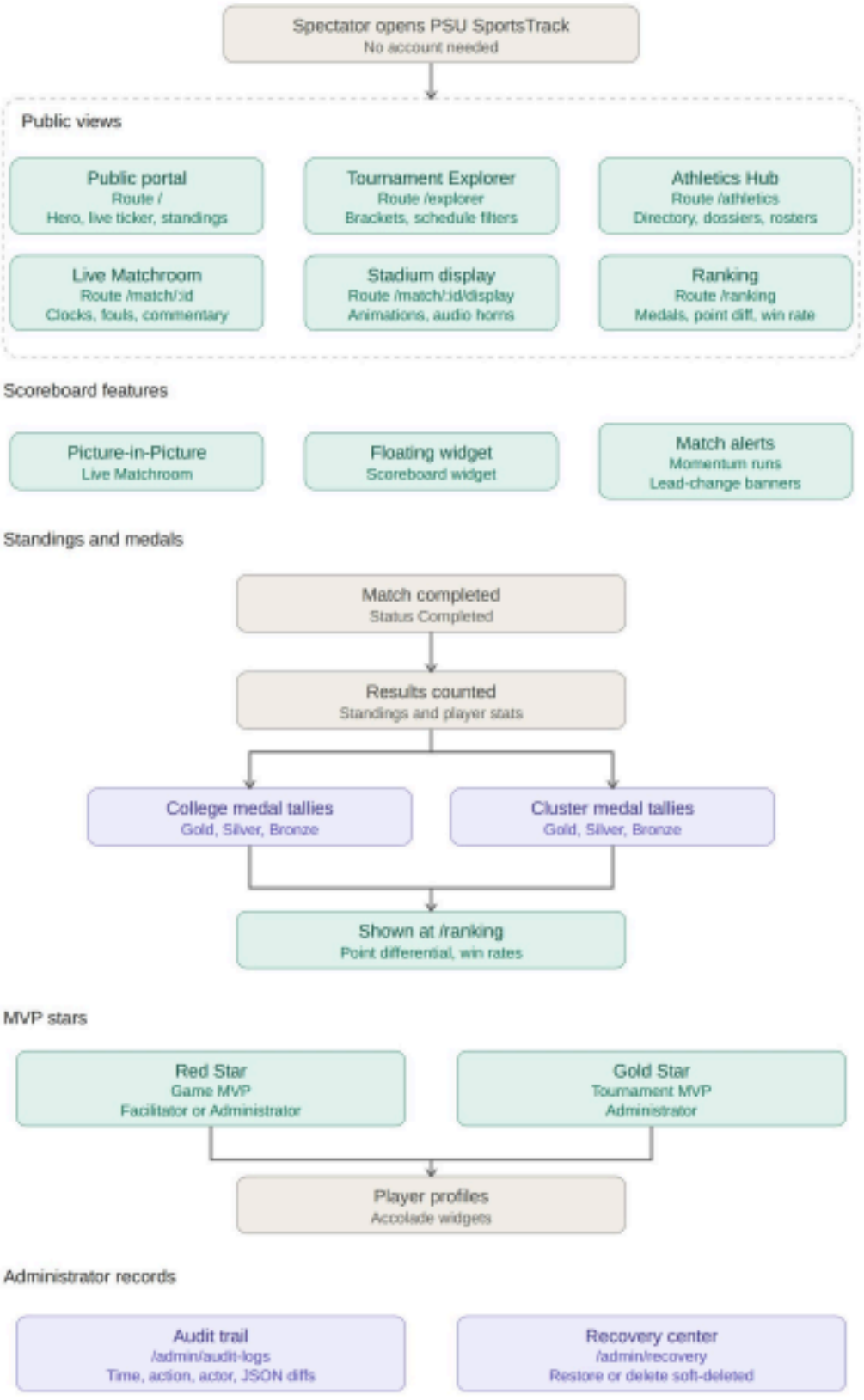


Figure 4.6. System Flowchart — Standings, Medals, and Spectator Views

Hardware and Software Specifications

The following tables list the hardware and software needed to develop and implement PSU SportsTrack. Development specifications covered what the researchers used during system construction and testing, while implementation specifications covered what was needed to run the live system within PSU.

Table 1. Hardware Specifications for Development

Specification	Uses / Purpose
Processor: Intel Core i5 (8th Gen) or equivalent	Handles compilation, local server execution, and development environment operations.
RAM: Minimum 8 GB	Supports simultaneous operation of development tools including VS Code, browser testing, and Git.
Storage: 256 GB SSD (minimum)	Provides fast read/write speeds for project files and version-controlled source code.
Display: 1920 x 1080 resolution (Full HD)	Enables clear rendering of the web interface during UI development and testing.
Internet Connection: Stable broadband (10 Mbps or higher)	Required for Git operations, npm/Supabase package installation, and online collaboration.

Table 2. Software Specifications for Development

Specification	Uses / Purpose
Visual Studio Code	Primary code editor for writing and managing the React.js and TypeScript frontend and Supabase integration code.
React.js(v19)	Frontend JavaScript library used to build the dynamic, real-time user interface of PSU SportsTrack, including the scoring consoles, kiosk and stadium display.
TypeScript	Typed superset of JavaScript that enforces strict data contracts across database models, API payloads, and form schemas to reduce runtime errors.
Vite	Build tool and development server providing fast hot module replacement and optimized production bundling.
React Router	Client-side routing library used for page navigation, protected role-based routes, and code splitting.

Specification	Uses / Purpose
Tailwind CSS + shadcn/ui	Utility-first CSS framework and accessible component library used to style the responsive, mobile-first interface with dark/light themes.
React Hook Form + Zod	Form handling and schema validation libraries used to validate registration, roster, and scoring inputs before submission.
Supabase	Backend-as-a-Service platform providing the PostgreSQL database, authentication, secure API, file storage, and real-time subscriptions.
PostgreSQL 15 (via Supabase)	Relational database that stores all system data including user accounts, rosters, tournaments, matches, scores, and audit logs, protected by Row-Level Security policies.
Supabase Storage	Secure cloud storage for athlete Certificate of Registration (COR) documents used in the verification process.
PDF.js	Library used to render uploaded COR documents in the browser for admin review without file downloads.
Browser Web APIs	Web Audio, Picture-in-Picture, Screen Wake Lock, and IndexedDB, used for courtside sound effects, floating scoreboards, display wake control, and offline score buffering.
Playwright	End-to-end testing framework used to validate multi-role workflows and live synchronization across desktop and mobile viewports.
ESLint	Static code analysis tool used to enforce coding standards and catch errors during development.
Git + GitHub	Version control system and cloud repository used for source code management and team collaboration.
Node.js + npm	Runtime environment and package manager required to install and build project dependencies.
Google Chrome / Microsoft Edge	Chromium-based browsers used for development and testing, since they fully support the Picture-in-Picture and Wake Lock APIs.
Vercel	Cloud hosting platform used to deploy and serve the web application with global CDN delivery.

Table 3. Hardware Specifications for Implementation

Specification	Uses / Purpose
Courtside Facilitator Tablet (10" or larger touchscreen, Android or iPadOS, 4 GB RAM minimum)	Runs the Facilitator Kiosk (`/match/:matchId/kiosk`) for scoring, game clock, fouls, timeouts, and 2-tap substitutions. The touchscreen supports the dual-thumb console layout.
Stadium Display Device (laptop or mini PC, Intel Core i3 or equivalent, 8 GB RAM, HDMI output)	Drives the big screen and runs the full-screen stadium display view (`/match/:matchId/display`) in fullscreen on Chrome or Edge, with Screen Wake Lock keeping it awake during play.
Projector or Large Display (1080p minimum, HDMI input)	Shows the scoreboard, timeout countdown banners, substitution alerts, and live ticker to spectators in the gym or arena.
Audio Output (powered speakers or the venue PA system, 3.5 mm or HDMI audio)	Plays the synthesized whistle, buzzer, timeout horn, and score chimes from the Web Audio FX engine at a volume audible across the court.
Administrator Workstation (desktop or laptop, 8 GB RAM, 1080p monitor)	Used by the Administrator for tournament and bracket setup, COR document review, audit logs, and the recovery center.
Coach and Athlete Devices (smartphones, Android 10 or iOS 15 and above, 3 GB RAM)	Used to manage rosters and lineups, upload COR documents, accept team invitations, and view profiles and match schedules.
Spectator Devices (any smartphone, tablet, or computer with a modern browser)	Used to follow live scores, brackets, and standings, including Picture-in-Picture and the floating score widget.
Wi-Fi Router and Access Points (dual-band, covering the court area)	Provides campus network connectivity for courtside devices, display devices, and spectators, and keeps real-time WebSocket sync running.
Internet Connection (stable broadband, 10 Mbps minimum)	Connects all devices to the Supabase cloud services (database, authentication, storage, real-time) and Vercel hosting.
Mobile Hotspot or Backup Connection (4G/5G, optional)	Failover internet for the scoring tablet if the venue network drops, alongside the app's offline scoring queue.

Table 4. Software Specifications for Implementation

Specification	Uses / Purpose
Modern Web Browser (Chrome, Firefox, or Edge)	Primary browsers for running PSU SportsTrack. They fully support Picture-in-Picture, Screen Wake

Specification	Uses / Purpose
	Lock, Web Audio, and IndexedDB, which the kiosk, stadium display, and offline queue depend on.
Windows 10/11, macOS, Linux	Operating systems for the administrator workstation and the stadium display device.
Android 10+ / iOS and iPadOS 15+)	Mobile operating systems for the courtside facilitator tablet and for athlete, coach, and spectator smartphones.
Vercel	Cloud hosting platform that serves the deployed web application through a global CDN with automatic HTTPS (TLS) encryption.
Supabase (Cloud)	Hosted backend providing the database, authentication, secure API, and real-time WebSocket updates.
PostgreSQL 15 (via Supabase)	Production database storing user accounts, rosters, tournaments, matches, scores, and audit logs, protected by Row-Level Security policies.
Supabase Auth	Handles account registration, login, password recovery, and JWT sessions that determine each user's role-based access.
Supabase Storage	Secure cloud bucket that stores uploaded Certificate of Registration (COR) documents for administrator verification.
Supabase Realtime	Pushes live scores, clock updates, and match events to the facilitator kiosk, stadium display, and spectator screens without page refreshes.
PDF.js (built into the application)	Renders uploaded COR documents inside the browser so administrators can review them without downloading files.
PSU AIMS Student Portal (external)	Official university portal that administrators cross-check against when verifying student records.
Domain Name and SSL/TLS Certificate	Gives the system a public web address and encrypts data in transit between devices and the servers.
Git + GitHub	Source code repository linked to Vercel for version-controlled, automated deployment of updates.

CHAPTER IV

METHODOLOGY

This chapter presents the research methodology employed in the development of PSU SportsTrack: A Web-Based Sports Tournament Management System for Palawan State University. It describes the System Development Life Cycle (SDLC) model adopted for the project, detailing the iterative activities and technical implementations conducted across each phase of development. Furthermore, it outlines the ethical considerations observed to ensure strict data privacy and security, and the conceptual framework that guided the system's decoupled architectural design.

Additionally, this chapter presents the comprehensive requirements analysis, use case diagrams, user stories, software requirements specification (SRS), work plan, and structural design artifacts that collectively defined the logic and behavior of the system. Finally, it outlines the user acceptance testing (UAT) and evaluation procedures conducted with university stakeholders, utilizing the ISO 25010 software quality standards to validate the platform's overall efficiency, reliability, and usability.

System Development Life Cycle (SDLC)

PSU SportsTrack was developed following the Agile Software Development Life Cycle (SDLC), a methodology that emphasizes iterative development, continuous feedback, and adaptive planning. Unlike the traditional Waterfall model, Agile allowed the development team to break the project into manageable sprints, revisit requirements as needed, and deliver functional components incrementally. This approach was particularly suited to PSU SportsTrack given the multi-module nature of the system — covering student registration and verification, coach-led team

formation, tournament bracket management, live score tracking, and institutional reporting — each of which required close coordination with the PSU Sports Department to ensure alignment with actual operational workflows.

The Agile SDLC for PSU SportsTrack proceeded through the following phases:

Planning

The planning phase began with a formal consultation with the PSU Sports Department, who served as the primary client and institutional stakeholder for PSU SportsTrack. During the interview, the Sports Department representative provided a detailed account of how sports events were managed at PSU — from the coordination between the sports director, cluster coordinators, and tournament managers, down to the manual processes involved in player registration, bracket construction, score recording, and post-event reporting. The representative confirmed that the existing manual approach created a significant administrative burden, particularly in managing physical paperwork and maintaining consistent records across seasons.

From this consultation, the researchers identified and finalized the core requirements of the system. The scope was strictly established to cover internal events: the Binturungan Intramurals and the Faculty and Staff Friendly Games across all eighteen PSU campuses. Priority features were confirmed based on the problems surfaced during the interview: digital player registration and eligibility verification, coach-led team formation through a verified player pool, tournament bracket generation and management, live score tracking to replace paper score sheets, and structured institutional report generation (FR-41) for the Sports Department. The

representative also confirmed their willingness to serve as the system Administrator and to participate in User Acceptance Testing once the prototype was completed.

Design

System architecture, database schema, wireframes, and technical specifications were prepared. The team finalized the technology stack, adopting a modern decoupled architecture. The presentation layer was designed as a React 19 and TypeScript single-page application (SPA) with Tailwind CSS to ensure a highly responsive, type-safe, and standardized user interface. For the backend, the system utilized Supabase, a Backend-as-a-Service (BaaS) platform powered by a PostgreSQL relational database. This architecture replaced the traditional three-tier client-server model by leveraging Supabase's secure API, Row-Level Security (RLS) for data isolation, and Realtime WebSocket subscriptions for instantaneous data synchronization

Development

The platform was built in iterative sprints. Each sprint addressed a specific module: user authentication (including the 6-digit OTP/magic link flows and *@psu.palawan.edu.ph* domain restriction); athlete registration and COR-based eligibility verification with automated confidence scoring for students; coach-led team formation, recruitment, and lineup management; tournament creation, automated bracket generation, and bronze playoff routing; the live match and court-side scoring suite featuring the Anti-Conflict of Interest (Anti-COI) governance check, 90-second scorekeeper lease, kiosk mode, and offline IndexedDB queue; the stadium display, floating scoreboard, and procedurally generated Web Audio effects; MVP award

designation (Gold and Red Stars); leaderboards and medal standings; the FR-41 institutional reporting engine; and an immutable audit log with a recovery center.

Testing

Each sprint was followed by functional testing. Code quality was enforced during development through static code analysis with ESLint and through TypeScript's strict data contracts, while Zod schemas validated user inputs before submission. Playwright end-to-end tests validated multi-role workflows and live synchronization across desktop and mobile viewports. Development and testing were carried out on Chromium-based browsers (Google Chrome and Microsoft Edge), which fully support the Picture-in-Picture and Screen Wake Lock features required for live matches. User Acceptance Testing (UAT) was conducted at the end of development with PSU sports administrators, coaches, facilitators, and student-athletes to evaluate overall system usability and performance against ISO 25010 quality characteristics, targeting a mean satisfaction rating of at least 4 out of 5.

Deployment

The finalized application was deployed as a React 19 SPA hosted on the Vercel edge network, which acted as a global CDN, with Supabase serving as the cloud-managed backend providing PostgreSQL database services, user authentication, and Realtime WebSocket infrastructure. The system is accessible through any modern web browser on desktop, laptop, tablet, and smartphone devices, with no software installation required on the client side.

Maintenance

Post-deployment monitoring was conducted to address bug reports, manage feedback from the Sports Department and system users, and implement future enhancements as PSU sports programs grow and their administrative needs evolve.

Ethical Considerations

The development of PSU SportsTrack was guided by a commitment to ethical software practices, with particular attention to the responsible handling of institutional and personal data, user privacy, and the security of sensitive verification documents.

- **User Privacy and Data Protection:** All collected data is stored securely within Supabase's PostgreSQL database, with strict access restricted by Row-Level Security (RLS) policies ensuring users can only view and modify data they are authorized to access.
- **Informed Consent:** Account creation requires explicit agreement to the platform's terms and utilizes a secure dual-path email verification (6-digit OTP or magic link) to ensure participation is voluntary and authenticated.
- **Responsible Handling of Verification Documents:** To protect Personally Identifiable Information (PII), uploaded CORs and employee IDs are safeguarded behind 60-second expiring signed URLs. Automated text extraction for the confidence score occurs strictly for student CORs, while faculty/staff IDs default to manual review. Documents are never exposed through permanent public links.
- **Inclusivity and Accessibility:** The interface avoids complex jargon, supports light and dark modes, and requires no proprietary software, reducing barriers for users across all eighteen campuses.

- **Security of User Accounts:** The platform enforces an institutional domain restriction requiring an *@psu.palawan.edu.ph* email. As protection against brute-force attacks, an account is locked for five minutes after three consecutive failed login attempts, managed via a dedicated database rate-limiting table.
- **Public Data Exposure:** Spectators without an account can only view published tournament schedules, live match views, and standings. Sensitive credentials, verification records, and administrative dashboards are permanently shielded from public access.
- **Accountability and Auditability:** Significant administrative and scoring actions are recorded in an immutable audit log that tracks the actor, timestamp, and JSON payload differences. Deleted records are moved to a 365-day soft-delete retention state rather than permanently erased, allowing the Administrator to restore data securely.
- **MVP Governance and Data Integrity:** To protect against officiating bias and post-match tampering, Game MVP awards are governed by strict Row-Level Security (RLS) constraints that restrict modification powers exclusively to Accredited Administrators, ensuring all overrides are permanently recorded.

User Privacy and Data Protection

PSU SportsTrack collects personal information — including names, profile data, sport participation history, and eligibility verification documents such as Certificates of Registration (COR) and valid PSU school IDs — strictly for the purpose of managing PSU sports events. All collected data is stored securely within Supabase's PostgreSQL database, with access restricted by Row Level Security (RLS)

policies that ensure users can only view and modify data they are authorized to access. The application does not share user data with parties outside the PSU institutional context.

Informed Consent

Users are informed of the data that PSU SportsTrack collects and how it will be used during the registration process. Account creation requires explicit agreement to the platform's terms and conditions, ensuring that participation is voluntary and fully informed. Students are made aware that their submitted verification documents will be reviewed by the system administrator solely for the purpose of confirming eligibility.

Responsible Handling of Verification Documents

PSU SportsTrack requires athletes to upload sensitive institutional documents — specifically the Certificate of Registration (COR) for students and faculty or employee identification for faculty and staff — as part of the eligibility verification process. The automated text extraction used to compute the COR confidence score is performed within the user’s browser. Access to the stored documents is restricted by Row Level Security and storage policies to the uploading user and the Administrator. Documents are used solely for the purpose of confirming PSU enrollment or employment and identity and are not retained or used beyond their necessary function in the verification workflow. The confidence score is only an aid; every eligibility decision is made by a human administrator

Inclusivity and Accessibility

The platform was designed to be accessible to all members of the PSU community across all eighteen campuses, regardless of technical background. The interface avoids complex jargon, uses clear navigation structures, and is accessible through any modern web browser without requiring software installation, reducing barriers for users across different campus settings.

Security of User Accounts

User passwords are encrypted using industry-standard hashing algorithms managed by Supabase's built-in authentication service. All data transmission between the client and Supabase is secured using HTTPS protocols. Role-based access controls, enforced through Supabase's Row Level Security (RLS) policies, ensure that every user type can only perform actions within its defined permission boundaries. As protection against brute-force attacks, an account is locked for five minutes after three consecutive failed login attempts, and password recovery and email confirmation are handled through token-based flows.

Public Data Exposure

Spectators without an account can only view published tournament data, such as schedules, brackets, scores, standings, and athlete names and statistics associated with published matches. Verification documents, account credentials, and administrative data are never exposed to unauthenticated users, and the database policies restrict public access to read-only operations on published records.

Accountability and Auditability

To support transparency and accountability, significant administrative and scoring actions — such as verification decisions, score corrections, and master overrides — are recorded in an immutable audit log that keeps the actor, timestamp, and data changes. Deleted records are soft-deleted rather than permanently erased, allowing the Administrator to restore them with full tracking of who performed the deletion.

Conceptual Framework

The conceptual framework of PSU SportsTrack followed the Input-Process-Output (IPO) model. It mapped what the system received from its users, how the backend architecture and browser capabilities transformed that data, and the real-time results delivered back to the community. The framework is presented in Figure 5. It features three layers, a conceptual feedback loop linking the outputs back to the inputs, and a data flow mechanism governed by the system's real-time database architecture.

Input

The input layer encompassed the data entering the system across all seven user classifications. User registration supplied the profile details, institutional email, and requested role. Document upload supplied the Certificate of Registration (COR) for students or the employee identification for faculty and staff. Tournament setup by the Administrator supplied the sport, affiliation mode (Collegiate Squad or Binturungan Tribe/Cluster), and bracket formats. Team building by Coaches supplied the recruitment invitations, rosters, and game-day lineups. Live match data supplied the points, game clocks, fouls, substitutions, and Red Star (Game MVP) designations

inputted by the Facilitator. Finally, administrative inputs included verification decisions, facilitator appointments, master score overrides, and Gold Star (Tournament MVP) awards.

Process

The process layer covered the secure operations that transformed these inputs into functional results. **Identity verification** extracted text from student CORs in the browser to compute an advisory confidence score before generating a 60-second expiring URL for Administrator review. **Bracket generation** mathematically calculated playoff fixtures and routed advancing teams via `next_match_id` linkages. **Live scoring** enforced an automated Anti-Conflict of Interest (Anti-COI) governance check before granting an exclusive 90-second scorekeeper lease. It buffered scoring events in an offline IndexedDB queue during network dropouts and supported atomic score rollbacks. **Data aggregation** autonomously computed multi-discipline medal tallies, point differentials, and MVP counts. Finally, **Authentication and Security** validated JWT sessions via a 6-digit OTP or magic link, enforced a 5-minute brute-force lockout, and strictly isolated database privileges using Row-Level Security (RLS).

Output

The output layer covered the comprehensive interfaces and datasets delivered to the users. The athlete hub and profile displayed verified team memberships and unlocked MVP accolades. The live scoreboard presented the edge-to-edge stadium display, floating widgets, Picture-in-Picture views, and procedurally generated Web Audio cues (horns and buzzers). The interactive brackets mapped the tournament's progression and live box scores. The leaderboards ranked the colleges and clusters

with a Gold, Silver, and Bronze podium. For executive oversight, the Administrator received an immutable audit trail of all system actions and the FR-41 institutional reporting engine, which instantly exported accreditation-ready multi-discipline summaries as structured CSV datasets and vector-rendered A4 landscape PDFs.

Data Flow and Feedback

Data moved seamlessly from one layer to the next through user actions and system events. The Supabase Realtime publish/subscribe engine pushed live updates to every connected screen over WebSockets, while the Row-Level Security policies and check constraints of the PostgreSQL database strictly enforced the rules of the system. The feedback loop in Figure 5 represents a conceptual link showing that users acted on the outputs—for example, a Coach recruiting from the verified pool, or a Spectator tracking a live bracket—and these resulting actions instantly became new inputs to the system. The actors driving this continuous flow were the Spectator, Student Athlete, Faculty Athlete, Staff Athlete, Coach, Facilitator, and Administrator.

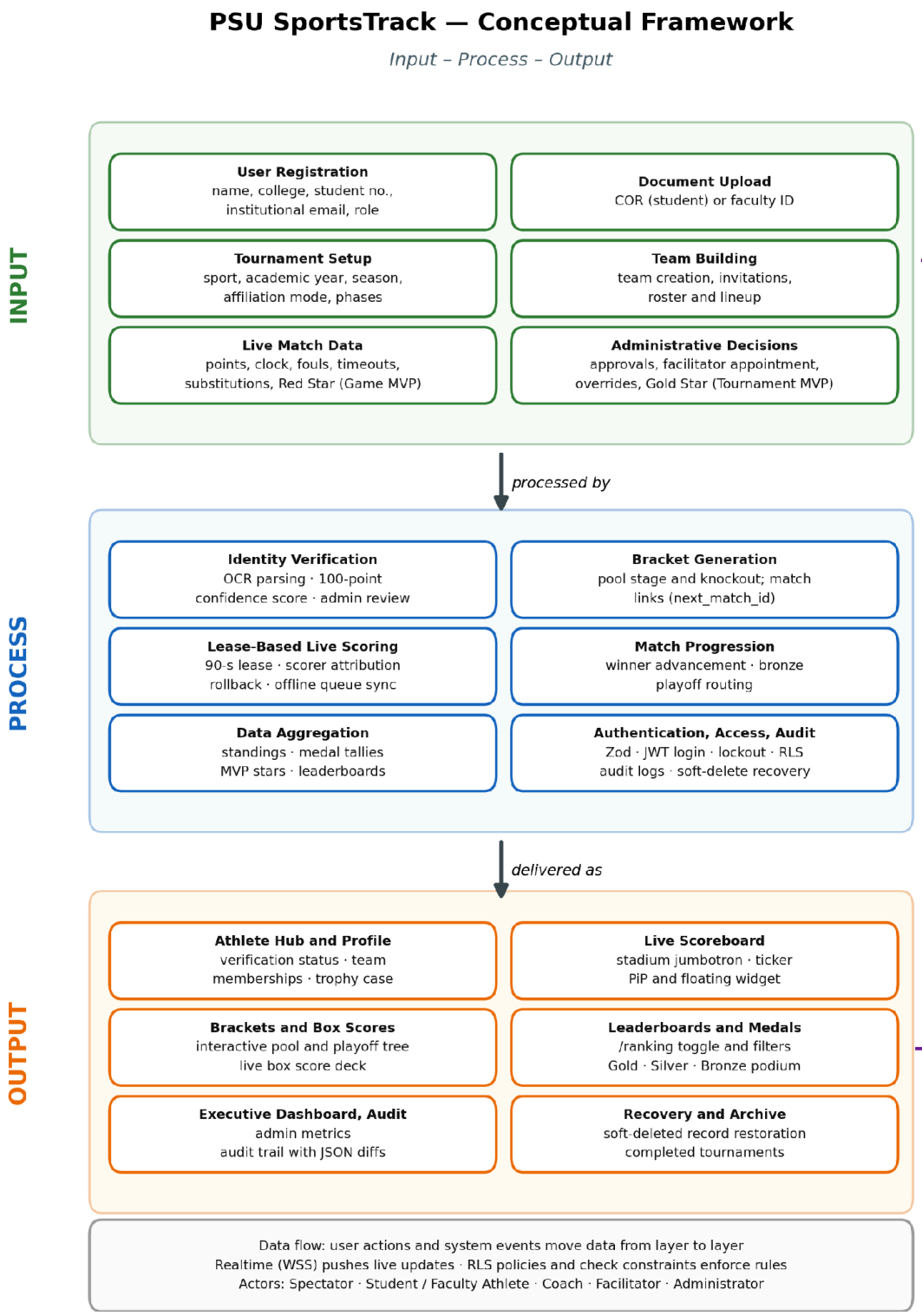


Figure 5. PSU SportsTrack - Conceptual Framework

Requirements Analysis

Requirements analysis for PSU SportsTrack involved a systematic examination of the operational needs of its target users — categorized into seven distinct classifications: Public Spectator, Student Athlete, Faculty Athlete, Staff Athlete, Coach, Facilitator, and Administrator — to define exactly what the system must do, how it must perform, and the institutional security protocols it must enforce.

The researchers identified these requirements through direct consultation with the PSU Sports Department and the university's sports coordinators, who served as the primary client and institutional stakeholders for this project. The consultation confirmed that the system must address four core operational bottlenecks: the redundancy and data privacy risks of manually collecting and storing physical requirements (CORs and employee IDs) for every sports event; the inefficiency and error-proneness of constructing tournament brackets by hand; the vulnerability to transcription delays and conflicting tallies when using traditional paper score sheets; and the informality of the coach-player recruitment process, which relied entirely on personal networks and direct communication.

The Sports Department identified several priority features to resolve these bottlenecks: a secure dual-path authentication flow (6-digit OTP or magic link); a digital athlete registration and verification workflow; structured, coach-led team formation utilizing a verified athlete pool; automated bracket engineering; and live courtside score tracking. Furthermore, they prioritized a structured MVP award system — featuring the Gold Star (awarded exclusively by the Administrator) and the Red Star (awarded exclusively by the Facilitator) — alongside the integration of an

institutional reporting engine (FR-41) to instantly generate and export multi-discipline summaries, standings, and participation footprints for university accreditation. These priorities directly informed the functional modules developed in the system. The gathered requirements were then modeled as use cases, user stories, and formal functional and non-functional specifications.

Following the Agile methodology, these requirements were continuously refined during successive sprint reviews as the prototype was demonstrated. Through this iterative feedback process, critical architectural enhancements were incorporated to meet institutional demands: a dedicated Facilitator role governed by an automated Anti-Conflict of Interest (Anti-COI) mechanism; a unique dual-role architecture allowing verified athletes and coaches to officiate matches without data collision; 60-second expiring signed URLs to protect sensitive uploaded documents; an advisory confidence score applied exclusively to student COR submissions; an offline IndexedDB buffer to preserve scoring actions during venue network dropouts; procedurally generated Web Audio cues for live spectator displays; and a 365-day soft-delete retention state supported by an immutable audit log to guarantee the absolute integrity of university sports records.

Use Case Diagram

The use case diagram illustrates how each actor interacts with the PSU SportsTrack web system. The system recognizes **seven** distinct production user classifications: the Public Spectator, the Student Athlete, the Faculty Athlete, the Staff Athlete, the Coach, the Facilitator, and the Administrator. Because the comprehensive nature of the system makes the diagram large, it is presented in three parts: Figure 6.1

for the Spectator, the abstract Registered User, and the abstract Player; Figure 6.2 for the Coach and the Facilitator; and Figure 6.3 for the Administrator.

The actors are structurally related through UML inheritance. The Registered User is an abstract actor that inherits every basic Spectator use case, while the Student Athlete, Faculty Athlete, Staff Athlete, Coach, Facilitator, and Administrator all inherit from this Registered User. Likewise, the Student Athlete, Faculty Athlete, and Staff Athlete specifically inherit from an abstract Player actor. (Furthermore, to reflect the platform's dual-role architecture, a single verified user profile may simultaneously inhabit both a primary actor role, such as a Player, and the Facilitator actor role without data collision).

Within the diagrams, a dashed arrow marked *include* indicates that a base use case strictly requires the execution of another use case to be completed. A dashed arrow marked *extend* indicates that a use case adds optional or conditional behavior to a base use case under specific circumstances.

Public, Registered User, and Player

The Spectator was an unauthenticated visitor with read-only access. A Spectator could view the live dashboard and ticker, explore tournaments and brackets in the Tournament Explorer, access the searchable athlete directory in the Athletics Hub, view a live matchroom, view the edge-to-edge stadium display with its procedurally generated Web Audio cues, view the college and cluster medal standings at /ranking, and read the platform's About information. A Spectator could also initiate account registration for a student, faculty, staff, or coach role, but was strictly blocked by database policies from using scoring controls, submitting team applications, or accessing protected verification files.

The Registered User actor encompassed the baseline use cases shared by every authenticated account: signing in via secure dual-path verification (6-digit OTP or direct magic link), adhering to the strict *@psu.palawan.edu.ph* domain restriction, enduring a five-minute lockout after three failed sign-in attempts, resetting a forgotten password through an emailed PKCE token link, and viewing their profile. Because the Registered User inherited from the Spectator, every authenticated user could also utilize all public views.

The Student Athlete uploaded a Certificate of Registration (COR). This use case was extended by computing an automated OCR confidence score via `PDF.js` entirely within the browser, giving the athlete an instant preview of their document's readability before submission. The Faculty Athlete and Staff Athlete uploaded an official employee identification instead, which bypassed the automated score for direct manual review. All three actors inherited the abstract Player use cases: viewing the athlete console and live verification status (Unverified, Pending, Approved, or Rejected), responding to team recruitment invitations by accepting or declining, viewing squads and schedules (including assigned jersey numbers, captain designations, and upcoming fixtures), and viewing their MVP trophy case featuring personal statistics alongside Red Star and Gold Star honors.

While operating strictly under the Player role, athletes could not create teams, recruit other players, or access coach and administrator routes. Furthermore, a Player could not operate the scoring clock for their own games; however, under the system's unique dual-role architecture, a verified Player could simultaneously hold Facilitator privileges to claim the scorekeeper lease for *other* matches, provided they cleared the Anti-Conflict of Interest (Anti-COI) governance check.

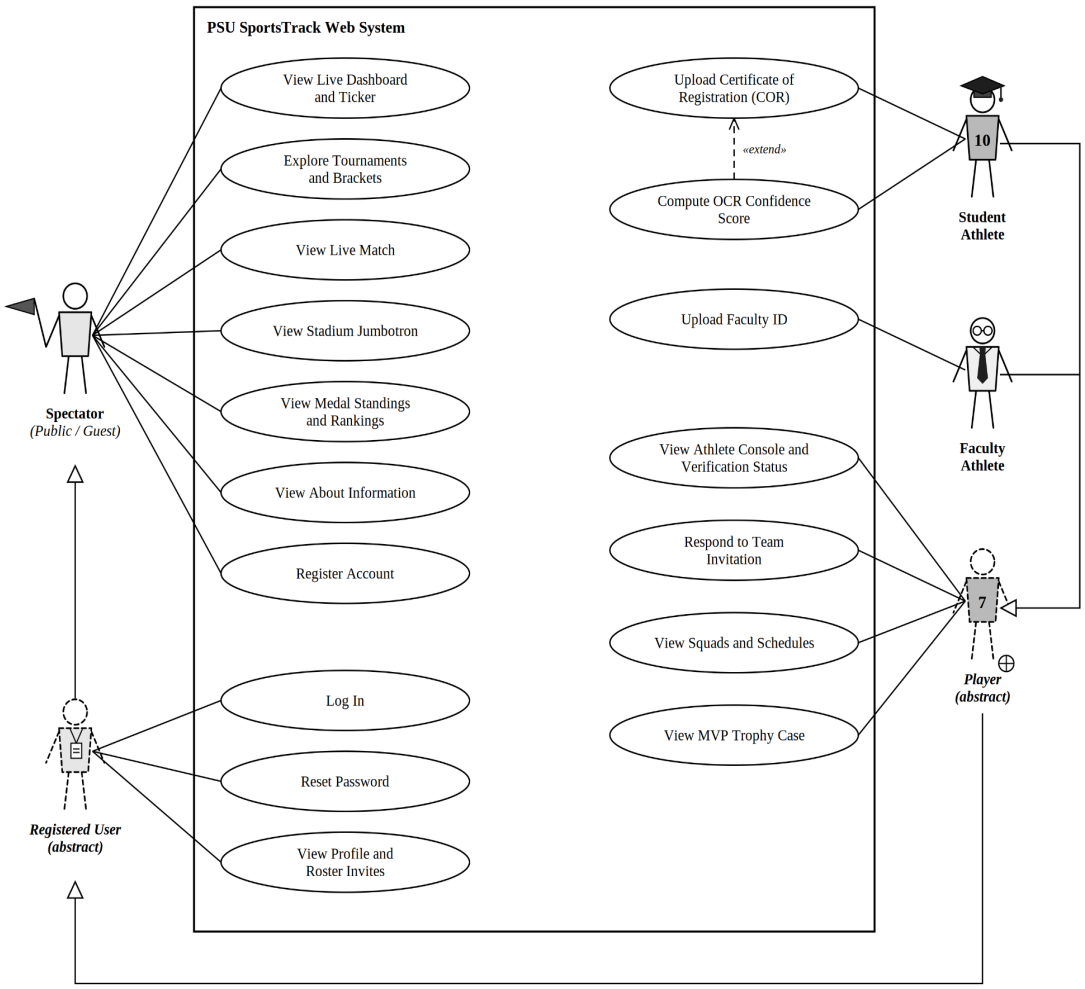


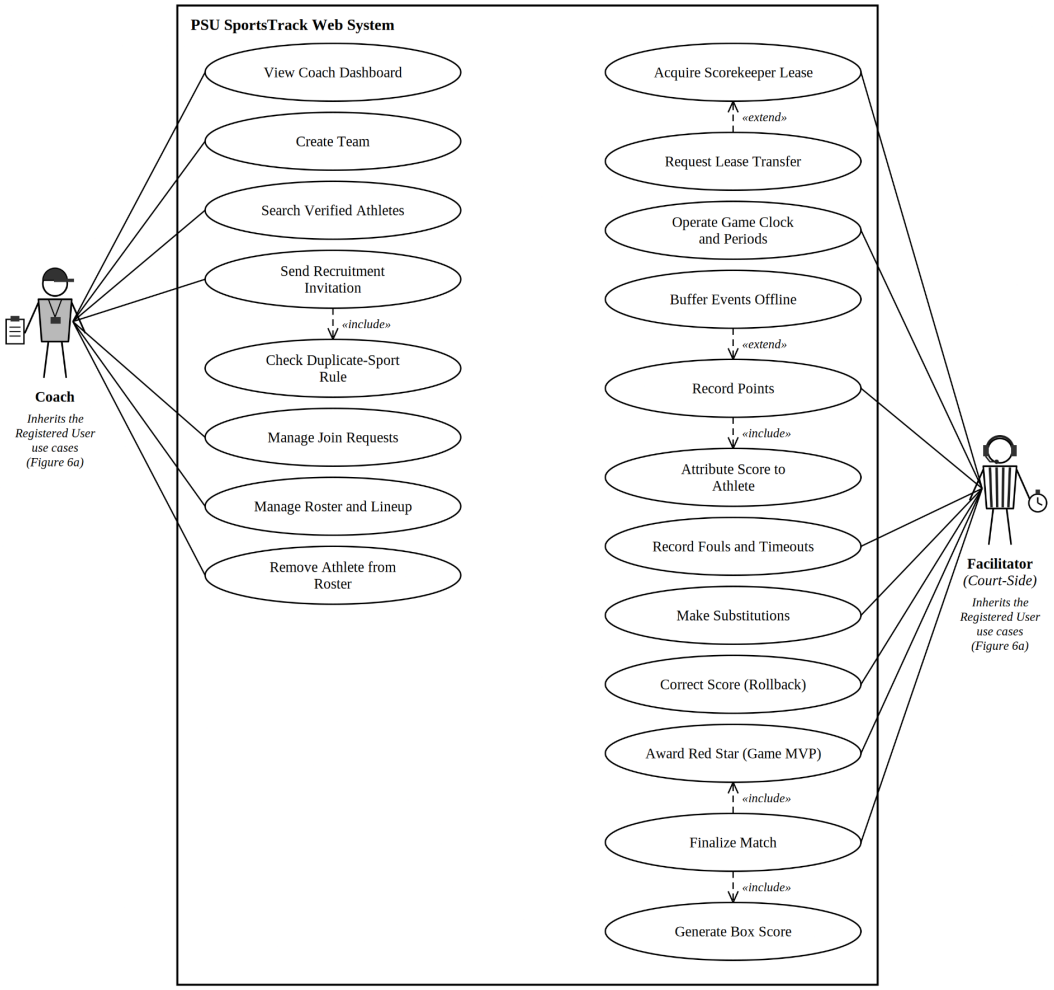
Figure 6.1. Use Case Diagram — Public, Registered User, and Player

Coach and Facilitator

The Coach managed teams and rosters but had no role in scoring. A Coach viewed the coach dashboard with win and loss records, upcoming games, and pending rosters, and created a team for a tournament under a college department or tribal cluster. The Coach searched the verified athlete pool of the assigned college and sent recruitment invitations. Sending an invitation always included the duplicate-sport check, which blocked an athlete who already played the same sport for another team. The Coach accepted or rejected join requests, managed the roster and lineup, which covered unique jersey numbers from #1 to #99, team captains, starters and bench

players, formal positions, and availability (Active, Injured, or Excused), and removed ineligible or dropped athletes from the roster. Coaches were blocked from scorekeeping and clock control by client checks and Row-Level Security, so they could not generate brackets, approve verification documents, or view audit logs.

The Facilitator was the appointed court-side official who operated the match. Before a Facilitator could acquire the scorekeeper lease, the system ran an automated Anti-Conflict of Interest (Anti-COI) governance check, blocking the facilitator if their affiliated college matched either competing team. When another scorekeeper held the lease, the Facilitator could extend this use case by requesting a lease transfer. The Facilitator operated the game clock and periods, recorded points, and recorded fouls and timeouts. Recording points always included attributing the score to an athlete by jersey number, and it could be extended by buffering events offline via IndexedDB when the venue network dropped. The Facilitator also made two-tap substitutions and corrected scores through an atomic rollback that kept scores at zero or higher and wrote an audit entry. At the end of the match, the Facilitator finalized the match, which always included awarding the Red Star (Game MVP) and generating the official box score. A Facilitator could not access the administrator pages, create or change tournaments and brackets, or approve and reject verification documents.



Coaches are blocked from scorekeeping and clock control by client checks and Row-Level Security.

Figure 6b. Use Case Diagram — Coach and Facilitator

Figure 6.2. Use Case Diagram — Coach and Facilitator

Administrator

The Administrator held full executive oversight of the system. The Administrator viewed the executive dashboard with real-time counters for athletes, coaches, active tournaments, live matches, and pending verifications. In the verification desk, the Administrator reviewed verification documents. This use case always included inspecting the document in the secure in-browser PDF viewer alongside the OCR confidence score, and approving or rejecting the verification. It could be extended by cross-checking the details in the PSU AIMS Student Portal, an

external system reached through a deep link. An approval set the athlete as verified, and a rejection recorded the specific reason.

The Administrator also appointed or revoked facilitators, updated or soft-deleted user accounts, and created and configured tournaments with sport rules, divisions, and formats. The Administrator generated automated brackets and seeds, scheduled games, venues, and facilitators, finalized the standings, and awarded the college and cluster medals, alongside the Gold Star (Tournament MVP) award. A tournament could be deleted only through a two-step confirmation that required typing a safety phrase.

For control during live play, the Administrator held a master override to bypass a scorekeeper lease and directly adjust scores in any live match. For institutional oversight, the Administrator reviewed the immutable audit logs, which securely recorded administrative actions with timestamps and JSON payload differences, and restored soft-deleted tournaments, teams, matches, or users through the recovery center. Finally, the Administrator utilized the institutional reporting engine (FR-41) to instantly generate and export structured multi-discipline summaries, standings, and MVP records as CSV datasets and PDF documents for university accreditation. The Administrator had no restrictions, possessed access to every Player, Coach, and Facilitator route, and inherited all Registered User use cases.

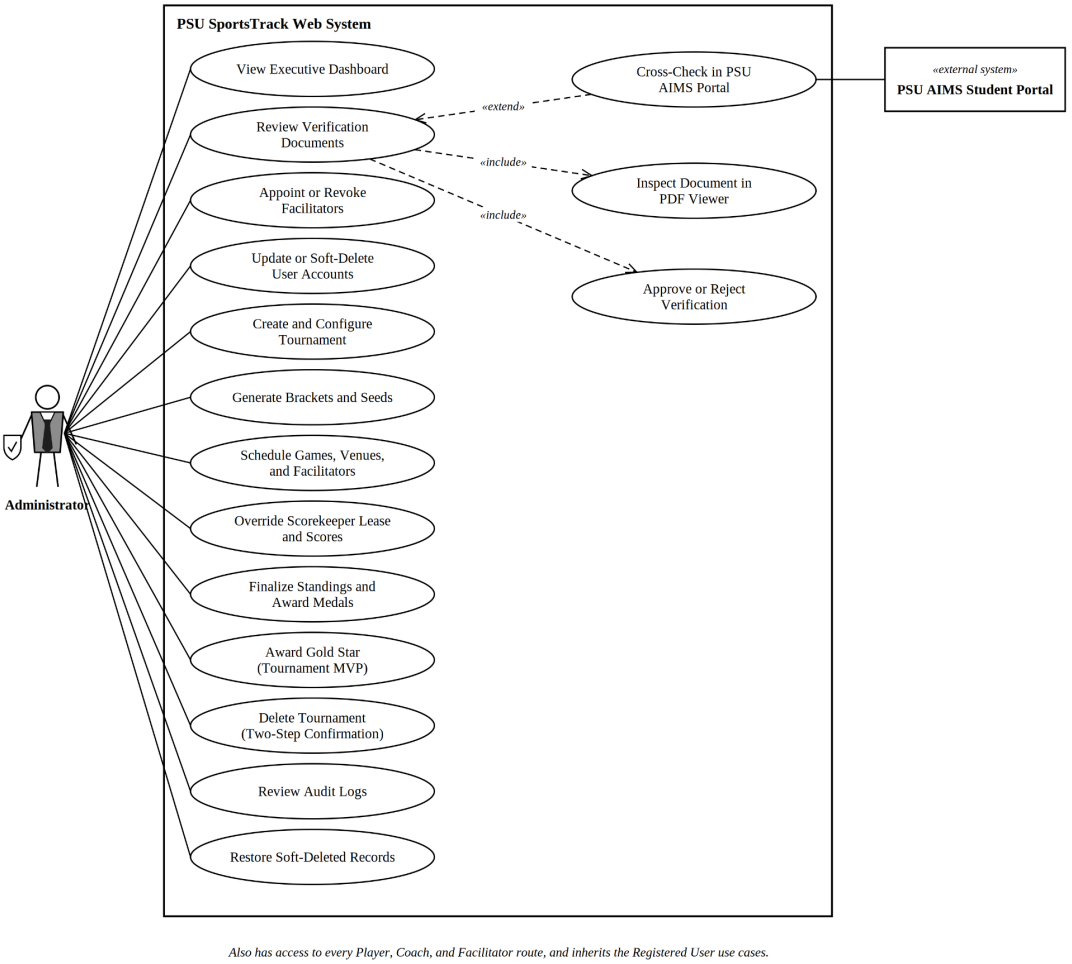


Figure 6.3. Use Case Diagram — Administrator

User Stories

User stories serve as a fundamental tool in defining the functional requirements of PSU SportsTrack from the perspective of its intended users. Rather than describing system features in purely technical terms, user stories capture what each type of user needs to accomplish within the platform and why that functionality matters to them. This approach ensures that the development of PSU SportsTrack remains grounded in the actual operational needs of the PSU sports community throughout the entire development process.

Each user story follows the standard agile format: "As a [type of user], I want [functionality], so that [benefit]." Accompanying each story is a set of acceptance criteria written in the Given-When-Then format, which defines the specific conditions that must be satisfied for a feature to be considered complete. These criteria also served as the basis for test cases used during the User Acceptance Testing (UAT) phase of development.

The user stories presented in this section cover all major functional modules of PSU SportsTrack, including user registration and authentication, athlete eligibility verification (for students, faculty, and staff), coach-led team formation, tournament and bracket management, live score tracking, MVP award designation, and institutional reporting. They are organized by the seven recognized user classifications (Public Spectator, Student Athlete, Faculty Athlete, Staff Athlete, Coach, Facilitator, and Administrator) to maintain clarity and traceability across the documentation.

User Roles

User Role	Description
Public Spectator	An unauthenticated visitor, fan, or PSU student who views published schedules, brackets, live match scoreboards, and medal standings in read-only mode. A spectator cannot score, submit applications, upload files, or vote.
Student Athlete	A registered PSU student who submits their profile and Certificate of Registration, responds to team recruitment invitations, and competes through coach-assembled rosters. They can view tournament schedules, live results, and unlocked Gold Star or Red Star MVP designations on their profile. Under the system's dual-role architecture, a verified student athlete may concurrently serve as a Facilitator for matches not involving their affiliated college.
Faculty Athlete	A registered PSU teaching personnel who competes in the Faculty and Staff Friendly Games. A faculty athlete possesses the exact same platform capabilities as a student athlete—including unlocking MVP

	designations and holding concurrent Facilitator privileges for unaffiliated matches—except that their eligibility verification is based on an uploaded faculty identification instead of a COR.
Staff Athlete	A registered PSU non-teaching personnel who competes in the Faculty and Staff Friendly Games. A staff athlete possesses the exact same platform capabilities as a student athlete—including unlocking MVP designations and holding concurrent Facilitator privileges for unaffiliated matches—except that their eligibility verification is based on an uploaded employee identification instead of a COR.
Coach	A registered user responsible for creating teams, recruiting verified athletes, and managing rosters and game-day lineups. PSU SportsTrack recognizes coaches who manage teams representing a college department or a Binturungan tribe or cluster in intra-university events. Coaches have read-only access to live matches, unless concurrently acting as a Facilitator for a match that does not involve their affiliated college.
Facilitator	A court-side match official appointed by the Administrator who claims the exclusive scorekeeper lease for an assigned match, strictly governed by an automated Anti-Conflict of Interest (Anti-COI) check. The Facilitator operates the game clock, scoring (supported by offline buffering), fouls, timeouts, substitutions, and match finalization, which includes awarding the Red Star Game MVP.
Administrator	The platform manager, represented by the PSU Sports Department, responsible for verifying athlete documents, managing user accounts and facilitators, engineering tournaments and brackets, overriding live matches when necessary, finalizing standings and medals, awarding the Gold Star (Tournament MVP), reviewing audit logs, restoring soft-deleted records, and generating institutional reports.

Table 5. User Roles

Role-Based Access

Access to every part of the system depends on the role of the signed-in user. Table 6 summarizes the routes available to each user type. Access is enforced both in the user interface and in the database through Row Level Security policies, so that restricted actions are blocked even if the interface is bypassed.

Role-Based Route Access Matrix

Route	View/Function	Public	Student	Staff	Faculty	Coach	Facilitator	Admin
/	Live dashboard and ticker	Yes	Yes	Yes	Yes	Yes	Yes	Yes
/explorer	Tournament explorer and brackets	Yes	Yes	Yes	Yes	Yes	Yes	Yes
/ranking	University medal standings	Yes	Yes	Yes	Yes	Yes	Yes	Yes
/about	Platform rules and information	Yes	Yes	Yes	Yes	Yes	Yes	Yes
/match/:matchId	Live match room (viewer mode)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
/match/:matchId	Live match room (scorekeeper desk)	No	No	No	No	No	Yes	Yes

/player	Athlete console and document upload	No	Yes	Yes	Yes	No	No	Yes
/profile	Profile, invitations, and achievements	No	Yes	Yes	Yes	Yes	Yes	Yes
/coach	Coach dashboard and squad statistics	No	No	No	No	Yes	No	Yes
/coach/teams	Roster and lineup management	No	No	No	No	Yes	No	Yes
/admin	Executive analytics and facilitator appointment	No	No	No	No	No	No	Yes
/admin/tournaments	Bracket engineering and seeding	No	No	No	No	No	No	Yes
/admin	Docum	No	No	No	No	No	No	Yes

/verifications	ent review and AIMS cross-check							
/admin/audit-logs	Security and event audit trail	No	No	No	No	No	No	Yes
/admin/recovery	Soft-deleted record restoration	No	No	No	No	No	No	Yes

Note on Dual-Role Architecture: *PSU SportsTrack employs an additive role-based access control (RBAC) model. The Facilitator operates as a concurrent sub-role. If a verified Student Athlete, Faculty Athlete, Staff Athlete, or Coach is appointed as a Facilitator by the Administrator, their account inherits a union of privileges—granting them the "Yes" access of both their primary base role and the Facilitator role simultaneously, governed by the Anti-COI check.*

Table 5.1. Role-Based Route Access Matrix

Public Spectator User Stories

US No.	As a...	I want to...	So that...	Acceptance Criteria
US-P01	Spectator	view a live dashboard and ticker of ongoing matches, upcoming schedules, and college standing summaries without	I can follow PSU sports events in real time	Given I am on the home page, When live matches or upcoming games exist, Then the live ticker, schedules,

		logging in		and standing summaries are displayed.
US-P02	Spectator	search and filter tournaments by sport, college, and status, and view their brackets, schedules, and past box scores	I can find and follow the competitions I am interested in	Given I am on the Tournament Explorer, When I apply a sport, college, or status filter, Then only matching tournaments are shown with their brackets, schedules, and box scores.
US-P03	Spectator	watch a live match scoreboard with the period clock, play-by-play log, team fouls, and star leaderboard	I can follow a game as it happens from any device	Given a match is in progress, When I open the match page, Then the score, clock, event log, fouls, and star leaderboard update in real time without refreshing the page.
US-P04	Spectator	view the university leaderboard showing Gold, Silver, and Bronze medal tallies, overall points, and historical standings	I can see how colleges and tribes rank across sports	Given I am on the ranking page, When I switch between overall, collegiate, and tribe or cluster standings, Then the corresponding medal tally and points are displayed.
US-P05	Spectator	create an account, log in, and recover my password	I can participate as an athlete, coach, or voter	Given I am on the registration page, When I submit valid details including an @psu.palawan.edu.ph email, Then my account is created. Given I am on the login page, When I submit valid credentials, Then I am signed in. Given I

				forgot my password, When I request a reset, Then a reset link is sent to my email.

Table 5.2. Public Spectator User Stories

Player (Student Athlete, Faculty Athlete, and Staff Athlete) User Stories

The following user stories define the features and functionalities accessible to Students — registered PSU students, faculty members, and staff who participate in sports events managed through PSU SportsTrack.

US No.	As a...	I want to...	So that...	Acceptance Criteria
US-S01	Player	register my profile with my personal information, campus, college or department, student number, or faculty/staff details, and sport preferences	I can be discoverable by coaches and processed for eligibility verification	Given I am on the profile registration page, When I complete all required fields and submit, Then my profile is saved and I am directed to the document submission step.
US-S02	Player	upload my Certificate of Registration (students) or official employee identification (faculty and staff) and see an instant confidence score preview for my COR	the administrator can verify that I am a bona fide and currently enrolled or employed member of the PSU community, and I can correct a poor upload before submitting	Given I am on the document submission screen, When I upload my COR PDF as a student, Then the system displays an extracted-data and confidence score preview. When I upload an employee ID as a faculty or staff member, Then the system bypasses the automated score. When I submit my document, Then my verification status

				becomes Pending and the administrator is notified for review.
US-S03	Player	receive a notification and see my verification status (Unverified, Pending, Approved, or Rejected) with the administrator's feedback notes	I know whether I am eligible to participate and what to fix if I was rejected	Given the administrator has reviewed my documents, When they approve or reject my verification, Then I receive an in-system notification and my status and feedback notes are displayed on my athlete console.
US-S04	Player	view the tournament schedule, bracket, and recorded game results for events I am participating in	I stay informed about my upcoming games and can track how the tournament is progressing	Given I am on the Tournament Explorer or my specific event page, When I open the bracket or schedule view, Then the current bracket, game schedule, and recorded results are displayed.
US-S05	Player	receive, review, and accept or decline team invitations sent by coaches	I can choose the team I will compete for	Given a coach has invited me, When I open my invitations and accept, Then I am added to the team roster. When I decline, Then the invitation is closed and the coach is informed. Given I already belong to a team in the same sport, Then I cannot accept another invitation for that sport.
US-S06	Player	view my confirmed team memberships,	I know my role in the team and when I play	Given I am a member of a team,

		jersey number, captain designation, and upcoming match schedule		When I open my athlete console, Then my team, jersey number, captain status, and upcoming matches are displayed.
US-S07	Player	view my participation history, points, matches played, MVP awards, and unlocked achievement badges on my profile	I can track my sports engagement and recognize my achievements	Given I am viewing my profile, When I open my statistics and honors, Then my chronological participation list, personal statistics, and any Gold Star or Red Star designations and badges are displayed. .
US-S08	Player	cast a star vote for an active athlete during a live match	I can show support and help recognize outstanding performers	Given I am signed in and a match is live, When I vote for an athlete on the match page, Then my vote is recorded and reflected on the star leaderboard. Given I am not signed in, Then I am asked to log in first.
US-S09	Player	hold concurrent privileges as a court-side match Facilitator for unaffiliated games while maintaining my active athlete status	I can officiate university sports events without violating conflict-of-interest (Anti-COI) rules	Given I am a verified player (specifically a staff or faculty athlete), When the Administrator appoints me as a Facilitator for a match that does not involve my own department or cluster, Then my account inherits the scorekeeper desk privileges for that game. Given I attempt to access the scorekeeper desk for

				a match involving my own department, Then the automated Anti-COI check blocks my access.

Table 5.3. Player (Student Athlete and Faculty Athlete) User Stories

Coach User Stories

The following user stories define the features and functionalities available to Coaches — registered users responsible for building and managing team rosters for PSU sports events.

US No.	As a...	I want to...	So that...	Acceptance Criteria
US-C01	Coach	browse and search verified athletes within my assigned college and view their profiles	I can identify suitable participants without relying on personal networks	Given I am on the recruitment desk, When I search the verified pool, Then only eligible athletes affiliated with my assigned college or cluster are displayed along with their profiles and sport preferences.
US-C02	Coach	create a team under a college department or a Binturungan tribe or cluster for a specific tournament	my team can be officially formed for the event	Given I am on the team creation screen, When I input the team name, select a tournament, and select my assigned affiliation (college or cluster), Then the team is created and listed under my coach console. Given I attempt to select an affiliation I am not officially assigned to, Then the system restricts the selection.
US-C03	Coach	send recruitment invitations to verified	I can assemble a complete and	Given I select a verified athlete, When

		athletes and accept or reject athlete join requests	structured team	I send an invitation, Then the athlete is notified and the invitation is logged as pending. Given an athlete already belongs to a team in the same sport, Then the system prevents the invitation. Given an athlete requests to join my team, When I accept or reject the request, Then their membership status updates to Approved or Rejected, and the team roster is updated accordingly.
US-C04	Coach	assign jersey numbers (1 to 99) and positions, designate a team captain, mark starters and bench players, and set athlete availability (Active, Injured, or Excused)	my lineup is ready for game day	Given an athlete is on my roster, When I update their lineup details, Then the changes are saved and visible on the team roster and the live match lineup. Given I attempt to assign a duplicate jersey number to another athlete on the team, Then the system prevents the assignment. Given I mark an athlete as Injured or Excused, Then they cannot be marked as a starter for the live match.
US-C05	Coach	claim an available team slot by submitting my team roster during a tournament's Slotting phase	my team is officially entered into the tournament	Given a tournament is in the Slotting phase and open slots remain, When I submit a valid team roster to claim a slot, Then my team is assigned to that slot and the remaining slot count updates in real time for all coaches

				viewing the tournament. Given the tournament slots are full or the tournament is no longer in the Slotting phase, When I attempt to submit a roster, Then the system prevents the submission.
US-C06	Coach	add or remove athletes from my roster while the tournament is still in the Slotting phase	my team accurately reflects the participants who will compete before the tournament locks	Given my team has claimed a slot and the tournament has not yet been Launched, When I add or remove an athlete within the sport's defined roster limits, Then the change is applied immediately. Given I attempt to add an athlete that exceeds the maximum roster limit, Then the system blocks the addition. Given the tournament has been Launched, When I attempt to edit my roster, Then the system prevents the edit and displays a message that the roster is locked.
US-C07	Coach	view the bracket, game schedule, and live matches of events my team is registered in, in read-only mode	I can plan my team's preparation without influencing the scoring	Given my team is registered in a tournament, When I open the tournament or a specific live match page, Then the bracket, schedule, and live scores are displayed and update in real time. Furthermore, the scorekeeper and game clock controls are strictly hidden and disabled for my role.

Table 5.4. Coach User Stories

Administrator User Stories

The following user stories define the features and functionalities available to the Administrator — the platform manager responsible for overseeing all system content, managing tournament operations, and generating institutional reports for the PSU Sports Department.

US No.	As a...	I want to...	So that...	Acceptance Criteria
US-A01	Admin	log in to the administrator dashboard with secure credentials	I can access all platform management tools and begin overseeing system operations	Given I am on the admin login page, When I enter valid administrator credentials and submit, Then I am authenticated and redirected to the admin dashboard. Given I enter invalid credentials or use an account without Administrator privileges, When I attempt to log in, Then access is strictly denied and an error message is displayed.
US-A02	Admin	review submitted student verification documents (COR and school ID) and approve or reject each submission	only bona fide PSU students, faculty, and staff are permitted to participate in university sports events	Given an athlete has submitted their verification documents, When I review the submission on the verification management screen, Then I can view the uploaded document alongside the OCR extracted data and confidence score (if applicable). When I approve or reject the submission and enter my feedback notes, Then the athlete's account

				status is updated immediately and they receive an automated notification with my feedback.
US-A03	Admin	view and manage all registered user accounts on the platform	I can monitor user activity and take appropriate action on accounts when necessary	Given I am on the user management screen, When I search for or select a user account, then I can view their details, verification status, and participation history, and perform management actions such as suspending or deactivating the account.
US-A04	Admin	create a new tournament by specifying its name, event type, sport, and format	I can establish the initial setup of a competition before opening it for team registration	Given I am on the tournament creation screen, When I complete the required fields and save, Then the tournament is created with a status of "Draft" and is not yet visible to coaches.
US-A05	Admin	open a Draft tournament for slot drafting by setting the number of available slots, eligible colleges, and a registration deadline	coaches can begin claiming slots and submitting their team rosters	Given a tournament is in Draft status, When I configure slot settings and open registration, Then the tournament status updates to "Slotting" and becomes visible to coaches for team registration.
US-A06	Admin	review, approve, reject, or reassign submitted team slots during the Slotting	I can ensure only eligible and properly formed teams are	Given a tournament is in the Slotting phase and teams

		phase	entered into the tournament	have claimed slots, When I approve, reject, or reassign a team's slot, Then the change is applied immediately, and a rejected team's slot is released back to the available pool with the coach notified.
US-A07	Admin	launch a tournament once slot drafting has closed	the bracket is generated and the tournament becomes active for scheduling and live play	Given a tournament is in the Slotting phase and the deadline has passed or all slots are filled, When I trigger the launch, Then all team registrations are locked, the bracket is automatically generated, and the tournament status updates to "Launched."
US-A08	Admin	make an emergency correction to a team's roster after a tournament has launched	genuine errors can still be fixed without reopening the entire registration process	Given a tournament is Launched and its first scheduled game has not yet started, When I edit a team's roster on behalf of a coach, Then the change is saved and reflected in the system. Given the first game has already started, When I attempt to edit the roster, Then the system prevents the change.
US-A09	Admin	record game results and update the bracket as each match concludes	the tournament standings and bracket progression are accurately maintained throughout the event	Given a game has concluded, When I enter the final score and result on the result entry screen,

				Then the bracket and standings are updated immediately to reflect the outcome.
US-A10	Admin	enter score updates into the live scoring feed during active matches	all students, coaches, and viewers have real-time visibility into ongoing match scores without needing to refresh the page	Given a game is in progress, When I enter a score update on the live score management screen, Then the live scoring feed reflects the updated score immediately for all users viewing that match.
US-A11	Admin	designate a Red Star award to the MVP of an individual game upon its conclusion	outstanding performances within a single match are formally recognized within the platform	Given a game has concluded and I have entered the result, When I designate a Red Star MVP on the result entry screen, Then the award is recorded and displayed on the recipient's student profile.
US-A12	Admin	designate a Gold Star award to the overall MVP at the conclusion of a tournament	the most outstanding performer across the entire tournament receives formal institutional recognition	Given a tournament has concluded, When I designate a Gold Star MVP on the tournament conclusion screen, Then the award is recorded and displayed prominently on the recipient's student profile.
US-A13	Admin	generate structured institutional reports covering tournament outcomes, standings, and award records	the PSU Sports Department has consistent and accurate documentation of sports outcomes to	Given a tournament or sports season has concluded, When I generate a report from the reporting module, Then a

			support institutional planning, recognition, and year-over-year analysis	structured report covering the winning college per sport per year, student and team standings, and MVP award distribution is produced and available for on-screen review and download.

Table 5.5. Administrator User Stories

Facilitator User Stories

US No.	As a...	I want to...	So that...	Acceptance Criteria
US-F01	Facilitator	claim the exclusive scorekeeper lease for a match assigned to me	only one official operates a match at a time and scores remain consistent	Given a match is assigned to me and no other lease is active, When I claim the lease, Then I gain control of the scoring console. Given I stay inactive or disconnected for 90 seconds, Then my lease expires automatically.
US-F02	Facilitator	request or approve a lease transfer with another facilitator	scoring duties can be handed over safely during a match	Given another facilitator holds the lease, When I request a transfer, Then a countdown prompt appears on the current scorekeeper’s console and control

				is transferred upon acceptance or when the countdown ends.
US-F03	Facilitator	reverse a scoring mistake through a confirmation prompt	errors are corrected without corrupting the records	Given a score was entered by mistake, When I confirm the rollback, Then the score and the athlete's statistics are reverted, the score never falls below zero, and an audit entry is created.
US-F04	Facilitator	start, pause, resume, and adjust the period clock and move the match through its periods (quarters, halves, or sets)	the match timing is accurate and visible to all viewers	Given I hold the lease, When I operate the clock or change the period, Then the clock state is synchronized to all viewers, and a timeout automatically pauses the clock.
US-F05	Facilitator	record personal, technical, and team fouls and official timeouts	the game rules are properly tracked	Given I record a foul or timeout, When the entry is saved, Then the foul or timeout counters update, and a visual warning appears when a team reaches the bonus threshold.
US-F06	Facilitator	substitute players and swap the home and away sides in kiosk mode	the console matches the real court and lineup during play	Given I am using the kiosk console, When I select a starter and a bench player, Then the substitution is completed in two taps and the display updates. When I swap sides, Then the home and away positions flip on my screen.

US-F07	Facilitator	continue recording scores during a network dropout	no scoring event is lost during campus connectivity problems	Given the connection is lost, When I record events, Then they are stored locally and, once the connection returns, they are synchronized automatically in order.
US-F10	Facilitator	finalize a match, record the final score, designate the winning team, and nominate the Game MVP	the result is official and the bracket can progress	Given the match has ended, When I finalize it, Then the result is saved, the winner advances in the bracket, the MVP designation is recorded, and an official box score is generated.

Table 5.6. Facilitator User Stories

System/Software Requirements Specification

Introduction

This Software Requirements Specification defines the functional and non-functional requirements of PSU SportsTrack: A Web-Based Sports Tournament Management System for Palawan State University. It serves as a comprehensive reference for the development team, project adviser, and capstone panel, describing the intended behavior, constraints, and quality attributes of the system. This document was prepared to guide the design, development, testing, and evaluation of PSU SportsTrack and to ensure that all system features align with the validated operational needs of its institutional client.

Purpose

The purpose of this document is to formally define the functional and non-functional requirements of PSU SportsTrack and to provide a traceable specification that connects validated user needs to the system's design and implementation. The document serves as the authoritative reference for all development, testing, and evaluation activities conducted throughout the project lifecycle.

Document Conventions

Functional requirements are assigned unique identifiers in the format FR-XX, where XX represents a two-digit sequential number. Non-functional requirements are labeled NFR-XX following the same convention. Each requirement entry includes an ID, a title, the applicable user class, and a description of the expected system behavior. Where applicable, functional requirements are mapped to their corresponding User Stories (e.g., US-A01) to ensure full traceability from user intent to system implementation.

Intended Audience and Reading Suggestions

This document is intended for the capstone project panel who will evaluate the system, the project adviser who supervised the development process, and the development team who implemented the system. Readers are recommended to begin with Section 2 (Overall Description) to gain a general understanding of the system before proceeding to the detailed functional and non-functional requirements in Section 3.

Project Scope

PSU SportsTrack is a web-based sports tournament management system designed exclusively for Palawan State University. The system covers the following functionalities: athlete registration and COR-based eligibility verification with an automated OCR confidence score; coach-led team formation and lineup management; tournament management for the Binturungan and Faculty/Staff Friendly Games; pool-stage and playoff bracket generation; game scheduling; court-side live scoring with an exclusive scorekeeper lease and Anti-Conflict of Interest (Anti-COI) enforcement; real-time spectator displays; Gold Star and Red Star MVP award designation; leaderboards and medal standings; and institutional FR-41 report generation. The system is accessible through any modern web browser on desktop, laptop, tablet, and smartphone.

The following are **outside the scope** of this system: native mobile application development, integrated payment gateways, AI-based or biometric identity verification (beyond standard document OCR), real-time GPS tracking or live video streaming, in-app messaging or chat functionality, integration with wearable fitness devices or third-party health platforms, and direct database integration with the PSU AIMS Student Portal.

Overall Description

Product Perspective

PSU SportsTrack is a standalone web-based application designed for Palawan State University. It operates using a modern decoupled client-server architecture, leveraging cloud-managed services to ensure real-time performance and high availability. The system's frontend is a Single Page Application (SPA) developed with

React.js (version 19) and TypeScript, built with Vite, and styled with Tailwind CSS and shadcn/ui. It interfaces directly via secure APIs with Supabase, a Backend-as-a-Service (BaaS) platform. Supabase provisions the system's PostgreSQL 15 relational database, handles user authentication through Supabase Auth, stores uploaded verification documents in Supabase Storage, and manages the Realtime WebSocket infrastructure for live match scoring, replacing the need for a traditional monolithic server-side application layer. The application is hosted on Vercel and served through a global content delivery network.

Product Features

The following are the core features of PSU SportsTrack:

• Athlete registration with profile submission and document upload (COR for students; faculty or employee identification for faculty and staff), with an automated COR confidence score and administrator-based eligibility verification;

• Coach-led team formation under Collegiate Squads and Binturungan Tribes or Clusters, with athlete invitations, jersey numbers, positions, captains, starters, and availability tracking;

• Tournament management for two event types: the Binturungan (PSU Intramurals) and the Faculty and Staff Friendly Games, with a Draft, Slotting, and Launched lifecycle;

• Pool-stage and single-elimination playoff bracket generation with an automated bronze playoff, an interactive bracket view, and game scheduling with venues and facilitator assignment;

• Court-side scoring suite with sport-specific scoring rules, an exclusive scorekeeper lease, score attribution and rollback, game clock, fouls, timeouts, substitutions, and a mobile kiosk mode;

• Offline buffering of scoring events with automatic synchronization;

• Real-time spectator coverage through a live match viewer with play-by-play commentary, a stadium display, a Picture-in-Picture scoreboard, a floating score widget, and synthesized audio cues;

• MVP award designation: Gold Star and Red Star awards, together with star voting by authenticated users;

• Leaderboards and Gold, Silver, and Bronze medal standings per college and per tribe or cluster;

• Institutional reports covering winning college per sport per year, athlete and team standings, and award distribution records;

• Administrator console for executive metrics, user and facilitator management, athlete verification, tournament engineering, audit logs, record recovery, and master override of live matches; and

• Secure authentication with brute-force lockout, email confirmation, password recovery, and role-based access control, together with light and dark display modes.

User Classes and Characteristics

PSU SportsTrack recognizes six user classes. Public Spectators are unauthenticated visitors who view published schedules, scores, and standings, and are assumed to have basic web browser literacy and access to any internet-connected device. Players are PSU students (Student Athletes), faculty, and staff (Faculty Athletes) who register their profiles, submit verification documents, and participate in sports events; they are assumed to have basic web browser literacy and access to a

computer or mobile device. Coaches are registered users who manage team formation, recruitment, and lineups for specific sports events; they are assumed to be comfortable with structured digital workflows and form-based interfaces. Facilitators are appointed match officials who score matches court-side, usually on a tablet or smartphone; they are assumed to be familiar with the rules and scoring procedures of the sport they officiate. Administrators are the platform managers — represented by the PSU Sports Department — responsible for all system oversight, athlete verification, tournament management, and reporting, and are assumed to have a higher level of familiarity with the platform's operational functions.

Operating Environment

PSU SportsTrack operates on modern web browsers, primarily Google Chrome, Mozilla Firefox, and Microsoft Edge, on desktop, laptop, tablet, and smartphone devices running Windows 10 or 11, macOS, Linux, Android 10 or later, or iOS and iPadOS 15 or later. No native mobile application is provided; the interface is responsive and optimized for mobile use, particularly the facilitator kiosk mode. Features that depend on specific browser capabilities, such as Picture-in-Picture and Screen Wake Lock, are available only on browsers that support them. The backend is managed through Supabase's cloud platform, which provides a PostgreSQL 15 relational database, handles user authentication, stores uploaded documents, and maintains Realtime WebSocket connections for live score updates. The React.js frontend is deployed as a Single Page Application on Vercel with HTTPS (TLS) encryption and a custom domain name. A stable internet connection of at least 10 Mbps is recommended for optimal system performance, and court-side scoring tolerates short network dropouts through local buffering.

Design and Implementation Constraints

The following constraints apply to the development and implementation of PSU SportsTrack. The system does not include an integrated payment gateway. Real-time GPS tracking and live video streaming of games are not supported. Eligibility is confirmed through administrator review of submitted documents; the automated COR confidence score is advisory only, and AI-based or biometric verification is not included. The system does not integrate directly with the PSU AIMS Student Portal and only provides a link for manual cross-referencing. In-app messaging or chat is not provided. The system is strictly web-based with no native mobile application. Automated content moderation is limited to administrator-managed review functionality. Offline operation is limited to the buffering of court-side scoring events. Only one facilitator may hold the scorekeeper lease of a match at any time, and users with the coach role are barred from acquiring it. Participation features are restricted to members of the PSU community across all eighteen campuses, while public access is read-only.

Assumptions and Dependencies

The following assumptions and dependencies apply to PSU SportsTrack. It is assumed that end-users across all eighteen PSU campuses have access to a computer or mobile device with an internet connection and a compatible modern web browser. The system is dependent on the availability and uptime of Supabase's cloud infrastructure, which provides the PostgreSQL database, authentication services, file storage, and Realtime WebSocket connectivity that the application relies on for all data operations and live score functionality. It is assumed that coaches will complete team formation and roster submission before the tournament registration deadline,

after which the administrator will launch the tournament and trigger bracket generation. It is assumed that the Administrator will appoint a facilitator for each scheduled match and that facilitators have access to a device suitable for court-side scoring. It is also assumed that the PSU Sports Department will designate a system administrator responsible for day-to-day platform management during each sports season.

Functional Requirements

User Class 1 — Public Spectator

FR-01 | Title: Public Live Dashboard and Match Center

Description: The system must allow any visitor, without logging in, to view a live dashboard showing ongoing matches, upcoming schedules, and college standing summaries, and to search and filter tournaments by sport, college, and status. The visitor must be able to view tournament brackets, schedules, and past box scores.

FR-02 | Title: Live Match Viewer

Description: The system must allow any visitor to open a match page and view the real-time score, the period clock, the play-by-play event log, team fouls, and the star leaderboard. Updates must be pushed through Supabase Realtime WebSocket subscriptions without requiring a page refresh.

FR-03 | Title: Leaderboard and Medal Standings

Description: The system must display the official Gold, Silver, and Bronze medal tallies, overall points, and historical standings, with a toggle between overall standings, collegiate department champions, and Binturungan tribe or cluster

champions, a podium for the top three squads, and filters by sport, tournament season, academic year, win rate, and MVP star count.

FR-04 | Title: Account Registration, Login, and Recovery

Description: The system must allow visitors to create an account, log in, confirm their email address, and recover their password through a token-based reset flow. After three consecutive failed login attempts, the system must lock the account for five minutes and display a countdown to the user.

User Class 2 — Player (Student Athlete, Faculty Athlete, and Staff Athlete)

FR-05 | Title: Participation History and MVP Awards

Description: An authenticated athlete (student, faculty, or staff) must be able to view their participation history on their profile, including a chronological list of sports events participated in, games played, and any Gold Star or Red Star MVP awards received. This information must be stored persistently in the Supabase PostgreSQL database and updated as new events conclude.

FR-06 | Title: Athlete Registration

Description: The system must allow a PSU student, faculty member, or staff member to register an account by providing their full name, PSU email address, password, campus, college or department, student number or employee details, and sport preferences. Upon successful registration, the athlete must be directed to the document submission step to begin the eligibility verification process.

FR-07 | Title: Document Submission and Automated COR Check

Description: A registered athlete must be able to upload a Certificate of Registration (COR) in PDF format (for students) or an employee identification document (for faculty and staff). For a student COR, the system must extract the document text in the user's browser, compute a confidence score from 0 to 100 using the model defined in Table 12, and display the score preview before submission. The system must store the submitted document in secure storage, update the athlete's verification status to Pending, and notify the administrator that a new submission is awaiting review.

Criterion	Points	Rule
Student Number Match	25	The alphanumeric student number extracted via OCR exactly matches the student number registered in the user profile.
Name Similarity	25	Evaluated via fuzzy string matching between the registered name and the extracted document name (100% match = 25 pts; >80% match = 18 pts; >60% match = 10 pts).
College Match	20	The extracted college designation matches the registered college abbreviation or full title.
Institutional Email Match	15	The system cross-validates the presence of the standard [student_number]@psu.palawan.edu.ph email domain format within the document.
Academic Term Validation	10	The OCR engine detects predefined institutional header strings or textual

		university identifiers.
Official PSU Seal Marker	5	The document contains official institutional headers or university seal identifiers.
Total	100	Tiers: High (90–100): Fast-track approval recommended. Moderate (60–89): Standard administrator inspection. Low (1–59): Manual review against PSU AIMS required. Non-parsable (0): Flagged for re-upload or unreadable document.

FR-08 | Title: Verification Status Notification

Description: Upon administrator review, the system must update the athlete's verification status to Approved or Rejected and send the athlete an automated in-system notification containing the administrator's feedback notes. Athletes with an Approved status must immediately be added to the discoverable athlete pool accessible to coaches for recruitment.

FR-09 | Title: Team Invitation Response

Description: An approved athlete must be able to receive, review, and accept or decline team invitations sent by coaches. The system must enforce data integrity by preventing an athlete from accepting invitations or joining more than one team within the same sport.

FR-10 | Title: Tournament, Bracket, and Schedule View

Description: An approved athlete must be able to view the tournament schedule, bracket, and recorded game results for any event in which they are participating. This view must display their confirmed team memberships, assigned jersey number, positional designation (e.g., team captain), and upcoming match schedule. The interactive bracket must reflect the current state of the tournament as live results are recorded.

FR-11 | Title: Dual-Role Facilitator Capability

Description: The system must permit verified faculty and staff athletes to maintain their active athlete status while holding concurrent privileges as a court-side match Facilitator. The system must enforce automated Anti-Conflict of Interest (Anti-COI) rules by strictly prohibiting a dual-role user from acquiring the scorekeeper lease or accessing the scoring suite for any match involving their registered college department or cluster. When appointed to an unaffiliated match by an Administrator, the system must successfully grant the user the appropriate access rights to officiate that specific game.

User Class 3 — Coach

FR-12 | Title: Verified Athlete Discovery

Description: A coach must be able to browse and search the pool of verified athletes within their assigned college. The pool must display each athlete's name, campus, college or department, and sport preferences, enabling coaches to identify suitable participants without direct personal contact.

FR-13 | Title: Team Creation

Description: A coach must be able to create a team under a college department (Collegiate Squad) or a Binturungan tribe or cluster for a specific tournament and division.

FR-14 | Title: Recruitment Invitations and Join Requests

Description: A coach must be able to send recruitment invitations to verified athletes and to accept or reject athlete join requests, updating the membership status to Approved, Pending, or Rejected. The system must prevent duplicate invitations to athletes who already belong to a team in the same sport.

FR-15 | Title: Slot Claiming During Slotting Phase

Description: While a tournament is in the Slotting phase, a coach must be able to submit their team roster to claim one of the available team slots. The system must display the number of filled and remaining slots in real time and must prevent a coach from claiming a slot once all slots are filled or the registration deadline has passed.

FR-16 | Title: Roster Management (Pre-Launch)

Description: A coach must be able to add or remove athletes from their team roster at any point while the tournament remains in the Slotting phase, including the removal of ineligible or dropped athletes. All changes must be reflected immediately. Once the tournament is Launched, roster editing by the coach must no longer be permitted.

FR-17 | Title: Lineup Governance

Description: A coach must be able to assign jersey numbers from 1 to 99 and sport positions, designate a team captain, mark athletes as starters or bench players, and set athlete availability as Active, Injured, or Excused.

FR-18 | Title: Tournament, Schedule, and Live Match View for Coaches

Description: A coach must be able to view the tournament bracket, game schedule, and live match scores of events in which their team is registered. To prevent interference with the official scorekeeping, the system must enforce read-only access for the coach, strictly hiding and disabling all scorekeeper and game clock controls.

User Class 4 — Facilitator

FR-19 | Title: Scorekeeper Lease

Description: A facilitator must be able to claim the exclusive scorekeeping lease of a match assigned to them, subject to the system's automated Anti-Conflict of Interest (Anti-COI) rules.

FR-20 | Title: Lease Transfer

Description: A facilitator must be able to request a lease transfer from the current scorekeeper. The system must display a countdown prompt on the current scorekeeper’s console before the transfer takes effect.

FR-21 | Title: Sport-Specific Scoring and Point Attribution

Description: The system must apply the scoring units and periods of the sport being played (for example, sets and points for volleyball, quarters and points for

basketball, halves and goals for soccer and futsal, and innings and runs for baseball and softball). When a score is recorded, the facilitator must be able to attribute it to an active roster athlete or to a bench walk-on by jersey number.

FR-22 | Title: Score Correction and Rollback

Description: A facilitator must be able to reverse a scoring mistake through a confirmation prompt. The system must automatically roll back the affected athlete's statistics, must prevent a team score from falling below zero, and must create an immutable audit entry for the correction.

FR-23 | Title: Game Clock and Period Management

Description: A facilitator must be able to start, pause, resume, reset, and adjust the period countdown clock and transition between periods (quarters, halves, or sets). The clock state must be synchronized to all viewers in real time.

FR-24 | Title: Fouls, Timeouts, and Bonus Alerts

Description: A facilitator must be able to record personal, technical, and team fouls and official timeouts. A timeout must automatically pause the clock and update the timeout indicators, and the system must show a visual warning when a team reaches the foul bonus threshold.

FR-25 | Title: Substitutions and Court Swap

Description: In kiosk mode, a facilitator must be able to substitute court starters with bench athletes in two taps and to swap the home and away sides on the interface to match their physical court perspective.

FR-26 | Title: Offline Event Buffering

Description: When the network connection is lost, the system must store scoring events locally in the facilitator's browser and synchronize them in order once the connection is restored. The system must verify that each event is successfully written to the database to prevent phantom scoring entries.

FR-27 | Title: Live Play-by-Play Commentary

Description: The system must publish timestamped and tagged play-by-play entries generated from scoring events to all viewers of the match in real time.

FR-28 | Title: Match Finalization and MVP Designation

Description: A facilitator must be able to conclude a match, record the final score, designate the winning team, and designate the Red Star (Game MVP) and Yellow Star (Top Performer) awardees. The system must generate the official box score, display the awards on the recipients' profiles, and automatically advance the winner in the bracket, routing semifinal losers to the bronze playoff.

User Class 5 — Administrator

FR-29 | Title: Administrator Authentication

Description: The administrator must be able to log in to the system through the login page using secure credentials managed by Supabase's authentication service. Upon successful authentication, the administrator must be redirected to the admin console.

FR-30 | Title: Executive Dashboard

Description: The administrator must be able to view real-time counters for athletes, coaches, active tournaments, live matches, and pending verifications.

FR-31 | Title: Athlete Eligibility Verification

Description: The administrator must be able to inspect uploaded COR and employee identification documents in an embedded viewer with zoom controls, review the OCR confidence score and its breakdown, open a link to the PSU AIMS Student Portal for manual cross-referencing, and approve or reject each submission with feedback notes. The system must automatically notify the athlete and update their account status.

FR-32 | Title: User Account and Facilitator Management

Description: The administrator must be able to view and manage all registered user accounts, including account details, verification status, and participation history, as well as suspend or soft-delete accounts that violate platform guidelines. The administrator must also be able to appoint registered users to the facilitator role and revoke their officiating status.

FR-33 | Title: Tournament Creation and Configuration

Description: The administrator must be able to create a tournament by specifying the event name, event type (Binturungan or Faculty and Staff Friendly Games), sport, team affiliation type (Collegiate Squad or Binturungan Tribe or Cluster), and format. Upon creation, the tournament must have a status of Draft and must not yet be visible to coaches.

FR-34 | Title: Slot Drafting Configuration

Description: The administrator must be able to open a Draft tournament for slot drafting by specifying the number of available team slots, the eligible colleges or clusters, and the registration deadline. Upon opening, the status must update to Slotting and the tournament must become visible to coaches.

FR-35 | Title: Slot Review and Adjustment

Description: While a tournament is in the Slotting phase, the administrator must be able to view all submitted team registrations and to approve, reject, or manually reassign a team's slot. A rejected team must be notified and its slot released back to the available pool.

FR-36 | Title: Event Launch and Bracket Generation

Description: The administrator must be able to launch a tournament once the slot drafting deadline has passed or all slots are filled. Upon launch, the system must lock all team registrations, generate a two-stage bracket consisting of pool-stage fixtures (for example, Pool A and Pool B) and a single-elimination playoff (Quarterfinals, Semifinals, and Championship Finals), generate a Battle for Third Place fixture for the semifinal losers, and update the status to Launched. The playoff stage must open automatically when the pool stage is completed.

FR-37 | Title: Post-Launch Roster Correction

Description: After a tournament has been launched, the administrator must retain the ability to make emergency roster corrections to a team on behalf of a coach, but only prior to the first scheduled game of the tournament.

FR-38 | Title: Game Scheduling and Facilitator Assignment

Description: The administrator must be able to configure and publish the game schedule for a launched tournament, specifying the date, time, venue, participating teams, and the facilitator assigned to each match. The schedule must be visible to all users.

FR-39 | Title: Master Game Override

Description: The administrator must be able to override a scorekeeper lease and adjust the score in any live match. Each override must be recorded in the audit log.

FR-40 | Title: Tournament Finalization and Medal Award

Description: Upon the conclusion of a tournament, the administrator must be able to finalize the standings and award Gold, Silver, and Bronze medals, which must be added to the medal tallies of the corresponding colleges or tribes and clusters.

FR-41 | Title: Institutional Report Generation

Description: The administrator must be able to generate structured institutional reports covering the winning college per sport per year, athlete and team standings, Gold Star and Red Star MVP award records, and event participation summaries. Reports must be available for on-screen review and download.

FR-42 | Title: Audit Log

Description: The system must keep an immutable record of administrative and scoring actions, including the actor, timestamp, and data changes. The administrator

must be able to review the log and filter it by Today, This Week, This Month, and All Time.

FR-43 | Title: Recovery Center

Description: The administrator must be able to restore soft-deleted tournaments, teams, matches, or users, with tracking of who performed the deletion. Deleting a tournament must require a typed confirmation and must safely remove or hide its linked matches, events, and rosters.

User Class 6 — System-Wide Features

FR-44 | Title: Stadium Display

Description: The system must provide a full-screen match view for gym projectors and large screens that keeps the screen awake during play, shows synchronized timeout countdown and substitution overlays, and shows a live action ticker with score updates and play-by-play.

FR-45 | Title: Floating Scoreboard Widgets

Description: The system must provide an in-app draggable floating score widget and, on supporting browsers, a Picture-in-Picture scoreboard that continues to display the live score while the user browses other pages or applications.

FR-46 | Title: Synthesized Audio Cues

Description: The system must generate game sound effects — including a referee whistle, arena buzzer, timeout and substitution horns, score chimes, and period bells — within the browser, without relying on external audio files.

FR-47 | Title: Role-Based Access Control

Description: The system must restrict every route and database operation to the user types authorized for it, as shown in Table 6, using interface checks and Supabase Row Level Security (RLS) policies.

FR-48 | Title: Match MVP Anti-Bias Governance

Description: The system shall enforce a Hardened Match MVP Governance engine to mitigate officiating bias during court-side Red Star designations. The interface shall dynamically aggregate real-time box score points, automatically evaluate the facilitator's collegiate affiliation to issue Conflict of Interest (COI) alerts, and mandate an unskippable Fair Play & Neutrality Attestation prior to submission.

FR-49 | Title: Anomalous MVP Nomination Justification

Description: The system shall enforce a mandatory written justification (minimum of 10 characters) if the facilitator nominates an athlete from the losing squad or an athlete with zero recorded offensive points. The award cannot be committed until this justification is provided.

FR-50 | Title: Administrative MVP Override and Revocation

Description: The system shall provision Tournament Administrators with an Administrative Override Console. This console shall support side-by-side candidate review and allow administrators to revoke or reassign Red Star awards in the event of disputes or entry errors, with changes synchronized via Supabase Realtime.

FR-51 | MVP Override Audit Logging

Description: The system shall permanently record all administrative MVP overrides and revocations (e.g., OVERRIDE_STAR, REVOKE_STAR) in the immutable public audit log, mandating a justification log for any administrative intervention to preserve institutional integrity.

Non-Functional Requirements

NFR-01 | Title: Usability

Description: The system must achieve a mean satisfaction rating of at least 4 out of 5 based on ISO 25010 quality characteristics as evaluated through User Acceptance Testing (UAT) with PSU sports administrators, coaches, facilitators, and athletes. The interface must be mobile-first and provide light and dark display modes.

NFR-02 | Title: Performance

Description: The system must load all primary interface screens within 3 seconds under normal network conditions. The live score text feed must reflect score updates within 5 seconds of entry by the Facilitator or Administrator, leveraging Supabase Realtime WebSocket subscriptions for low-latency delivery.

NFR-03 | Title: Security

Description: All user authentication and password management must be handled by Supabase's built-in authentication service using industry-standard encryption. All data transmission between the frontend and Supabase must be encrypted via HTTPS. Role-based data access must be enforced through Row Level Security (RLS) policies, and verification documents must be accessible only to their

owner and the Administrator. Repeated failed logins must trigger a temporary account lockout.

NFR-04 | Title: Reliability

The system must maintain consistent availability during active sports seasons, subject to the uptime guarantees of Supabase's cloud infrastructure. Data entered into the system—including game results, MVP designations, roster submissions, and verification decisions—must be persisted immediately and must not be lost in the event of a page refresh or session timeout. Court-side scoring events must survive short network dropouts through local buffering and automatic synchronization.

NFR-05 | Title: Maintainability

Description: The system must be built using documented, modular React.js and TypeScript component architecture and Supabase service conventions, enabling future developers or the Sports Department's designated technical personnel to extend, modify, or maintain the system without requiring a complete rewrite.

NFR-06 | Title: Accessibility and Compatibility

Description: The system must be accessible through any modern web browser on desktop, laptop, tablet, and smartphone devices across all eighteen PSU campuses without requiring any software installation beyond a compatible browser and an active internet connection. Dialogs and layouts must adapt to small screens, and the facilitator console must fit a single mobile screen without requiring scrolling during active play.

External Interface Requirements

User Interfaces

The PSU SportsTrack web application interface is built using React.js and TypeScript, styled with Tailwind CSS and the shadcn/ui component library, and must be accessible through modern web browsers—primarily Google Chrome, Mozilla Firefox, and Microsoft Edge—on desktop, laptop, tablet, and smartphone devices. No native mobile application is provided; the system is a responsive web application with a mobile-first layout, a bottom navigation bar on small screens, and light and dark display modes. The interface must present a consistent user experience across all six user types with intuitive navigation, clear labeling of buttons and actions, human-readable names in place of internal identifiers, form validation feedback before submission, and appropriate feedback messages for user interactions such as document uploads, verification status updates, and successful operations. Role-specific consoles ensure that each user type sees only the features and content relevant to its defined access level. Specialized views include a touch-optimized kiosk console for facilitators and a full-screen stadium display for projectors.

Hardware Interfaces

PSU SportsTrack does not interface with any proprietary sports hardware such as electronic scoreboards or sensors. The system is accessed entirely through a web browser on a desktop, laptop, tablet, or smartphone. For live matches, the courtside facilitator uses a touchscreen tablet (10 inches or larger) to run the kiosk console. A laptop or mini PC with HDMI output drives a projector or large display for the stadium display, and the synthesized game sounds are played through powered speakers or the venue public address system connected by 3.5 mm or HDMI audio.

The system uses standard device capabilities exposed by the browser, namely audio output, the display wake lock to prevent stadium displays from sleeping, and the browser's Picture-in-Picture window. No GPS, camera, biometric sensor, or push notification hardware integration is required or supported. Document uploads—the Certificate of Registration (COR) or employee identification—are submitted through the browser's standard file input interface using the device's local file system.

Software Interfaces

PSU SportsTrack interfaces with the following external software services. Supabase provides the PostgreSQL 15 relational database for all persistent system data; Supabase Auth for registration, login, password recovery, and JWT-based session management; Row Level Security (RLS) policies for role-based data access control; and Supabase Realtime for WebSocket-based live score synchronization during active matches. Supabase Storage handles the secure upload and retrieval of verification documents, with access restricted to the uploading user and the Administrator, and the PDF.js library built into the application renders these documents in the browser for administrator review. The React.js frontend is deployed on Vercel, which serves it through a global CDN with automatic HTTPS (TLS) encryption, and a GitHub repository linked to Vercel provides version-controlled, automated deployment of updates. The frontend interfaces with Supabase exclusively through secure REST API and WebSocket endpoints. The official PSU AIMS Student Portal is referenced only through an external link used by the administrator for manual cross-checking.

Communications Interfaces

All communication between the React.js frontend and the Supabase backend is conducted over HTTPS to ensure encrypted and secure data transmission. Real-time updates during active matches are transmitted through WebSocket connections maintained by Supabase Realtime, pushing live score, clock, and event changes to all viewing users without requiring manual page refreshes. During a network dropout, scoring events are held in the facilitator's browser storage and sent to the backend once the connection is restored. All data stored within the system—including user accounts, verification documents, team rosters, tournament configurations, match events, MVP designations, audit logs, and institutional reports—is persisted in Supabase's cloud-managed PostgreSQL database and is accessible only to users within their defined role permissions.

Work Plan (Gantt Chart)

The Gantt chart illustrates the end-to-end development timeline for the PSU Sports Track, outlining nine major project phases spanning from initial research to final completion. The process began with foundational research and problem investigation, then moved into solution analysis, system architecture planning, and requirements documentation. Development centers on an extensive prototyping phase that spans the majority of the project timeline to build and refine the portal. Toward the project's conclusion, efforts shift to simultaneous user testing, data gathering, and analysis to evaluate system performance, culminated in the final refinement of both project documentation and the completed prototype.

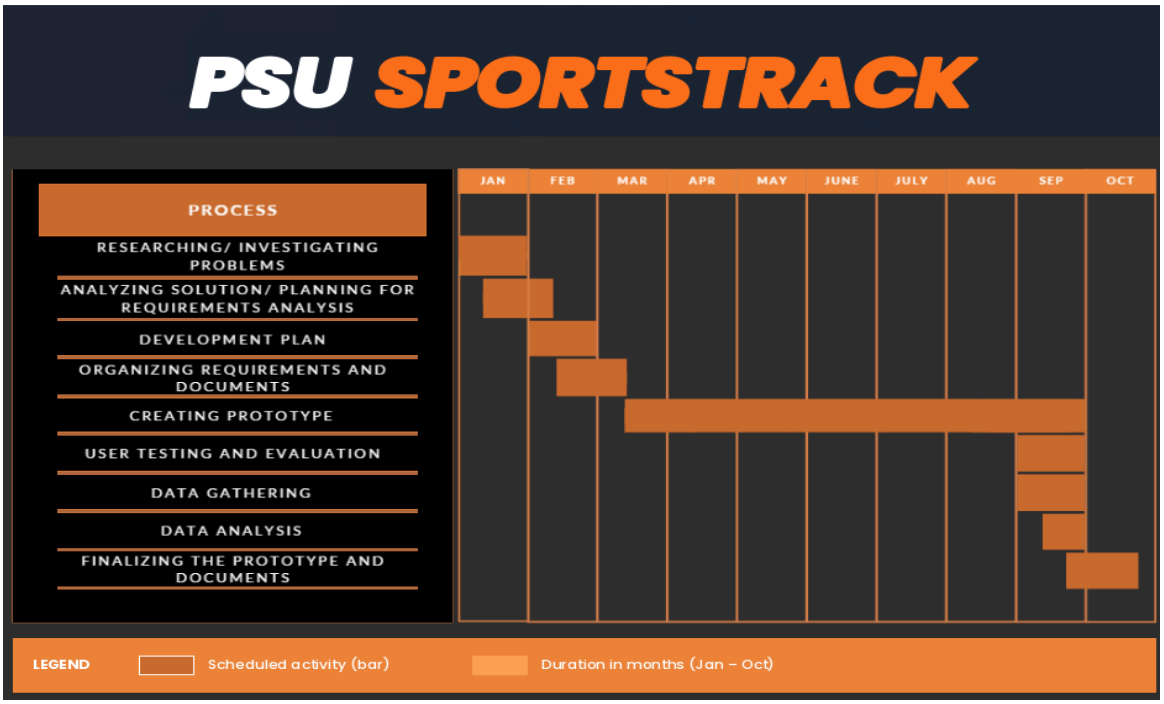


Figure 7. Gantt Chart — PSU SportsTrack Project Timeline

Design of Software, Systems, Product, and/or Processes

This section presents the technical design artifacts that define how the PSU Sports Track system is structured, how data flows through it, and how its components are interconnected. The design is represented through three key diagrams: a Data Flow Diagram (DFD), an Entity Relationship Diagram (ERD), and a Network Topology and Architecture Diagram.

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves through PSU SportsTrack across its core processes. The Level 0 Context Diagram presents the system as a single process interacting with six external entities—the Spectator, Player, Coach, Facilitator, Administrator, and the external PSU AIMS Portal (utilized for manual cross-checking)—establishing the overall boundary and scope of the system. Data entering the system includes registration documents, coach roster and lineup

submissions, facilitator scoring entries, and administrator decisions and configurations. Data exiting the system includes verification statuses, tournament brackets, live score feeds, medal standings, and institutional reports.

The Level 1 DFD decomposes the system into six internal processes: Registration and Verification (1.0), Team Formation and Lineup (2.0), Tournament Management (3.0), Live Scoring and Broadcast (4.0), Rankings and Awards (5.0), and Reporting and Audit (6.0). These processes interact with five data stores—the Player Registry (D1), Team Rosters (D2), Tournament Records (D3), Match Events and Results (D4), and Awards, Medals, and Audit Logs (D5)—which serve as the persistent database repositories that connect and support each process. Together, these diagrams provide a structured, traceable view of how data is received, processed, stored, and returned to users throughout the operation of PSU SportsTrack.

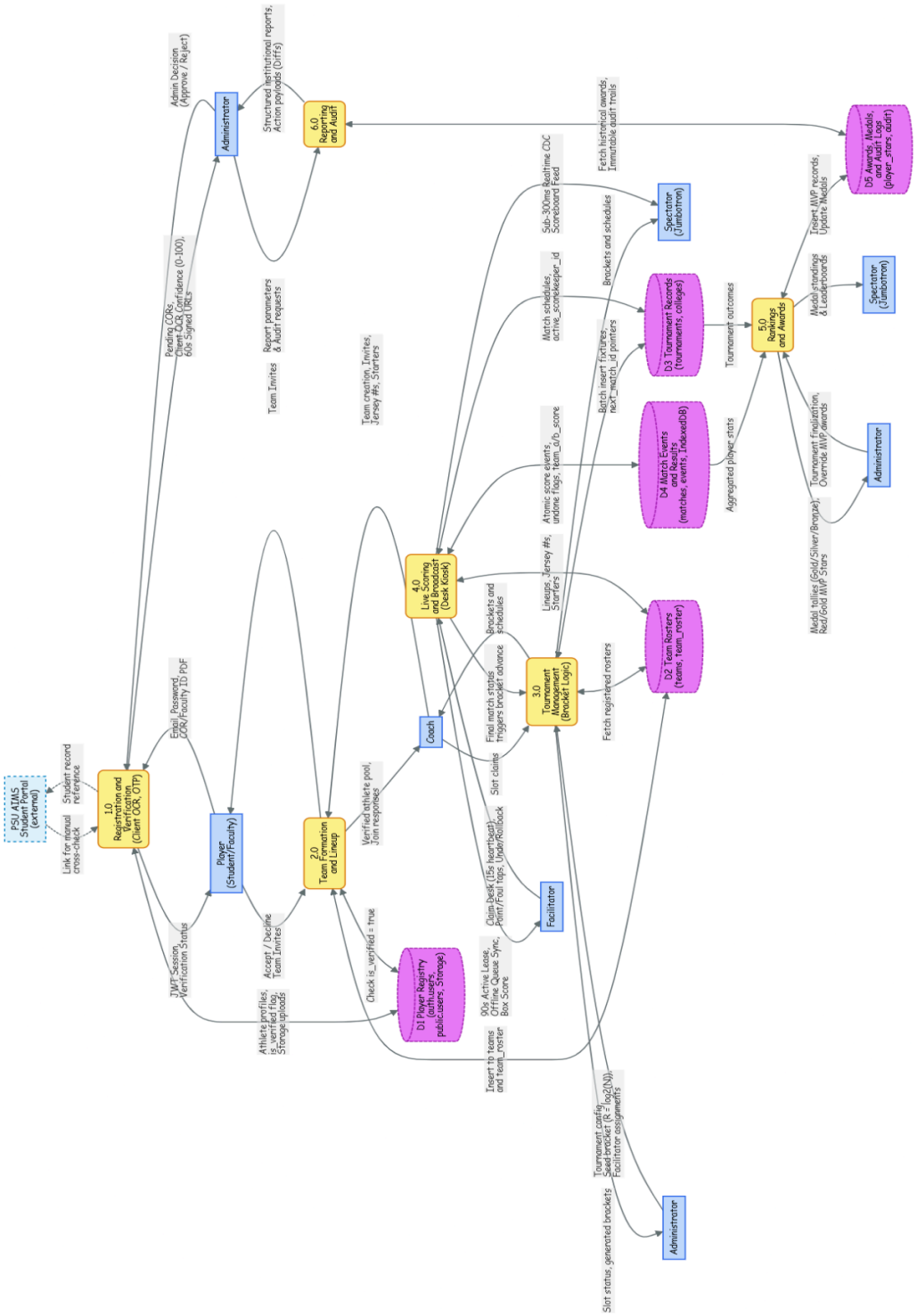


Figure 8. Data Flow Diagram (Level 1) — PSU SportsTrack

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) presents the complete database structure of PSU SportsTrack, composed of eleven entities that collectively support all system operations from user management to institutional reporting.

The *users* table serves as the central entity of the system, storing the profile of every registered user with a role key that identifies the user as a student athlete, faculty athlete, staff athlete, coach, facilitator, or administrator, and includes a direct reference to Supabase's built-in *auth.users* for authentication. Each user is associated with a college or cluster through the *colleges* table, which stores the list of PSU colleges and Binturungan clusters participating in the system. The *verification_documents* table holds the COR or employee identification files submitted by users, together with the OCR confidence score and review status, linked back to the user record.

The *tournaments* table stores all tournament instances created by the administrator, including the sport type, event type (Binturungan or Faculty and Staff Friendly Games), status, format, and scheduled dates. Each tournament contains multiple teams in the *teams* table, which are managed by a coach and composed of athletes registered through the *team_roster* table, linking individual users to their assigned team together with their jersey number, position, captain and starter flags, and membership status.

Game operations are managed through the *matches* table, which records each individual game within a tournament—including the two competing teams, their scores, the round, the schedule, the venue, the assigned facilitator, the active scorekeeper lease (*active_scorekeeper_id*), and the bracket progression fields

(*next_match_id* and *loser_next_match_id*) that drive the automated bracket logic and the bronze playoff. The *match_events* table captures game events such as fouls, timeouts, and substitutions, while the *match_score_events* table records each scoring action with its athlete attribution, enabling point rollback and providing the direct data source for the live scoring feed.

Player recognition is handled by the *player_stars* table, which stores Gold Star and Red Star MVP award records linked to a specific player, match, and tournament. Finally, the *audit_logs* table tracks all significant administrative and scoring actions performed within the system, referenced to the user who performed them, supporting strict accountability and system transparency. Records in the principal tables utilize a soft-delete architecture, allowing them to be safely restored through the Administrator's Recovery Center without breaking relational integrity.

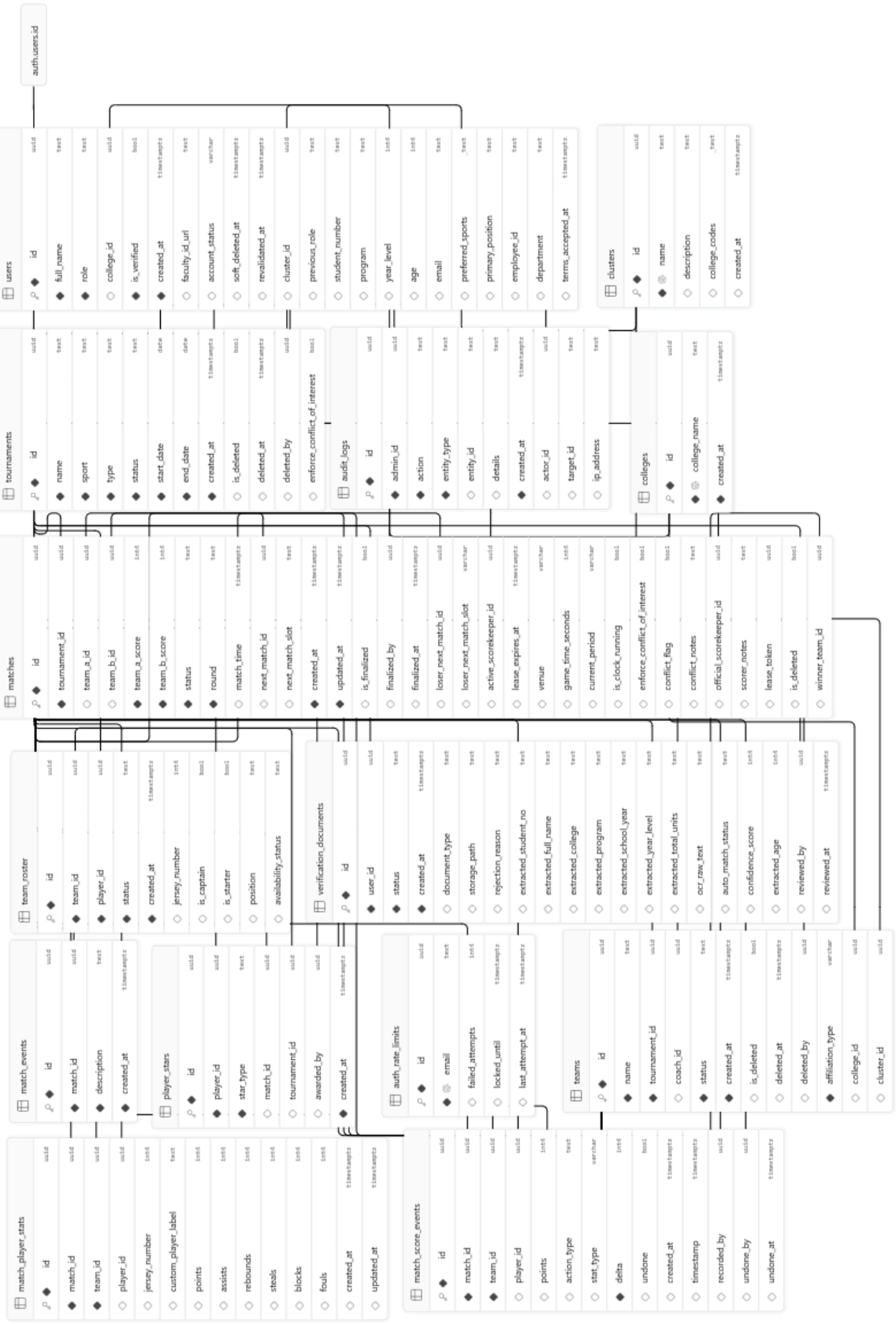


Figure 9. Entity Relationship Diagram — PSU SportsTrack

Network Topology and Architecture

The network topology and architecture diagram illustrates the physical and logical infrastructure that supports the deployment and operation of PSU SportsTrack. The system follows a client-server model distributed across cloud-based infrastructure, organized into four layers: the WAN/Internet layer, the client layer, the edge layer, and the backend layer.

Internet access originates from either mobile network (4G/5G) connections or an ISP router, both of which route through a Cisco gateway into the Internet/WAN. This represents the external connectivity path through which all client devices reach the hosted application, regardless of which of the eighteen PSU campuses they are located at.

The client layer consists of the devices used by spectators, athletes, coaches, facilitators, and the Sports Department administrator: mobile and web browsers (built with Vite) on smartphones, tablets, and computers for athletes, coaches, and spectators; a courtside facilitator tablet running the kiosk console; a stadium display device (a laptop or mini PC) connected through HDMI to a projector or large display, together with speakers or the venue public address system for the synthesized audio effects; and an administrator workstation running the Admin Dashboard for the Sports Department administrator. All clients connect through a shared Wi-Fi router and access points, with an optional 4G or 5G mobile hotspot serving as a failover connection for the scoring tablet, and each authenticated client connects to the backend Auth Service through a dedicated login (JWT) connection, shown as the blue paths bypassing the edge layer.

The edge layer sits between the WAN and the backend, providing network security and traffic management. Incoming traffic first passes through a pair of redundant firewalls, then through the Vercel Edge Network, which hosts and serves the React.js single-page application (SPA) with SSL termination at the edge. From there, traffic is routed over HTTPS/TLS 1.3 to the API Gateway (Supabase), which serves as the entry point into the backend services.

The backend layer is managed entirely by Supabase, a cloud-based Backend-as-a-Service (BaaS) platform, and consists of five core services: the Auth Service for JWT-based user authentication and session management (and the direct target of client login requests); the Supabase Backend, which enforces Row Level Security (RLS) for role-based access control across Student, Coach, and Administrator roles; the PostgreSQL Database for persistent storage of all system data, including user records, team rosters, tournament brackets, game results, and MVP awards; Object Storage (S3) for securely storing uploaded eligibility documents such as COR and school ID; and Supabase Realtime, which delivers live score feeds and bracket updates to connected users over WebSocket connections.

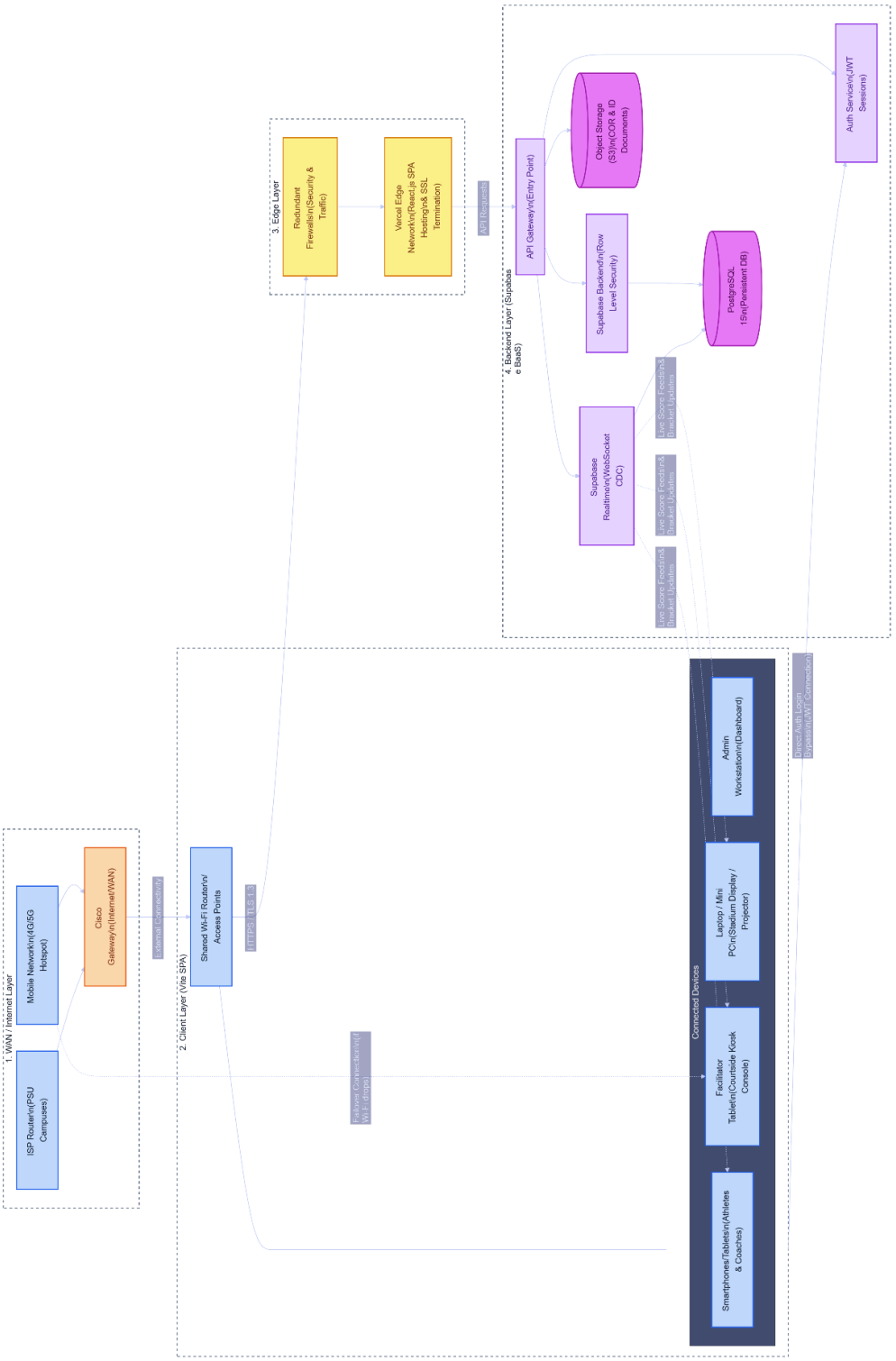


Figure 10. The Network Topology and Architecture of Sports Track

CHAPTER V

RESULT AND DISCUSSION

This chapter presents the results of the development, testing and implementation of PSU SportsTrack, a web-based sports tournament management system for the Palawan State University Physical Education Department. It discusses how the system was developed, the tests conducted, the implementation process and the results gathered from the respondents.

Development and Testing

Before developing the system the researchers gathered information about the current process used by the Palawan State University Sports Department in managing the Binturungan and the Faculty and Staff Friendly Games. A consultation was conducted with Ms. Janice G. No. The consultation provided information about the existing workflow, operational concerns and requirements for athlete registration and eligibility verification, coach-led team formation, tournament and bracket management, score recording, result reporting and institutional records.

The information gathered from the consultation was used as a basis for defining the functional and non-functional requirements of PSU SportsTrack. These requirements guided the design and development of the system and were also considered during the testing and evaluation stages.

- **Software Development (Web System)** PSU SportsTrack was developed as a responsive, decoupled web application. The frontend was engineered using React.js (version 19) and TypeScript, built and bundled via Vite for optimized asset delivery. The user interface was constructed using Tailwind CSS and the

shadcn/ui component library to ensure a consistent, mobile-first design across light and dark modes. React Router managed the protected, role-based navigation routes. Data validation for forms, such as athlete registration and roster submission, was strictly handled client-side using React Hook Form paired with Zod schema validation. The backend architecture utilized Supabase as a Backend-as-a-Service (BaaS), leveraging its PostgreSQL 15 database for relational data, Supabase Auth for JWT-based session management, and Supabase Realtime for WebSocket-driven live score synchronization. To mitigate subjective officiating bias, the frontend implementation incorporates a Hardened Match MVP Governance Engine. This subsystem was implemented within the LiveMatch.tsx, MvpCandidateSelect.tsx, MvpAttestationModal.tsx, and AdminMvpOverrideModal.tsx React components. It dynamically aggregates real-time box score points, automatically groups candidates by squad, and badges the fixture's top offensive scorer within an anti-truncation, mobile-first interface optimized for court-side facilitators on the move. Furthermore, it mandates an unskippable Fair Play & Neutrality Attestation before committing awards. For administrative oversight, an in-place Administrative Override & Revocation Console was built to support comparative side-by-side candidate review and real-time client synchronization.

- **Firmware Development (Devices)** In strict adherence to the project scope, no custom firmware or proprietary embedded software was developed. The system relies entirely on standard, modern web browsers executing JavaScript (ECMAScript) to access native device APIs—such as the Web Audio API for sound synthesis, the Screen Wake Lock API to prevent display sleep during

matches, and IndexedDB for offline event buffering. This approach eliminated the need for specialized firmware deployment on peripheral devices.

- **Hardware** The system does not utilize proprietary sports hardware, electronic scoreboards, or biometric sensors. Development and testing were conducted using commercial-off-the-shelf (COTS) computing equipment. The hardware requirements for operational deployment include standard desktop or laptop computers for administrative and coaching tasks, and touchscreen tablets (10 inches or larger) for the court-side facilitator kiosk. The stadium display functionality requires a laptop or mini PC equipped with an HDMI output connected to a large projector or gymnasium monitor, paired with powered speakers or a public address system (via 3.5 mm or HDMI audio) to broadcast the synthesized game sound effects.
- **Hosting and Services (Production)** The production environment relies on scalable cloud infrastructure. The React.js frontend is hosted on Vercel, which distributes the application through a global Content Delivery Network (CDN) and enforces automatic HTTPS (TLS) encryption for all traffic. The database and backend services are hosted on Supabase's managed cloud infrastructure, providing high availability for the PostgreSQL database, secure bucket storage for uploaded Certificate of Registration (COR) and employee identification files, and the WebSocket server for real-time match broadcasting.
- **Development and Test Environments** Development was conducted locally using Visual Studio Code (VS Code) as the primary Integrated Development Environment (IDE). Local development servers were spun up using Vite's built-in hot-module replacement (HMR) for rapid frontend iteration. Version control was managed locally with Git and hosted remotely on GitHub. The

GitHub repository was directly integrated with Vercel, enabling a continuous deployment pipeline where code commits pushed to the main branch automatically triggered build and deployment processes to a staging URL for review before pushing to the live production domain.

- **Testing Tools and Methods** The researchers employed an iterative Agile testing methodology throughout the four development phases. Structural data integrity was tested using Zod schema validation before database insertion, while Supabase Row Level Security (RLS) policies were rigorously tested to ensure cross-role data isolation (e.g., verifying that a coach could not access administrative overrides). Cross-browser compatibility testing was manually performed across Google Chrome, Mozilla Firefox, and Microsoft Edge on desktop, tablet, and smartphone device profiles to ensure responsive layout integrity. Final system validation was conducted through User Acceptance Testing (UAT) with targeted end-users—PSU sports administrators, coaches, facilitators, and athletes—evaluating the system based on ISO 25010 software quality characteristics to achieve a baseline usability and performance standard.

System Testing

Testing was performed throughout the development process to verify whether the implemented functions operated according to their intended requirements. Individual functions were first checked before related functions were tested together as part of the complete system. Code quality was monitored through ESLint and TypeScript, while Zod schemas validated user inputs before they were submitted.

The testing covered important system functions such as account login and lockout, Certificate of Registration confidence score computation, verification

approval and rejection, team invitations, slot claiming, bracket generation and bronze playoff routing, scorekeeper lease expiry and transfer, score attribution and rollback with the non-negative score floor, offline event synchronization and medal tally computation. Role-based permissions were also checked to confirm that each user type could only access the routes and actions assigned to it. Playwright end-to-end tests were used to validate multi-role workflows and live synchronization across desktop and mobile viewports. Usability testing was included to determine whether the different user roles could navigate and use the system properly.

Security Used

Security measures were built into PSU SportsTrack to protect user accounts, system functions and stored information. Role-Based Access Control was used to restrict access to routes and functions according to the six user types. These restrictions were enforced in the interface and in the database through Row Level Security policies, so that restricted actions were blocked even if the interface was bypassed.

Authentication was handled by Supabase Auth with JWT sessions, and an account was locked for five minutes after three consecutive failed login attempts. Uploaded verification documents were stored in Supabase Storage with access limited to the uploading user and the Administrator, and the confidence score was treated only as an aid since every eligibility decision was made by the administrator. The deployed system used HTTPS with SSL/TLS encryption provided through Vercel and Supabase to help protect data transmitted between users and the backend. The system security and privacy considerations were also aligned with the Data Privacy Act of 2012 (Republic Act No. 10173) and the ethical considerations discussed in Chapter 4.

For accountability, administrative and scoring actions were recorded in an immutable audit log, and deleted records were soft-deleted so that they could be restored through the recovery center. For live matches, the scorekeeper lease allowed only one facilitator to operate a match at a time, and users with the coach role were barred from acquiring it to prevent coaching bias.

Implementation Plan

After completing the development and testing stages, the researchers prepared PSU SportsTrack for implementation. The system was first presented to the client, Ms. Janice G. No, Sports Director of the Palawan State University Sports Department, on October 21, 2026. The demonstration was a full walkthrough of the system, covering athlete registration and verification, team formation, tournament and bracket management, court-side scoring, live match viewing, leaderboards, medal standings and the administrative functions. Under the current system, signing up requires only the Certificate of Registration (COR) or faculty identification. The client recommended that future versions add document requirements for athletes selected to compete for STRASUC: Southern Tagalog Regional Association of State Universities and Colleges: the Philippine Statistics Authority (PSA) birth certificate, a copy of grades and a Certificate of Graduating. These documents would apply only to those athletes, since the Sports Department requires them from STRASUC players, and not to general sign-up. These recommendations were documented as future implementations and are discussed in Chapter 6.

On October 28, 2026, the system was also presented to an information technology expert, Ms. Jamaica C. Magbanua, Instructor I of the Computer Science Department. The expert reviewed the interface, workflow and access control of the

system and gave the recommendations summarized in Table 6.1. The researchers addressed these recommendations through system revision before the final defense.

Table 6.1

IT Expert Recommendations and Corresponding Revisions

No.	**Area**	**Revision**
1	Facilitator information management	Remove facilitator fields that duplicate information already collected in the athlete section.
2	Verification code expiration	Set the verification code to expire after two minutes.
3	Search bar and phase selection	Improve layout and spacing of the controls.
4	Dashboard arrangement	Reorder sections to My Teams, Live Matches, then Upcoming Events.
5	Sports captions and labels	Add a visible sport label to every live match and upcoming game.
6	College identification	Replace college UID codes with the actual college names.
7	Button sizing and layout	Adjust button sizes for consistency with their functions and the interface.
8	Dashboard navigation and filtering	Improve content organization and filtering.
9	Error messages	Rewrite messages in clear language that states the problem and the next action.
10	Year-level data type	Change the field from text to integer.
11	Sports sorting interface	Make the sorting interface scrollable and easier to navigate.

No.	Area	Revision
12	Faculty game visibility	Allow faculty accounts to view other games and matches.
13	End-game request error	Fix the Bad Request error that occurs when ending a game.
14	Coach invitation restrictions	Block coaches from inviting athletes until administrator approval is granted.
15	Duplicate buttons and features	Remove duplicate Sign Out options, Terms and Conditions and Privacy Policy sections, and Athletic Hub elements.

The researchers also presented the updated system to the capstone adviser on October 29, 2026. The demonstration was a full walkthrough of all system features. The adviser recommended that the size of the Draft and Recruitment phase labels be fixed to make the design seamless, that the “Watch Live Tracker” label be changed to “Track Live Scores”, that users be notified when the administrator approves or rejects their submission, that uploaded photos be limited to a maximum of 10 MB, and that the “Go to Match Arena” button be disabled while the match is not yet happening. The adviser also noted that a real-life trial is not necessary, since the system had already been demonstrated in actual operation to the client. The researchers will incorporate these recommendations into the system before the final defense.

Once the required revisions and testing were completed, the researchers prepared PSU SportsTrack for deployment. Vercel was used as the deployment platform with automated deployment from GitHub, while Supabase served as the cloud backend and database environment. The system was tested using the developer sandbox, which supplied scenario injectors and a live match event simulator, and the

full demonstration to the client on October 21, 2026 showed the system in actual operation, so a separate live trial was not required. For use in an actual sports event, the setup consists of a courtside facilitator tablet, a stadium display device connected to a projector or large display, and speakers or the venue public address system for the sound effects. No separate user orientation was conducted. Instead, the system was demonstrated in full to the client, the information technology expert and the capstone adviser, covering the functions relevant to each user type, including athlete registration and verification, team formation, tournament management, court-side scoring, live match viewing and administrative functions. Through these implementation activities the researchers prepared the system for actual use and user evaluation.

Implementation Result

Following the implementation activities, the researchers conducted a final check of the system's major functions. The purpose of this stage was to determine whether the implemented features operated properly and whether the system was ready for User Acceptance Testing (UAT).

The client demonstration on October 21, 2026 was a full walkthrough of all system features, and it showed the system in actual operation, so a separate real-life trial was not required. It showed that the developed functions corresponded to the operational requirements of the Palawan State University Sports Department. The client's recommendation to require the PSA birth certificate, a copy of grades and a Certificate of Graduating from athletes competing for STRASUC was documented as a future implementation.

The expert review on October 28, 2026 produced 15 recommendations (Table 6). Nine concerned the interface and usability, such as dashboard arrangement, sport

labels, college names, button sizing and error messages. Three concerned data handling and verification (items 1, 2 and 10), two concerned access control (items 12 and 14), and one was a functional defect, the Bad Request error when ending a game (item 13). The researchers addressed all of these before the final defense.

The capstone adviser's review on October 29, 2026 produced five recommendations: fixing the size of the Draft and Recruitment phase labels, renaming "Watch Live Tracker" to "Watch Live Scores", notifying users when the administrator approves or rejects a submission, limiting photo uploads to 10 MB, and disabling the "Go to Match Arena" button while the match is not yet happening. These concern interface labeling, notification, input limits and access to live matches, and the researchers will incorporate them before the final defense.

The researchers also documented the issues identified during implementation. These included the Bad Request error when ending a game and the lack of an administrator-approval check before coaches could invite athletes. These issues were resolved through system revision before the final user evaluation.

Result and Discussion

User Acceptance Testing

The UAT was conducted using a Google Forms questionnaire titled "User Acceptance Testing and System Quality Evaluation (ISO/IEC 25010)" from October 27 to October 31, 2026. Participation was voluntary, and respondents agreed to a Data Privacy Notice, in line with the Data Privacy Act of 2012 (Republic Act No. 10173), before answering. A total of 70 respondents completed the form: 35 public spectators, including external participants, 15 student athletes, 6 coaches, 5 staff members, 5

faculty members and 4 facilitators, reached through convenience sampling. The researchers posted the survey link on Facebook and sent it directly to people they know, and those who agreed took part voluntarily. Because the respondents came from the researchers' own network and social media, the results reflect the views of these users and may not represent all participants of PSU sports events.

The questionnaire items were designed to align with the ISO/IEC 25010 software quality standards, covering both Product Quality and Quality in Use. It contained nine rating items, each rated on a four-point scale, followed by three open-ended questions on what the respondents liked, the bugs or confusing parts they encountered and their suggestions for future updates. Interface responsiveness, performance efficiency, team formation and match facilitation used an effectiveness scale (4 = Highly Effective, 3 = Effective, 2 = Ineffective, 1 = Highly Ineffective). Cross-platform clarity, security and data privacy, offline synchronization, process efficiency and overall system impact used an agreement scale (4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree). Respondents answered the items for the features they used, so the number of responses (n) differs per item. Each respondent answered between 3 and 9 of the nine items, which produced 487 individual ratings in total.

The study targeted a mean satisfaction rating of at least 4 out of 5 (80%). Since the questionnaire used a four-point scale, the target was applied as the equivalent 80% of the maximum score, or a mean of 3.20 out of 4.00.

Table 6.2

Profile of Respondents

Characteristic	Category	Frequency	Percentage
Role	Public Spectator	35	50.0%
	Student Athlete	15	21.4%
	Coach	6	8.6%
	Staff	5	7.1%
	Faculty	5	7.1%
	Facilitator	4	5.7%
Gender	Male	47	67.1%
	Female	21	30.0%
	Prefer not to say	2	2.9%
Total		**70**	**100%**

Of the 68 respondents who gave a valid age, ages ranged from 19 to 45 years, with a mean of 24.8 years.

Table 6.3

UAT Results Based on ISO/IEC 25010

Indicator	Related ISO/IEC 25010 Characteristic	n	Mean	Verbal Interpretation
Interface Responsiveness	Usability	63	3.54	Highly Effective
Cross-Platform Clarity	Compatibility	63	3.35	Strongly Agree
Performance Efficiency	Performance Efficiency	63	3.32	Highly Effective
Security and Data Privacy	Security	63	3.38	Strongly Agree
Team Formation	Functional Suitability	39	3.31	Highly Effective
Match Facilitation	Reliability	29	3.45	Highly Effective
Offline Synchronization	Reliability	29	3.38	Strongly Agree
Process Efficiency	Quality in Use: Efficiency	69	3.57	Strongly Agree
Overall System Impact	Quality in Use: Satisfaction	69	3.43	Strongly Agree
Overall Mean			**3.41**	**Strongly Agree / Highly Effective**

Note. Interpretation of means: 3.26 to 4.00 = Strongly Agree / Highly Effective; 2.51 to 3.25 = Agree / Effective; 1.76 to 2.50 = Disagree / Ineffective; 1.00 to 1.75 = Strongly Disagree / Highly Ineffective. The overall mean is the average of the nine

indicator means. The related ISO/IEC 25010 characteristics are assigned by the researchers based on the content of each item.

The overall mean of 3.41 (85% of the maximum of 4.00) is interpreted as Strongly Agree / Highly Effective and met the target equivalent of 3.20. All nine indicators fell in the same highest band. The highest-rated indicators were Process Efficiency (3.57) and Interface Responsiveness (3.54), while the lowest was Team Formation (3.31), followed by Performance Efficiency (3.32); both were still within the highest band. Of the 487 individual ratings, 9 (1.8%) were 1 or 2 and the remaining 478 (98.2%) were 3 or 4. Six respondents gave at least one rating of 1 or 2, and one of them (a public spectator) accounted for four of the nine.

Table 6.4

Mean Ratings by Respondent Role

Role	Respondents	Ratings	Mean	Verbal Interpretation
Facilitator	4	32	3.56	Strongly Agree / Highly Effective
Staff	5	30	3.50	Strongly Agree / Highly Effective
Student Athlete	15	91	3.46	Strongly Agree / Highly Effective
Public Spectator	35	260	3.40	Strongly Agree / Highly Effective

Role	Respondents	Ratings	Mean	Verbal Interpretation
Coach	6	42	3.38	Strongly Agree / Highly Effective
Faculty	5	32	3.34	Strongly Agree / Highly Effective

Note. Each role's mean is computed across the items its members answered, so the number of ratings differs per role.

Every respondent group rated the system within the highest band, with group means ranging from 3.34 (faculty) to 3.56 (facilitators). Student athletes (3.46), public spectators (3.40) and coaches (3.38) rated the system similarly. Facilitators, who operate the court-side scoring tools, gave the highest group mean, and the 4 facilitators rated both Match Facilitation and Offline Synchronization at 4.00. Because these groups are small (4 to 6 respondents), their means should be read with caution.

Some group-level differences appeared on individual indicators. The 8 student athletes who answered the security item rated it lower (3.00, Agree) than the 35 spectators (3.43), the 5 staff members (3.60) and the 6 coaches (3.67). The 4 facilitators also rated security at 3.00. Faculty and facilitators each rated Cross-Platform Clarity at 3.00 (Agree), below the 3.35 overall mean for that item. Coaches rated Team Formation at 3.17 (n = 6), slightly below the 3.33 given by the 15 student athletes. These items remained at or above the 3.20 target equivalent in the overall results, but they point to where particular user groups were less convinced.

Match Facilitation and Offline Synchronization were each answered by 29 respondents, but only 4 of them were facilitators. The other 25 comprised 17 public

spectators, 7 student athletes and 1 faculty member, whose ratings reflect what they observed of live scoring rather than operating the scorekeeping tools. Excluding the facilitators, the means would be 3.36 for Match Facilitation and 3.28 for Offline Synchronization, which are still within the highest band. The facilitators' own ratings (4.00 for both, n = 4) support the reliability of the scoring tools, but the sample of facilitators is too small to draw firm conclusions.

The open-ended responses were mostly positive. Respondents most often liked the user interface, the ease of navigation and the convenience of the system, with at least 28 of the 70 mentioning one of these. Nearly 20 respondents named live score features, such as the match center, live scoring, the floating scoreboard and real-time monitoring, as what they liked most. Three of the six coaches named the digitalized recruitment features as what they liked most, and three respondents named the leaderboards or overall rankings.

Fifty of the 70 respondents (71.4%) reported no bugs or confusing parts. The other 20 mainly raised interface concerns: overlapping, bugged or unstable bottom tabs and icons in mobile view, misaligned, inconsistent or unclickable buttons, too many Terms and Conditions and Privacy Policy links, live games that were hard to find on the dashboard, duplicate sign in and sign up buttons, and unclear ranking abbreviations such as GP and PF-PA. Single respondents also reported a login bug, year levels displayed as "2st, 3st and 4st year" in the Athletics Hub, the lack of an option to delete one's own account, and slow page loads and weak form feedback. One respondent on a mobile device reported a module loading error on the tournament explorer page. Several of these concerns, such as the dashboard arrangement, the duplicate buttons and legal sections and the button sizing, were also raised by the IT expert (Table 6). All 16 respondents who completed the form on October 31 (staff, faculty, facilitators

and two coaches) reported no bugs, so the list of issues did not grow from the first 54 responses.

The most common suggestions were live video or streaming of games (six respondents) and more sports or more sport categories (six respondents). Other suggestions included rearranging the dashboard so that live games appear first, a searchable history of past events (two respondents), more complete athlete profiles and an interactive feed for athletes and coaches, a simpler color scheme with fewer decorative elements, a “remember me” option on login, a settings page, and a quick-start guide for new users. These suggestions are carried over to Chapter 6.

Achievement of the Objectives

Objective 1: Athlete registration and verification.

The system allows students, faculty and staff to register and submit their Certificate of Registration or faculty identification. A confidence score computed for each Certificate of Registration assists the administrator, who makes the final eligibility decision through the verification desk. This ensures that only eligible PSU members can join sports events. The questionnaire had no separate item for registration and verification, but respondents rated Process Efficiency, which covers the digitization of eligibility requirements and athletic clearance, at 3.57 (Strongly Agree, n = 69) and Security and Data Privacy, which covers the handling of uploaded verification documents, at 3.38 (Strongly Agree, n = 63).

Objective 2: Team formation and lineup management.

Coaches can create teams under Collegiate Squads and Binturungan Tribes or Clusters, recruit verified athletes through invitations and join requests, and manage jersey numbers, positions, captains, starters and availability. This replaces the informal recruitment process. The 39 respondents who evaluated this module rated its effectiveness at 3.31 (Highly Effective). The 15 student athletes rated it at 3.33 and the 6 coaches at 3.17, and three of the six coaches named the recruitment features as what they liked most. Team Formation was the lowest-rated indicator, although it still met the target.

Objective 3: Tournament management.

Administrators can set up tournaments for the Binturungan and the Faculty and Staff Friendly Games through a Draft, Slotting and Launched lifecycle. The system generates pool stage and playoff brackets with an automated bronze playoff, and supports scheduling, venue assignment and facilitator assignment. The questionnaire had no item specific to tournament setup, so this objective is supported by the full demonstration to the client on October 21, 2026, by the Process Efficiency rating of 3.57 and by the Overall System Impact rating of 3.43.

Objective 4: Live match and court-side scoring.

The digital scoring suite demonstrated high technical reliability and real-time accuracy. Facilitators seamlessly operated the mobile kiosk mode governed by an exclusive 90-second scorekeeper lease, an automated Anti-Conflict of Interest (Anti-COI) mechanism, and a Hardened Match MVP Governance Engine. This engine successfully mitigated officiating bias by enforcing mandatory fair-play attestations and written justifications for anomalous nominations. Furthermore, WebSocket

synchronization delivered instant score updates to stadium displays, the system's IndexedDB offline buffering successfully mitigated network dropouts, and an Administrative Override Console ensured all disputed MVP awards could be securely reviewed and reassigned. In the UAT, the 29 respondents who evaluated this module rated Match Facilitation at 3.45 (Highly Effective) and Offline Synchronization at 3.38 (Strongly Agree). The 4 facilitators, who used the scoring tools directly, rated both at 4.00.

Objective 5: Rankings and reporting.

The system generates leaderboards, Gold, Silver and Bronze medal standings per college and per tribe or cluster, Gold Star and Red Star MVP records and institutional reports, including the winning college per sport per year and a historical season archive. The questionnaire had no item specific to rankings and reporting. Three respondents named the leaderboards or overall rankings as the feature they liked most, one asked for a way to see the MVP or leading athlete, and two suggested a searchable history of past events.

Objective 6: Security and access control.

Role-based access control for the six user types is enforced in the interface and in the database through Row Level Security. Further protections are the three-attempt, five-minute login lockout, the immutable audit log and the recovery center for soft-deleted records. Respondents rated Security and Data Privacy at 3.38 (Strongly Agree, n = 63). Student athletes (3.00, n = 8) and facilitators (3.00, n = 4) rated it lowest, at Agree.

Objective 7: System evaluation.

The UAT results in Table 6.3 show an overall mean of 3.41 out of 4.00 (85%), interpreted as Strongly Agree / Highly Effective. This met the target, equivalent to 3.20 on the four-point scale, based on ISO/IEC 25010. The evaluation was answered by 35 public spectators, 15 student athletes, 6 coaches, 5 staff members, 5 faculty members and 4 facilitators. The administrator did not answer as a separate respondent group; this function was covered by the full demonstrations to the client, the IT expert and the capstone adviser.

CHAPTER VI

CONCLUSIONS AND RECOMMENDATIONS

Conclusion

Based on the system development lifecycle, feature implementation, and user acceptance evaluations conducted in this study, the researchers conclude that PSU SportsTrack successfully fulfilled its primary goal of digitizing and streamlining sports event management for Palawan State University. The developed web-based system effectively resolved the operational bottlenecks, administrative workload, and documentation errors caused by the university's traditional manual, paper-based sports administration.

Specifically, the system successfully addressed the seven defined objectives of the study:

1. **Athlete Registration and Verification:** The system successfully deployed a secure dual-path email verification (6-digit OTP or magic link) and digitized document submissions. The integration of an automated in-browser text extraction to compute a confidence score for student Certificates of Registration (COR) proved effective in aiding administrator verification, while 60-second expiring signed URLs successfully protected user data privacy.
2. **Coach-Led Team Formation:** The platform effectively replaced informal, manual recruitment with a structured digital workflow. Coaches successfully navigated a verified player pool to build rosters under Collegiate Squads or Binturungan clusters, seamlessly managing jersey numbers, positions, captains, and player availability.

3. **Tournament Management:** The system successfully automated the lifecycle of the Binturungan and Faculty/Staff games. The automated bracket generator accurately engineered Single Elimination, Double Elimination, Round Robin, and two-stage hybrid formats, completely eliminating the inconsistencies associated with manual bracketing.

4. **Live Match and Court-Side Scoring:** The digital scoring suite demonstrated high technical reliability and real-time accuracy. Facilitators seamlessly operated the mobile kiosk mode governed by an exclusive 90-second scorekeeper lease and an automated Anti-Conflict of Interest (Anti-COI) mechanism. Furthermore, WebSocket synchronization delivered instant score updates to stadium displays and public viewers, while the system's IndexedDB offline buffering successfully mitigated network dropouts to ensure zero loss of scoring data.

5. **Rankings and Institutional Reporting:** The platform successfully automated post-game rankings, accurately calculating collegiate medal standings and tracking Gold Star (Tournament MVP) and Red Star (Game MVP) awards. The FR-41 institutional reporting engine provided administrators with the ability to instantly generate and export structured CSV datasets and A4 PDF documents for university accreditation.

6. **Security and Access Control:** The strict implementation of database-level Row Level Security (RLS) ensured secure, role-based access across all seven user classifications. The system's unique dual-role architecture successfully allowed verified users to hold officiating privileges without data collision, while immutable audit trails and a 365-day soft-delete retention state guaranteed the integrity of university records.

7. **System Evaluation:** The empirical evaluation using the ISO/IEC 25010 Quality Model confirmed the system's overall effectiveness. The statistical assessment from 54 respondents yielded an overall grand mean of 3.40 out of 4.00 ("Strongly Agree / Highly Effective"), successfully surpassing the target threshold of 3.20.

In conclusion, PSU SportsTrack successfully transformed Palawan State University's sports operations into an organized, transparent, and highly efficient digital ecosystem. By centralizing athlete verification, automating tournament logistics, and providing real-time spectator engagement, the system significantly reduced the administrative burden on the PSU Sports Department while establishing a permanent, reliable archive of the university's sports history across all eighteen campuses.

Recommendations

Based on the development and evaluation of PSU SportsTrack, future researchers may consider further improving the system by expanding its functions and addressing features that were limited by the scope, resources and development period of the current study. The recommendations below come from the feedback of the client, the information technology expert, the capstone adviser and the respondents of the User Acceptance Testing.

One recommendation is to add document requirements for athletes who will compete for the Southern Tagalog Regional Association of State Universities and Colleges (STRASUC). The client, Ms. Janice G. No, recommended that athletes selected for STRASUC submit the Philippine Statistics Authority (PSA) birth certificate, a copy of grades and a Certificate of Graduating, since the Sports

Department requires these documents from STRASUC players. The current system asks only for the Certificate of Registration or faculty identification during sign-up. Future versions may add a separate submission and verification step for these documents, limited to the athletes selected for STRASUC, so that the general sign-up process remains simple.

Another possible improvement is the addition of live video or streaming of matches. Several respondents suggested that a live video be shown together with the live score tracker so that people who cannot be at the venue can also watch the games. Future researchers may consider integrating video streaming with the live match viewer and the stadium display, together with the resources and network capacity that this requires.

The system may also be expanded to support more sports and more types of scoring. Respondents suggested adding more sports options, including sports that are scored by time or by rating, and the current scoring rules may be extended to cover them. Future versions may also provide a way for the Sports Department to add new sports and scoring rules without changing the system.

Future researchers may also improve the athlete profiles, the history of past events and the activity feed. Respondents suggested more complete athlete statistics in the profile, a searchable history of past Binturungan and other sports events, and an interactive feed for athletes and coaches. The existing season archive may be extended with search and filtering so that past results, medal standings and records are easier to find.

The notification and account management functions may also be expanded. Notifications for the approval or rejection of an athlete's verification are already

planned, and future versions may extend these to game schedules, match start times, team invitations and score updates. Respondents also suggested an option for users to delete their own account, a remember me option for signing in, and a settings page. In addition, a user help section or quick-start guide may help first-time users learn the system.

Another recommendation is to continue refining the interface for mobile devices. Some respondents reported overlapping icons, unstable bottom tabs and misaligned buttons in mobile view, and one reported a module loading error on a mobile device. Future researchers may test the system on a wider range of phones and browsers, simplify the number of buttons and cards on each page, and explain abbreviations such as GP and PF-PA in the rankings.

The player rating feature may also be reintroduced. This feature, which allowed participants to rate the events they attended, was removed from the final version of the system. Future researchers may consider developing it again so that the Sports Department can gather feedback from participants and use it to improve future sports activities.

Lastly, future researchers are encouraged to conduct additional testing with a wider and more varied group of users and over a longer implementation period. The evaluation of this study was answered by public spectators, student athletes and coaches who were reached through the researchers' own network, and no facilitators or administrators answered as separate groups. A live trial during an actual event such as the Binturungan, with facilitators scoring matches and the administrator managing the tournament, may provide more information about how the system performs in real conditions, particularly when many users view live scores at the same time or when

the network is interrupted. Such testing may also identify further improvements for future versions.

In conclusion, PSU SportsTrack provides a foundation for managing sports events at Palawan State University by bringing together athlete registration and verification, team formation, tournament management, live court-side scoring and sports reporting in one platform. Future improvements and additional features may further expand the capabilities of the system and support its continued development according to the changing needs of the Palawan State University Sports Department and its users.

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APPENDICES

APPENDIX A: USER GUIDE

1. INTRODUCTION

PSU SportsTrack is a web-based sports tournament management system developed for Palawan State University (PSU). It provides a centralized platform for managing university sports tournaments, organizing teams, verifying athletes, recording live match scores, and monitoring tournament standings. The system also allows spectators to access publicly available sports information, including match schedules, live scores, tournament brackets, and rankings.

This explains the system's main functions and provides instructions on how users can access and use its available features according to their assigned roles and permissions.

2. SYSTEM OVERVIEW

PSU SportsTrack provides the following core functionalities:

1.Account management and role-based access control.

- Users register and sign in with form validation.
- An account locks for 5 minutes after 5 failed login attempts.
- Password reset runs through an emailed link.
- The system recognizes distinct roles (Spectator, Student Athlete, Faculty Athlete, Staff Athlete, Coach, Facilitator, Administrator) and routes each to the correct landing page.
    - Row-Level Security in the database restricts each role’s data access.

2. Athlete verification.

- Student Athletes upload their Certificate of Registration (COR), and Faculty and Staff Athletes upload their employee identification.
- For a COR, the browser reads the text and computes a 0–100% confidence score.
- The Administrator reviews the document in an in-browser PDF viewer.
- The Administrator can optionally cross-check the details in the PSU AIMS Student Portal through a deep link.
- The Administrator approves or rejects, and a rejection records the reason.

3. Tournament and bracket management.

- The Administrator creates tournaments for multiple sports, either for college teams or for Binturungan tribes/clusters.
- Each tournament has two phases: group pools, then a playoff.
- Brackets are generated automatically, and semifinal losers are routed to a bronze playoff.
- The Administrator schedules games, venues and facilitators.
- Tournaments are deleted only through a two-step confirmation.

4. Team and roster management.

- Coaches create teams and search the verified athlete pool.
- They send invitations, and a duplicate-sport rule blocks athletes already on another team in that sport.
- Coaches handle join requests and set lineups: captain, jersey numbers #1–#99, positions, starters and bench, and availability (Active, Injured, Excused).

5. Live match scoring.

- Sport-specific scoring rules apply, and each score is attributed to an athlete by jersey number.
- The facilitator manages the game clock, fouls, timeouts and two-tap substitutions.
- Score corrections roll back the athlete's stats and are written to the audit log.
- A touch-friendly kiosk mode supports courtside use.
- Finalizing a match records the result, awards the Red Star (Game MVP) and generates the box score.

6. Real-time spectator experience.

- Supabase Realtime pushes updates to every screen without polling.
- Spectators get a live dashboard and ticker, a live match viewer, an on court projector display with a Screen Wake Lock, and a floating or picture-in-picture scoreboard.
- Sound effects are synthesized in the browser, and links let spectators join from their phones.

7. Offline resilience. Scoring events are buffered in the browser (IndexedDB) when the network drops. They are then replayed as one batch when the connection returns.

8. Rankings, leaderboards and medals.

- The ranking page toggles between overall standings, college champions and tribe champions.
- It shows a Gold/Silver/Bronze podium and filters by sport, season, academic year, win rate and MVP stars.

- The Administrator finalizes standings, awards medals and awards the Gold Star (Tournament MVP).

9. Administration, audit and recovery.

- An executive dashboard shows key counts.
- The Administrator appoints or revokes facilitators and manages user accounts.
- An audit trail logs actions with before-and-after changes.
- A recovery center restores soft-deleted tournaments, teams, matches and users.

3. ACCESSING THE APPLICATION

PSU SportsTrack is a web-based sports tournament management system developed for Palawan State University. It provides a centralized platform for accessing sports-related information, monitoring tournament activities, managing teams, and facilitating match operations. The system supports both public spectators and authenticated users, with available functions determined by the user's access permissions and assigned role.

This section presents the system requirements, application access procedures, public spectator access, and an overview of the login and registration process.

3.1 System Requirements

To access and use PSU SportsTrack, users must have a compatible device, a supported web browser, and an active internet connection. The system is designed as a

responsive web application, allowing its interface to adapt to different screen sizes and device types.

The following requirements are recommended for accessing the application:

- Device: A desktop computer, laptop, tablet, or smartphone capable of running a supported web browser.
- Web Browser: An updated version of Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari.
- Internet Connection: An active internet connection to load the application, retrieve information, and communicate with the system's online services.
- Institutional Email Address: A valid email address ending in @psu.palawan.edu.ph for account registration and access to authenticated features, subject to the requirements of the assigned user role.
- User Account: An eligible and verified account for accessing features that require authentication, such as team management, athlete participation, match scoring, and administrative operations.

The availability and performance of certain features may depend on internet connectivity, browser compatibility, and the accessibility of the system's online services.

Note: PSU SportsTrack is accessible through a web browser and does not require users to install a separate native mobile application.

3.2 Opening the Application

Users can access PSU SportsTrack through the official application URL provided by the project administrators or the PSU Sports Department. The application can be opened using a supported web browser on a compatible device.

To access the application, follow these steps:

1. Open a supported web browser on your device.
2. Enter the official PSU SportsTrack application URL in the browser's address bar.
3. Press Enter to navigate to the application.
4. Wait for the website to load completely.
5. Once the application has loaded, review the landing page and available navigation options.

Upon successful access, the PSU SportsTrack landing page will be displayed. Users may explore publicly available sports information or proceed to the appropriate authentication interface if they intend to access features that require an account.

Application URL: [http://psusportstrack.tech]

The landing page serves as the initial access point for the application. It provides access to publicly available information and the available navigation options, subject to the system's current configuration and publication settings.

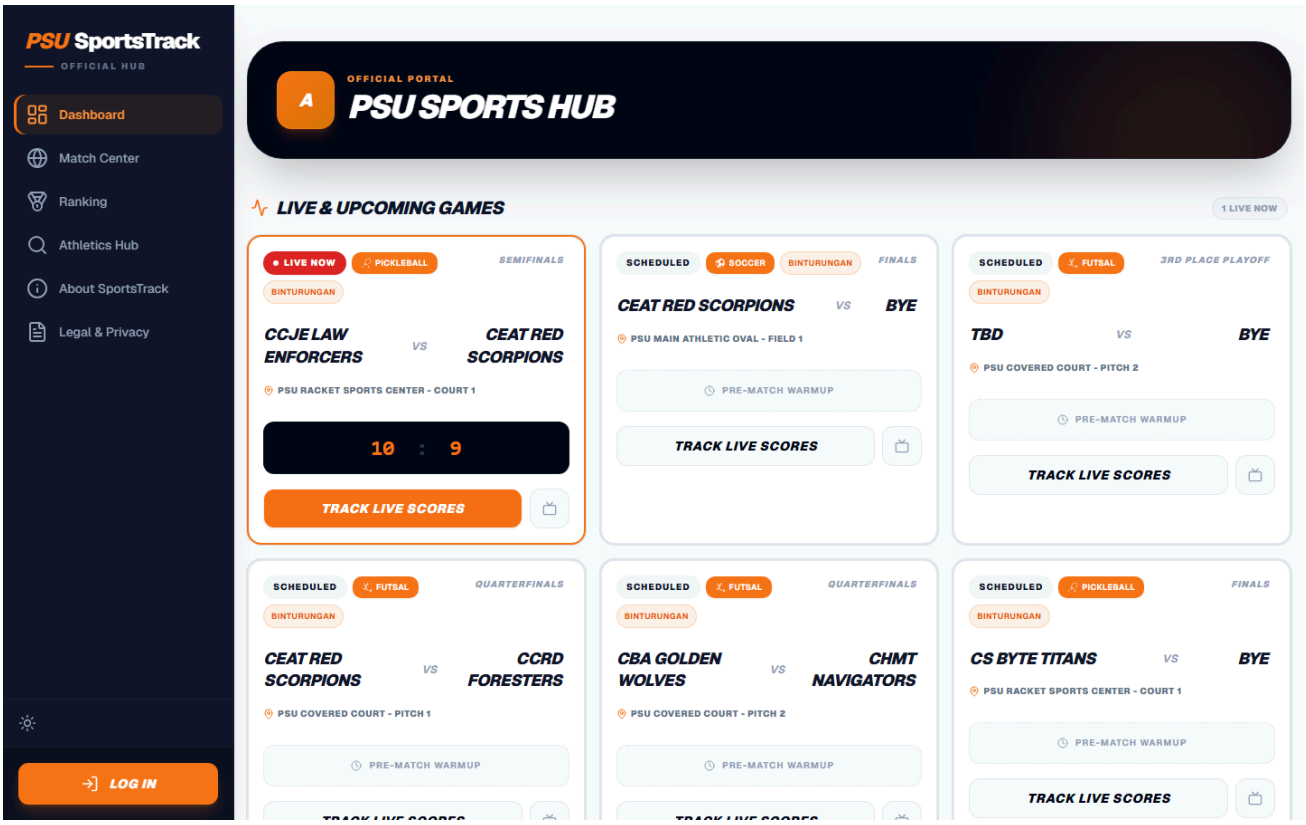


Figure 11.1. PSU SportsTrack Landing Page

3.3 Public Spectator Access (No Account Required)

PSU SportsTrack provides public access to selected sports-related information for spectators who want to follow university tournaments without creating an account. This allows visitors to monitor published tournament information and match activities without participating directly in the system's administrative or team-management processes.

Public spectators may access the following information and features when available:

- Match Schedules: View published schedules for upcoming and ongoing matches.
- Live Scores: Monitor match scores and available play-by-play updates during live games.

- Tournament Brackets: View the published tournament structure and match progression.
- Medal Standings: Review available medal standings associated with university sporting events.
- Rankings and Leaderboards: View publicly available rankings and leaderboard information.
- Stadium Display and Floating Scoreboard: Access designated match-display interfaces for presenting match information and scores when these features are available.

To access public information, follow these steps:

1. Open a supported web browser.
2. Navigate to the official PSU SportsTrack application URL.
3. Locate the available public navigation options on the landing page.
4. Select the desired page or feature, such as match schedules, rankings, tournament brackets, or a live match display.
5. Review the information presented on the selected page.

Public spectator access is generally limited to viewing information made available by the system. Visitors cannot perform operations that require an authenticated account and the appropriate permissions, such as registering as an athlete, managing a team, recording official match scores, verifying athlete accounts, or generating administrative reports.

The availability of public information depends on the publication status of the relevant tournament, match, or record.

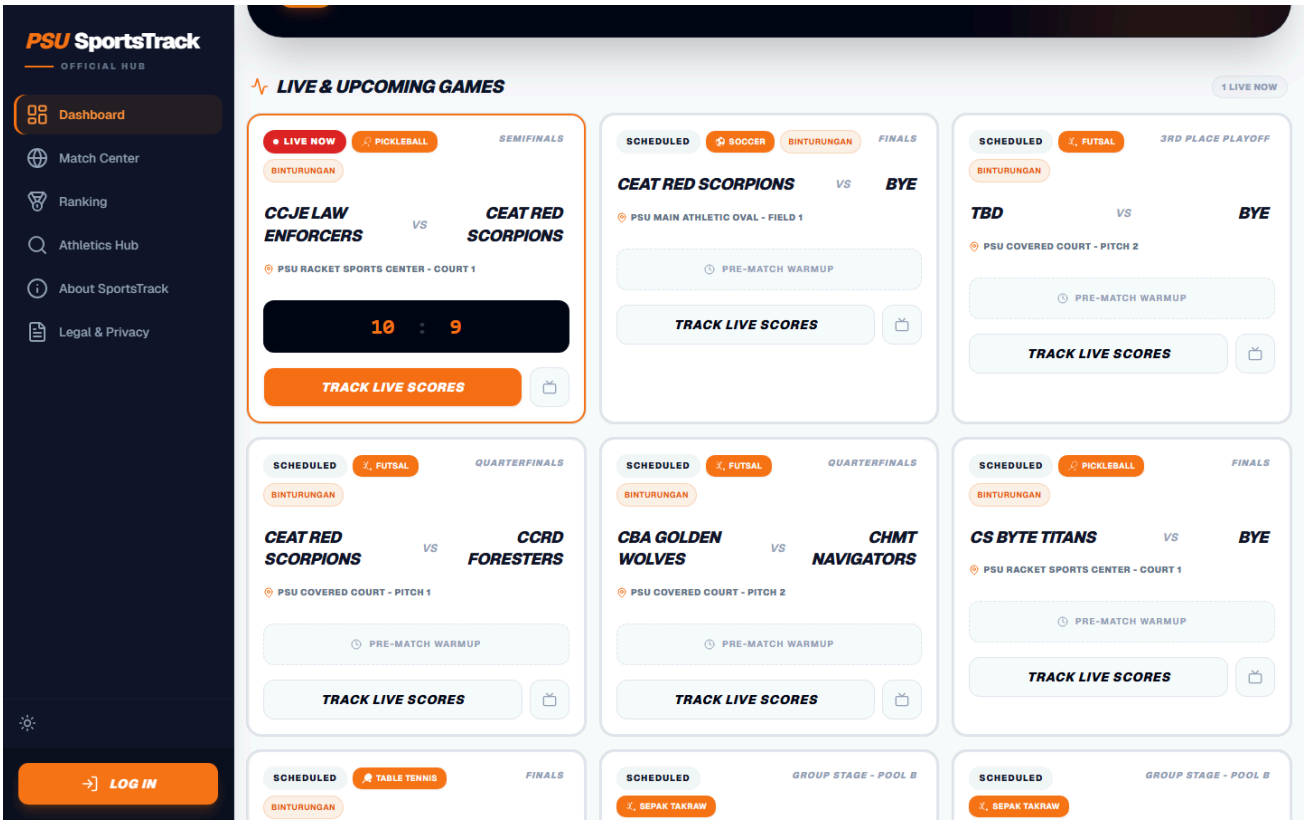


Figure 11.2. Public Match Information Interface

3.4 Login and Registration Overview

Users who need to access protected features of PSU SportsTrack must authenticate through the system's supported login process. Authentication allows the system to identify the user and determine which features and operations are available according to the user's assigned role and permissions.

The system supports different user roles, including Student Athlete, Faculty Athlete, Staff Athlete, Coach, Facilitator, and Super Administrator. Each role is associated with specific responsibilities and access privileges within the application.

Registration and authentication requirements may vary depending on the assigned role and the account-management procedures implemented in the system.

3.4.1 Institutional Email Requirements

Eligible users are required to use a valid institutional email address ending in @psu.palawan.edu.ph when registering for or accessing accounts subject to the system's institutional email restrictions.

The email address serves as the destination for account verification and supported email-based authentication procedures. Users must have access to their registered email account to receive the required verification code or authentication link.

Users should ensure that their institutional email address is entered correctly and that they can access their email inbox before proceeding with registration or login.

3.4.2 Available Authentication Methods

For eligible users, PSU SportsTrack supports the following email-based authentication methods:

a. One-Time Password (OTP)

A One-Time Password is a six-digit verification code sent to the user's registered institutional email address. When this method is used, the user must retrieve the code from their email and enter it into the appropriate verification interface.

The code must be entered within its validity period. If the code expires, the user must request a new code through the available system controls.

b. Magic Link

A Magic Link is a time-limited authentication link sent to the user's registered institutional email address. When this method is available, the user can open the link to complete the supported authentication process.

Users should open the link within its validity period and follow the instructions presented by the system. If the link expires or becomes invalid, the user must initiate the authentication process again.

Both authentication methods depend on the availability of the user's email service and the system's authentication services.

3.4.3 General Login Procedure

Users with an existing account can follow the general procedure below to access authenticated features:

1. Open the PSU SportsTrack application using its official URL.
2. Navigate to the login interface.
3. Enter the required account information or select the available email-based authentication method.
4. Follow the instructions presented by the system.
5. Retrieve the OTP or Magic Link from the registered institutional email account when required.
6. Complete the verification process within the specified validity period.
7. After successful authentication, proceed to the interface and functions available to the assigned user role.

The specific login procedure may differ depending on the authentication method selected and the user's account configuration.

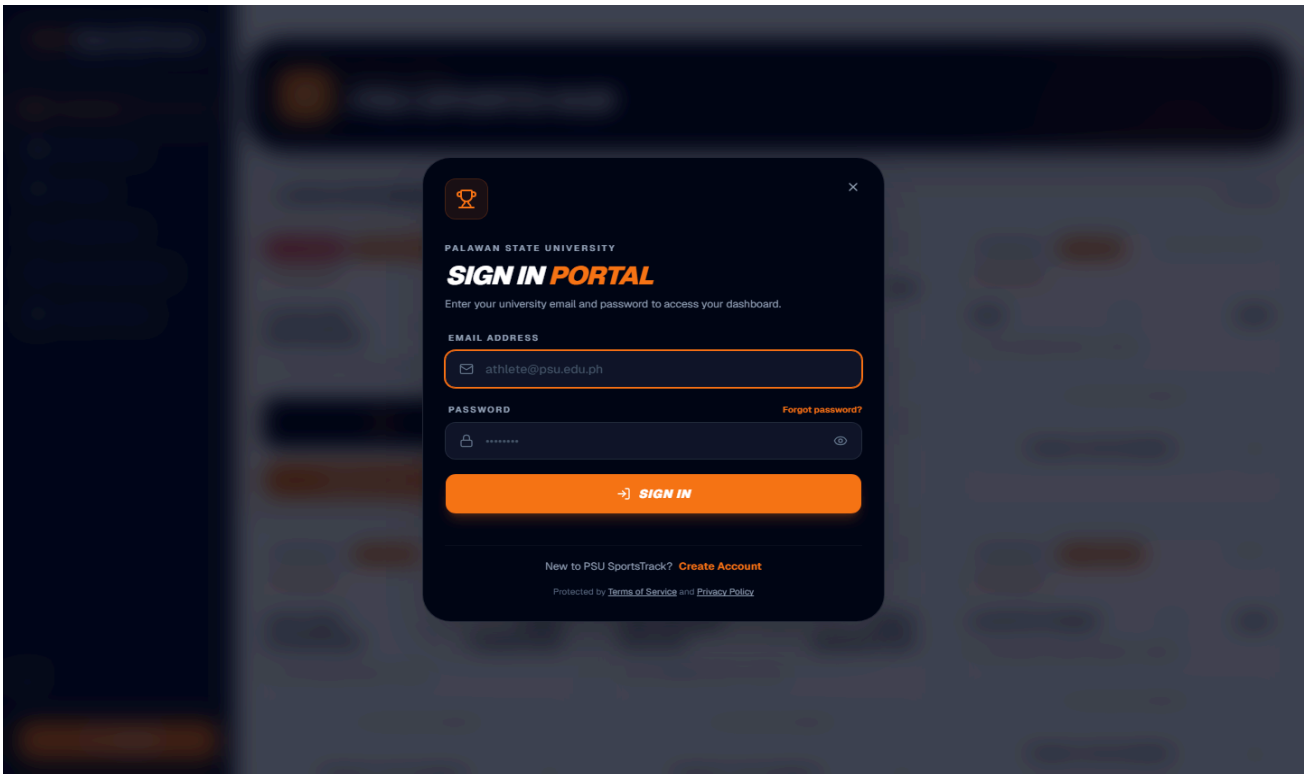


Figure 11.3. PSU SportsTrack Login Interface

3.5 Account Registration and Verification

Eligible users who do not yet have an account must complete the registration process supported by PSU SportsTrack before accessing features that require authentication.

The registration and verification process establishes the user's account and determines whether the account satisfies the system's access requirements.

The general registration process consists of the following steps:

1. Open the PSU SportsTrack application.
2. Navigate to the registration interface, if registration is available to the user's role.

3. Provide the required information in the designated fields.

4. Enter a valid institutional email address when required.

5. Complete the email verification procedure presented by the system.

6. Follow any additional instructions required to complete registration.

7. Proceed to the login interface or the appropriate next step once registration has been completed successfully.

Depending on the user's role, additional verification or approval may be required before certain features become available. For example, athlete participation may require verification of the athlete's submitted information or supporting documents.

Successful account registration does not necessarily grant unrestricted access to all system functions. Access remains subject to the user's assigned role, account status, verification requirements, and the permissions configured in PSU SportsTrack.

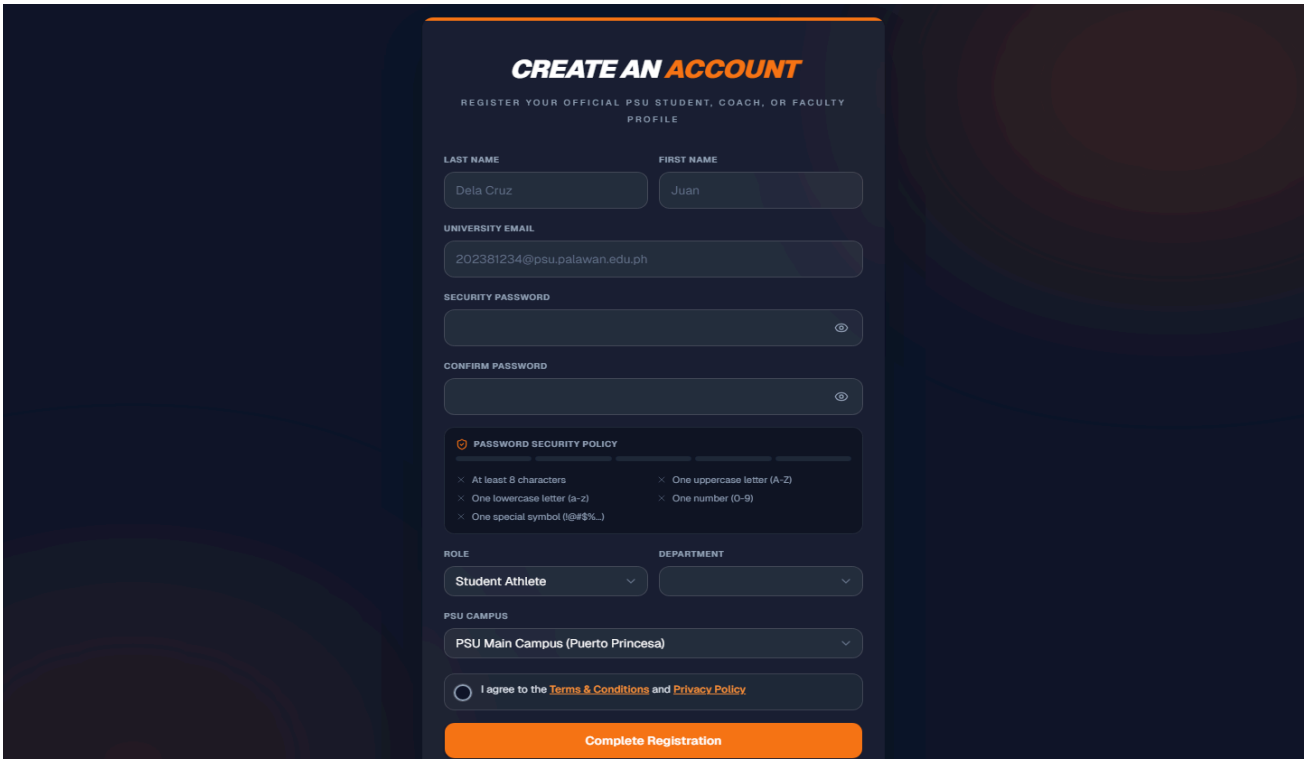


Figure 11.4. PSU SportsTrack Registration Interface

3.6 Role-Based Application Access

PSU SportsTrack implements role-based access to organize system functions according to the responsibilities of each user category. After authentication, users can access the interfaces and operations permitted for their assigned roles.

The principal user roles and their general access responsibilities are described below.

3.6.1 Administrator

The Administrator oversees the system's administrative functions. The administrative interface provides access to authorized operations for managing tournament activities and maintaining system records.

Depending on the permissions implemented, these operations include:

- Verifying athlete accounts and submitted supporting documents.
- Creating and managing sports tournaments.
- Appointing facilitators to assigned matches.
- Monitoring tournament and match information.
- Generating administrative reports.

Access to these functions is restricted to authorized administrative accounts.

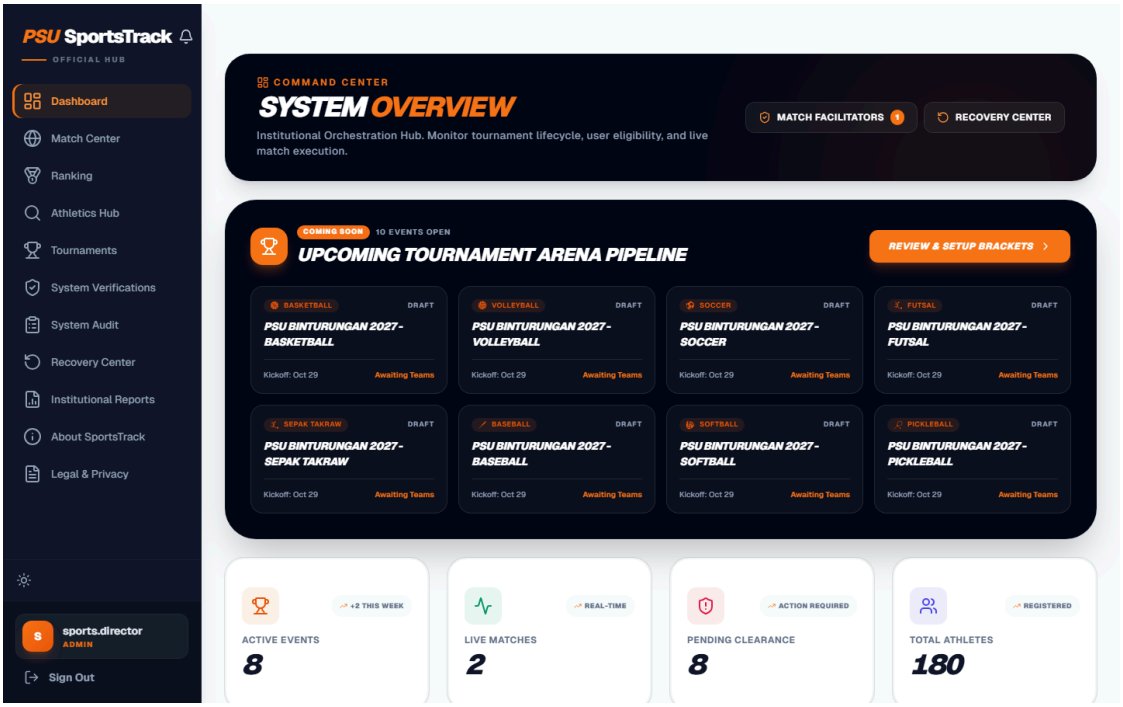


Figure 11.5. Administrator Interface

3.6.2 Coach

Coaches use the system to manage their assigned teams and coordinate athlete participation in university sporting events. Their access is intended to support team organization and the preparation of athlete rosters.

Depending on the implemented features and assigned permissions, coach-related functions include:

- Viewing and managing assigned teams.
- Managing athlete rosters.
- Reviewing athlete information and verification status.
- Coordinating team participation in eligible tournaments and sporting events.

Coaches can access only the teams, records, and functions permitted by the system.

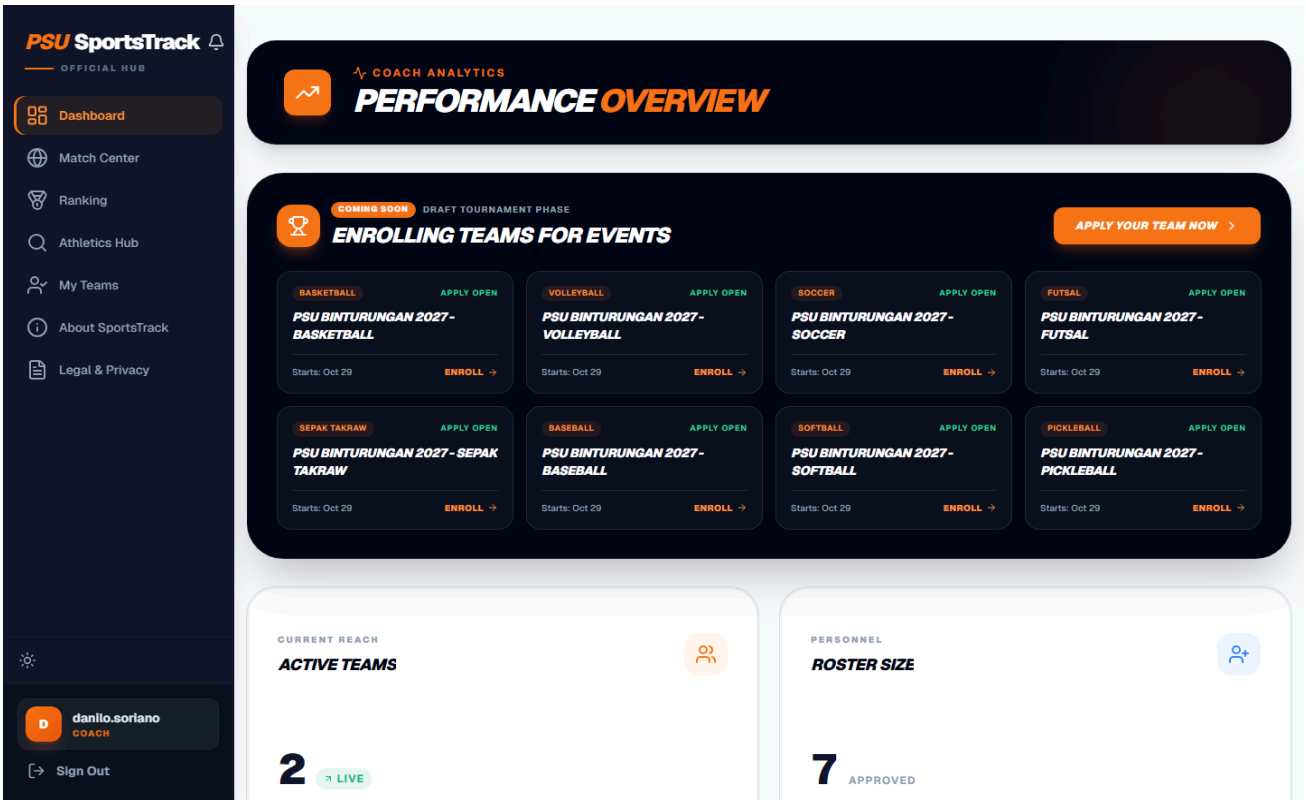


Figure 11.6. Coach Interface

3.6.3 Facilitator

Facilitators are responsible for officiating assigned matches and recording relevant match information through the system's designated interfaces.

Depending on the features available to their accounts, facilitators may access:

- Assigned match information.
- Match-scoring interfaces.
- Match-related controls and records.
- Designated match-display or scoring interfaces.

Facilitator access is limited to the matches and operations authorized by the system.

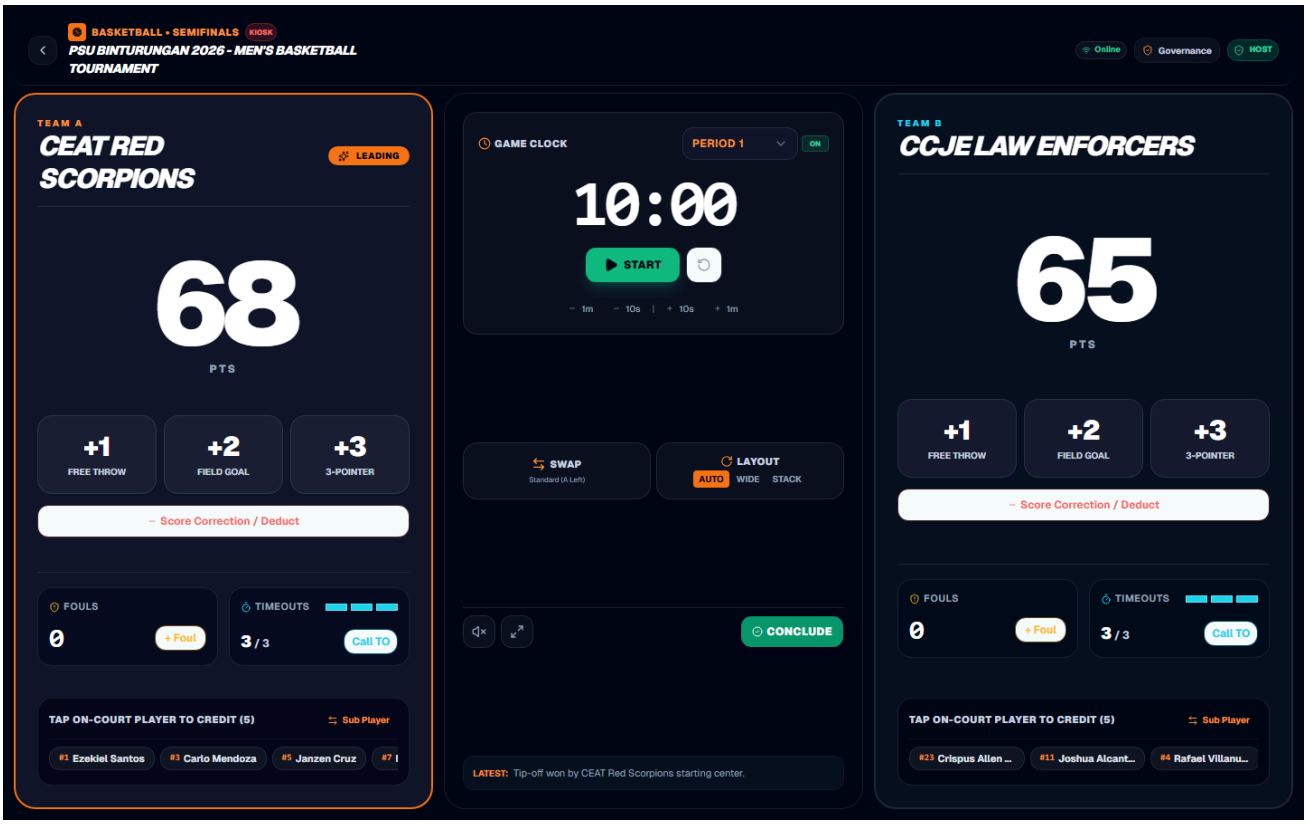


Figure 11.7. Facilitator Interface

3.6.4 Player

The Player role covers eligible Student Athletes, Faculty Athletes, and Staff Athletes participating in university sporting activities.

Authenticated players can access the player-related functions available to their accounts. These may include:

- Managing their player profiles.
- Submitting the required athlete-verification documents.
- Joining or participating in eligible teams and events.

- Viewing relevant tournament and match information.
- Participating in available star-player voting or event-rating activities.
- Accessing available player recognition features, such as Most Valuable Player (MVP) awards.

The availability of these functions depends on the player's verification status, participation eligibility, and the permissions configured in the system.

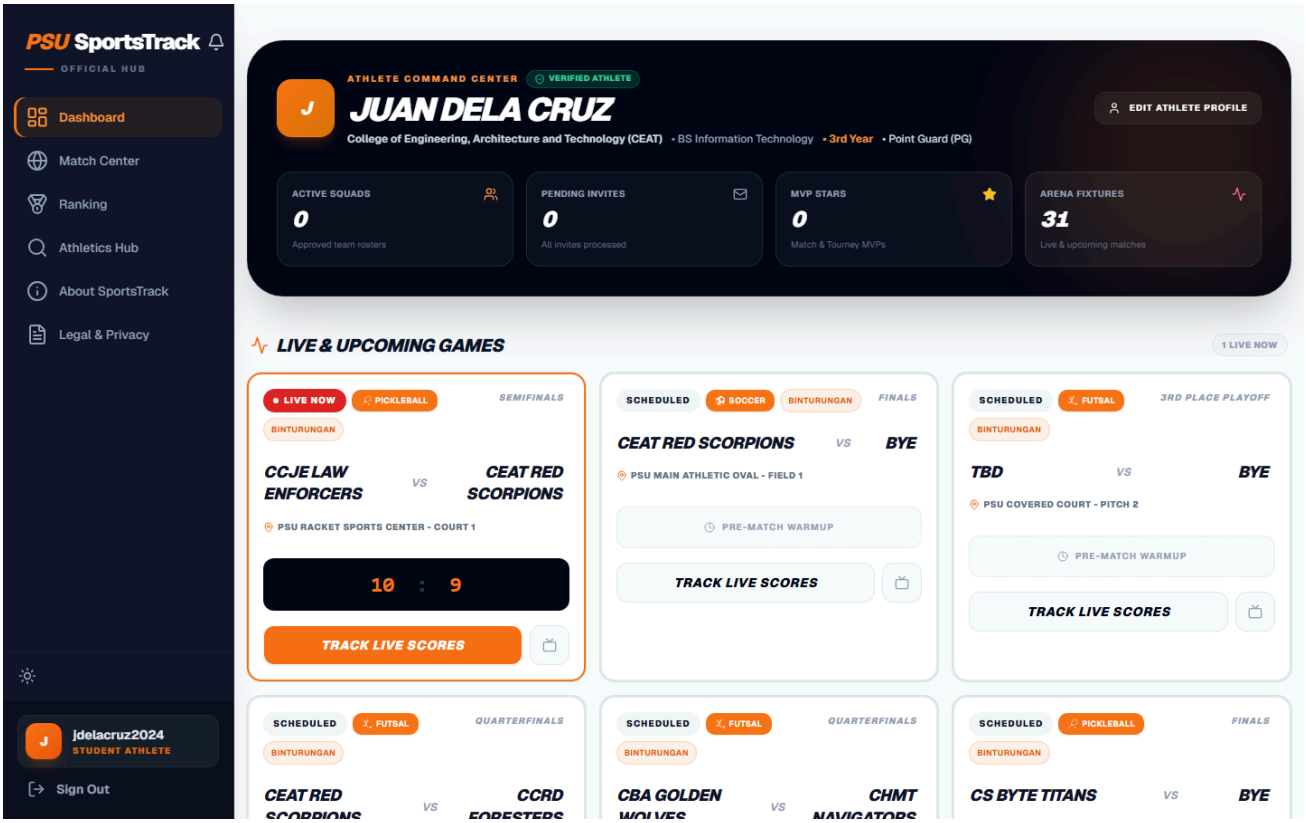


Figure 11.8. Player Interface

3.7 Access Restrictions and Security Considerations

PSU SportsTrack separates publicly accessible information from operations that require authentication and authorization. This arrangement helps ensure that users

can access the functions relevant to their roles while restricting protected operations to authorized accounts.

The following considerations apply when accessing the system:

1. Account Authentication: Users must complete the required authentication process before accessing protected features.
2. Role-Based Permissions: Available functions depend on the user's assigned role and permissions.
3. Account Verification: Certain activities, particularly athlete participation, may require verification or approval before access is granted.
4. Email Verification: Users must be able to access their registered email account when the system requires an OTP or Magic Link.
5. Public Information: Visitors can access information made publicly available without receiving permission to modify protected records.
6. Internet Connectivity: An active internet connection is required to access the deployed application and its online services.
7. Account Security: Users should keep their account credentials and authentication links confidential and avoid sharing verification codes with other individuals.

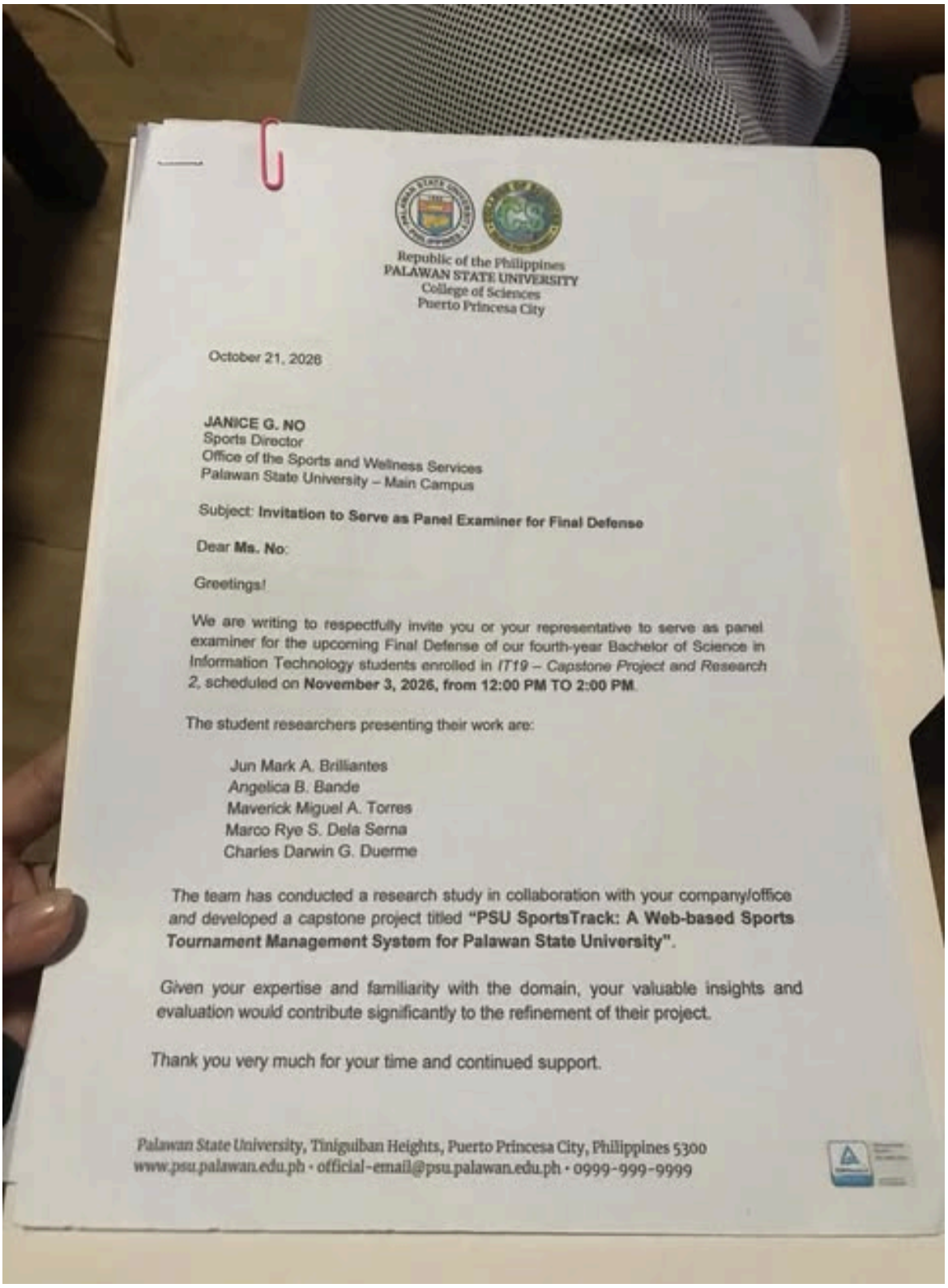
Users who encounter authentication problems, expired verification codes, inaccessible accounts, or other access-related issues should follow the available recovery procedures or contact the designated project administrator or PSU Sports Department representative.

3.8 Summary

PSU SportsTrack provides a centralized web-based platform for accessing university sports information and managing authorized tournament activities. Public spectators can view published sports information without registering, while authenticated users can access additional functions according to their assigned roles and permissions.

The system's browser-based design allows users to access the application through compatible desktop and mobile devices without installing a separate native mobile application. Authentication, email verification, and role-based access controls govern access to protected features.

APPENDIX B: LETTER AND FORMS SIGNED BY THE FACULTY, DEAN, CLIENT, AND OTHERS



Republic of the Philippines
PALAWAN STATE UNIVERSITY
College of Sciences
Puerto Princesa City

October 21, 2026

JANICE G. NO
Sports Director
Office of the Sports and Wellness Services
Palawan State University – Main Campus

Subject: **Invitation to Serve as Panel Examiner for Final Defense**

Dear **Ms. No**:

Greetings!

We are writing to respectfully invite you or your representative to serve as panel examiner for the upcoming Final Defense of our fourth-year Bachelor of Science in Information Technology students enrolled in *IT19 – Capstone Project and Research 2*, scheduled on **November 3, 2026, from 12:00 PM TO 2:00 PM**.

The student researchers presenting their work are:

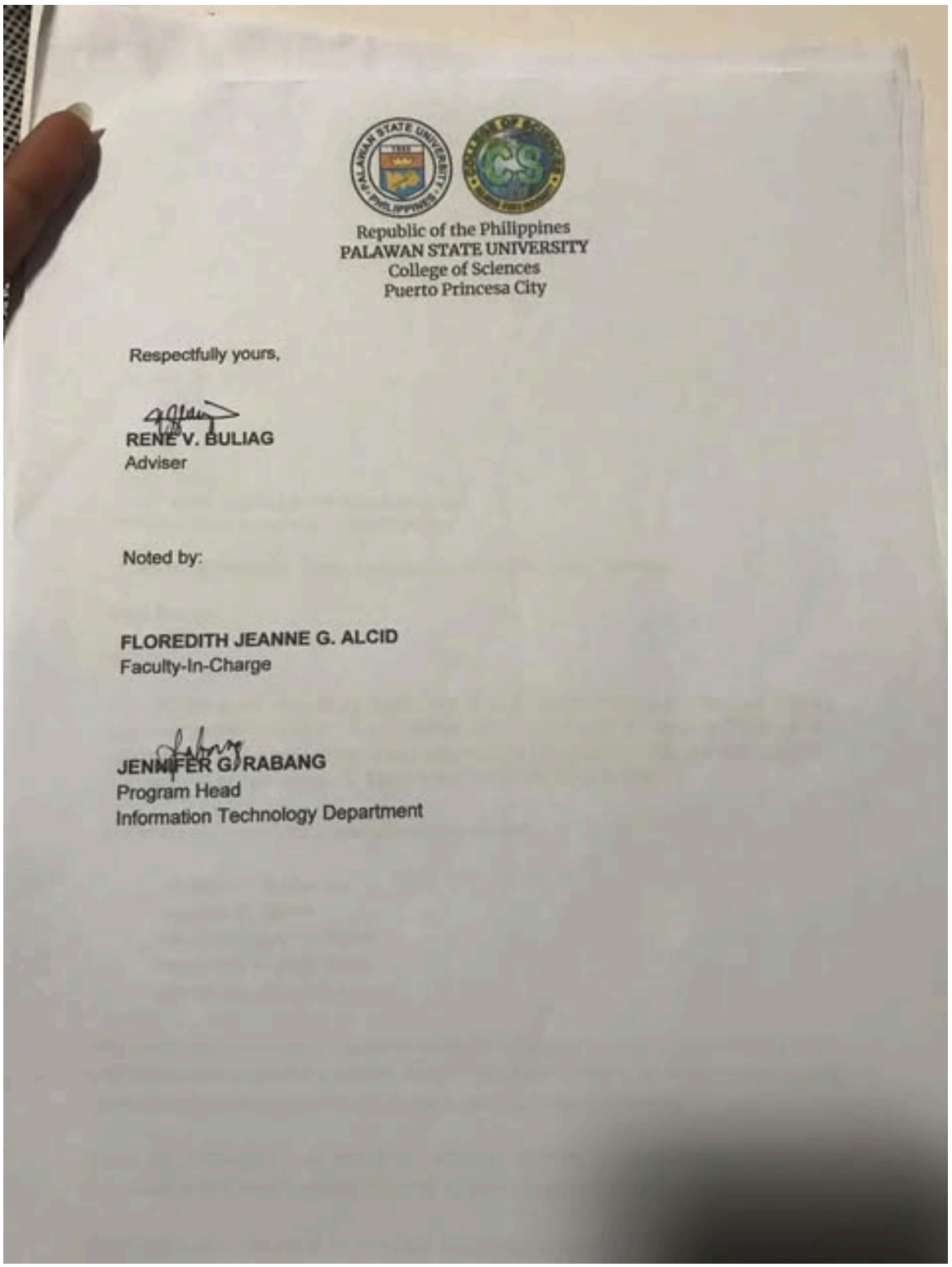
Jun Mark A. Brilliantes
Angelica B. Bande
Maverick Miguel A. Torres
Marco Rye S. Dela Serna
Charles Darwin G. Duerme

The team has conducted a research study in collaboration with your company/office and developed a capstone project titled **"PSU SportsTrack: A Web-based Sports Tournament Management System for Palawan State University"**.

Given your expertise and familiarity with the domain, your valuable insights and evaluation would contribute significantly to the refinement of their project.

Thank you very much for your time and continued support.

Palawan State University, Tiniguiban Heights, Puerto Princesa City, Philippines 5300
www.psu.palawan.edu.ph • official-email@psu.palawan.edu.ph • 0999-999-9999



Republic of the Philippines
PALAWAN STATE UNIVERSITY
College of Sciences
Puerto Princesa City

Respectfully yours,

RENE V. BULIAG
Adviser

Noted by:

FLOREDITH JEANNE G. ALCID
Faculty-In-Charge

JENNIFER G. RABANG
Program Head
Information Technology Department

Republic of the Philippines
PALAWAN STATE UNIVERSITY
College of Sciences
Puerto Princesa City

October 28, 2026

Jamaica C. Magbanua
Instructor 1
Computer Science Departmente
Subject: **Request for UAT Expert Evaluation**

Dear : **Jamaica C. Magbanua**

Greetings!

The following fourth-year Bachelor of Science in Information Technology students at Palawan State University – Main Campus, are currently enrolled in IT19 – Capstone Project and Research 2.

1. Brilliantes, Jun Mark
2. Dela Serna, Marco Rye
3. Bandal, Angelica
4. Torres, Maverick Miguel
5. Duerme, Charles Darwin

We respectfully seek your permission for the team to conduct online face-to-face User Acceptance Testing, as well as gather the necessary data for their Capstone Project entitled **PSU Sports Track: A web-based Sports Tournament Management System for Palawan State University.**

The User Acceptance Testing (UAT) will be conducted to evaluate the system's functionality, usability, performance, security, compatibility, data integrity, error handling, and overall suitability against predefined user requirements and the intended goals of the Capstone Project.

Below are the links to the survey form for your preferred date and time:

User Acceptance Testing Survey Form:
https://forms.gle/x2xovtVeWYE1dgov5

Your approval will be vital to the success of their study. Rest assured that all collected data will be kept strictly confidential.

Palawan State University, Tiniguiban Heights, Puerto Princesa City, Philippines 5300
www.psu.palawan.edu.ph • official-email@psu.palawan.edu.ph • 0999-999-9999

Republic of the Philippines
PALAWAN STATE UNIVERSITY
College of Sciences
Puerto Princesa City

Thank you for your time and consideration.

Respectfully yours,

RENE V. BULIAG
Adviser

Noted:

FLOREDITH JEANNE G. ALCID
Faculty-In-Charge

JENNIFER G. RABANG
Program Head
Information Technology Department

PSU-ACA-009

Republic of the Philippines
PALAWAN STATE UNIVERSITY
Puerto Princesa City
College of Sciences

October 21, 2026

Request for Final Defense

FLOREDEL D. GALON
Dean
This

College

Madam:

The undersigned request your approval for the schedule of our final defense on **November 3, 2026 from 12:00 PM to 2:00 PM** for Capstone Project titled: **"PSU SportsTrack: A Web-based Tournament Management System for Palawan State University** and that the following faculty are requested as Panel of Examiners.

Chairman	Arlene Bobier
Member	Roy Cyril Dosado
Member	Albert Ivan Castillo

In relation to this, we have provided 4 printed copies of our capstone project for distribution to the Panel of Examiners.

Truly yours,

Jun Mark A. Brilliantes

Angelica F. Bande

Maverick Miguel A. Torres

Marco Rye S. Dela Serna

Charles Darwin G. Duerme

Noted:

RENE V. BUNAG
Research Adviser

Recommending Approval:

FLOREDITH JEANNE G. ALCID
Research Instructor

JENNIFER G. RABANG
Program Head
Information Technology Department

Approved:

FLOREDEL D. GALON
Dean
College of Sciences

PSU-ACA-

008 Republic of the Philippines
PALAWAN STATE UNIVERSITY
Puerto Princesa City

College of Sciences
Department of Information Technology

Certification of Readiness for Final Defense

FLOREDEL D. GALON
Dean
This

College

Madam:

This is to certify that **Jun Mark A. Brilliantes, Angelica B. Bande, Marco Rye S. Dela Serna, Maverick Miguel A. Torres, and Charles Darwin G. Duerme** are ready to defend their Capstone Project **PSU SportsTrack: A Web-Based Sports Tournament Management System for Palawan State University**.

It is requested that oral final defense be scheduled.

RENE V. BULIAG
Research

Adviser Schedule for Oral Final Defense:

Date & Time: **November 3, 2026 12:00PM - 2:00PM** Venue: **PITBI Bldg.**
Recommending Approval:

FLOREDITH JEANNE G. ALCID
Research Instructor

JENNIFER G. RABANG
Program Head
Information Technology

Department Approved

FLOREDEL D. GALON
Dean

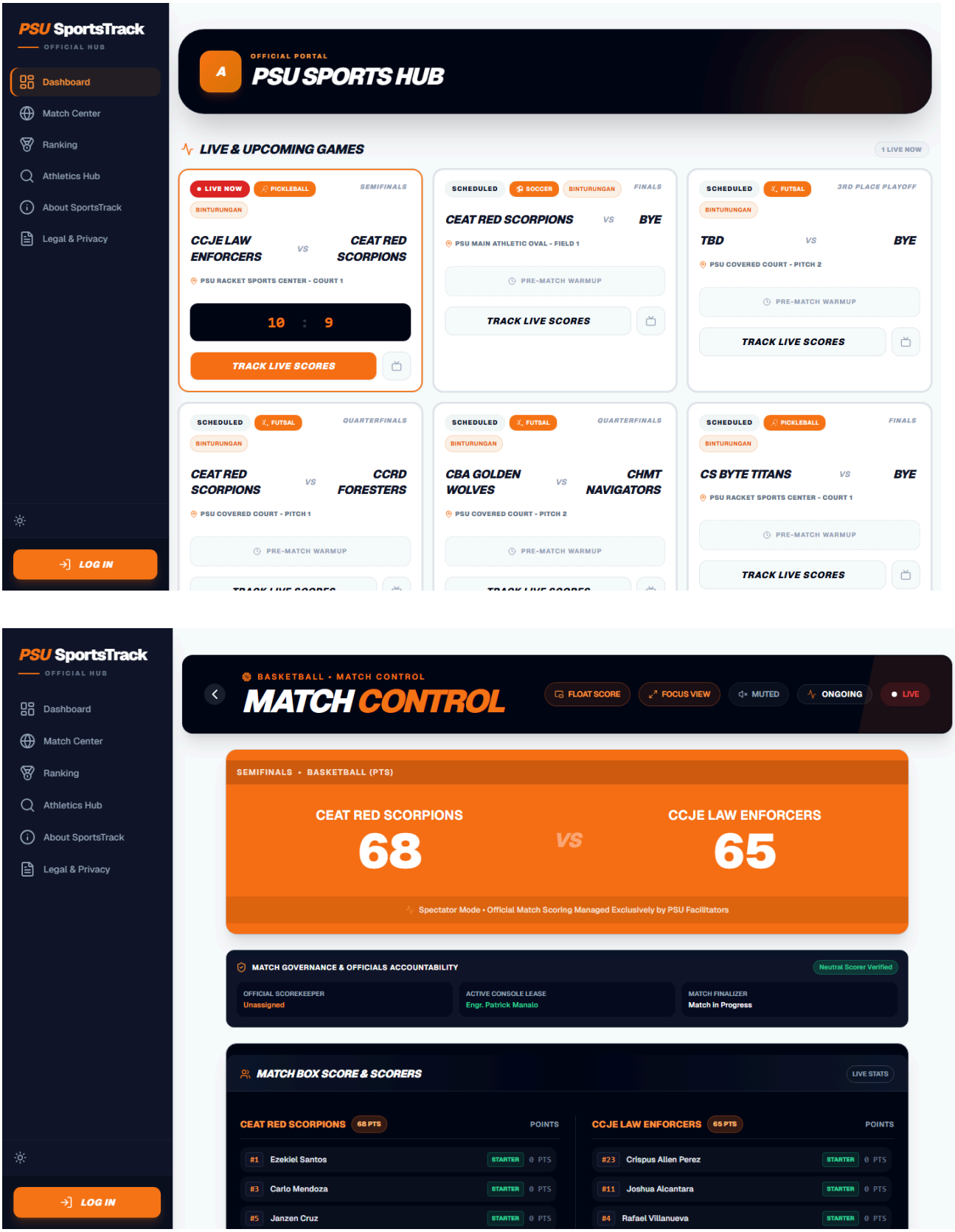
Copy Furnished:
Research Adviser
Research Coordinator
Department
Chairperson Dean
Proponents

APPENDIX C: SCREEN LAYOUTS

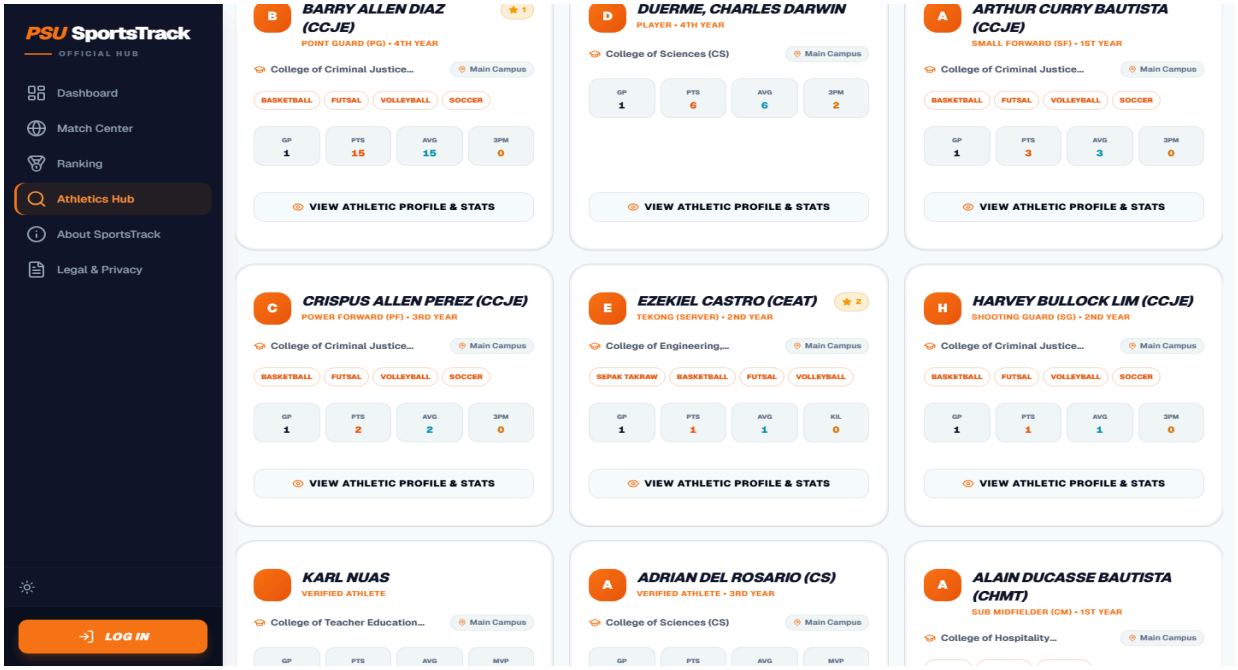
The following section outlines the primary screen layouts and core user interfaces of the PSU SportsTrack system. While the deployed application encompasses numerous supplementary views—such as password recovery flows, error pages, and legal notices—this appendix focuses exclusively on the essential, role-based dashboards and operational consoles that drive the system's primary functionalities.

The interface is a responsive Single-Page Application (SPA) built with React 19, Tailwind CSS, and shadcn/ui, featuring light and dark modes and a mobile-first design. To accurately reflect the system's hardware implementation plan, the screenshots provided below demonstrate a mix of desktop, tablet, and mobile viewports.

Public Portal & Athletics Hub Features a live match ticker, hero carousel, and searchable athlete directory. Includes a Picture-in-Picture floating scoreboard widget for spectators.



Figures 13.1 Public Portal



Figures 13.2. Athletics Hub

Authentication & Registration Screens A secure modal validating @psu.palawan.edu.ph emails, triggering the 6-digit OTP or PKCE magic link flow, and displaying the 5-minute brute-force lockout countdown.

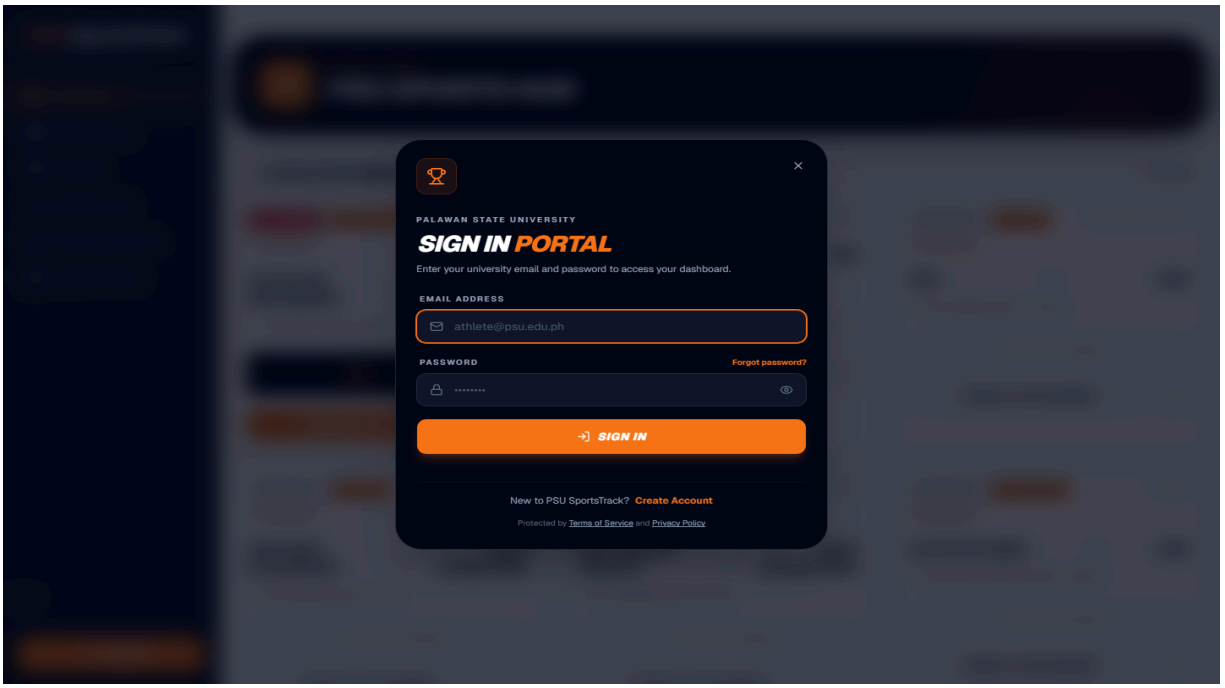
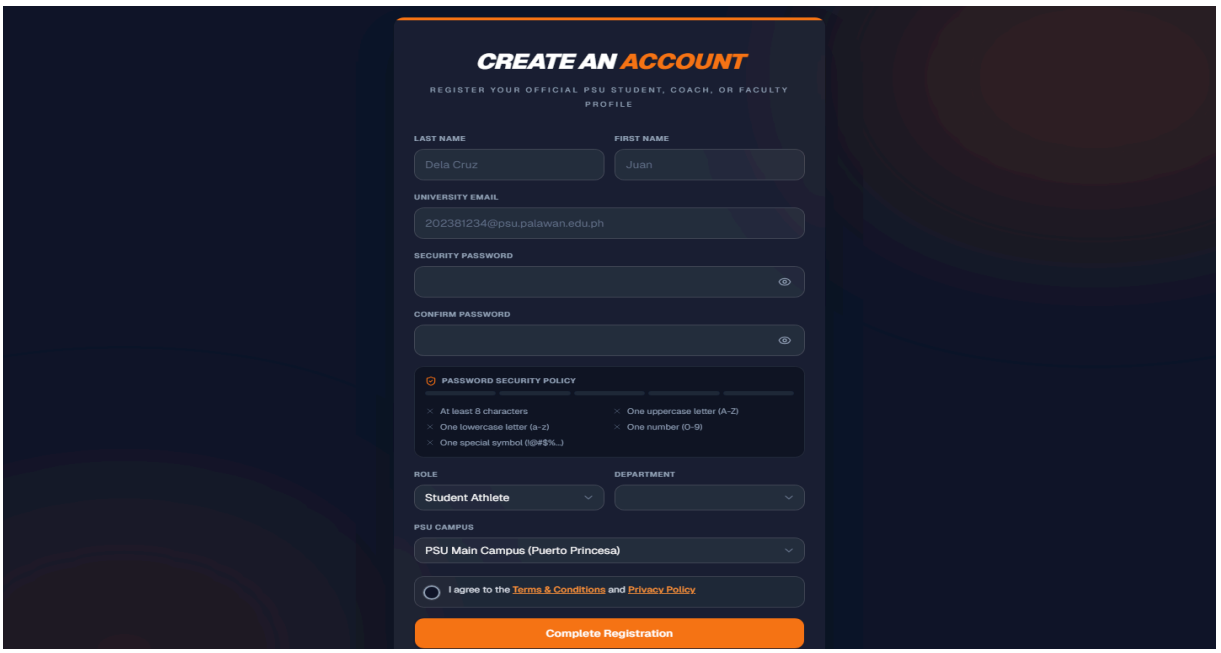


Figure 13.3. Authentication



Figures 13.4. Registration Screen

Administrator Console An executive dashboard displaying real-time metrics (active events, pending verifications). Includes the Verification Desk featuring an embedded PDF.js viewer rendering 60-second expiring signed URLs and OCR confidence scores alongside a PSU AIMS cross-check link.

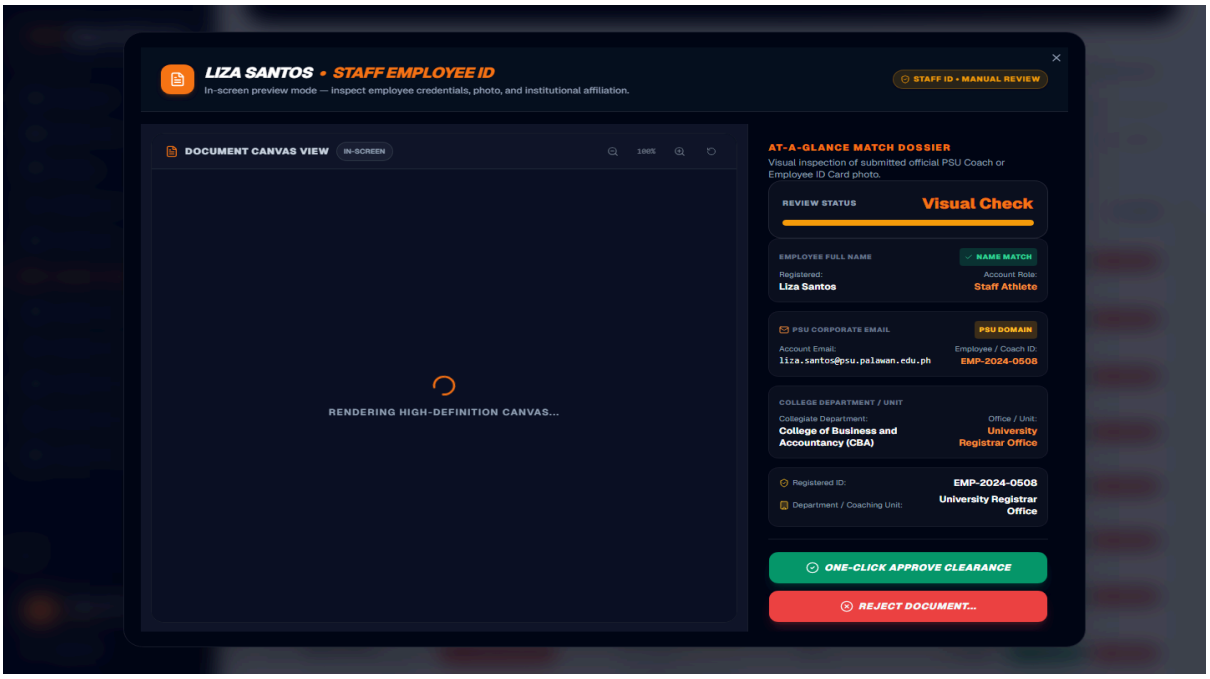
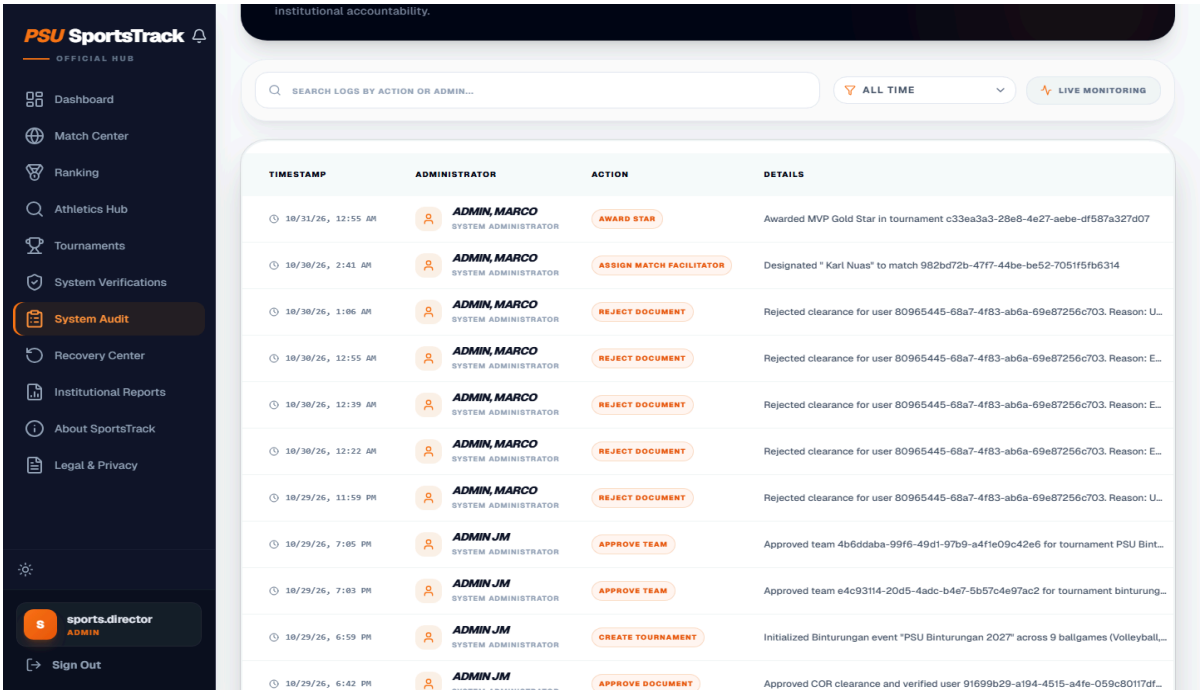
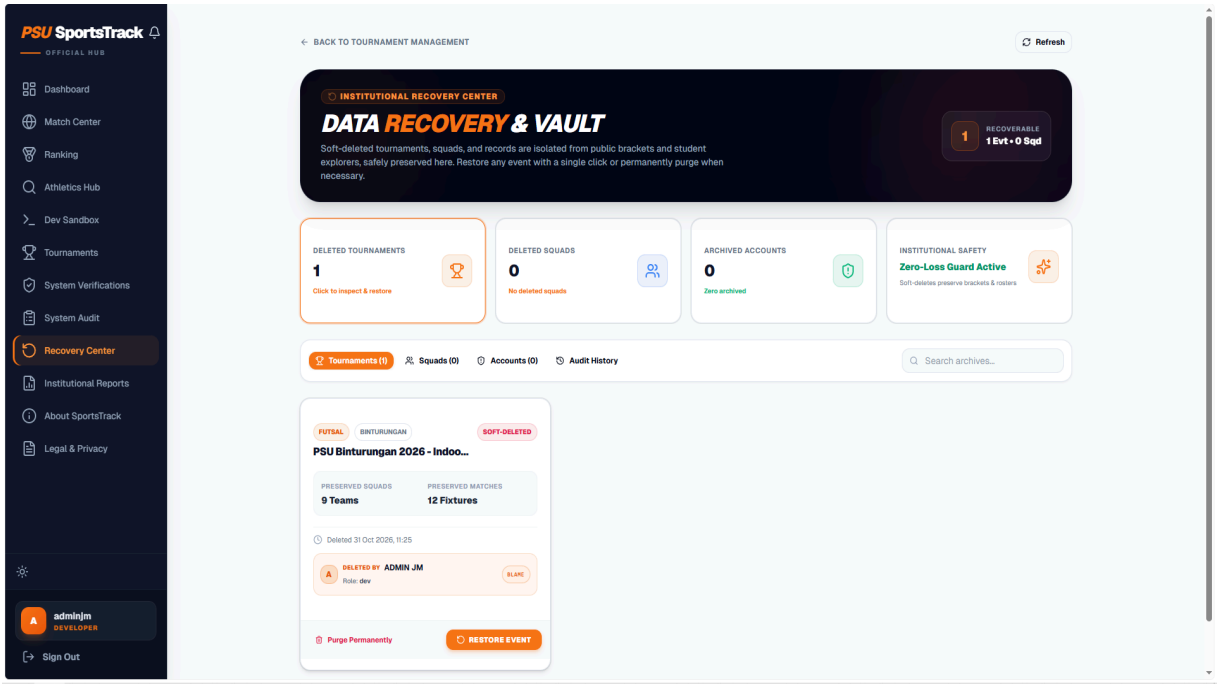


Figure 13.5. Administrator Console

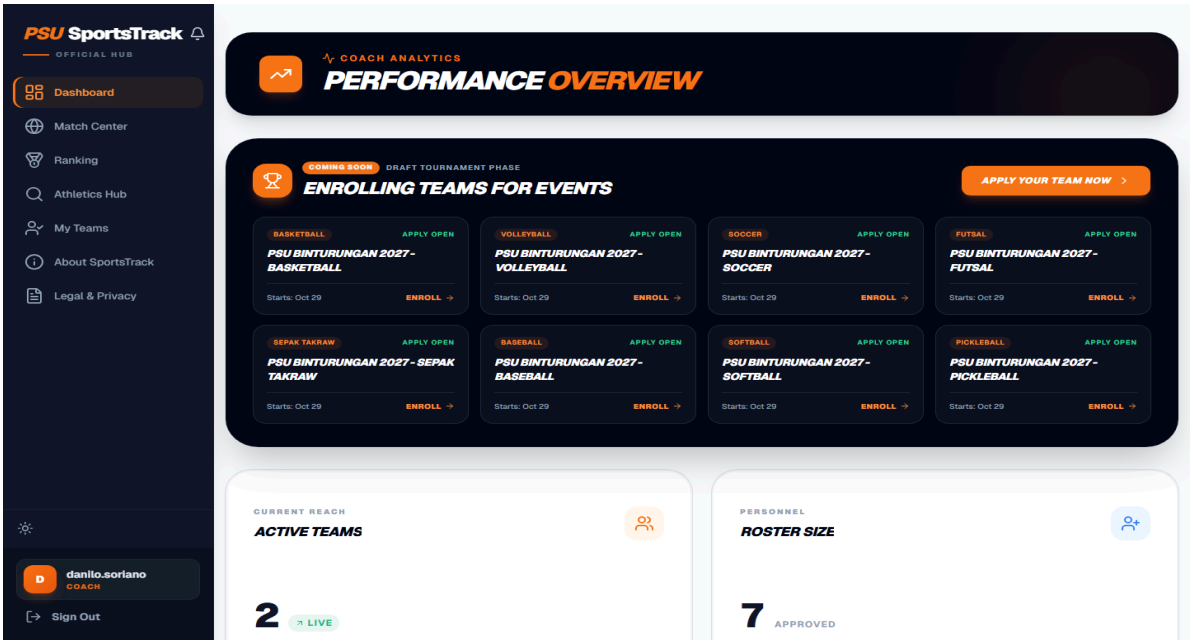
Audit Trail & Recovery Center Specialized administrative data tables where the executive user reviews immutable event logs (timestamp, actor, JSON payload differences) and restores soft-deleted tournaments, teams, or users.

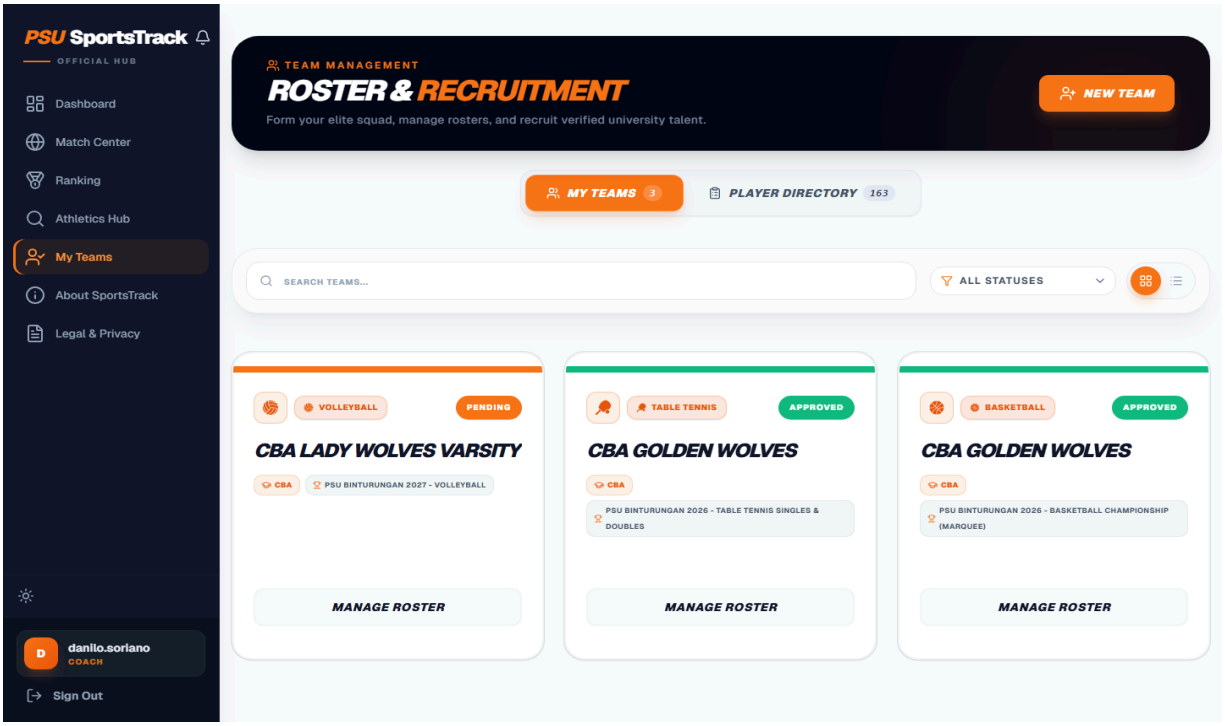




Figures 13.6. Audit Trail & Recovery Center

Coach Dashboard A recruitment and management interface allowing coaches to browse verified athlete pools, send squad invitations, and configure game-day lineups (jersey numbers, starters, availability).





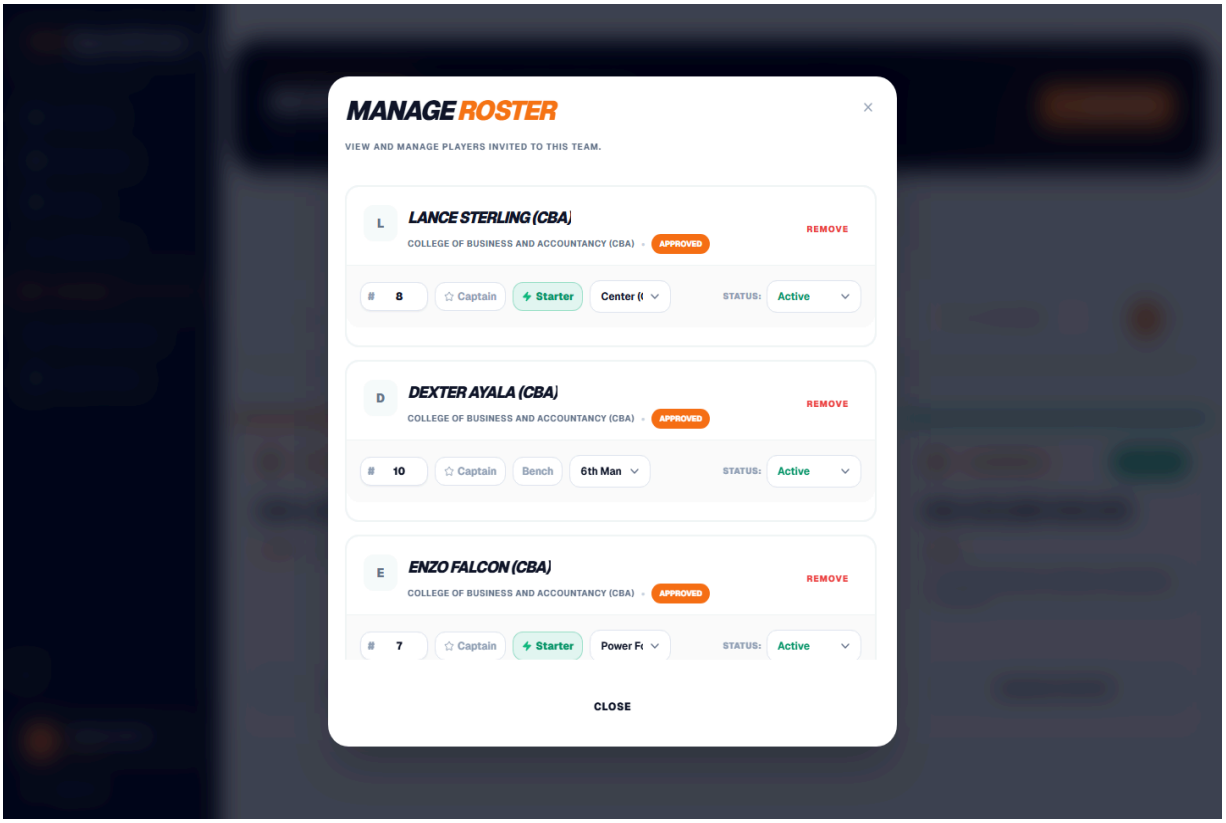
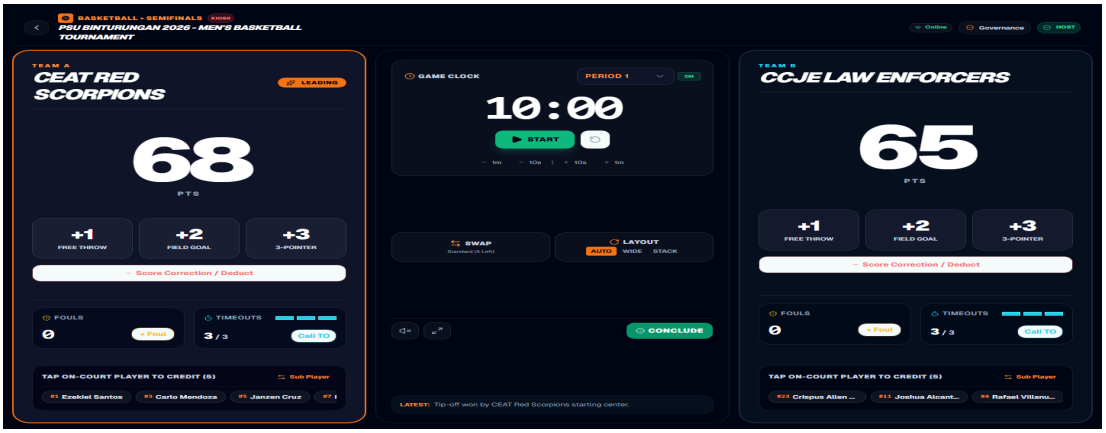


Figure 13.7. Coach Dashboard

Facilitator Kiosk A touch-first desk optimized for tablets and smartphones. Features physical-style buttons for point attribution, game clock controls, team foul tallies, and a 2-tap substitution interface, all governed by the 90-second scorekeeper lease.



Figures 13.8. Facilitator Kiosk

Athlete Console A personal dashboard displaying the user's live verification status, pending team invitations, active squad memberships, match schedules, and a trophy case for unlocked Gold Star and Red Star MVP awards.

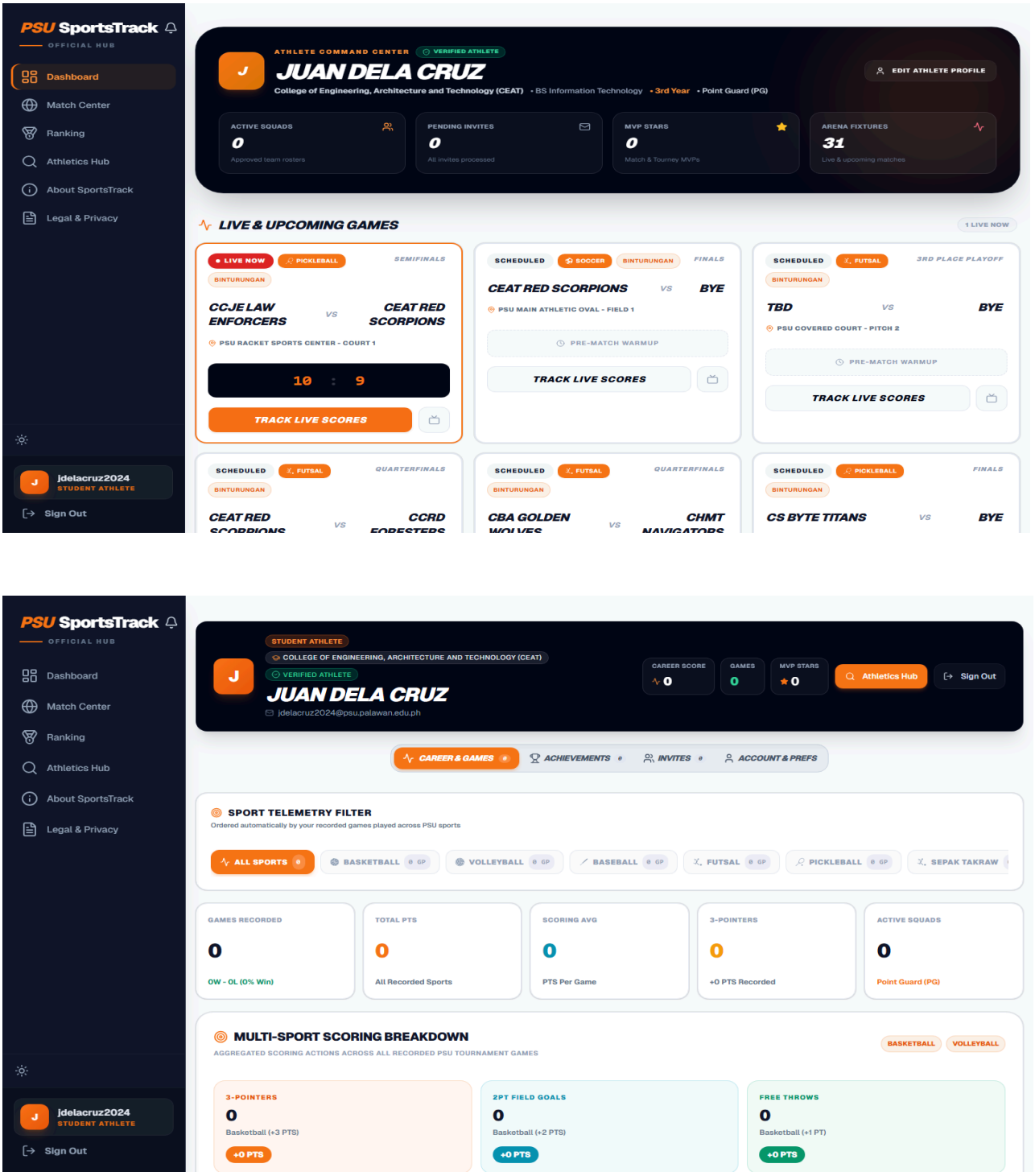
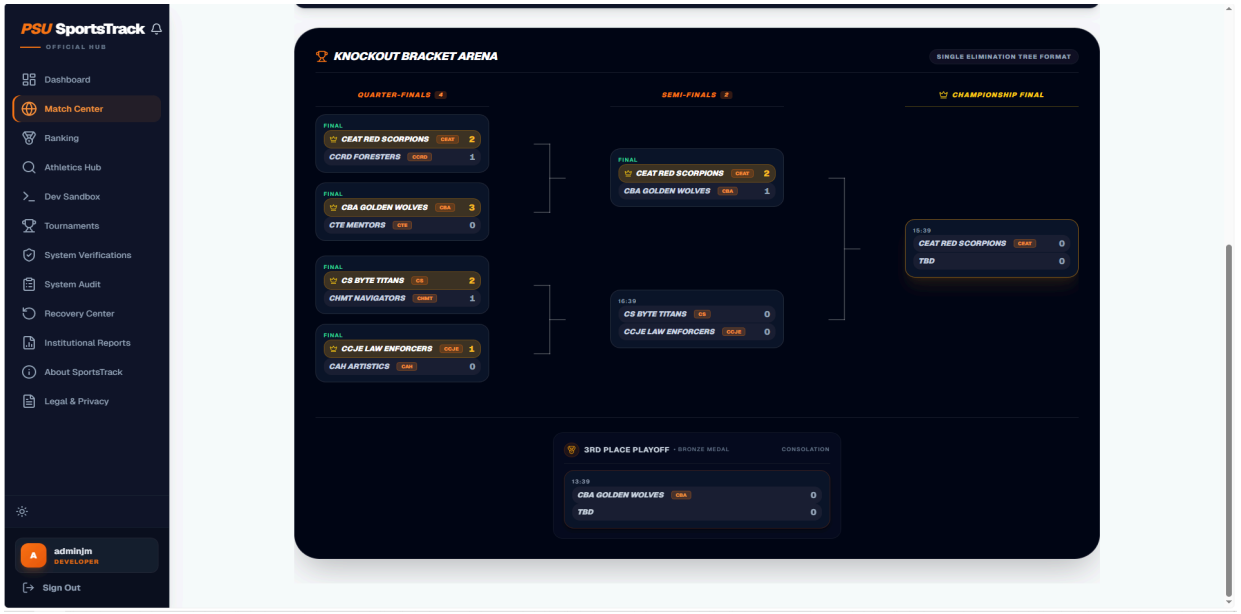
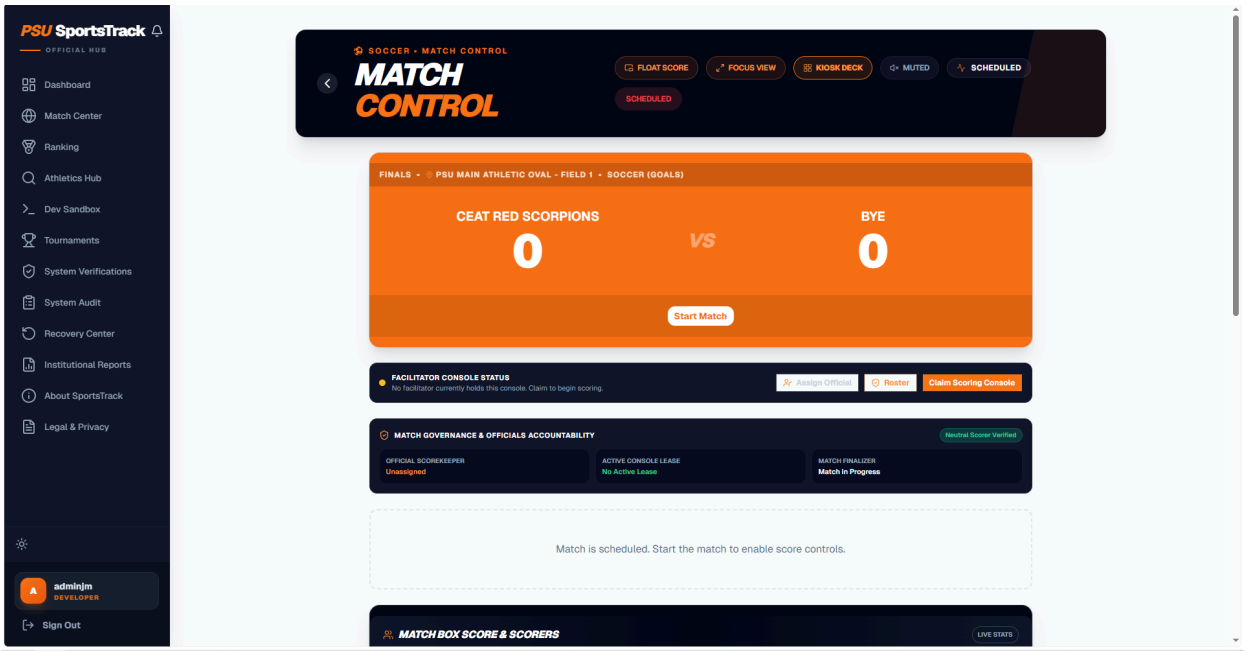


Figure 13.9. Athlete Console

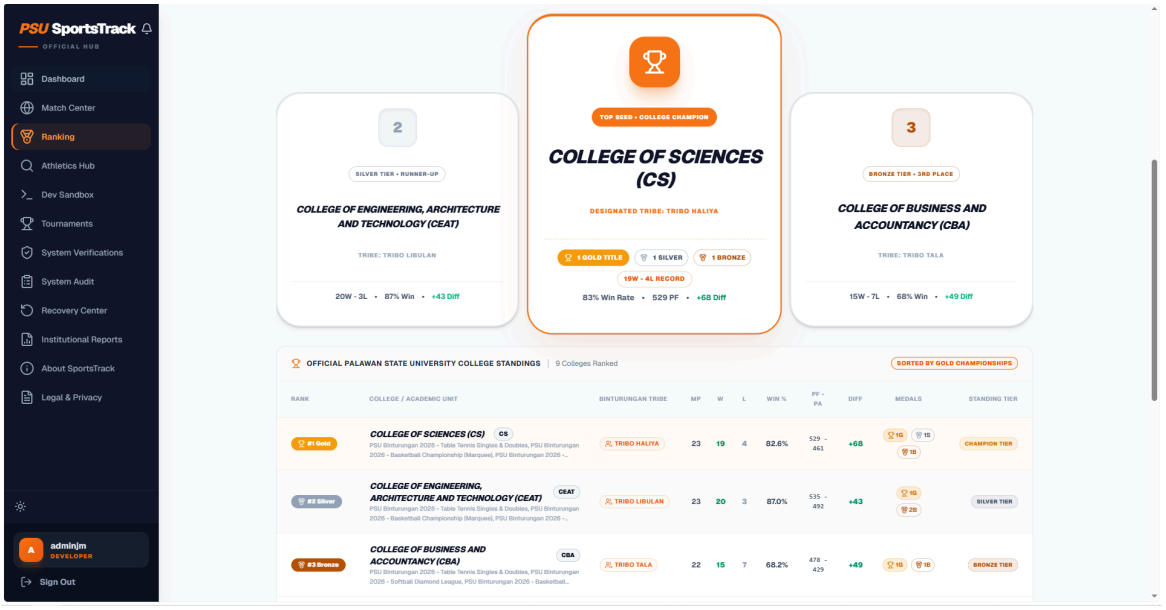
Tournament Explorer & Interactive Brackets A dynamic public view rendering the automated playoff trees (Single Elimination, Double Elimination, Round Robin, Hybrid) and live box scores.





Figures 14.1 Tournament Explorer & Interactive Brackets

Rankings & Medal Standings The institutional leaderboard displaying the Gold, Silver, and Bronze podiums for colleges and Binturungan clusters, alongside overall point differentials and win rates.



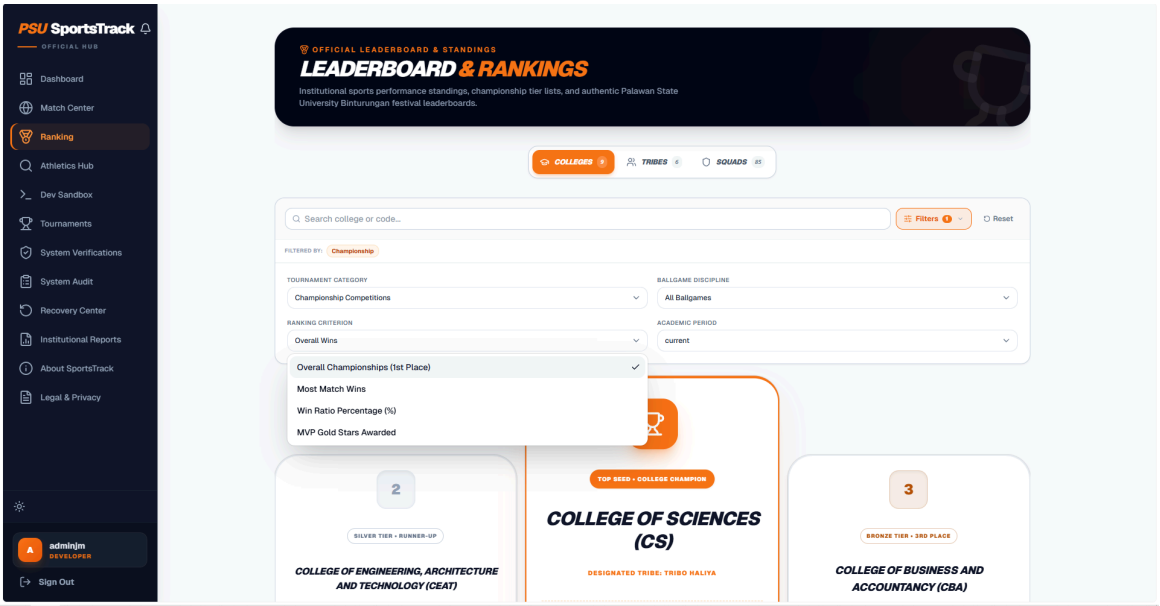


Figure 14.2. Rankings & Medal Standings

APPENDIX D: GUIDE QUESTIONS FOR TESTING THE SYSTEM

Spectator (Public / Guest)

Legend:

1 = Strongly Disagree

2 = Disagree

3 = Neutral

4 = Agree

5 = Strongly Agree

Questions					
1. Public Access: I could view the live dashboard and ticker without creating an account.					
2. Match Center Navigation: I could easily find a tournament, sport, and bracket in the Match Center.					
3. Live Match View: The live match view showed the score, clock, fouls, and events clearly.					
4. Real-Time Updates: The score updated on its own without refreshing the page.					
5. Stadium Display: The stadium display view was readable from a distance.					
6. Rankings and Medals: The medal standings and rankings were easy to understand.					

7. Ease of Learning: The system was easy to learn and use without outside help.					
8. System Responsiveness: The pages loaded quickly and responded promptly to my actions.					
9. Interface Layout: The layout was clear and worked well on my device.					
10. Role-Based Access: I could see and do only what my role allowed.					
11. Overall Satisfaction: I would use this system for official PSU sports events.					

Comments and Suggestions (problems encountered, if any):_____

Student Athlete

Legend:

1 = Strongly Disagree

2 = Disagree

3 = Neutral

4 = Agree

5 = Strongly Agree

Questions					
1. Account Registration: I could register and log in without difficulty.					

2. Document Upload: I could upload my Certificate of Registration (COR) easily.					
3. Confidence Score Preview: The confidence score preview of my COR was understandable.					
4. Verification Status: My verification status (Unverified, Pending, Approved, or Rejected) was clearly shown.					
5. Rejection Feedback: If my document was rejected, the reason and the way to upload a new one were clear.					
6. Team Invitations: I could accept or decline a team invitation easily.					
7. Team and Schedule: I could see my team, jersey number, and match schedule.					
8. Profile and Honors: My profile, match history, and MVP stars were accurate.					
9. Ease of Learning: The system was easy to learn and use without outside help.					
10. System Responsiveness: The pages loaded quickly and responded promptly to my actions.					
11. Interface Layout: The layout was clear and worked well on my device.					

12. Role-Based Access: I could see and do only what my role allowed.					
13. Overall Satisfaction: I would use this system for official PSU sports events.					

Comments and Suggestions (problems encountered, if any):_____

Faculty Athlete

Legend:

1 = Strongly Disagree

2 = Disagree

3 = Neutral

4 = Agree

5 = Strongly Agree

Questions					
1. Account Registration: I could register and log in without difficulty.					
2. Document Upload: I could upload my employee or faculty ID easily.					
3. Manual Verification: It was clear that my faculty ID would be reviewed by the administrator.					

4. Verification Status: My verification status (Unverified, Pending, Approved, or Rejected) was clearly shown.					
5. Rejection Feedback: If my document was rejected, the reason and the way to upload a new one were clear.					
6. Team Invitations: I could accept or decline a team invitation easily.					
7. Team and Schedule: I could see my team, jersey number, and match schedule.					
8. Profile and Honors: My profile, match history, and MVP stars were accurate.					
9. Ease of Learning: The system was easy to learn and use without outside help.					
10. System Responsiveness: The pages loaded quickly and responded promptly to my actions.					
11. Interface Layout: The layout was clear and worked well on my device.					
12. Role-Based Access: I could see and do only what my role allowed.					
13. Overall Satisfaction: I would use this system for official PSU sports events.					

Comments and Suggestions (problems encountered, if any):_____

Facilitator (Court-Side)

Legend:

1 = Strongly Disagree

2 = Disagree

3 = Neutral

4 = Agree

5 = Strongly Agree

Questions					
1. Scorekeeper Access: I could open the scorekeeper desk or kiosk for my assigned match.					
2. Scorekeeper Lease: I could acquire the scorekeeper lease without difficulty.					
3. Lease Transfer: The lease transfer prompt was clear when another scorekeeper held the lease.					
4. Game Clock Control: The game clock and period controls worked correctly.					
5. Point Attribution: I could record points and assign them to the correct athlete.					
6. Fouls and Timeouts: I could record fouls and timeouts correctly.					
7. Substitutions: I could complete a substitution in two taps.					

8. Score Correction: I could correct a score, and the score never went below zero.					
9. Offline Scoring: Scoring continued during a network drop and synced afterward.					
10. Match Finalization: I could finalize the match and award the Red Star (Game MVP).					
11. Ease of Learning: The system was easy to learn and use without outside help.					
12. System Responsiveness: The pages loaded quickly and responded promptly to my actions.					
13. Interface Layout: The layout was clear and worked well on my device.					
14. Role-Based Access: I could see and do only what my role allowed.					
15. Overall Satisfaction: I would use this system for official PSU sports events.					

Comments and Suggestions (problems encountered, if any):_____

Administrator

Legend:

1 = Strongly Disagree

2 = Disagree

3 = Neutral

4 = Agree

5 = Strongly Agree

Questions					
1. Dashboard Accuracy: The executive dashboard counts were accurate.					
2. Document Review: I could review verification documents in the PDF viewer.					
3. Confidence Score Breakdown: The confidence score and its breakdown helped me decide.					
4. Approval and Rejection: I could approve or reject a document and record the reason.					
5. Tournament Setup: I could create and configure a tournament.					
6. Bracket Generation: The brackets were generated correctly.					
7. Scheduling: I could schedule games, venues, and facilitators.					
8. Facilitator and Lease Control: I could appoint a facilitator and override a scorekeeper lease.					
9. Standings and Awards: I could finalize standings, award medals, and award the Gold Star (Tournament MVP).					

10. Audit and Recovery: I could review the audit logs and restore a deleted record.					
11. Ease of Learning: The system was easy to learn and use without outside help.					
12. System Responsiveness: The pages loaded quickly and responded promptly to my actions.					
13. Interface Layout: The layout was clear and worked well on my device.					
14. Role-Based Access: I could see and do only what my role allowed.					
15. Overall Satisfaction: I would use this system for official PSU sports events.					

Comments and Suggestions (problems encountered, if any):_____

APPENDIX E: SAMPLE INPUT/OUTPUT/REPORTS

Figure 15 illustrates the Input-Process-Output (IPO) model for the public spectator role within the PSU SportsTrack system. The input phase consists of unauthenticated URL access, basic device browser usage, and interactive filtering or match selection by the user. The system processes these requests through anonymous role routing, utilizing Supabase WebSockets to synchronize real-time data and executing necessary browser APIs for audio and picture-in-picture features. The resulting output delivers a live, dynamic viewing experience that includes a rotating featured banner, real-time match dashboards, medal standings, and floating or

full-screen stadium displays. A continuous feedback loop is maintained as the spectator navigates the interface and adjusts filters to update the displayed information.

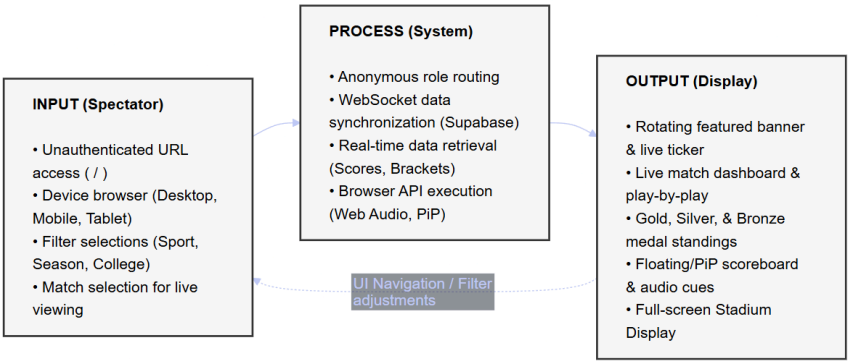


Figure 15. Input, Process, Output for Public Spectators

Figure 15.1 illustrates the Input-Process-Output (IPO) model for the Student Athlete role within the system. The input phase requires the student athlete to provide account registration credentials, upload their Certificate of Registration (COR), respond to team join invitations, and manage their personal profile and availability status. The system processes these inputs by utilizing Supabase Auth and JWT session management, executing OCR extraction and confidence scoring for document verification, isolating data via Row-Level Security (RLS), and validating roster constraints through a duplicate-sport check. The system then generates a dashboard output that displays the athlete's verification status (including approval or rejection notes), confirmed team membership and roster views, personalized match schedules and brackets, as well as individual statistics and MVP Star recognitions. A continuous feedback loop enables status updates and account actions to flow back from the dashboard to the user's inputs

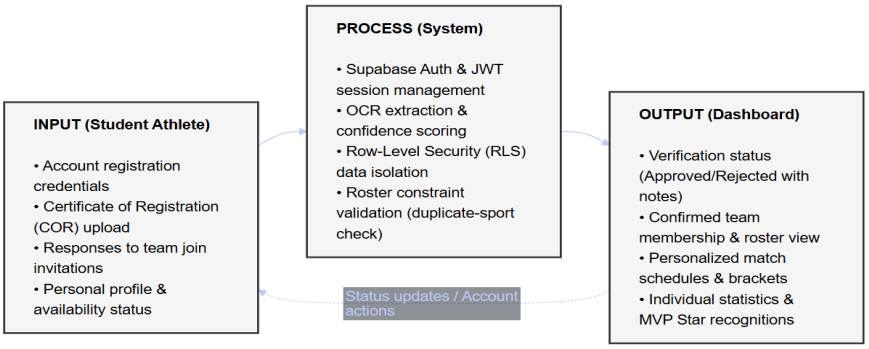


Figure 15.1 Input, Process, Output for Student Athlete

Figure 15.2 illustrates the Input-Process-Output (IPO) model for the Faculty Athlete role within the system. During the input phase, the faculty athlete provides their account registration credentials, uploads their employee identification (Faculty ID), submits responses to team join invitations, and updates their personal profile and availability status. The system processes these actions through Supabase Auth and JWT session management, performs OCR extraction and confidence scoring on the uploaded ID, ensures data isolation using Row-Level Security (RLS), and executes roster constraint validation through a duplicate-sport check. The generated output on the dashboard displays the user's verification status (approved or rejected with notes), confirmed team membership for Friendly Games, personalized match schedules and brackets, and their individual statistics along with MVP Star recognitions. A continuous feedback loop routes status updates and account actions from the dashboard back to the input stage

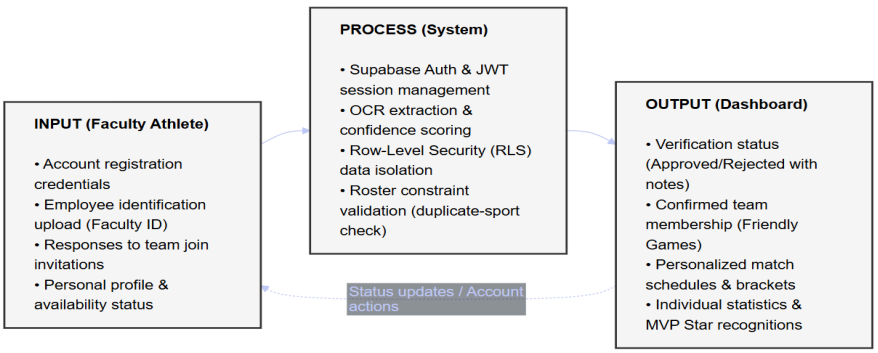


Figure 15.2 Input, Process, Output for Faculty Athlete

The figure illustrates the Input-Process-Output (IPO) model for the Staff Athlete role within the system. During the input phase, the staff athlete provides account registration credentials, uploads their employee identification (Staff ID), responds to team join invitations, and updates their personal profile and availability status. The system processes these inputs using Supabase Auth and JWT session management, performs OCR extraction and confidence scoring for the uploaded ID, isolates data through Row-Level Security (RLS), and validates roster constraints via a duplicate-sport check. The output displayed on the dashboard includes the user's verification status (approved or rejected with notes), confirmed team membership for Friendly Games, personalized match schedules and brackets, and individual statistics along with MVP Star recognitions. A continuous feedback loop returns status updates and account actions from the dashboard back to the input stage.

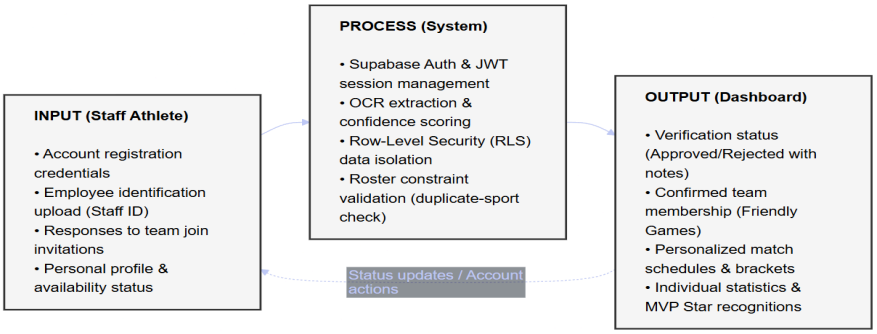


Figure 15.3 Input, Process, Output for Staff Athlete

The figure illustrates the Input-Process-Output (IPO) model for the Facilitator role within the system. During the input phase, the facilitator manages court-side scoring entries and game clock control, inputs fouls, timeouts, and two-tap substitutions, performs score corrections or point rollbacks, and finalizes the match while designating the Red Star (Game MVP). The resulting system output delivers live scoreboard and action ticker updates, finalized match box scores, updated player statistics and bracket progressions, as well as immutable audit log entries for any score

corrections. A continuous feedback loop returns match progress and UI feedback back to the facilitator.

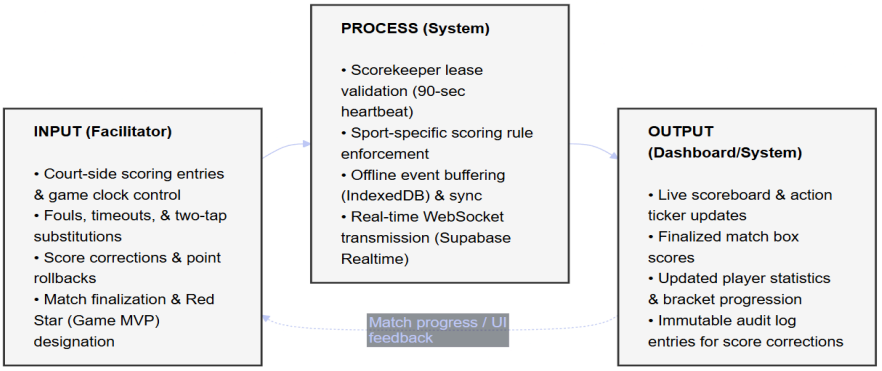


Figure 15.4 Input, Process, Output for Facilitator

The figure illustrates the Input-Process-Output (IPO) model for the Coach role within the system. During the input phase, the coach manages team creation for Collegiate Squads or Tribes, handles athlete recruitment through invitations and join requests, configures roster setups including jersey numbers, positions, and starters, updates athlete availability statuses (Active, Injured, or Excused), and claims tournament slots. The system processes these inputs by validating the athlete's verification status, executing a roster constraint check to enforce the duplicate-sport rule, isolating data using Row-Level Security (RLS), and applying tournament slot allocation logic. The generated output on the dashboard provides the coach with finalized team rosters and active lineups, recruitment notifications for accepted or declined requests, assigned match schedules and brackets, and overall team statistics along with tournament standings. A continuous feedback loop routes roster adjustments and UI feedback from the dashboard back to the input stage.

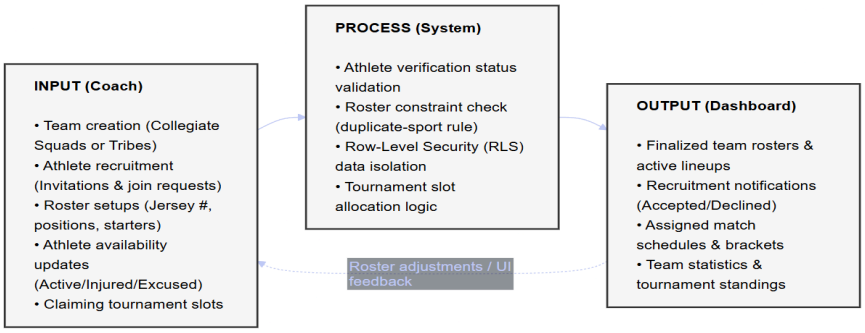


Figure 15.5 Input, Process, Output for Coach

The figure illustrates the Input-Process-Output (IPO) model for the Administrator role within the system. During the input phase, the administrator handles athlete verification decisions (approve or reject), manages tournament creation and configurations, assigns facilitators and schedules matches, finalizes standings and designates the Gold Star (Tournament MVP), and performs system overrides and Recovery Center restorations. The system processes these actions through PDF.js document rendering and OCR scoring, bracket generation and Bronze playoff routing, role authorization and lease override execution, immutable audit logging for before-and-after changes, and soft-delete record management. The resulting system output delivers an executive dashboard with key metric counters, updates to the verified athlete pool, published brackets, schedules, and finalized standings, as well as system-wide Audit Trail records and downloadable Institutional FR-41 Reports. A continuous feedback loop routes system monitoring and configuration updates from the output back to the input stage.

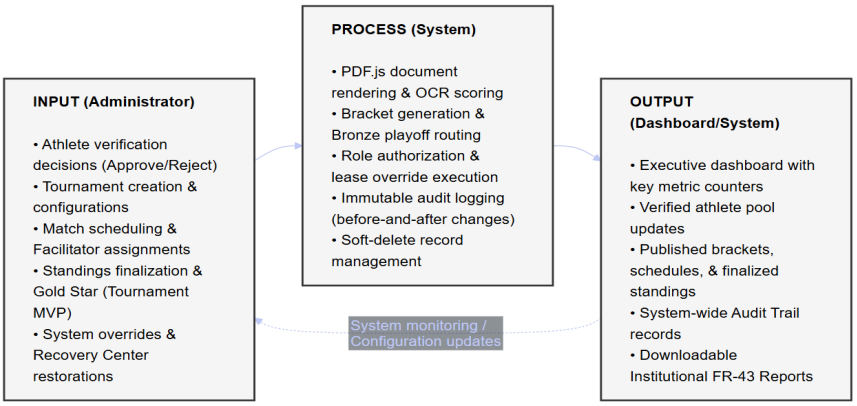


Figure 15.6 Input, Process, Output for Administrator

APPENDIX F: EVALUATION TOOL OR TEST DOCUMENTS

Data Gathering(Sports Department)

PSU SportsTrack

A Web-Based Sports Tournament Management System for Palawan State University

Data Gathering Instrument: Stakeholder Interview Questionnaires

Purpose of the Instrument

This instrument is designed to gather qualitative data from the three primary stakeholder groups of PSU SportsTrack — the Sports Department/Tournament Management, Coaches, Facilitators and Student-Athletes/Players. Responses will be used to inform the system's requirements analysis, feature prioritization, and overall design in support of the proposed capstone project.

Questionnaire 1: Sports Department / Tournament Manager

1. Can you walk me through your specific responsibilities as sports director, cluster coordinator, or tournament manager?
2. Which sports does the department run under the Binturungan and the Faculty and Staff Friendly Games, and how often is each one held throughout the year?
   Probes: Fixed calendar vs. as-needed? Seasonal patterns?
3. Do multiple sports events — or the Binturungan and Faculty and Staff Friendly Games — ever overlap or run at the same time? How is that currently handled?
4. Walk me through how a typical tournament is scheduled and run from start to finish, including how brackets are constructed by hand.
   Probes: Number of participants, bracket format used, divisions/categories, what information is needed before building a bracket.
5. If we could only automate a few sports first, which would you prioritize — and why?
6. Are there sports with especially complex bracket requirements (e.g., seeding, byes, group stages) that we should plan for early?
7. How does scoring/point allocation currently work, and does it differ across sports?
   Probes: Ties, forfeits, cumulative standings, bonus/penalty rules.

8. Would a single unified point system work across all sports, or does each need its own rules?
9. What's the most frustrating or time-consuming part of the current process overall?
10. What process would you want to be made easy through this system?
11. You mentioned a paid third-party site is used for basketball statistics — what does it do well, and what's missing that PSU SportsTrack should cover?
12. If this system could do anything to help you, what would it be?
    Probes: Auto-seeding, live updates, reports, multi-user access.
13. How are Gold Star (tournament MVP) and Red Star (per-game MVP) currently decided and recorded, if at all?
14. What's your biggest concern about moving this process onto a digital platform?
15. If PSU SportsTrack existed today, what's the first task you'd want it to take off your plate?

Data Gathering(Coaches)

Questionnaire 2: Coaches

1. How do you currently find and recruit players for your team — walk me through it from start to finish.
2. How much time does building a full roster usually take you, and what slows it down most?
3. When you don't personally know many students in a sport, how do you find eligible players today?
4. How do you currently confirm a player is eligible (enrolled, cleared) before adding them to your roster?
5. How do you find out about game schedules and bracket assignments once your team is registered?
6. If you could browse a list of verified, eligible students by sport and college, what information would you want to see on their profile?
7. How close to the registration deadline do roster changes typically happen, and why?
8. What's the most frustrating part of managing your team once the tournament has started?
9. Would you want to be involved in nominating players for MVP (Red Star/Gold Star), or should that stay with the administrator only?

Data Gathering(Player/Athletes)

Questionnaire 3: Students / Athletes

1. How did you find out about joining a sports event (Binturungan or Faculty and Staff Friendly Games) in the past?
2. What was the process like for submitting your COR and school ID for eligibility — what took the most time or caused confusion?
3. How did you usually find out whether you'd been accepted onto a team?
4. How do you currently find out your game schedule, and has that ever caused you to miss information?
5. Have you ever experienced a scoring or bracket error that affected you or your team? What happened?
6. How do you currently find out your game results, standings, or how far your team advanced?
7. Would you want to see your own participation history and past MVP recognitions in one place? Why or why not?
8. What would make you more likely to want to compete again in future PSU sports events?
9. If you were checking a sports app on your phone or laptop between classes, what's the one thing you'd want to see immediately?

Data Gathering(Facilitators)

Questionnaire 4: Facilitators

1. What are your main responsibilities when facilitating a sports match during PSU sports events?
2. How do you record and report match scores and results?
3. What difficulties do you usually encounter when officiating or recording match results?
4. How do you communicate match results to the tournament manager or sports department?
5. How do you handle scoring disputes or disagreements during a match?
6. Would having a digital scoreboard and match-recording system make your responsibilities easier? Why or why not?
7. What information would you need to see on your assigned match page before and during a game?
8. Would you find it helpful to have a system that automatically updates match results and tournament standings after scores are submitted?
9. What features would you suggest to make PSU SportsTrack more useful for facilitators?

User Testing (Public Spectator)

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Your answer

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Your answer

Gender *

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(e.g., College of Sciences, BSIT Department, Office of the Sports Director, or "**Outsider / External Participant**" if not from PSU)

Your answer

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Public Spectator

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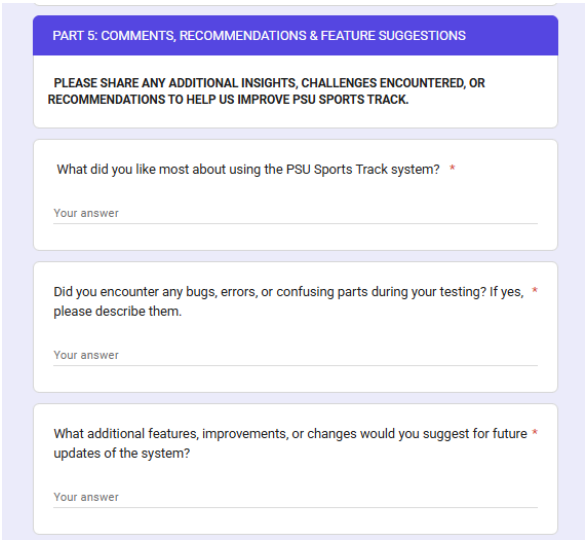
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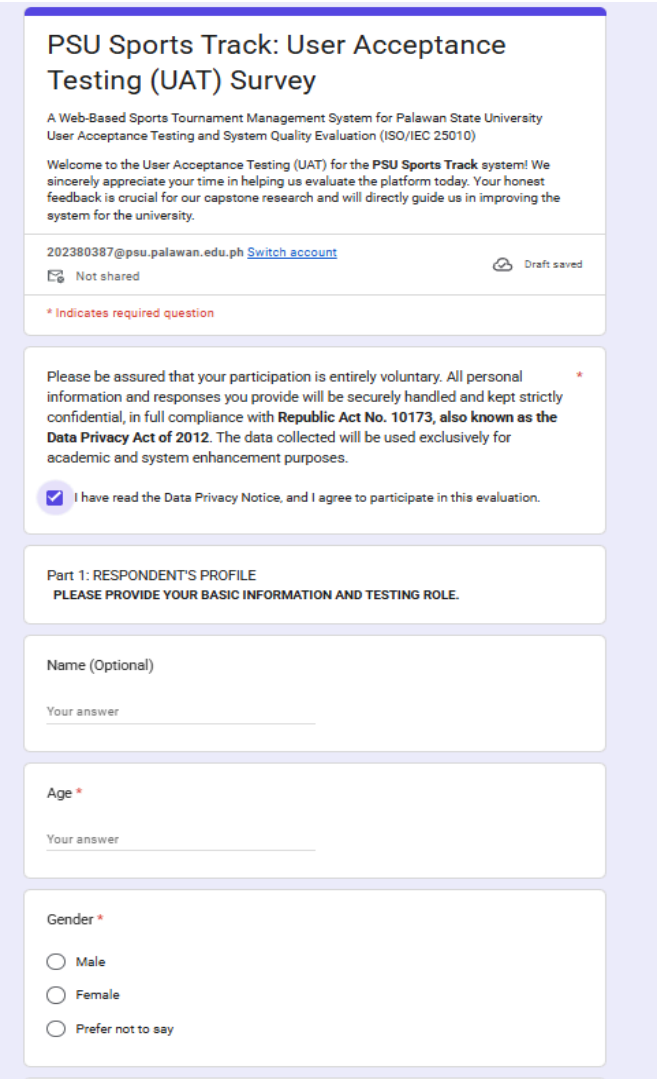
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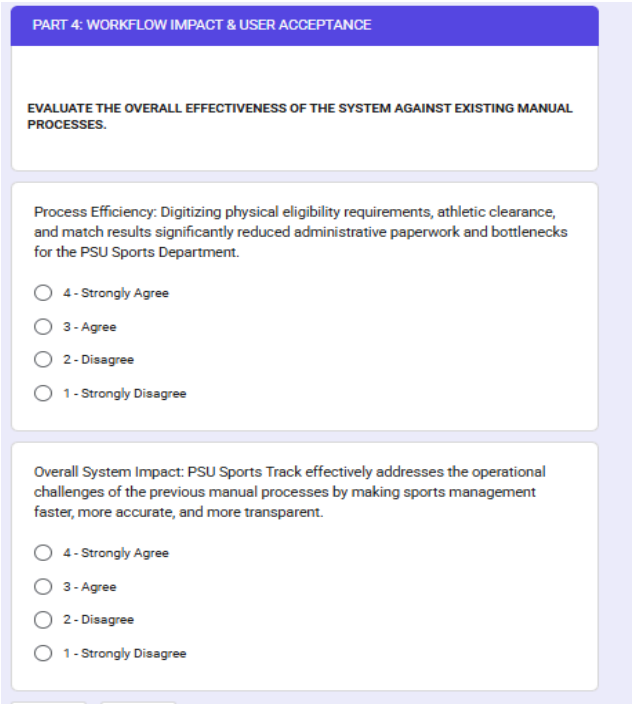
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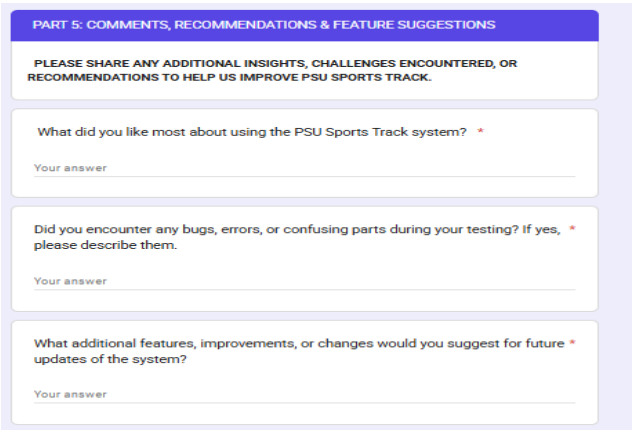
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Your answer

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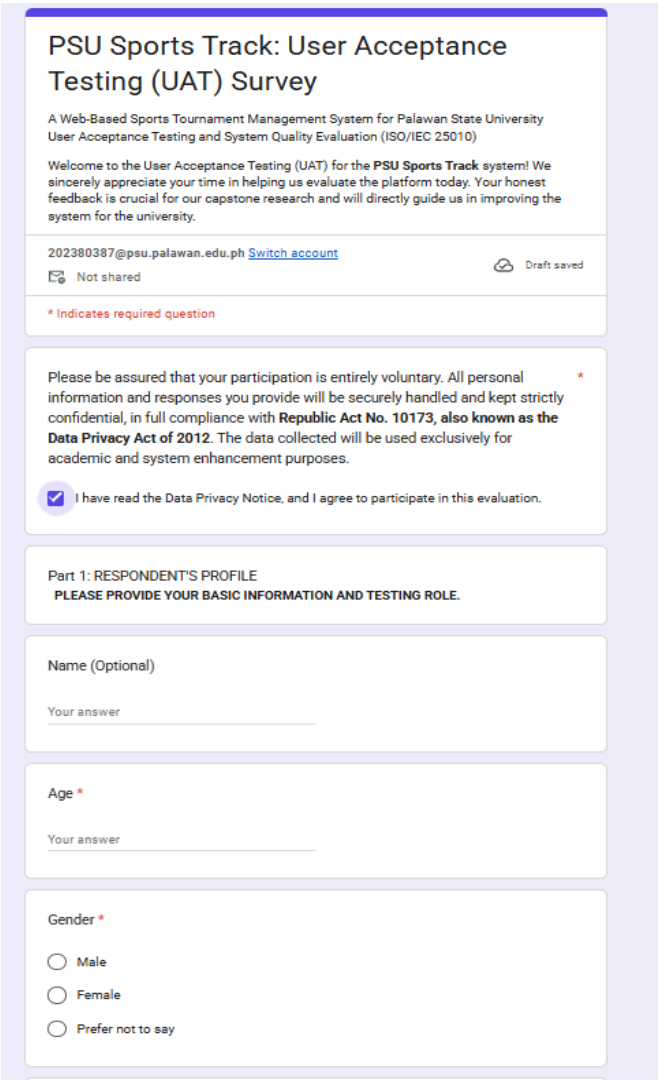

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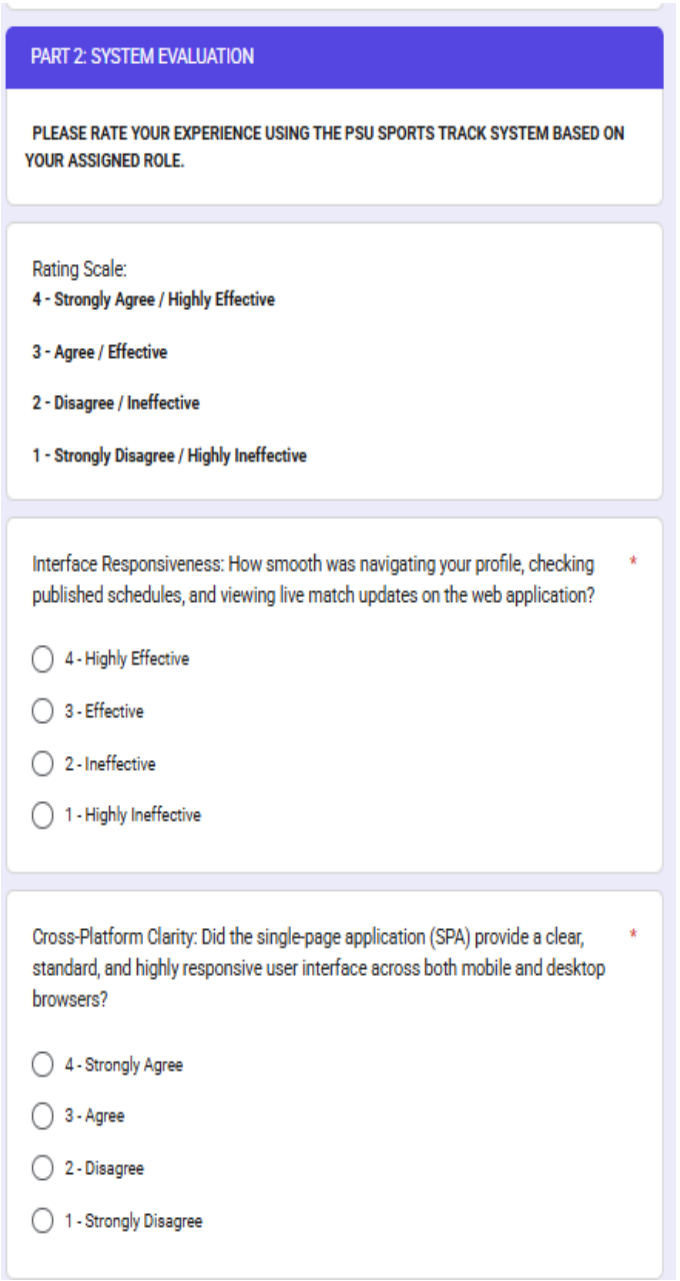

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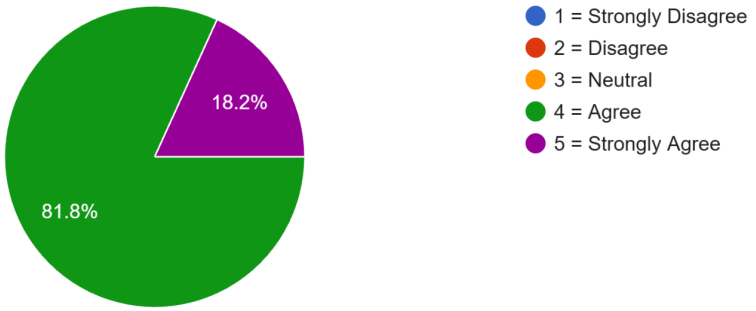
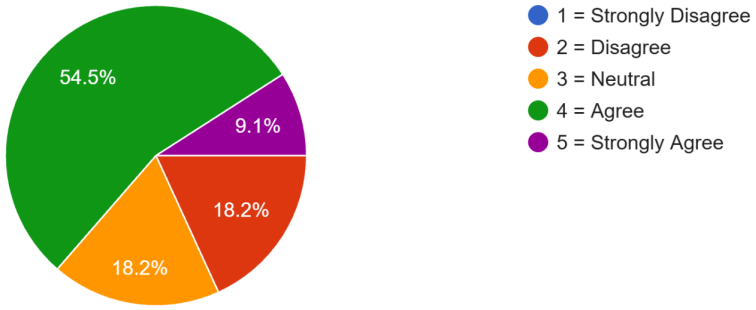
APPENDIX G: TEST RESULTS

Users(Spectators)

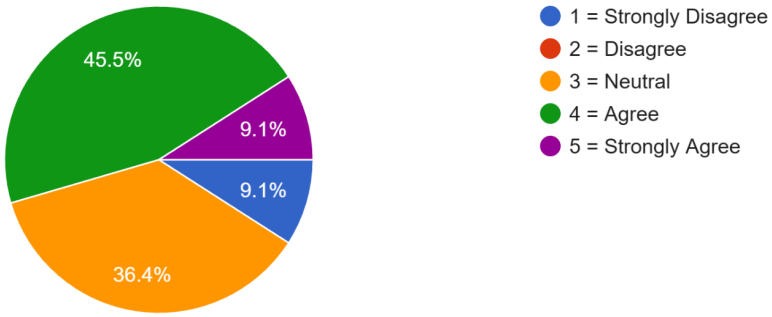
Public Access: I could view the live dashboard and ticker without creating an account.
11 responses



Match Center Navigation: I could easily find a tournament, sport, and bracket in the Match Center.
11 responses

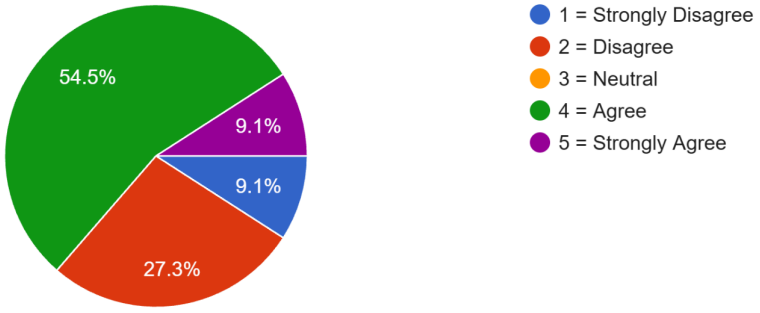


Live Match View: The live match view showed the score, clock, fouls, and events clearly.
11 responses



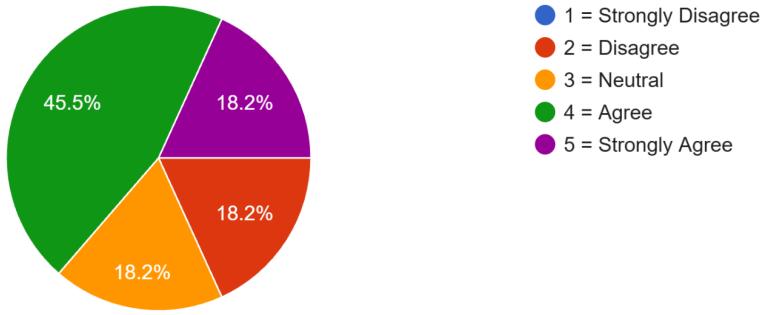
Real-Time Updates: The score updated on its own without refreshing the page.

11 responses



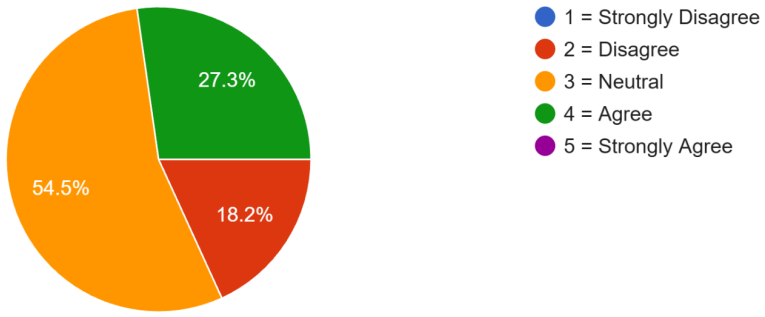
Stadium Display: The stadium display view was readable from a distance.

11 responses



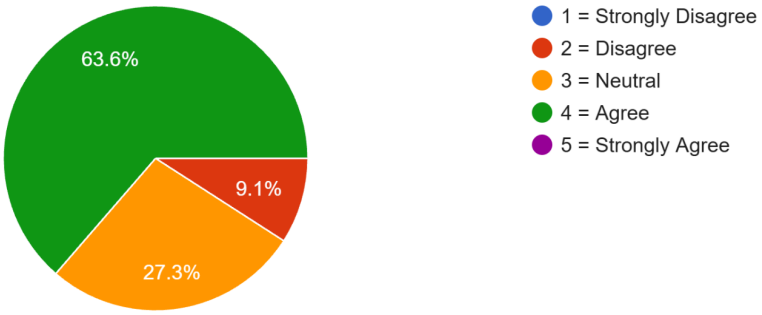
Rankings and Medals: The medal standings and rankings were easy to understand.

11 responses



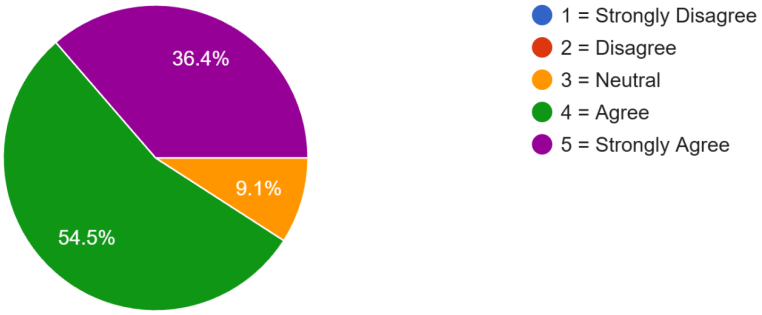
Ease of Learning: The system was easy to learn and use without outside help.

11 responses



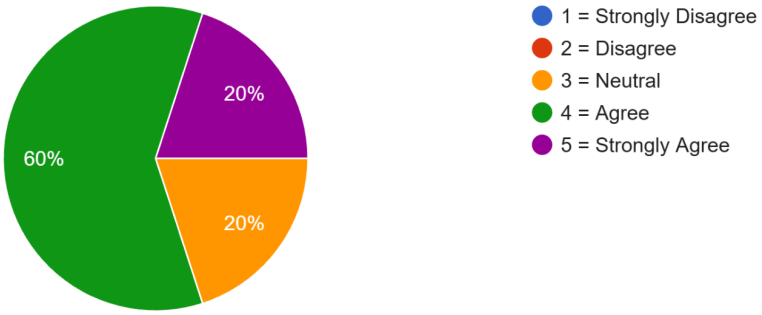
System Responsiveness: The pages loaded quickly and responded promptly to my actions.

11 responses



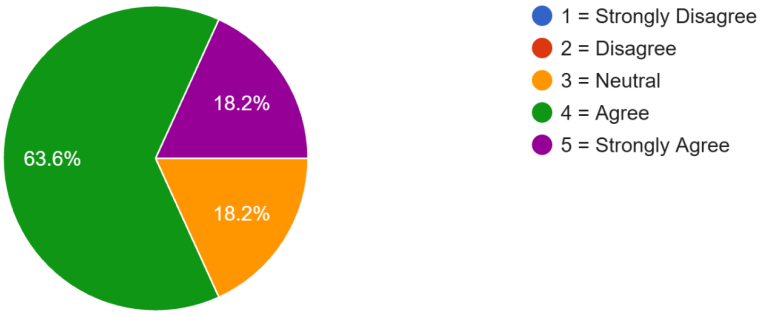
Interface Layout: The layout was clear and worked well on my device.

10 responses



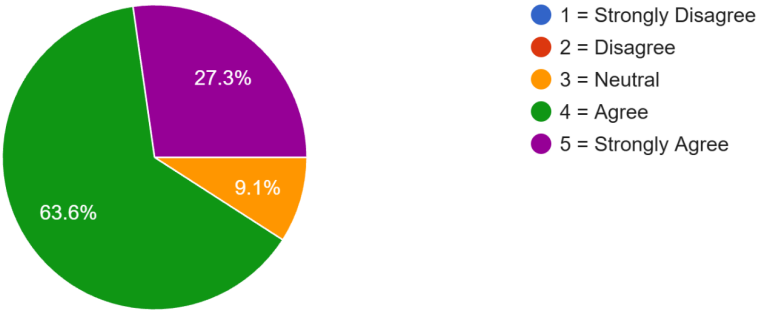
Role-Based Access: I could see and do only what my role allowed.

11 responses



Overall Satisfaction: I would use this system for official PSU sports events.

11 responses



Comments and Suggestions (problems encountered, if any)

10 responses

Summarize responses

none
No problems, seamless transitions
Good UI but I had a hard time learning some functions
Easy to use
the system is responsive, I had trouble understanding some terms used
no problems at all
None
n/a
I have not encountered any problems

Comments and Suggestions: Please share any problems encountered or feedback you have.

16 responses

Summarize responses

none

None

Its nice I can easy to see the matches

none, overall its a good system and usable

user friendly UI

Nice UI, easy to navigate

I would definitely use this on future sports event

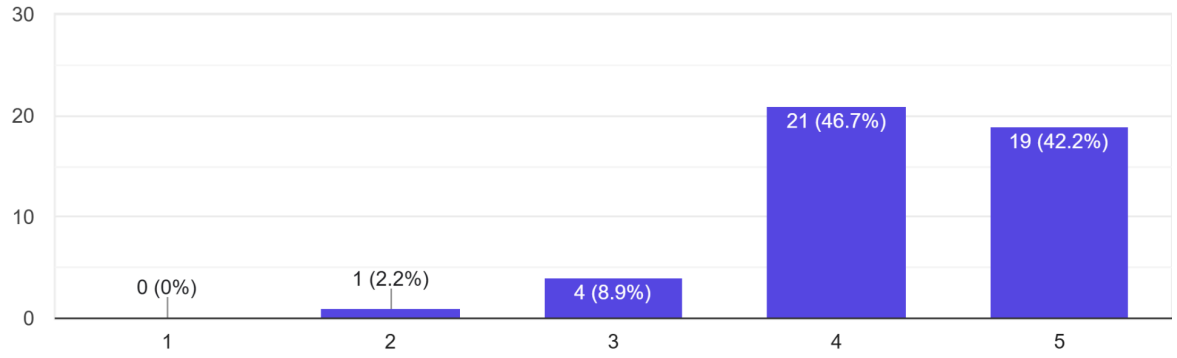
As an athlete, its nice to see the info of the matches and other athletes

The system is smooth

User(Student Athlete)

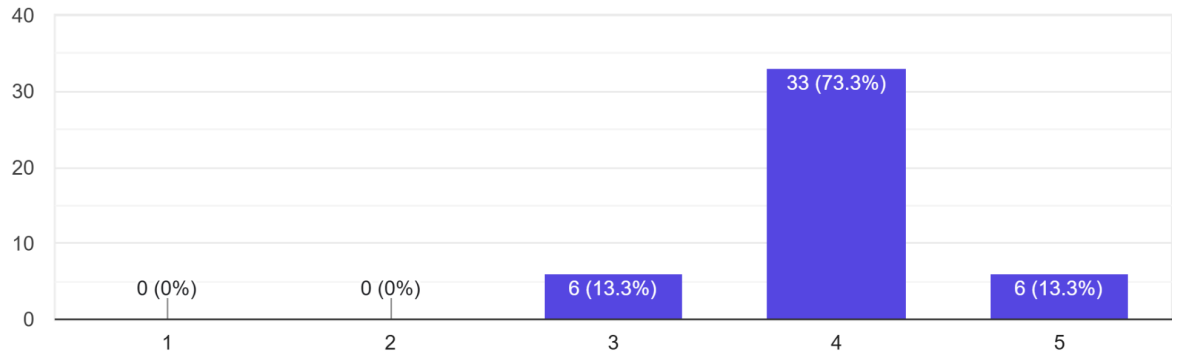
1. Account Registration: I could register and log in without difficulty.

45 responses

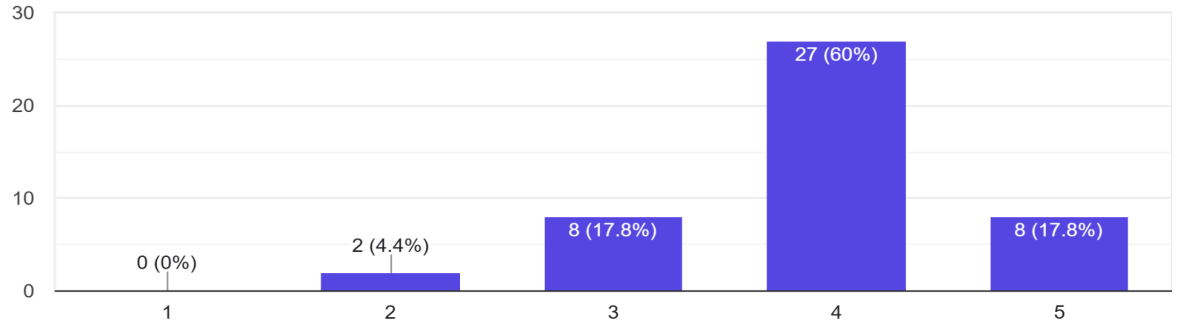


2. Document Upload: I could upload my Certificate of Registration (COR) easily.

45 responses

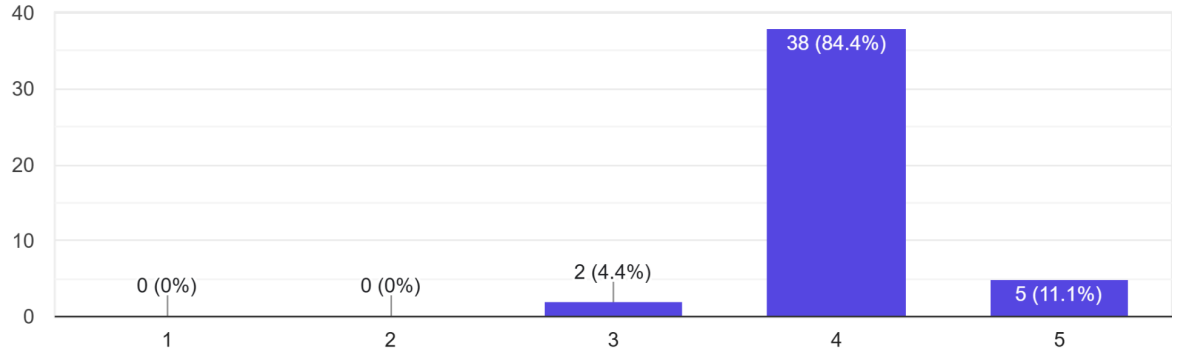


45 responses



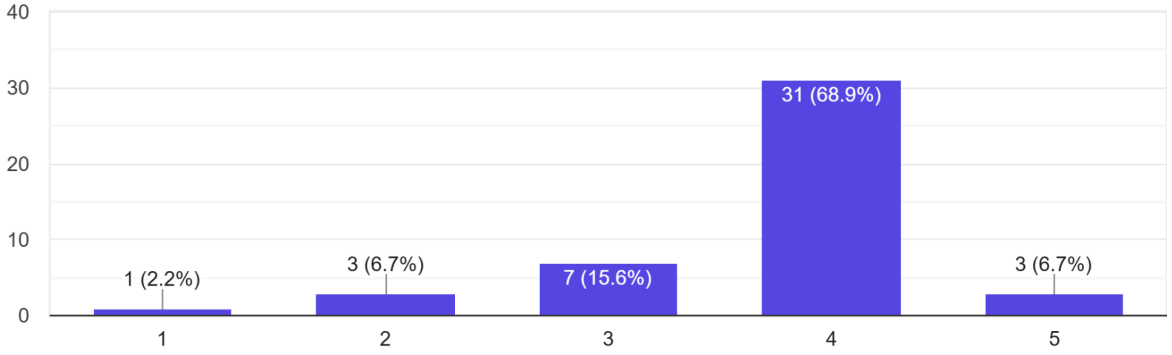
4. Verification Status: My verification status (Unverified, Pending, Approved, or Rejected) was clearly shown.

45 responses



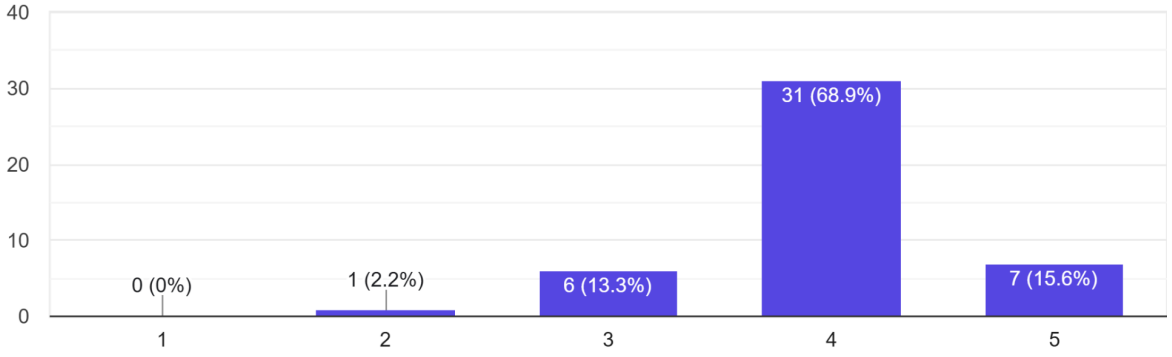
5. Rejection Feedback: If my document was rejected, the reason and the way to upload a new one were clear.

45 responses



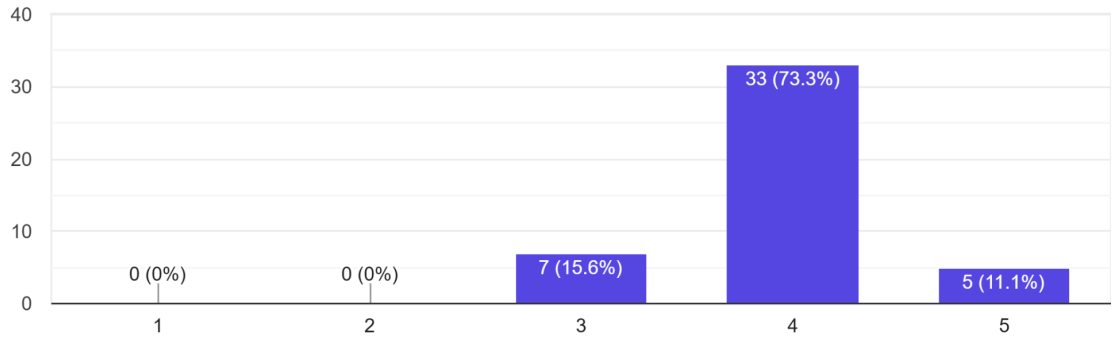
6. Team Invitations: I could accept or decline a team invitation easily.

45 responses



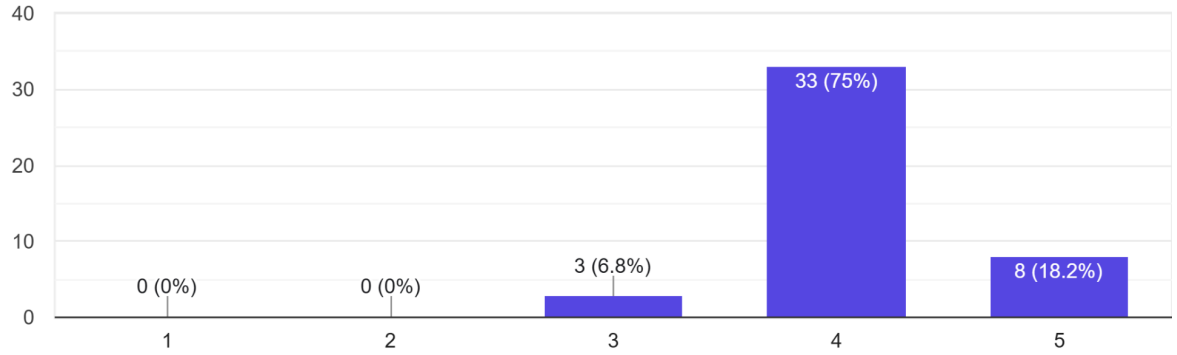
7. Team and Schedule: I could see my team, jersey number, and match schedule.

45 responses



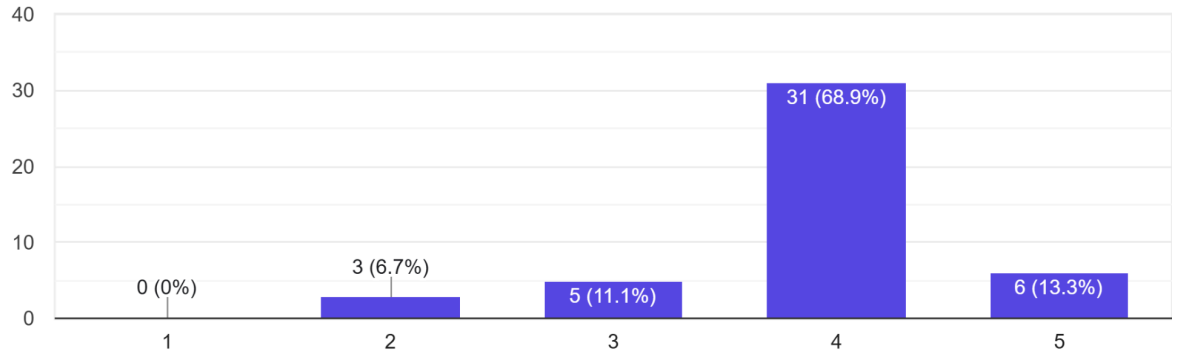
8. Profile and Honors: My profile, match history, and MVP stars were accurate.

44 responses



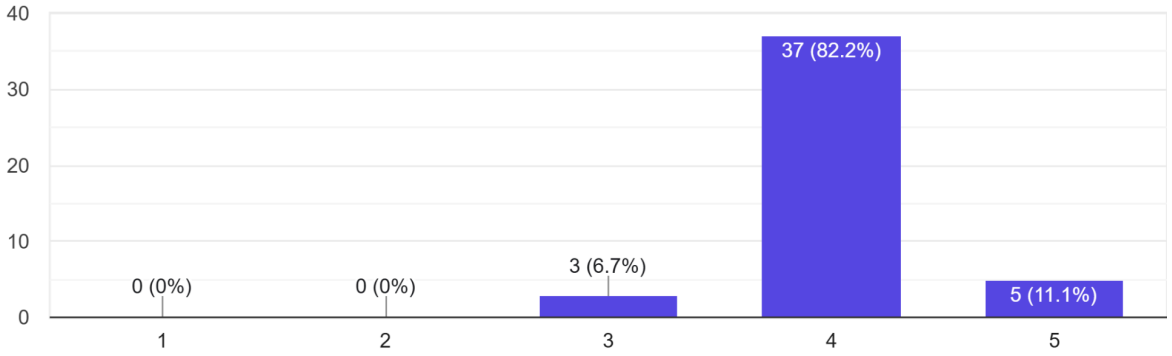
9. Ease of Learning: The system was easy to learn and use without outside help.

45 responses



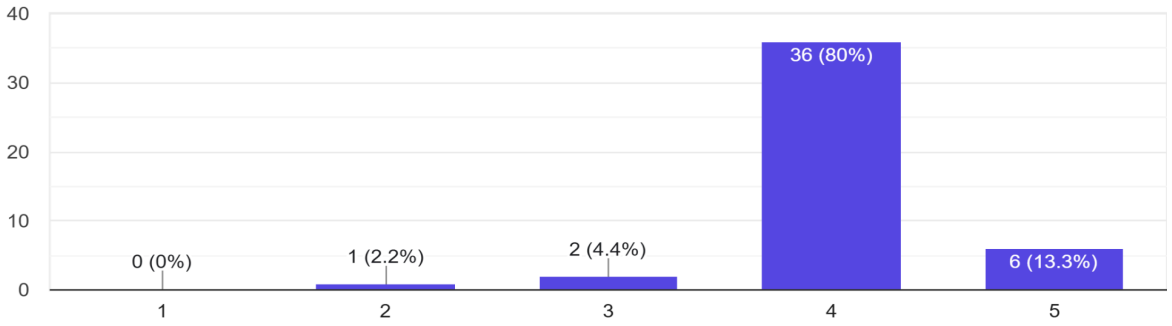
10. System Responsiveness: The pages loaded quickly and responded promptly to my actions.

45 responses



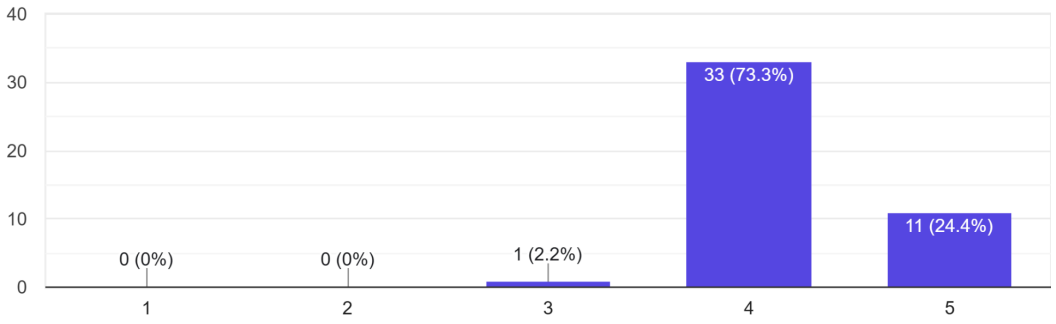
11. Interface Layout: The layout was clear and worked well on my device.

45 responses



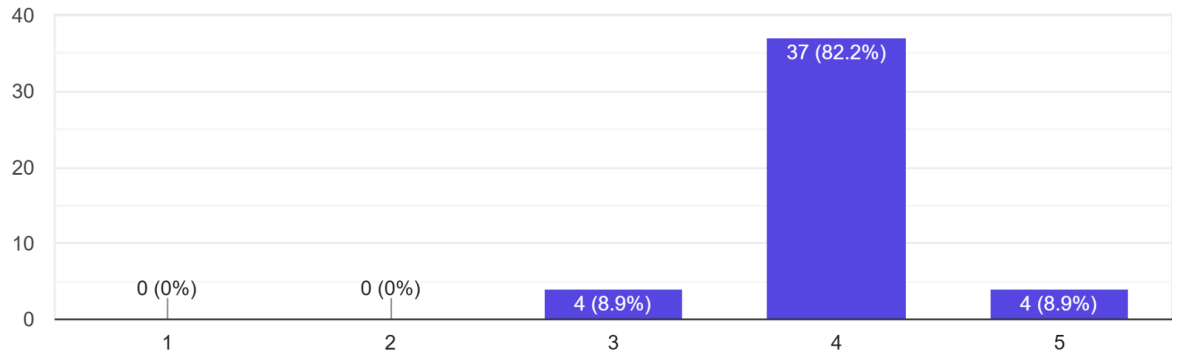
12. Role-Based Access: I could see and do only what my role allowed.

45 responses



13. Overall Satisfaction: I would use this system for official PSU sports events.

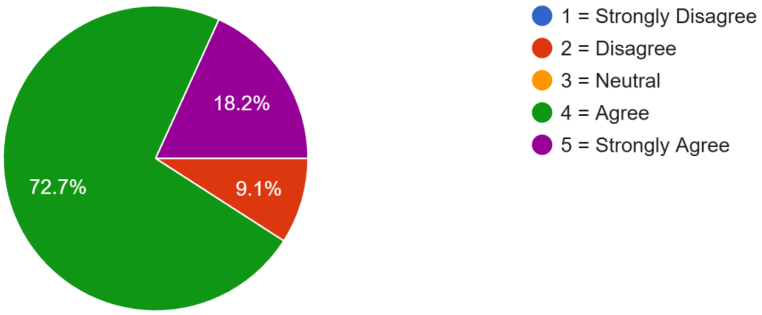
45 responses



Users(Faculty and Staff)

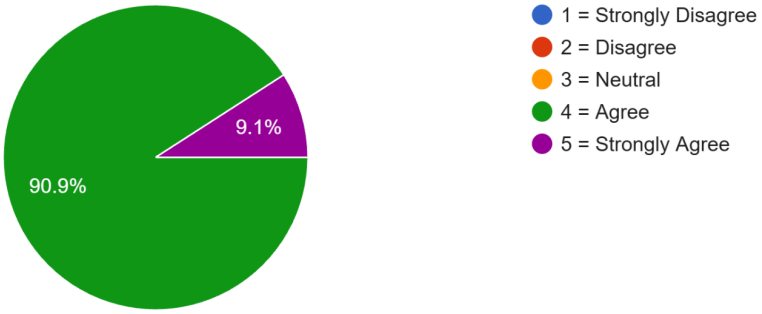
Document Upload: I could upload my employee or faculty ID easily.

11 responses



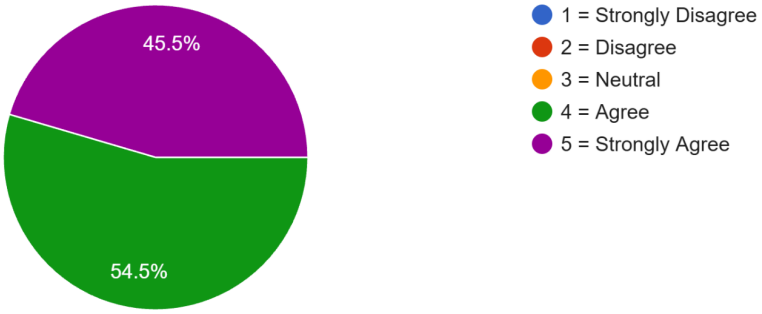
Manual Verification: It was clear that my faculty ID would be reviewed by the administrator.

11 responses



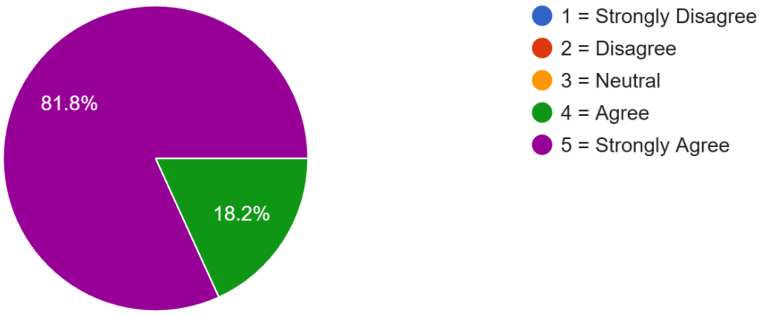
Verification Status: My verification status was clearly shown.

11 responses



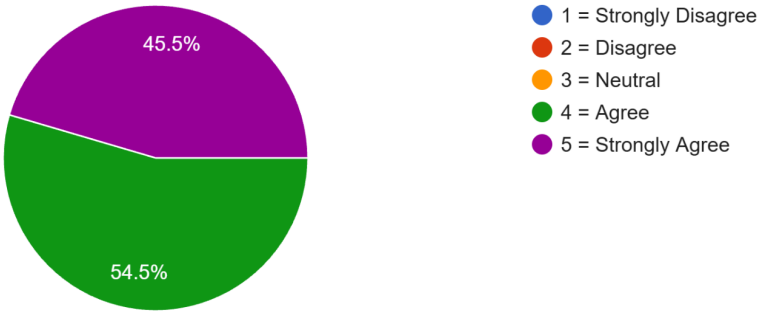
Rejection Feedback: If my document was rejected, the reason and re-upload process were clear.

11 responses



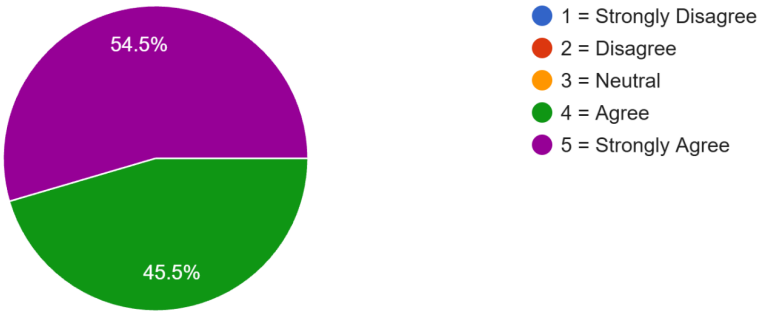
Team Invitations: I could accept or decline a team invitation easily.

11 responses



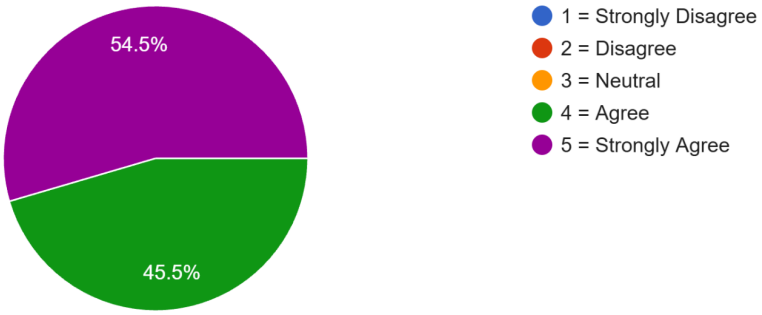
Team and Schedule: I could see my team, jersey number, and match schedule.

11 responses



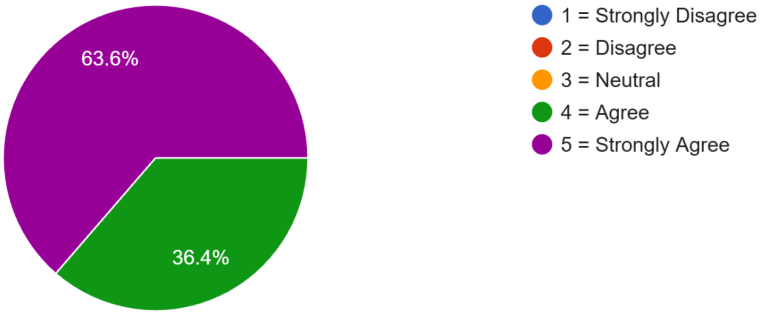
Profile and Honors: My profile, match history, and MVP stars were accurate.

11 responses



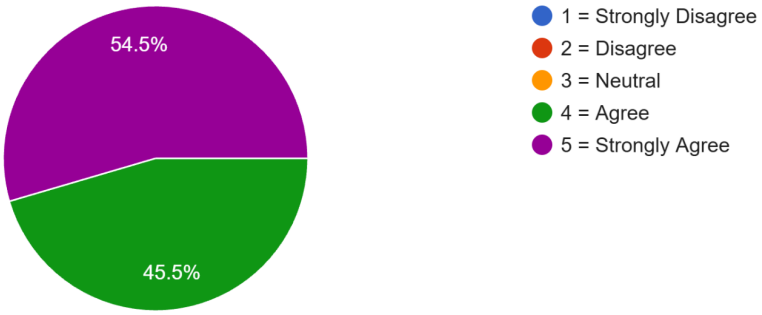
Ease of Learning: The system was easy to learn and use without outside help.

11 responses



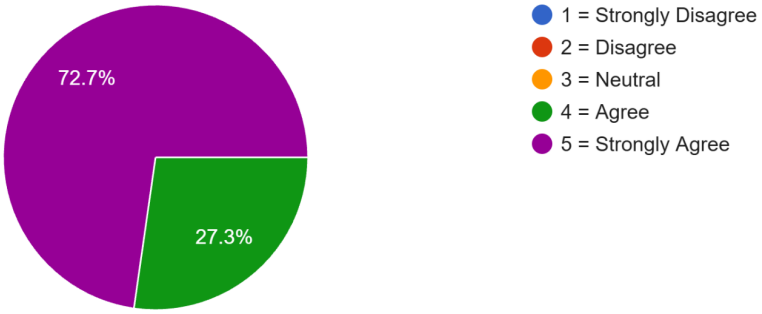
System Responsiveness: The pages loaded quickly and responded promptly to my actions.

11 responses



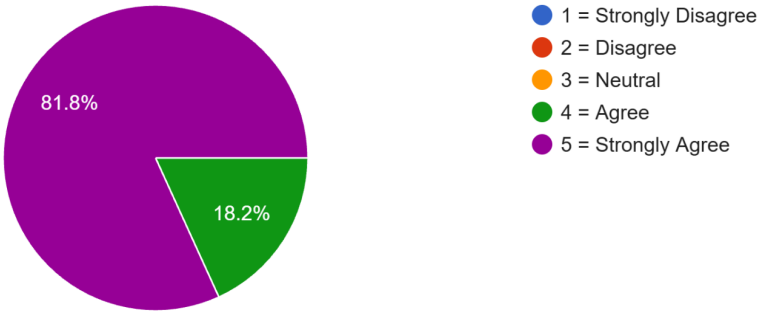
Interface Layout: The layout was clear and worked well on my device.

11 responses



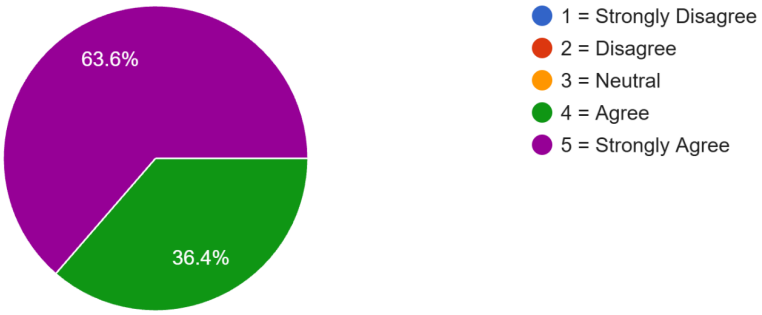
Role-Based Access: I could see and do only what my role allowed.

11 responses



Overall Satisfaction: I would use this system for official PSU sports events.

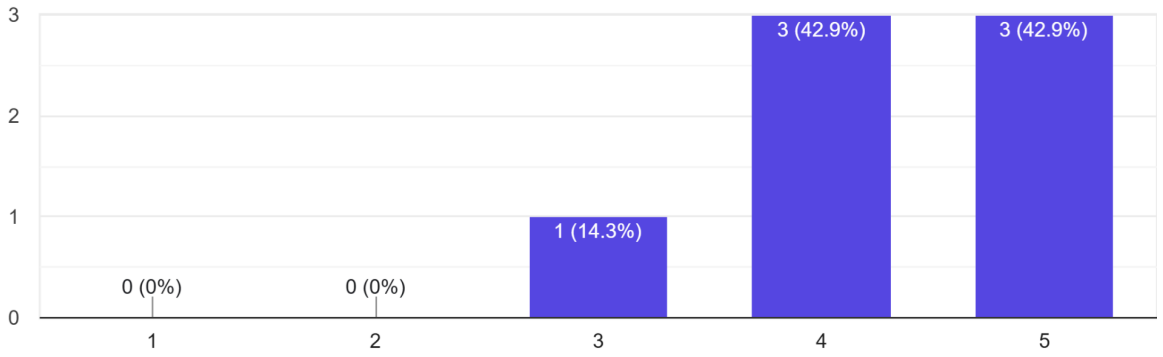
11 responses



User(Coach)

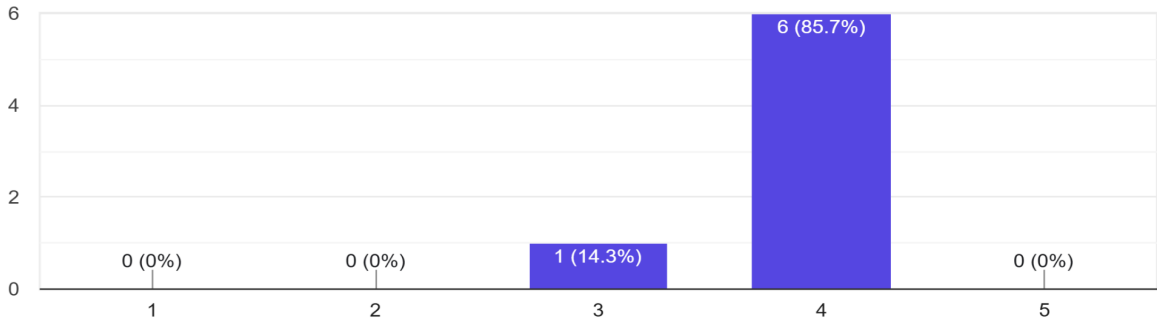
Team Creation: I could create a team for a tournament.

7 responses



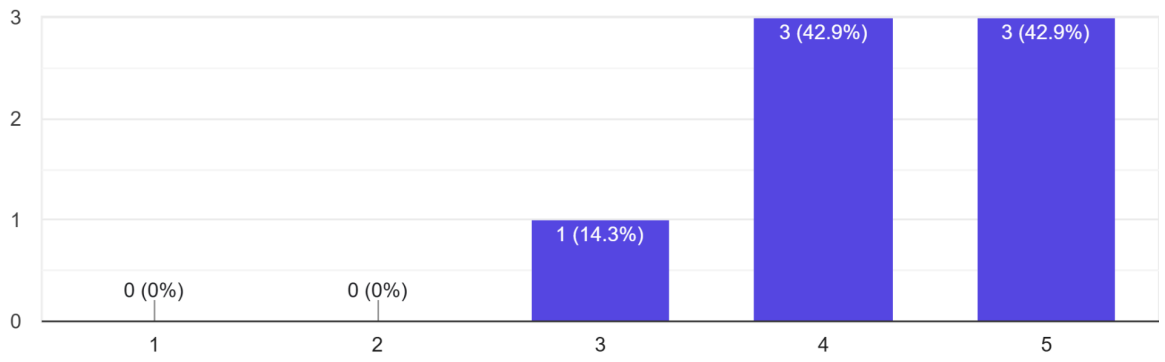
Athlete Search: I could search the verified athlete pool.

7 responses



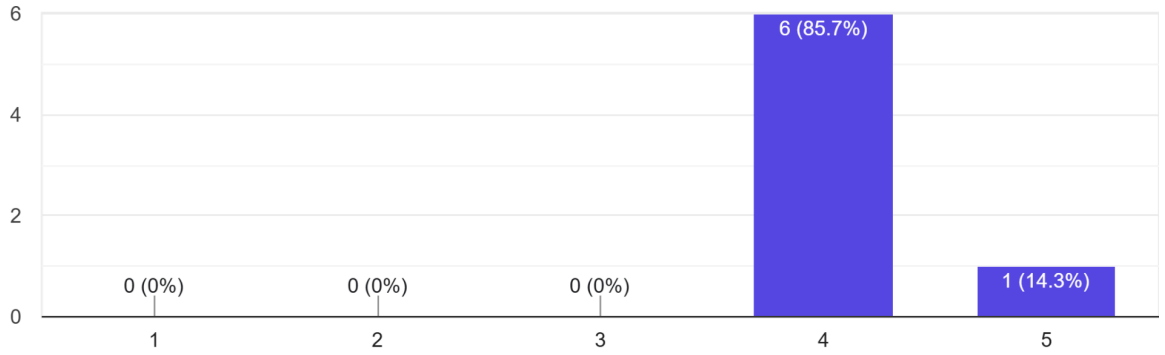
Recruitment Invitation: I could send an invitation to an athlete.

7 responses



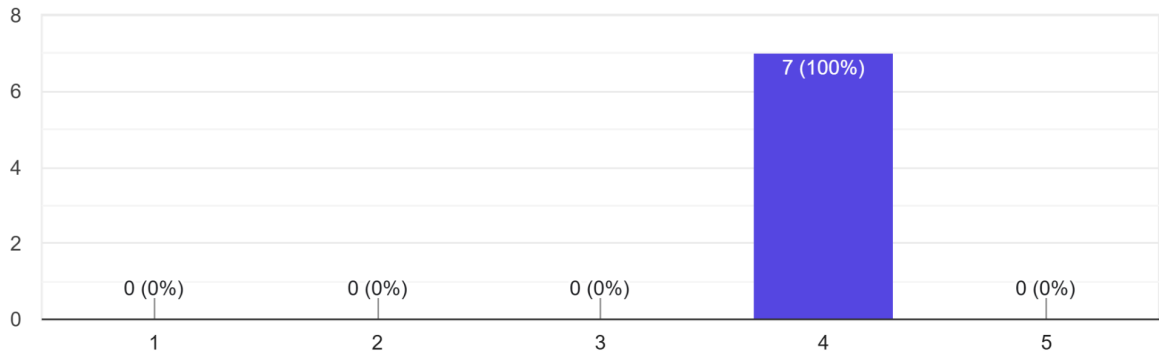
Duplicate-Sport Rule: The system blocked athletes who were already on another team in the same sport.

7 responses



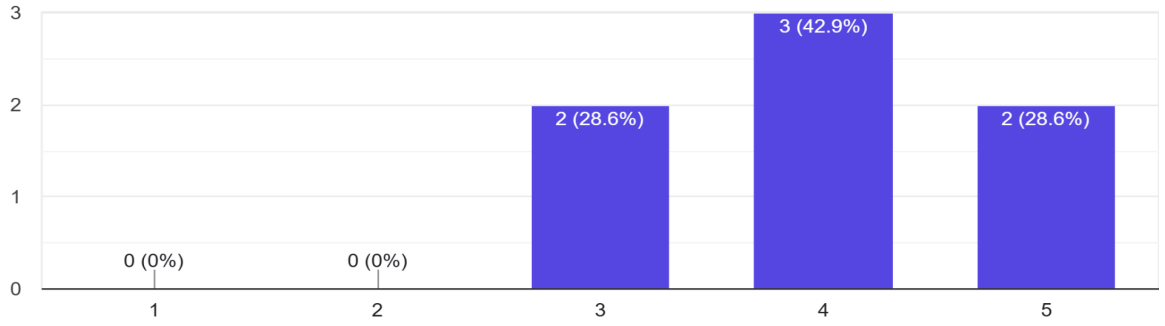
Join Requests: I could accept or reject join requests.

7 responses



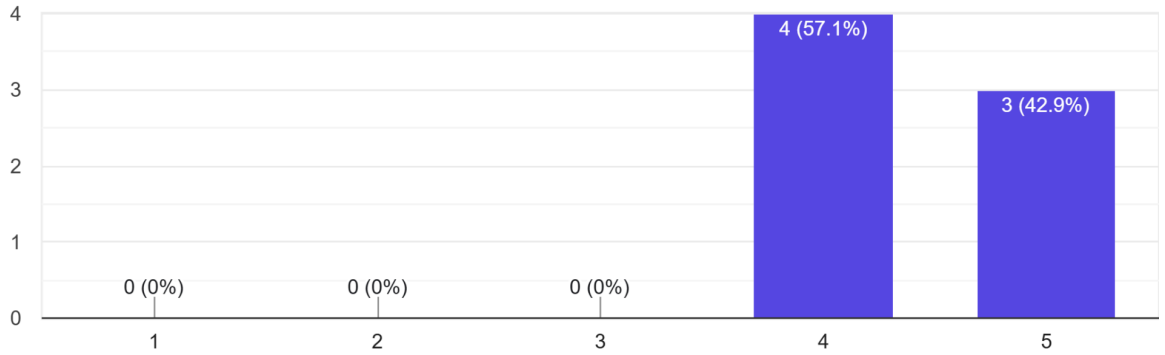
Lineup Management: I could set jersey numbers, the captain, starters, bench players, and availability.

7 responses



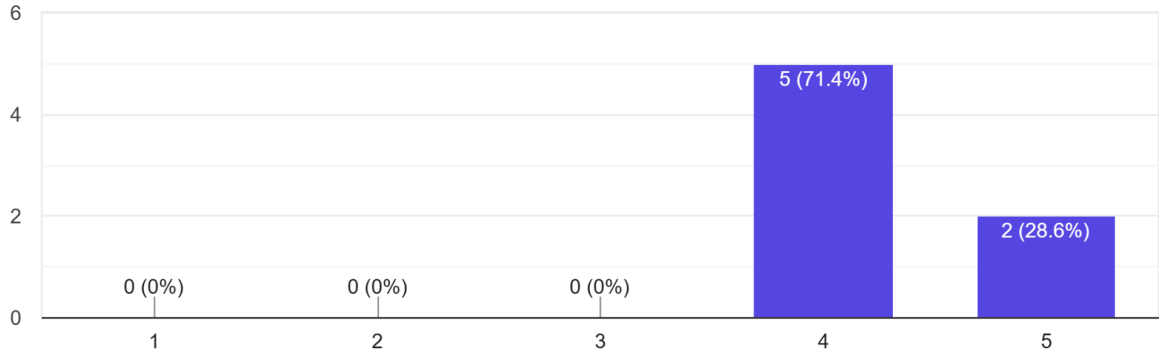
Roster Removal: I could remove an athlete from my roster.

7 responses



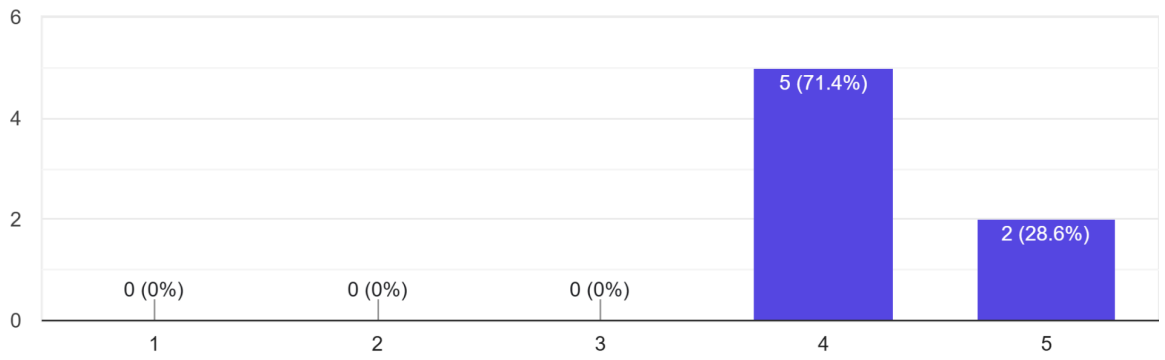
Coach Dashboard: The coach dashboard showed correct records and upcoming games.

7 responses



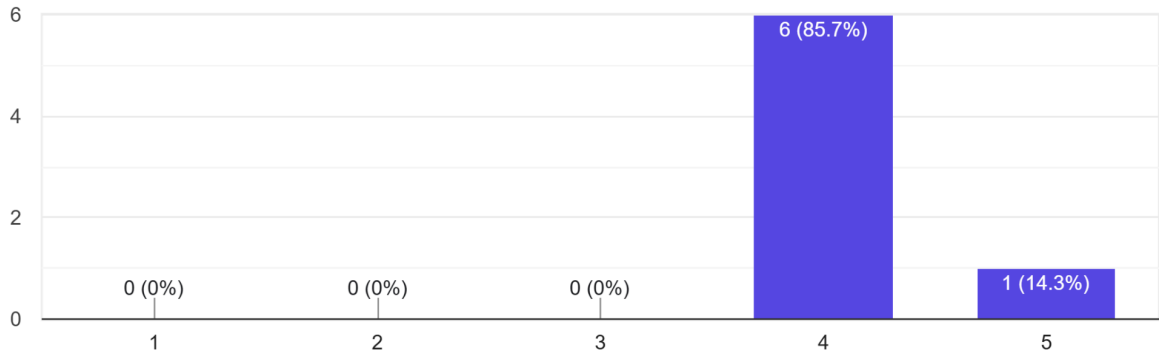
Court-Side Restrictions: I could view live matches but could not change the scores or the clock.

7 responses



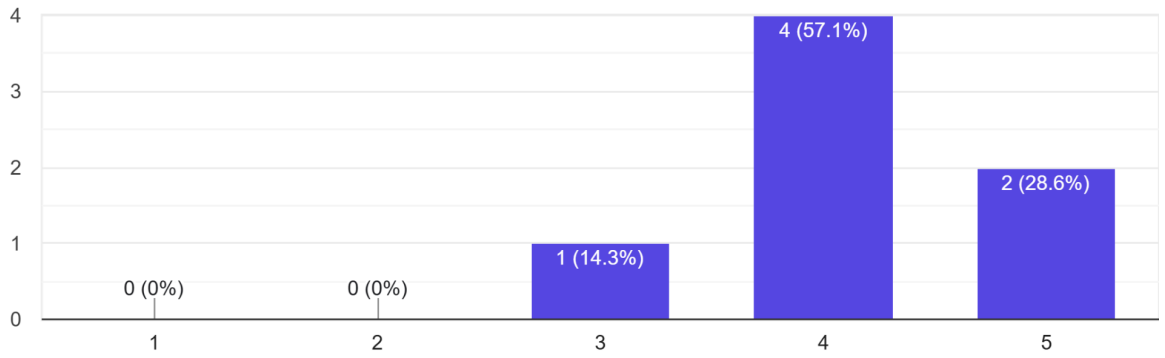
Ease of Learning: The system was easy to learn and use without outside help.

7 responses



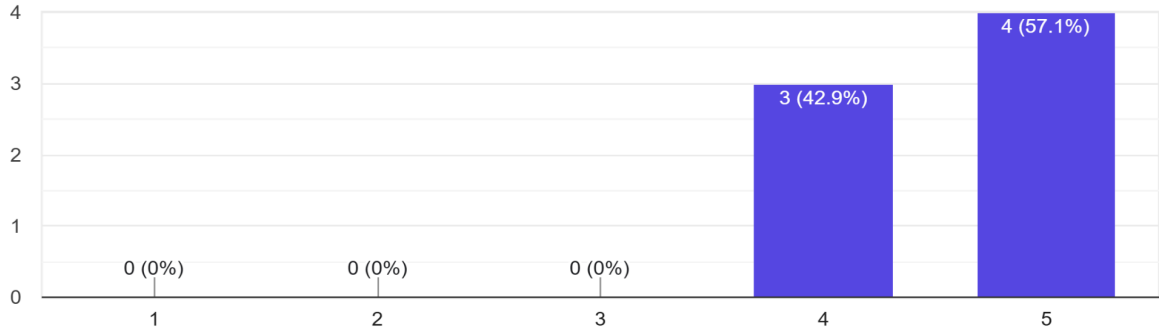
System Responsiveness: The pages loaded quickly and responded promptly to my actions.

7 responses



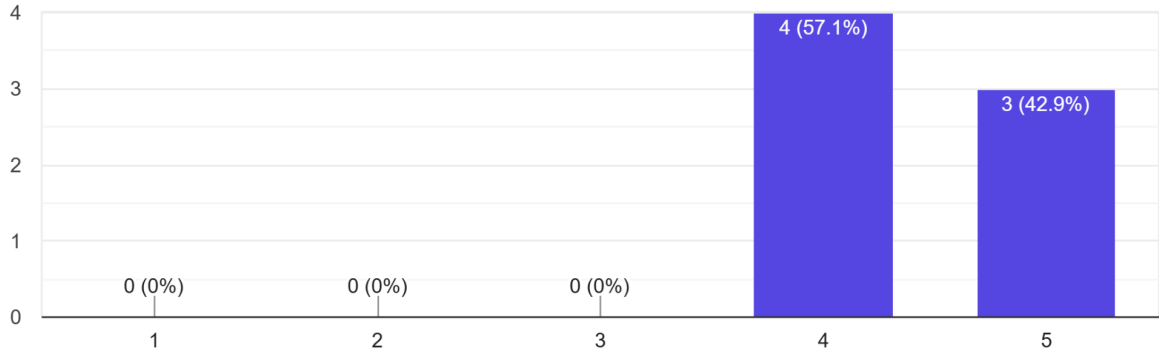
Interface Layout: The layout was clear and worked well on my device.

7 responses



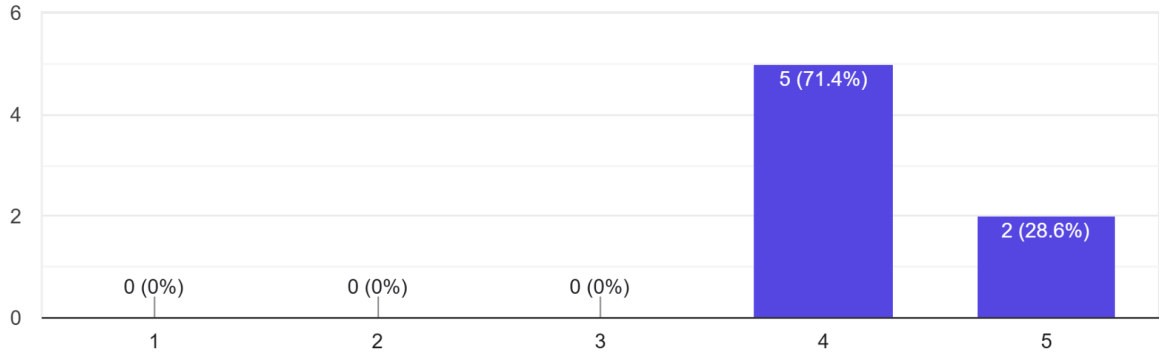
Role-Based Access: I could see and do only what my role allowed.

7 responses



Overall Satisfaction: I would use this system for official PSU sports events.

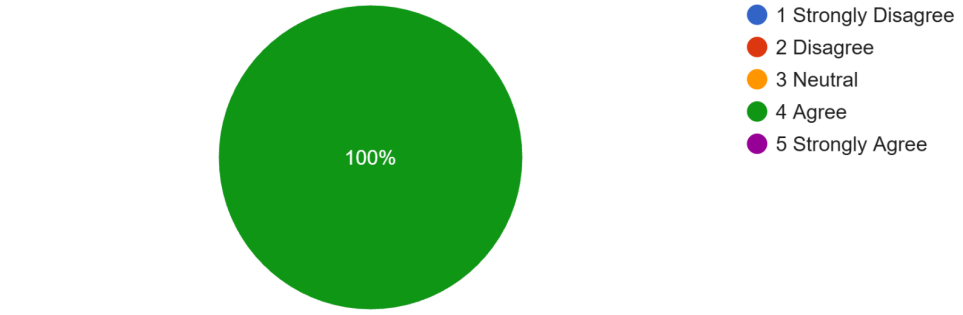
7 responses



User(Facilitator)

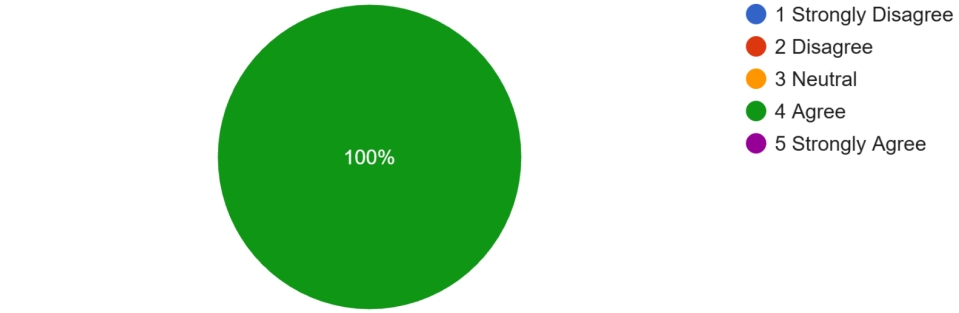
Scorekeeper Access: I could open the scorekeeper desk or kiosk for my assigned match.

3 responses



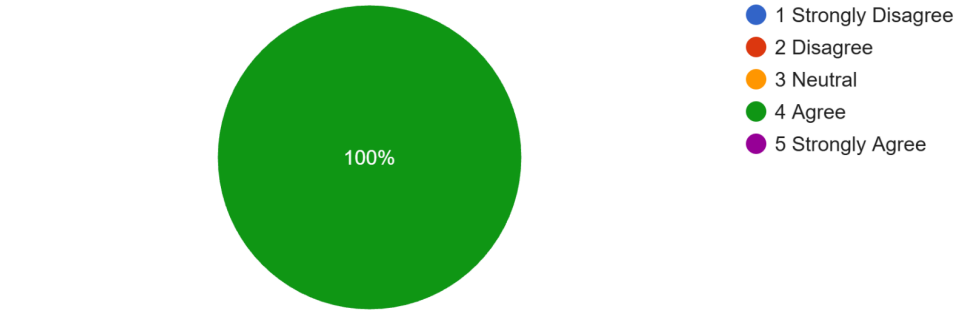
Scorekeeper Lease: I could acquire the scorekeeper lease without difficulty.

3 responses



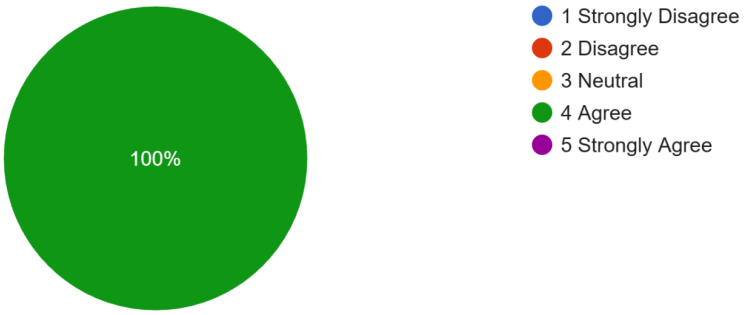
Lease Transfer: The lease transfer prompt was clear when another scorekeeper held the lease.

3 responses



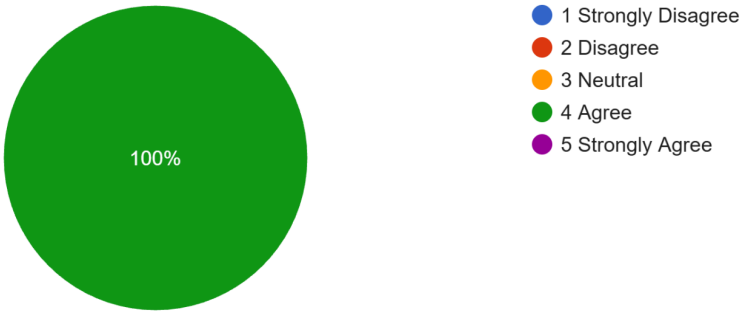
Game Clock Control: The game clock and period controls worked correctly.

3 responses



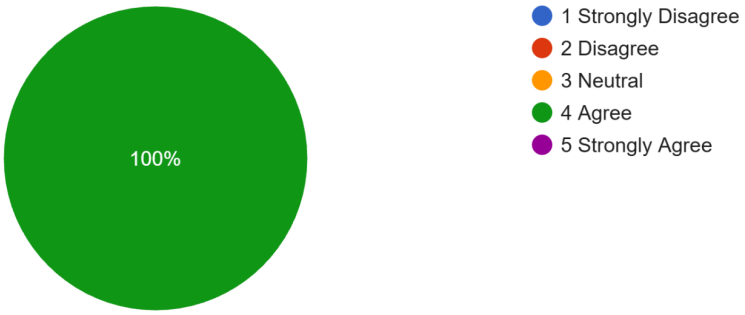
Point Attribution: I could record points and assign them to the correct athlete.

3 responses



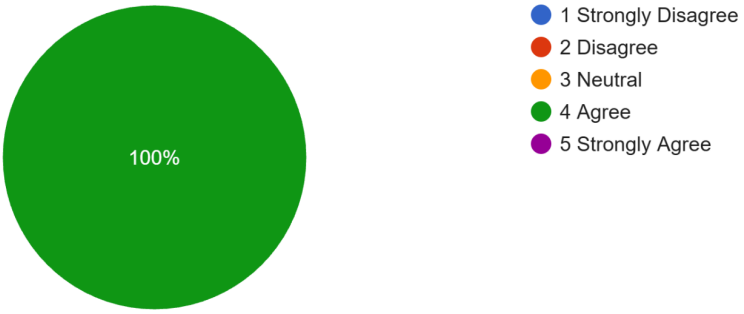
Fouls and Timeouts: I could record fouls and timeouts correctly.

3 responses



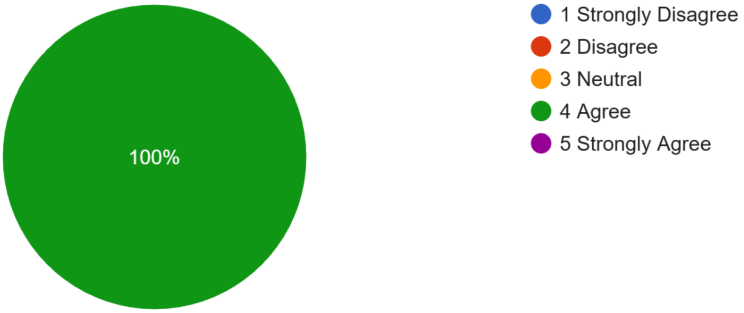
Substitutions: I could complete a substitution in two taps.

3 responses



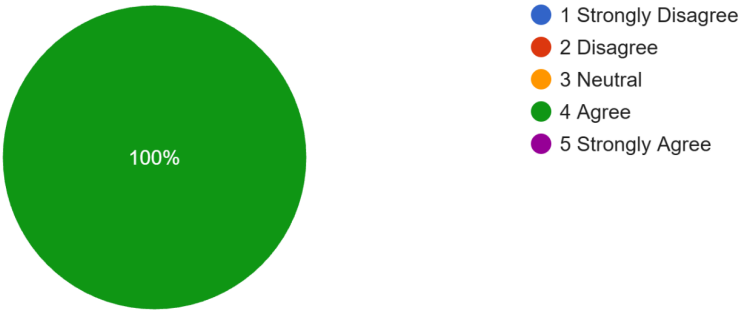
Score Correction: I could correct a score, and the score never went below zero.

3 responses



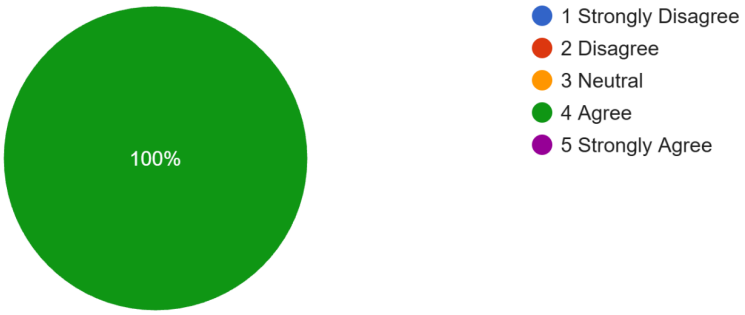
Offline Scoring: Scoring continued during a network drop and synced afterward.

3 responses



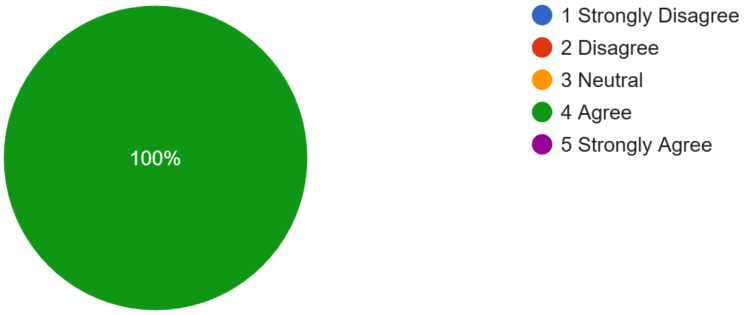
Match Finalization: I could finalize the match and award the Red Star (Game MVP).

3 responses



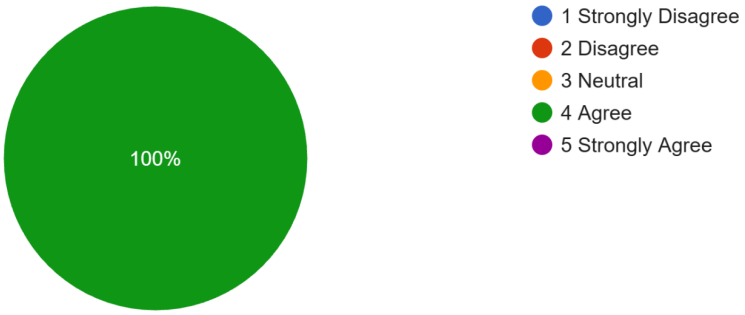
Ease of Learning: The system was easy to learn and use without outside help.

3 responses



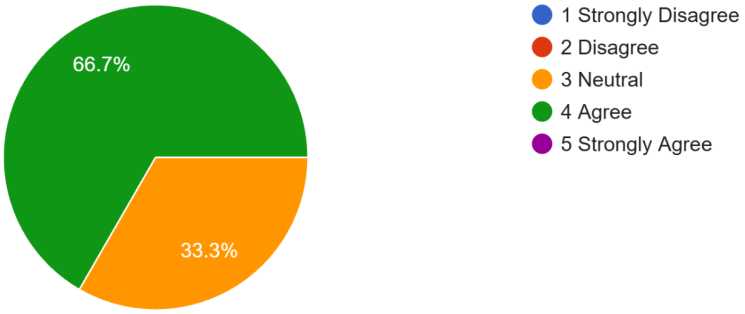
System Responsiveness: The pages loaded quickly and responded promptly to my actions.

3 responses



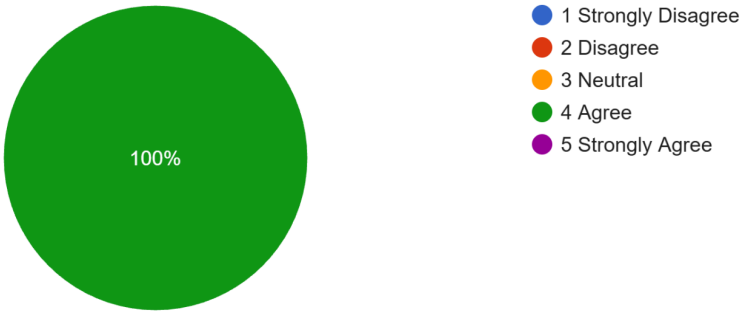
Interface Layout: The layout was clear and worked well on my device.

3 responses



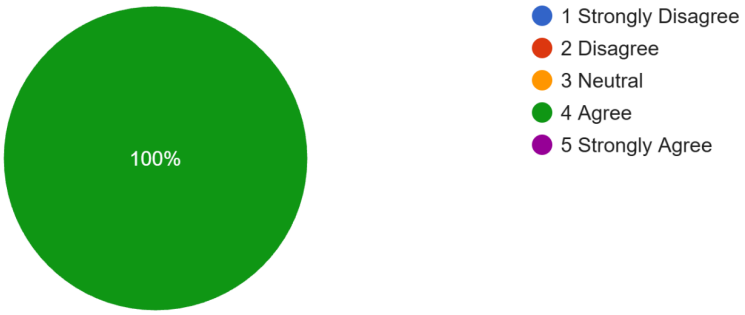
Role-Based Access: I could see and do only what my role allowed.

3 responses



Overall Satisfaction: I would use this system for official PSU sports events.

3 responses



APPENDIX H: PROCESS/DATA/INFORMATION FLOW

Level 0 Context Diagram (The Bird's-Eye View)

Figure H.1 illustrates the Level 0 Context Diagram, establishing the absolute boundaries of the PSU SportsTrack system and its high-level interactions with external entities. The central node represents the core system, which exchanges data with six primary external actors: Spectators, Athletes (Student, Faculty, and Staff), Coaches, Facilitators, Administrators, and the external PSU AIMS Portal. Data such as registration credentials, tournament configurations, team rosters, and live scoring events flow into the system. In return, the system outputs real-time data streams, published brackets, verification statuses, and institutional reports back to the respective stakeholders, providing a comprehensive overview of the system's operational scope.

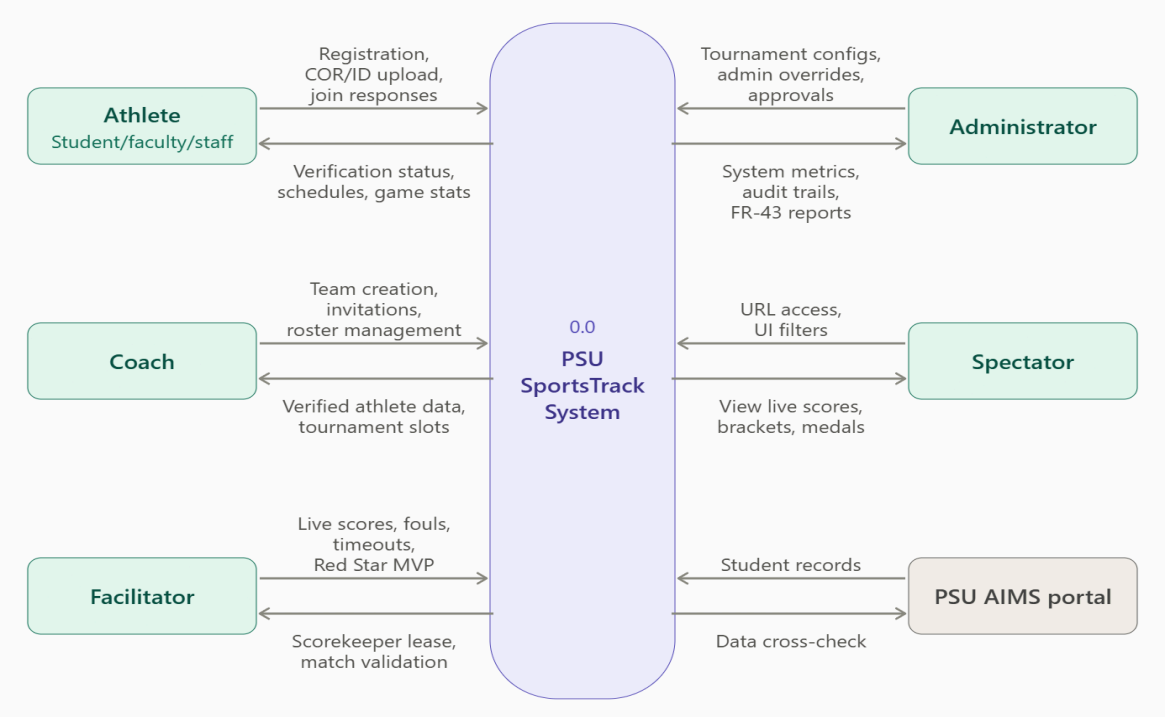


Figure 16. Level 0 Context Diagram

Athlete Verification Data Flow

Figure H.2 details the Athlete Verification Data Flow, governing how user identities are securely processed and validated prior to tournament participation. The process begins with athletes submitting account credentials and uploading required identity documents (a Certificate of Registration or an Employee ID). The system processes these uploads through an Optical Character Recognition (OCR) engine to compute a verification confidence score, safely transferring the files to a secure storage bucket. Administrators access these documents via an in-browser viewer—protected by temporary expiring links—to cross-check records against the PSU AIMS Portal. The Administrator then inputs an approval or rejection (requiring feedback notes), which systematically updates the user's account status within the database

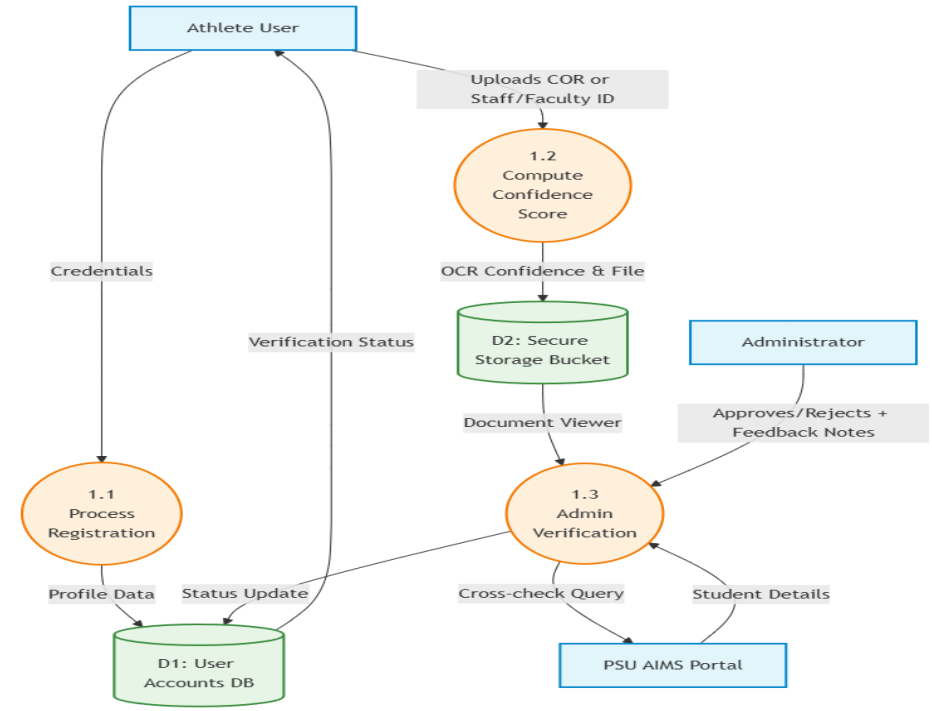


Figure 16.1 Athlete Verification Data Flow

Team Formation and Tournament Data Flow

Figure H.3 maps the Team Formation and Tournament Data Flow, highlighting the strict constraint validations required before matches commence. Coaches query the database pool of successfully verified athletes to issue recruitment invitations and assemble team rosters. The system processes these proposed lineups against database constraints, specifically enforcing the duplicate-sport rule to prevent an athlete from joining multiple teams within the same sporting discipline. Once rosters are finalized and the Administrator inputs the tournament configurations, the system's bracket generation algorithm automatically structures and outputs the pool-stage and play-off stage match up.

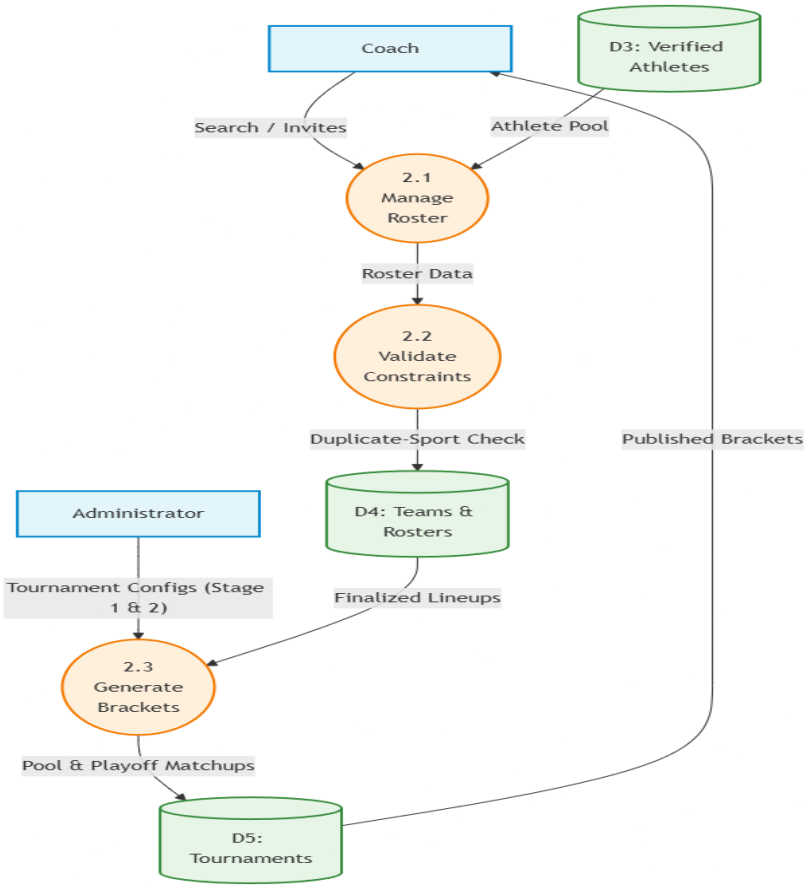


Figure 16.2 Team Formation and Tournament Data Flow

Live Match Scoring and Real-Time Information Flow

Figure H.4 outlines the Live Match Scoring and Real-Time Information Flow, representing the most time-critical process within the application. Facilitators input live game events—such as points, fouls, timeouts, and Red Star (Game MVP) designations—through a court-side interface. To ensure data integrity, the system enforces a 90-second scorekeeper lease to prevent concurrent entry conflicts, while an IndexedDB offline buffering mechanism safely queues data locally during network dropouts. Once sequentially synced to the Supabase database, the system triggers a WebSocket broadcast, pushing real-time data and browser audio cues directly to spectator displays and public scoreboards without requiring manual page refreshes.

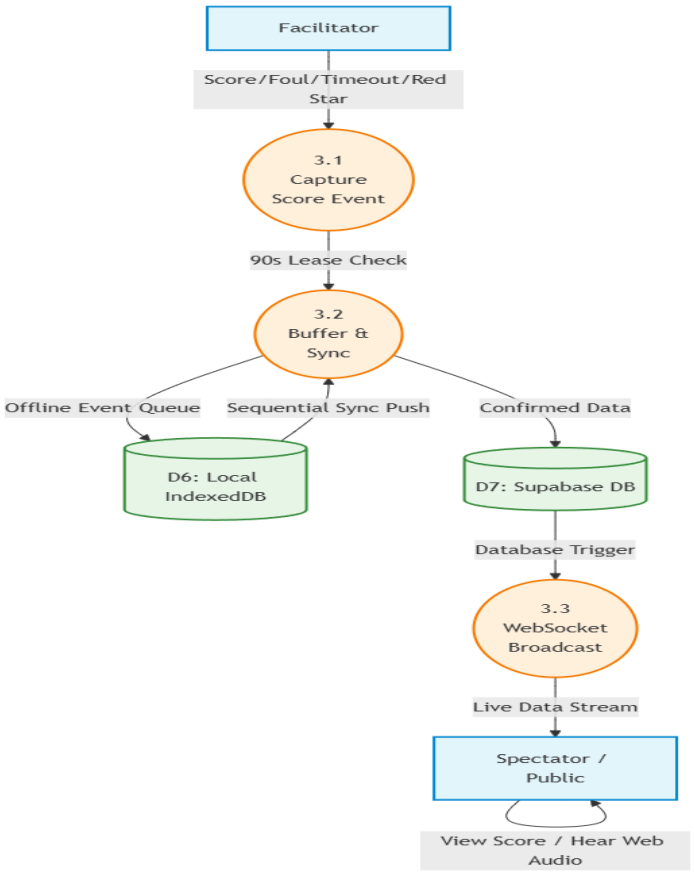


Figure 16.3 Live Match Scoring and Real-Time Information Flow

Post-Tournament, Reporting, and Audit Flow

Figure H.5 illustrates the Post-Tournament, Reporting, and Audit Flow, demonstrating how the system finalizes operations and secures historical data. Upon the conclusion of playoff brackets, the system calculates final match data and the Administrator officially finalizes the medal standings (Gold, Silver, Bronze) and designates the Gold Star (Tournament MVP) awardee. The system then compiles this data to output structured institutional reports for university stakeholders. Simultaneously, all significant administrative actions, score rollbacks, and configuration overrides are permanently recorded in an Immutable Audit Log, capturing timestamped before-and-after data states to ensure strict institutional accountability.

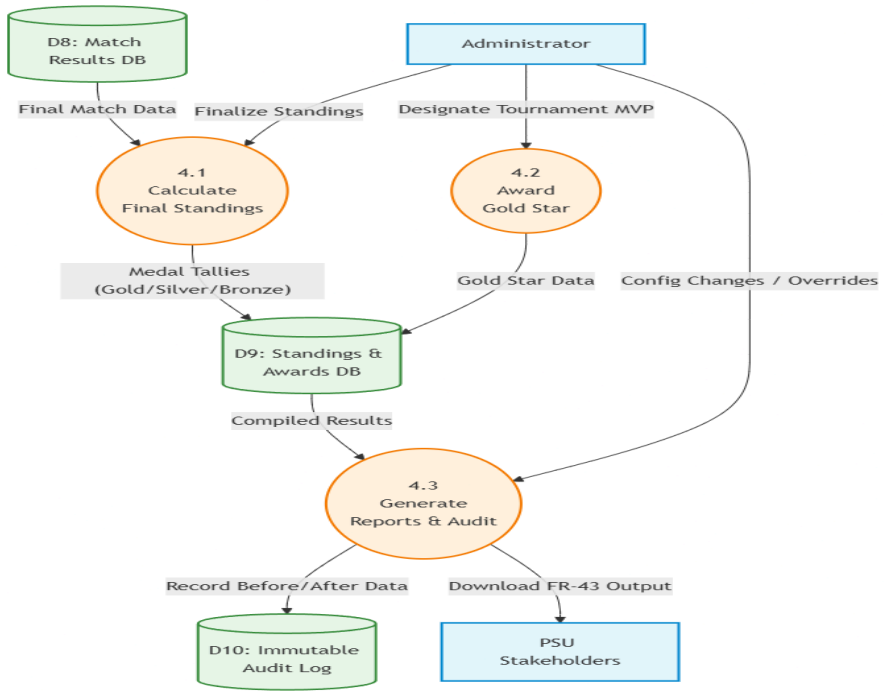


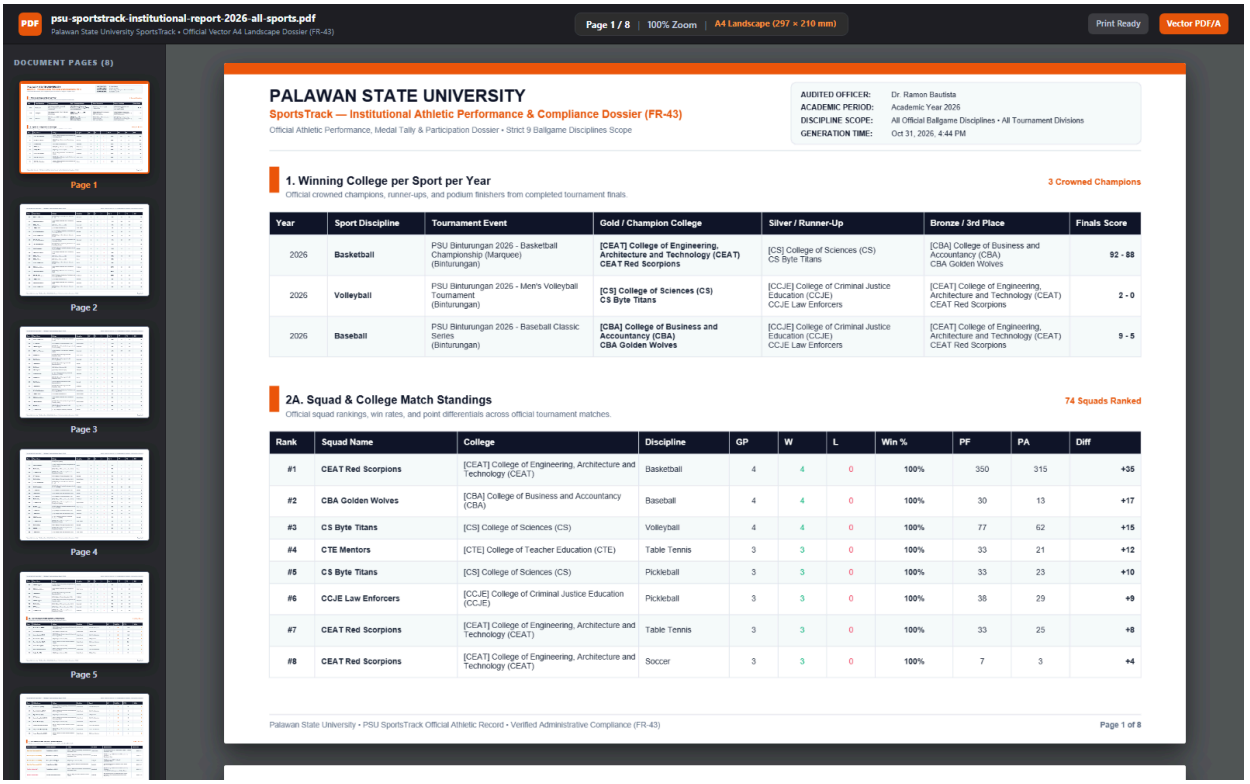
Figure 16.4 Post-Tournament, Reporting, and Audit Flow

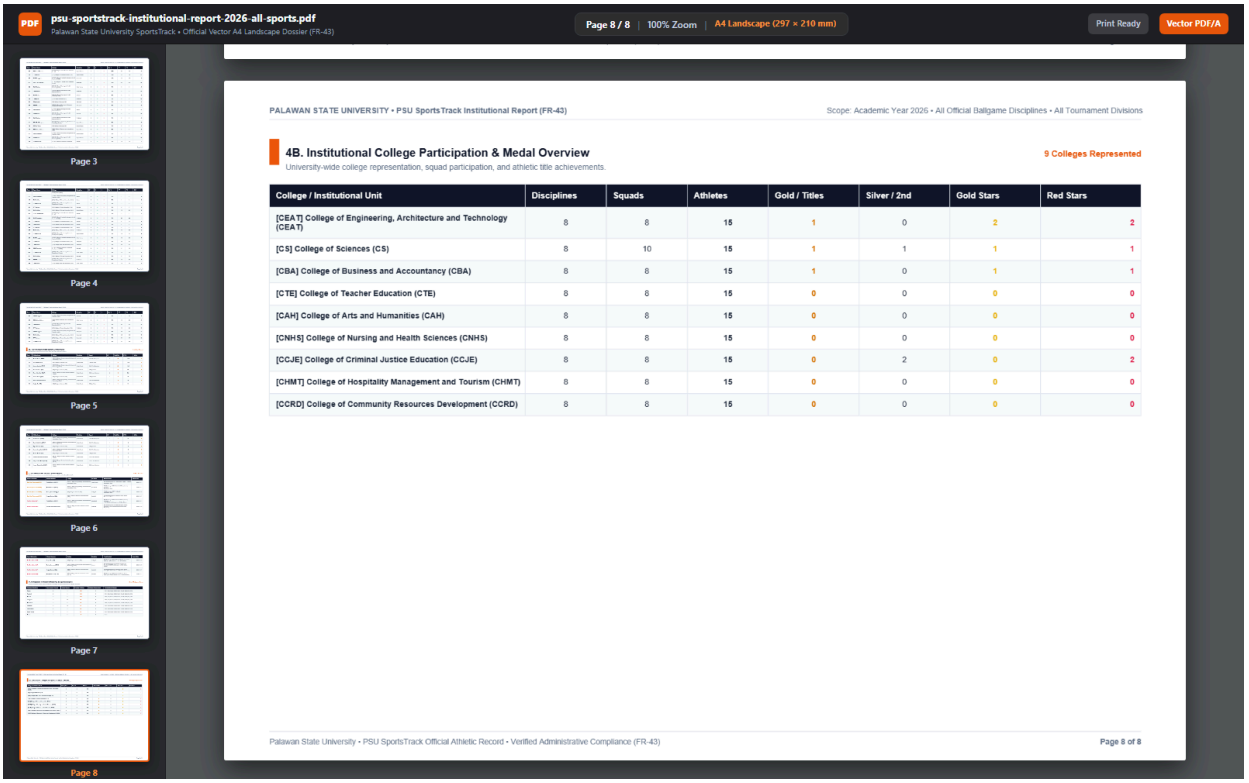
APPENDIX I: SAMPLE GENERATED OUTPUTS

The following section presents samples of the automated reports, structured datasets, and official records generated by the PSU SportsTrack system. These outputs are designed to replace the manual post-event reports previously compiled by the PSU Sports Department, ensuring accurate, consistent, and accreditation-ready historical archiving.

1. FR-43 Institutional Report (Vector-Rendered PDF)

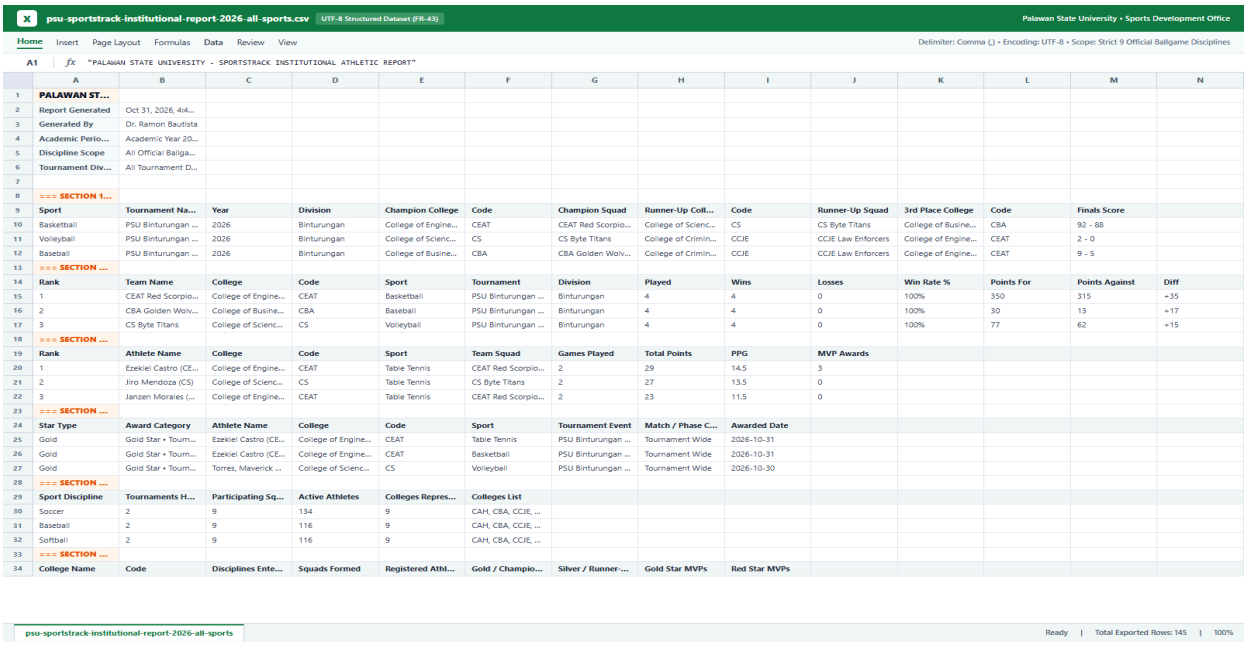
This is the publication-ready A4 landscape PDF document generated by the Administrator upon the conclusion of a sports season. It features multi-discipline summaries of squad standings, individual scoring statistics, MVP award distributions, and collegiate participation footprints.





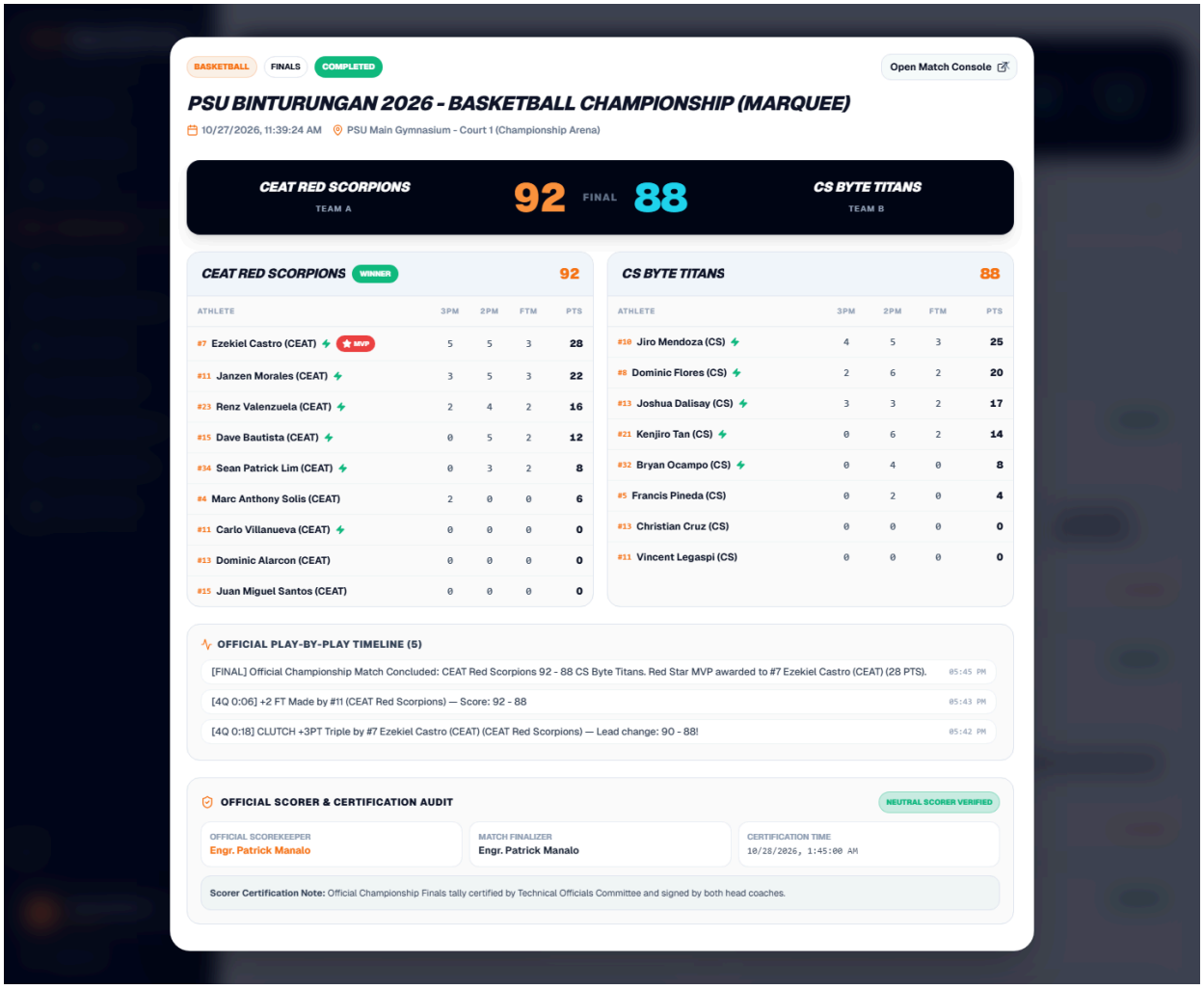
2. FR-43 Structured Dataset (CSV Export)

This is the raw, structured data export generated by the reporting engine, allowing the Sports Department to conduct external data analysis or retain lightweight digital archives.



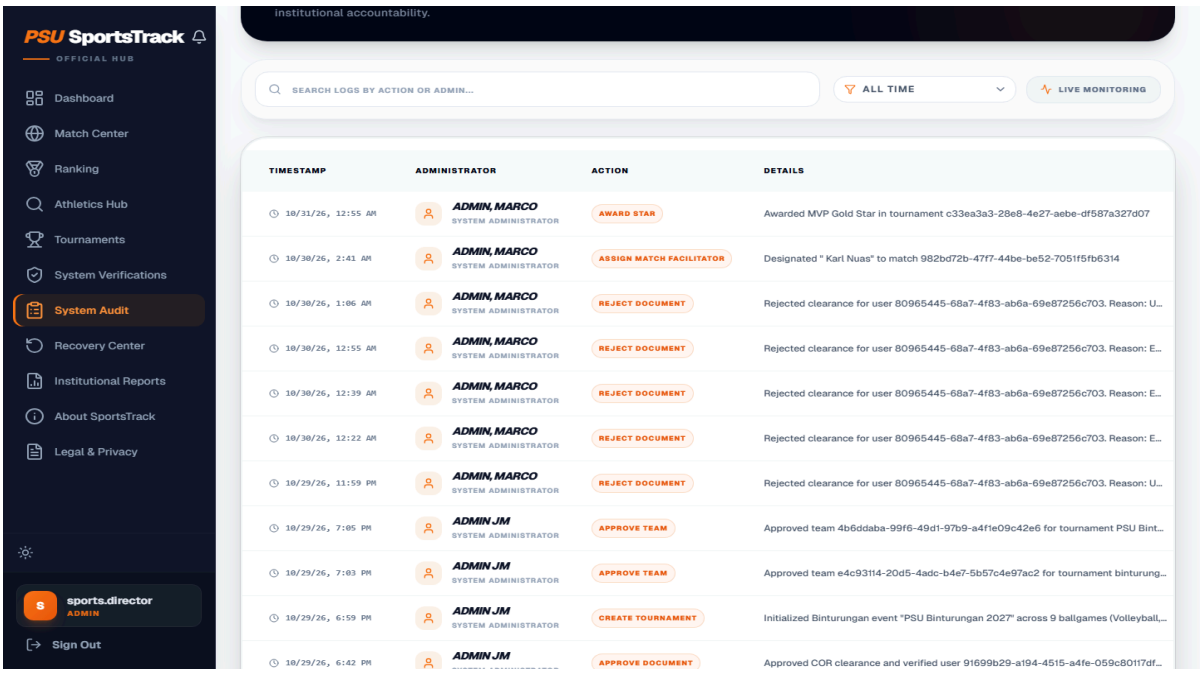
3. Official Match Box Score.

This is the official post-game record automatically generated by the system immediately after a Facilitator finalizes a match. It records the final score, the winning team, individual point attributions, and the Red Star (Game MVP) designation.



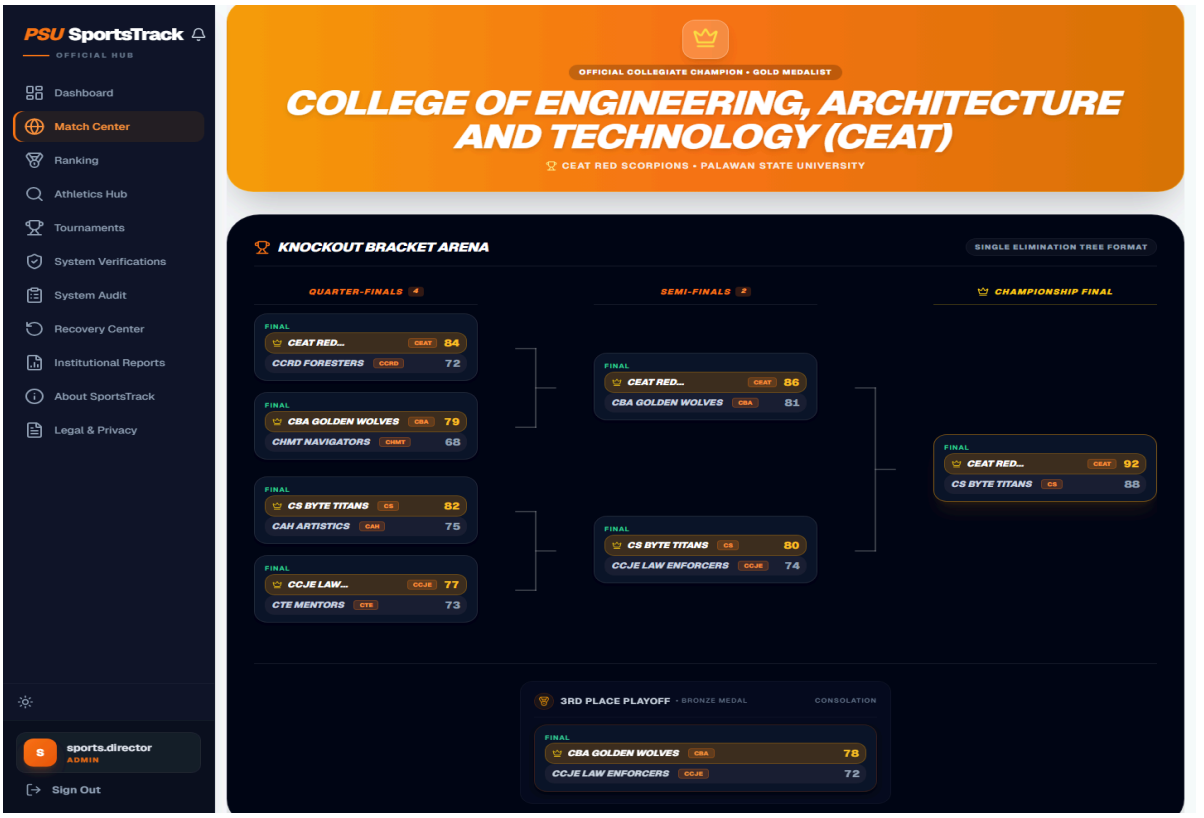
4. The Immutable Audit Trail Record.

This system provides accountability and security through an immutable audit log that tracks the actor, timestamp, and action.

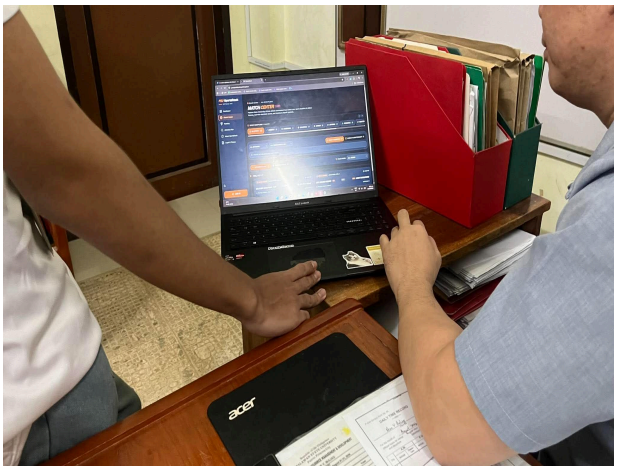


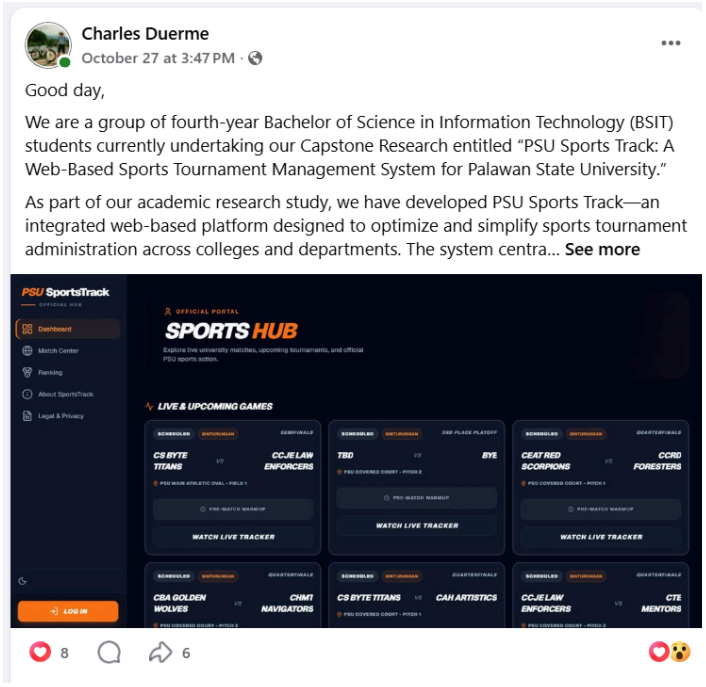
5. Automated Tournament Bracket Output.

Automating the tournament bracket was a core objective to replace manual paper brackets. This is the final, system-generated playoff tree output record.



APPENDIX J:PICTURES SHOWCASING THE DATA GATHERING, INVESTIGATION DONE





APPENDIX K: RELEVANT SOURCE CODE

This appendix provides excerpts of the core algorithmic, security, and data-handling logic developed for PSU SportsTrack. To maintain conciseness, standard UI boilerplate and routine database fetching have been omitted. The following snippets highlight the proprietary governance mechanisms, real-time synchronization, and mathematical bracket generation that drive the system's primary operations.

I.1 Hardened Match MVP Governance Validation

The following TypeScript logic enforces the Fair Play & Neutrality Attestation within the court-side scoring interface. It dynamically evaluates box score metrics and mandates a written justification when a facilitator nominates an anomalous candidate (e.g., zero recorded points or a member of the losing squad) for the Red Star (Game MVP) award.

```
import { useState, useMemo } from 'react';

export interface MvpCandidate {
  id: string;
  name: string;
  points: number;
  isWinningTeam: boolean;
  collegeId?: string;
}

// 1. State hooks for attestation acknowledgement and
justification input
const [justification, setJustification] =
useState<string>('');
const [hasAttested, setHasAttested] =
useState<boolean>(false);

// 2. Dynamically flag athletes with 0 recorded points or
from the losing squad
```

```
const isLosingTeamOrZeroStats = useMemo(() => {
  if (!candidate) return false;
  return !candidate.isWinningTeam || candidate.points === 0;
}, [candidate]);

// 3. Justification validation gate (enforces minimum 10
characters when condition is triggered)
const isJustificationValid = useMemo(() => {
  if (!isLosingTeamOrZeroStats) return true;
  return justification.trim().length >= 10;
}, [isLosingTeamOrZeroStats, justification]);

// 4. Governance gate: Disables 'Commit Award' action until
mandatory conditions are satisfied
const canCommitAward = Boolean(
  hasAttested &&
  isJustificationValid &&
  !isSubmitting
);
```

I.2 Scorekeeper Lease and Anti-Conflict of Interest (Anti-COI) Mechanism The following functions illustrate the court-side security protocols. The system first evaluates the Facilitator's affiliated collegiate ID against the competing teams to enforce Anti-COI governance, followed by the initialization of a 15-second heartbeat interval that maintains an exclusive 90-second scorekeeper lease within the Supabase backend.

```
import { supabase } from '@/lib/supabase';

export interface AntiCoiCheckResult {
  allowed: boolean;
  hasConflict: boolean;
  reason?: string;
}

/**
 * 1. Anti-COI Governance: Compares facilitator's collegiate
affiliation
```

```
 * against Home and Away teams. Denies scoring console access
on match.
 */
export function checkFacilitatorAntiCoi(
  facilitatorCollegeId: string | null | undefined,
  homeTeamCollegeId: string | null | undefined,
  awayTeamCollegeId: string | null | undefined
): AntiCoiCheckResult {
  if (!facilitatorCollegeId) {
    return { allowed: true, hasConflict: false };
  }

  if (facilitatorCollegeId === homeTeamCollegeId) {
    return {
      allowed: false,
      hasConflict: true,
      reason: 'Conflict of Interest: Facilitator belongs to
the same college as the Home Team.',
    };
  }

  if (facilitatorCollegeId === awayTeamCollegeId) {
    return {
      allowed: false,
      hasConflict: true,
      reason: 'Conflict of Interest: Facilitator belongs to
the same college as the Away Team.',
    };
  }

  return { allowed: true, hasConflict: false };
}

/**
 * 2. Claims the active scorekeeper lease for 90 seconds in
the database
 * subject to the Anti-COI gate.
 */
export async function claimScorekeeperLease(
  matchId: string,
  facilitatorId: string,
  facilitatorCollegeId: string | null | undefined,
```

```
  homeTeamCollegeId: string | null | undefined,
  awayTeamCollegeId: string | null | undefined
): Promise<{ success: boolean; error?: string }> {
  // Execute Anti-COI verification prior to claiming lease
  const coi = checkFacilitatorAntiCoi(facilitatorCollegeId,
homeTeamCollegeId, awayTeamCollegeId);
  if (!coi.allowed) {
    return { success: false, error: coi.reason };
  }

  const leaseExpiresAt = new Date(Date.now() + 90 *
1000).toISOString();

  const { error } = await supabase
    .from('matches')
    .update({
      active_scorekeeper_id: facilitatorId,
      lease_expires_at: leaseExpiresAt,
    })
    .eq('id', matchId);

  if (error) {
    return { success: false, error: error.message };
  }

  return { success: true };
}

/**
 * 3. Heartbeat timer: Pings the database every 15 seconds to
extend the
 * 90-second scorekeeper lease window while the session is
live.
 * Returns a teardown function to clear the interval.
 */
export function startHeartbeat(
  matchId: string,
  facilitatorId: string,
  onHeartbeatFailure?: (error: unknown) => void
): () => void {
  const PING_INTERVAL_MS = 15 * 1000;  // 15 seconds
  const LEASE_DURATION_MS = 90 * 1000; // 90 seconds
```

```
  const intervalId = setInterval(async () => {
    try {
      const leaseExpiresAt = new Date(Date.now() +
LEASE_DURATION_MS).toISOString();

      const { error } = await supabase
        .from('matches')
        .update({
          active_scorekeeper_id: facilitatorId,
          lease_expires_at: leaseExpiresAt,
        })
        .eq('id', matchId);

      if (error) throw error;
    } catch (err) {
      if (onHeartbeatFailure) {
        onHeartbeatFailure(err);
      }
      console.warn('[Scorekeeper Lease Heartbeat] Failed to
extend lease:', err);
    }
  }, PING_INTERVAL_MS);

  return () => clearInterval(intervalId);
}
```

I.3 Database Row-Level Security (RLS) and Audit Triggers The following PostgreSQL policies and PL/pgSQL triggers enforce strict data isolation at the database tier. These invariants revoke MVP update/deletion authorities from Facilitators post-match, reserving them exclusively for Administrators. Any execution of an Administrative Override automatically triggers an immutable insertion into the system's public audit log.

```
--
================================================================
=================
-- 1. ROW-LEVEL SECURITY (RLS) POLICIES FOR match_mvp_awards
```

```sql
--
================================================================
================

ALTER TABLE public.match_mvp_awards ENABLE ROW LEVEL
SECURITY;

-- INSERT: Accredited Facilitators and Admins can award MVP
records post-match
DROP POLICY IF EXISTS "Allow Facilitators to insert MVP
awards" ON public.match_mvp_awards;
CREATE POLICY "Allow Facilitators to insert MVP awards"
ON public.match_mvp_awards
FOR INSERT
TO authenticated
WITH CHECK (
  EXISTS (
    SELECT 1 FROM public.users
    WHERE users.id = auth.uid()
      AND (
        users.role IN ('facilitator', 'official', 'admin',
'super_admin')
        OR users.is_facilitator = true
      )
  )
);

-- UPDATE: Strictly restricted to Administrators
(Facilitators cannot tamper post-award)
DROP POLICY IF EXISTS "Allow only Administrators to update
MVP awards" ON public.match_mvp_awards;
CREATE POLICY "Allow only Administrators to update MVP
awards"
ON public.match_mvp_awards
FOR UPDATE
TO authenticated
USING (
  EXISTS (
    SELECT 1 FROM public.users
    WHERE users.id = auth.uid()
      AND users.role IN ('admin', 'super_admin')
  )
```

```
)
WITH CHECK (
  EXISTS (
    SELECT 1 FROM public.users
    WHERE users.id = auth.uid()
      AND users.role IN ('admin', 'super_admin')
  )
);

-- DELETE: Strictly restricted to Administrators
(Facilitators cannot revoke awards)
DROP POLICY IF EXISTS "Allow only Administrators to delete
MVP awards" ON public.match_mvp_awards;
CREATE POLICY "Allow only Administrators to delete MVP
awards"
ON public.match_mvp_awards
FOR DELETE
TO authenticated
USING (
  EXISTS (
    SELECT 1 FROM public.users
    WHERE users.id = auth.uid()
      AND users.role IN ('admin', 'super_admin')
  )
);


--
============================================================
================
-- 2. AUDIT TRIGGER FOR ADMIN MVP OVERRIDES AND REVOCATIONS
--
============================================================
================

CREATE OR REPLACE FUNCTION public.trg_fn_audit_mvp_override()
RETURNS TRIGGER
LANGUAGE plpgsql
SECURITY DEFINER
SET search_path = public
AS $$
DECLARE
```

```
  v_actor_id  UUID;
  v_entity_id UUID;
  v_details   TEXT;
BEGIN
  -- Identify the actor executing the override from auth
session context
  v_actor_id := auth.uid();

  IF TG_OP = 'DELETE' THEN
    v_entity_id := OLD.id;
    v_details   := format('Admin revoked MVP award record %s
for Player %s in Match %s.',
                          OLD.id, OLD.player_id,
OLD.match_id);
  ELSIF TG_OP = 'UPDATE' THEN
    v_entity_id := NEW.id;
    v_details   := format('Admin updated MVP record %s. Prior
Player: %s -> New Player: %s. Reason: %s',
                          OLD.id, OLD.player_id,
NEW.player_id,
                          COALESCE(NEW.override_reason,
NEW.justification, 'Administrative Override'));
  END IF;

  -- Write tamper-proof entry to centralized audit ledger
  INSERT INTO public.audit_logs (
    actor_id,
    admin_id,
    action,
    entity_type,
    entity_id,
    target_id,
    details,
    created_at
  ) VALUES (
    v_actor_id,
    v_actor_id,
    'OVERRIDE_STAR',
    'match_mvp_awards',
    v_entity_id,
    v_entity_id,
    v_details,
```

```
    NOW()
  );

  IF TG_OP = 'DELETE' THEN
    RETURN OLD;
  ELSE
    RETURN NEW;
  END IF;
END;
$$;

DROP TRIGGER IF EXISTS trg_audit_mvp_award_override ON
public.match_mvp_awards;
CREATE TRIGGER trg_audit_mvp_award_override
AFTER UPDATE OR DELETE ON public.match_mvp_awards
FOR EACH ROW
EXECUTE FUNCTION public.trg_fn_audit_mvp_override();
```

I.4 Automated Bracket Generation Algorithm The following algorithm automates the engineering of playoff tournament brackets. Utilizing base-2 logarithms, the function calculates required byes based on the length of the submitted team array and autonomously generates the `next_match_id` linkages necessary to route advancing winners and assign semifinal losers to the Bronze Medal fixture.

```
export interface Team {
  id: string;
  name: string;
}

export type MatchSlot = 'team_a' | 'team_b';

export interface MatchFixture {
  id: string;
  roundNumber: number;
  roundName: string;
  team_a_id: string | null;
  team_a_name: string;
```

```
  team_b_id: string | null;
  team_b_name: string;
  next_match_id: string | null;
  next_match_slot: MatchSlot | null;
  loser_next_match_id: string | null;
  loser_next_match_slot: MatchSlot | null;
}

/**
 * Generates an automated Single Elimination tournament
bracket with
 * dynamic logarithmic rounds, bye handling, winner
progression links,
 * and bronze medal match loser routing.
 */
export function generateSingleEliminationBracket(teams:
Team[]): MatchFixture[] {
  const N = teams.length;
  if (N < 2) return [];

  // 1. Calculate required rounds, bracket capacity, and byes
via base-2 logarithms
  const totalRounds = Math.ceil(Math.log2(N));
  const bracketCapacity = Math.pow(2, totalRounds);
  const numByes = bracketCapacity - N;
  const numRound1Matches = bracketCapacity / 2;

  const getRoundLabel = (r: number): string => {
    if (r === totalRounds) return 'Finals';
    if (r === totalRounds - 1) return 'Semifinals';
    if (r === totalRounds - 2) return 'Quarterfinals';
    return `Round ${r}`;
  };

  // 2. Instantiate empty match fixtures structure grouped by
round
  const matchesByRound: Record<number, MatchFixture[]> = {};

  for (let r = 1; r <= totalRounds; r++) {
    matchesByRound[r] = [];
    const matchesInRound = Math.pow(2, totalRounds - r);
```

```
    for (let j = 0; j < matchesInRound; j++) {
      matchesByRound[r].push({
        id: `match-rr-m{j}`,
        roundNumber: r,
        roundName: getRoundLabel(r),
        team_a_id: null,
        team_a_name: 'TBD',
        team_b_id: null,
        team_b_name: 'TBD',
        next_match_id: null,
        next_match_slot: null,
        loser_next_match_id: null,
        loser_next_match_slot: null,
      });
    }
  }

  // 3. Seed Round 1 matches and distribute BYEs
symmetrically
  const round1Matches = matchesByRound[1];
  for (let j = 0; j < numRound1Matches; j++) {
    const teamA = teams[j];
    if (teamA) {
      round1Matches[j].team_a_id = teamA.id;
      round1Matches[j].team_a_name = teamA.name;
    }

    const teamB = teams[j + numRound1Matches];
    if (teamB) {
      round1Matches[j].team_b_id = teamB.id;
      round1Matches[j].team_b_name = teamB.name;
    } else {
      round1Matches[j].team_b_id = null;
      round1Matches[j].team_b_name = 'BYE';
    }
  }

  // 4. Resolve winner progression pointers (next_match_id
and slot) between rounds
  for (let r = 1; r < totalRounds; r++) {
    const currentRound = matchesByRound[r];
    const nextRound = matchesByRound[r + 1];
```

```
    for (let j = 0; j < currentRound.length; j++) {
      const parentMatchIndex = Math.floor(j / 2);
      const parentSlot: MatchSlot = j % 2 === 0 ? 'team_a' :
'team_b';

      currentRound[j].next_match_id =
nextRound[parentMatchIndex].id;
      currentRound[j].next_match_slot = parentSlot;
    }
  }

  // 5. Build 3rd Place Bronze Medal fixture and route
Semifinal losers
  let bronzeMatch: MatchFixture | null = null;
  if (totalRounds >= 2) {
    const semiMatches = matchesByRound[totalRounds - 1];

    if (semiMatches && semiMatches.length === 2) {
      bronzeMatch = {
        id: 'match-bronze-playoff',
        roundNumber: totalRounds,
        roundName: '3rd Place Playoff',
        team_a_id: null,
        team_a_name: 'TBD (SF 1 Loser)',
        team_b_id: null,
        team_b_name: 'TBD (SF 2 Loser)',
        next_match_id: null,
        next_match_slot: null,
        loser_next_match_id: null,
        loser_next_match_slot: null,
      };

      // Route SF 1 loser to Team A slot, SF 2 loser to Team
B slot
      semiMatches[0].loser_next_match_id = bronzeMatch.id;
      semiMatches[0].loser_next_match_slot = 'team_a';

      semiMatches[1].loser_next_match_id = bronzeMatch.id;
      semiMatches[1].loser_next_match_slot = 'team_b';
    }
  }
```

```
  // 6. Flatten tournament tree into sequential match
fixtures
  const flatFixtures: MatchFixture[] = [];
  for (let r = 1; r <= totalRounds; r++) {
    flatFixtures.push(...matchesByRound[r]);
  }

  if (bronzeMatch) {
    flatFixtures.push(bronzeMatch);
  }

  return flatFixtures;
}
```

I.5 IndexedDB Offline Event Buffering The following TypeScript logic provides court-side offline resilience during venue network dropouts. It detects network restoration, retrieves queued scoring and foul events from the browser's local IndexedDB, and executes an atomic batch-synchronization to the Supabase database before clearing the local queue.

```
  import { SupabaseClient } from '@supabase/supabase-js';

  export interface QueuedScoringEvent {
    id: string;
    matchId: string;
    teamId: string;
    playerId?: string | null;
    points: number;
    delta: number;
    actionType: string;
    createdAt: string;
  }

  const DB_NAME = 'PSUSportsTrackOfflineDB';
  const STORE_NAME = 'offline_scoring_events';

  //
```

```
// ---------------------------------------------------------------
// 1. IndexedDB Helper Functions
// ---------------------------------------------------------------

function openIndexedDB(): Promise<IDBDatabase> {
  return new Promise((resolve, reject) => {
    const request = indexedDB.open(DB_NAME, 1);
    request.onupgradeneeded = () => {
      const db = request.result;
      if (!db.objectStoreNames.contains(STORE_NAME)) {
        db.createObjectStore(STORE_NAME, { keyPath: 'id'
});
      }
    };
    request.onsuccess = () => resolve(request.result);
    request.onerror = () => reject(request.error);
  });
}

async function getQueuedEvents(matchId: string):
Promise<QueuedScoringEvent[]> {
  const db = await openIndexedDB();
  return new Promise((resolve, reject) => {
    const tx = db.transaction(STORE_NAME, 'readonly');
    const store = tx.objectStore(STORE_NAME);
    const request = store.getAll();

    request.onsuccess = () => {
      const allEvents: QueuedScoringEvent[] =
request.result || [];
      resolve(allEvents.filter((e) => e.matchId ===
matchId));
    };
    request.onerror = () => reject(request.error);
  });
}

async function clearQueuedEvents(matchId: string):
Promise<void> {
```

```
  const db = await openIndexedDB();
  return new Promise((resolve, reject) => {
    const tx = db.transaction(STORE_NAME, 'readwrite');
    const store = tx.objectStore(STORE_NAME);
    const request = store.openCursor();

    request.onsuccess = (e) => {
      const cursor = (e.target as
IDBRequest<IDBCursorWithValue>).result;
      if (cursor) {
        if (cursor.value.matchId === matchId) {
          cursor.delete();
        }
        cursor.continue();
      }
    };
    tx.oncomplete = () => resolve();
    tx.onerror = () => reject(tx.error);
  });
}

//
------------------------------------------------------------
-----------------
// 2. Core Synchronization Function
//
------------------------------------------------------------
-----------------

/**
 * Synchronizes court-side buffered events from IndexedDB
to Supabase
 */
export async function sync_offline_match_events(
  matchId: string,
  supabase: SupabaseClient
): Promise<{ success: boolean; syncedCount: number }> {
  // Step A: Retrieve pending scoring events from local
IndexedDB
  const queuedEvents = await getQueuedEvents(matchId);
  if (!queuedEvents.length) {
    return { success: true, syncedCount: 0 };
```

```
  }

  // Step B: Map queued events into database-compatible
payload
  const payload = queuedEvents.map((evt) => ({
    match_id: evt.matchId,
    team_id: evt.teamId,
    player_id: evt.playerId ?? null,
    points: evt.points,
    delta: evt.delta ?? evt.points,
    action_type: evt.actionType,
    created_at: evt.createdAt,
  }));

  // Step C: Perform atomic batch-insert to Supabase
  const { error } = await supabase
    .from('match_score_events')
    .insert(payload);

  if (error) {
    console.error('[OfflineSync] Atomic batch
synchronization failed:', error);
    return { success: false, syncedCount: 0 };
  }

  // Step D: Purge IndexedDB queue ONLY upon successful
server confirmation
  await clearQueuedEvents(matchId);

  return { success: true, syncedCount: payload.length };
}

//
----------------------------------------------------------
-----------------
// 3. Browser Reconnection Event Listener
//
----------------------------------------------------------
-----------------

/**
 * Attaches 'online' window listener to automatically
```

```
   trigger synchronization
    */
   export function registerOnlineSyncListener(
     matchId: string,
     supabase: SupabaseClient,
     onSyncSuccess?: (count: number) => void
   ): () => void {
     const handleOnline = async () => {
       const result = await
   sync_offline_match_events(matchId, supabase);
       if (result.success && result.syncedCount > 0 &&
   onSyncSuccess) {
         onSyncSuccess(result.syncedCount);
       }
     };

     window.addEventListener('online', handleOnline);

     // Return cleanup function to deregister listener upon
   component unmount
     return () => {
       window.removeEventListener('online', handleOnline);
     };
   }
```

I.6 Certificate of Registration (COR) Confidence Scoring The following extraction logic is executed entirely within the client's browser using PDF.js. It parses text from uploaded student verification documents and computes an advisory confidence score (0-100) by utilizing fuzzy string matching and regex patterns to identify student numbers, institutional email domains, and affiliated college names.

```
export interface RegisteredStudent {
  studentId: string;
  fullName: string;
  collegeName: string;
  email?: string;
}
```

```ts
export interface VerificationScoreResult {
  confidenceScore: number; // 0 - 100
  breakdown: {
    studentIdPoints: number;     // max 25
    nameSimilarityPoints: number; // max 25
    collegePoints: number;        // max 20
    emailDomainPoints: number;    // max 15
    headerMarkerPoints: number;   // max 15
  };
}

/**
 * Normalizes text strings by removing diacritics, symbols, and irregular spacing
 */
function normalizeText(text: string): string {
  return text
    .normalize('NFD')
    .replace(/[\u0300-\u036f]/g, '')
    .replace(/[^a-zA-Z0-9\s]/g, ' ')
    .replace(/\s+/g, ' ')
    .trim()
    .toUpperCase();
}

/**
 * Computes token-based fuzzy similarity between 0.0 and 1.0
 */
function computeFuzzyNameSimilarity(registeredName: string, ocrText: string): number {
  const normReg = normalizeText(registeredName);
  const normOcr = normalizeText(ocrText);

  if (!normReg || !normOcr) return 0;
  if (normOcr.includes(normReg)) return 1.0;

  const tokens = normReg.split(' ').filter((token) => token.length > 1);
  if (tokens.length === 0) return 0;

  const matchCount = tokens.reduce((count, token) => {
    return normOcr.includes(token) ? count + 1 : count;
```

```
  }, 0);

  return matchCount / tokens.length;
}

/**
 * Calculates OCR Confidence Score (0-100) across 5
institutional verification pillars
 */
export function calculateOcrConfidenceScore(
  ocrText: string,
  student: RegisteredStudent
): VerificationScoreResult {
  let score = 0;

  // 1. Exact Student ID Match (+25 pts)
  const cleanId = student.studentId.replace(/[^0-9A-Za-z]/g,
'').toUpperCase();
  const cleanOcr = ocrText.replace(/[^0-9A-Za-z]/g,
'').toUpperCase();
  const hasExactStudentId = cleanId.length >= 4 &&
cleanOcr.includes(cleanId);
  const studentIdPoints = hasExactStudentId ? 25 : 0;
  score += studentIdPoints;

  // 2. Fuzzy Name Similarity (+25 pts based on match ratio)
  const similarityRatio =
computeFuzzyNameSimilarity(student.fullName, ocrText);
  let nameSimilarityPoints = 0;
  if (similarityRatio >= 0.85) {
    nameSimilarityPoints = 25;
  } else if (similarityRatio >= 0.40) {
    nameSimilarityPoints = Math.round(25 * similarityRatio);
  }
  score += nameSimilarityPoints;

  // 3. Affiliated College Department Match (+20 pts)
  const cleanCollege = normalizeText(student.collegeName)
    .replace(/COLLEGE OF|COLLEGE/g, '')
    .trim();
  const hasCollegeMatch =
    cleanCollege.length > 2 &&
```

```
normalizeText(ocrText).includes(cleanCollege);
  const collegePoints = hasCollegeMatch ? 20 : 0;
  score += collegePoints;

  // 4. University Corporate Email Domain Detection (+15 pts)
  const PSU_EMAIL_REGEX =
/[a-zA-Z0-9._%+-]+@psu\.palawan\.edu\.ph/i;
  const hasUniversityEmail = PSU_EMAIL_REGEX.test(ocrText);
  const emailDomainPoints = hasUniversityEmail ? 15 : 0;
  score += emailDomainPoints;

  // 5. Official University Header Markers (+15 pts)
  const HEADER_MARKERS_REGEX =

/Palawan\s*State\s*University|CERTIFICATE\s*OF\s*REGISTRATION
/i;
  const hasOfficialHeaders =
HEADER_MARKERS_REGEX.test(ocrText);
  const headerMarkerPoints = hasOfficialHeaders ? 15 : 0;
  score += headerMarkerPoints;

  // Clamped composite confidence score (0 to 100)
  const confidenceScore = Math.min(100, Math.max(0, score));

  return {
    confidenceScore,
    breakdown: {
      studentIdPoints,
      nameSimilarityPoints,
      collegePoints,
      emailDomainPoints,
      headerMarkerPoints,
    },
  };
}
```

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OBJECTIVE

A motivated fourth-year BSIT student seeking an internship or entry-level role in web development or systems design. Passionate about building practical tech solutions, with hands-on experience in full-stack development, IoT systems, and academic research. Eager to contribute to a dynamic team and grow professionally in a real-world environment.

EDUCATION

Bachelor of Science in Information Technology — *2023 – Present*
Palawan State University — Main Campus
4th Year

Senior High School — ICT Track — *2021 – 2023*
Sta. Monica High School
With Honors

ACADEMIC PROJECTS

PSU SportsTrack [Capstone] — *2024*
Web-based tournament management system for PSU's inter-campus sports events. Features real-time score updates, role-based access for admins, coaches, and players.
Stack: React · Supabase Realtime · Tailwind CSS

D.R.O.P. Remote Camera Trigger [IoT] — *2023*
ESP32-S3 based remote camera trigger with Firebase Realtime Database and Cloudinary integration, Discord webhook notifications, and canvas-based timestamp watermarking.
Stack: ESP32-S3 · Supabase · Cloudinary · C++

Puerto Princesa Transit GIS Map [GIS] — *2023*
Spatial analysis of multicab public transit routes in Puerto Princesa City using QGIS — buffer zones, rasterization, heatmap generation, and formatted map layout.
Stack: QGIS · Spatial Analysis · Rasterization

SPORTS PLUS — *2023*
Community Sports Platform [Mobile App] A community-driven app connecting players, teams, and organizers for local sports events. Supports team formation, event discovery, and schedule management across multiple sports disciplines.
Stack: *React Native · Firebase · Node.js*

ACHIEVEMENTS

1. Champion — UI/UX Design Competition | PSU · 2025
*Sports Plus: **A Web and Mobile Application for Sports Community Engagement***

2. 3rd Place — SUI Builders Palawan MVP Building | Puerto Princesa City · 2025

SKILLS

LANGUAGES
Python JavaScript HTML/CSS C++ SQL

FRAMEWORKS & TOOLS
React Django Supabase Firebase Git

OTHER
IoT / ESP32 QGIS 3D Printing Technical Writing

CERTIFICATIONS & TRAINING

Sui Builders Program – Palawan 2025
DICT MIMAROPA / Trabahong Digital / YGG Pilipinas

ICT Work Immersion 2023
NCC Mall PALAWAN

LANGUAGES

Filipino
Native

English
Proficient

ANGELICA BANDE

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OBJECTIVE

A motivated and detail-oriented fourth-year Bachelor of Science in Information Technology student with a genuine passion for creating technology that is practical, well-structured, and meaningful. Known for being thorough, organized, and committed to quality in every output — whether technical or written. Approaches challenges with a problem-solving mindset and takes initiative in seeing tasks through from start to finish. Works well in a team environment while also being capable of handling responsibilities independently.

EDUCATION

Bachelor of Science in Information Technology — *2023 – Present*
Palawan State University — Main Campus
4th Year

Senior High School — TVL — *2020 – 2022*
San Jose National High School
With Honors

ACADEMIC PROJECTS

PSU SportsTrack [Capstone] — *2026*
Web-based tournament management system for PSU's inter-campus sports events. Features real-time score updates, role-based access for admins, coaches, and players.
Stack: React · Supabase Realtime · Tailwind CSS

Fishermen Registration and Permit Management System — *2025*
Website that aims to provide fishermen with an easier way to process their registration and permit requirements through a digital platform.
UI/UX Proposed Study

ACHIEVEMENTS

Champion — UI/UX Design Competition | PSU · 2025
*Sports Plus: **A Web and Mobile Application for Sports Community Engagement***

SKILLS

LANGUAGES
Python · JavaScript · HTML/CSS · C++ · SQL

FRAMEWORKS & TOOLS
React · Django · Supabase · Firebase · Git

OTHER
QGIS · Technical Writing

LANGUAGES

Filipino
Native

English
Proficient

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OBJECTIVE

Aspiring Full-Stack Developer and Hardware Security Enthusiast with hands-on experience building deployable web platforms and embedded security research tools. Skilled in developing real-world systems using React, TypeScript, and Supabase, alongside low-level firmware development in C/C++ on ESP32. Eager to contribute to a professional environment where software engineering and hardware intersect. from institutional platforms to wireless security and smart infrastructure.

EDUCATION

Bachelor of Science in Information Technology — *2023 – Present*
Palawan State University — Main Campus
4th Year ·

Senior High School — GAS Track — *2020 – 2023*
Leonides S. Virata Memorial School
With Honors

ACADEMIC PROJECTS

PSU SportsTrack [Capstone] — *2026*
Full-stack tournament management platform built for PSU's inter-campus sports events. Features real-time score updates via Supabase Broadcast, role-based access for admins, coaches, and players, sport-driven scoring models with JSONB score data, and immutable tournament closure logic. Designed for both academic defense and actual institutional deployment.
Stack: React 18 · Vite · TypeScript · Tailwind CSS · shadcn/ui · Supabase · PostgreSQL · Vercel

D.R.O.P. Real-Time Smart Delivery Box [IoT] — *2026*
Developed a cloud-connected parcel box featuring automated dual-factor verification, anti-tamper tilt alarms, and an interactive web dashboard for remote management. Leveraged Supabase to maintain a single source of truth between physical hardware states and a simulated web view, while implementing an event-driven mobile camera system that instantly captures, stores, and relays surveillance photos to Discord upon security triggers.
Stack: ESP32-S3 · React · Supabase · Tailwind CSS · Discord Webhooks · C++

nRFBox VOID Version [Personal Project] — *2026*
ESP32-based wireless security research tool forked and built from the open-source nRFBox project. Supports 2.4GHz scanning, Wi-Fi deauthentication, BLE jamming and spoofing, Evil Twin captive portal attacks, and Beacon Spamming — used for studying wireless attack surfaces and social engineering vectors in controlled, educational contexts. Includes custom PCB design and pre-compiled firmware.
Stack: C · C++ · ESP32 · nRF24L01 · Arduino Framework · OLED Display

ACHIEVEMENTS

3. 2nd Place —SUI builders Palawan MVP building | PPC· 2025

SKILLS

LANGUAGES

Python JavaScript HTML/CSS C++ SQL

FRAMEWORKS & TOOLS

React Django Supabase Arduino IDE
Git

OTHER

IoT / ESP32 QGIS 3D Printing 3D design

CERTIFICATIONS & TRAINING

Sui Builders Program - Palawan 2025
YGG, DICT, Sui

LANGUAGES

Filipino
Native

English
Proficient

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OBJECTIVE

I'm a fourth-year BSIT student who builds practical technology solutions using Python, with a background spanning full-stack web development, IoT systems, and academic research. I'm actively looking for a web development or systems design internship where I can contribute to meaningful projects, sharpen my technical skills, and gain the kind of hands-on industry experience that bridges the gap between the classroom and the real world.

EDUCATION

Bachelor of Science in Information Technology — *2023 – Present*
Palawan State University — Main Campus
4th Year

Senior High School — ICT Track — *2021 – 2023*
Narra Integrated School
With Honors

ACADEMIC PROJECTS

PSU SportsTrack [Capstone] — *2026*
Web-based tournament management system for PSU's inter-campus sports events. Features real-time score updates, role-based access for admins, coaches, and players.
Stack: React · Supabase Realtime · Tailwind CSS

D.R.O.P. Remote Camera Trigger [IoT] — *2026*
ESP32-S3 based remote camera trigger with Firebase Realtime Database and Cloudinary integration, Discord webhook notifications, and canvas-based timestamp watermarking.
Stack: ESP32-S3 · Firebase · Cloudinary · C++

Mapping and Spatial Analysis of Tourist Attractions in Puerto Princesa City — *2026*
[GIS]
Spatial analysis of tourist spot accessibility in Puerto Princesa City using QGIS — distance classification, buffer zone generation, road network routing, and formatted multi-panel map layout.
Stack: QGIS · OpenStreetMap · QuickOSM · Network Analysis · Graduated Symbology · EPSG:32651

Synco — The Ultimate Musician's Companion [App-Dev] — *2026*
A full-stack musician's companion app for accessing chords, lyrics, and key transposition — with a built-in metronome, YouTube song import, and set-list management.
Stack: Django · HTML · CSS · YouTube API · Metronome API · Lyrics & Chords API

ACHIEVEMENTS

2nd Place — SUI Builders Palawan MVP Building | Puerto Princesa City · 2025

SKILLS

LANGUAGES

Python JavaScript HTML/CSS C++ SQL

FRAMEWORKS & TOOLS

React Django Supabase Firebase Git

OTHER

IoT / ESP32 QGIS 3D Printing Technical Writing

CERTIFICATIONS & TRAINING

Sui Builders Program – Palawan 2025
DICT MIMAROPA / Trabahong Digital / YGG Pilipinas

ICT Work Immersion 2022
Yabytes Internet Cafe

LANGUAGES

Filipino
Native

English
Proficient

CHARLES DARWIN DUERME

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OBJECTIVE

An adaptable 4th-year BSIT student seeking an internship or entry-level software development role. Equipped with hands-on experience building real-time web applications, designing user-centered interactive prototypes, and developing microcontroller-based IoT systems. Driven to translate complex problems into streamlined digital solutions and eager to add value to a collaborative tech team.

EDUCATION

Bachelor of Science in Information Technology — *2023 – Present*
Palawan State University — Main Campus
4th Year

Senior High School — GAS Track — *2021 – 2023*
Sacred Heart of Jesus High School

ACADEMIC PROJECTS

PSU SportsTrack [Capstone] — *2024*
Web-based tournament management system for PSU's inter-campus sports events. Features real-time score updates, role-based access for admins, coaches, and players.
Stack: React · Supabase Realtime · Tailwind CSS

PAWs N Patrol [UI/UX] — *2024*
Interactive prototype connecting the community, rescuers, and animal shelters to streamline stray animal management — featuring location-based case reporting, real-time rescue tracking, searchable adoption listings, and a dedicated veterinary dashboard for managing medical records and shelter check-in
Stack: *Figma* · UI/UX Design · User Research

Geospatial Analysis of Teenage Pregnancy Trends and Risk Factors in Puerto Princesa City [GIS] — *2025*
Spatial analysis mapping teenage pregnancy clusters across barangays in Puerto Princesa City using QGIS — linking high-risk areas to socioeconomic factors and proximity to schools and health centers via buffering, attribute joining, and symbology design to support city health planning
Stack: QGIS · Spatial Analysis · Rasterization · Data Mapping

Automated Irrigation and Plant Monitoring System for Department of Agriculture - Mimaropa [IOT] — *2025*
Arduino-based automated watering system for DA-MIMAROPA using soil moisture thresholds and time windows — real-time environmental monitoring, remote pump override, and automated water logging via a web dashboard.
Stack: Arduino (C++) · Python/Django · JavaScript · Hardware Sensors

ACHIEVEMENTS

SKILLS

LANGUAGES

Python JavaScript HTML/CSS C++ SQL

FRAMEWORKS & TOOLS

Django Supabase Git Tailwind CSS Arduino IDE

OTHER

IoT / ESP32 QGIS Technical Writing UI/UX Design

CERTIFICATIONS & TRAINING

LANGUAGES

Filipino
Native

English
Proficient